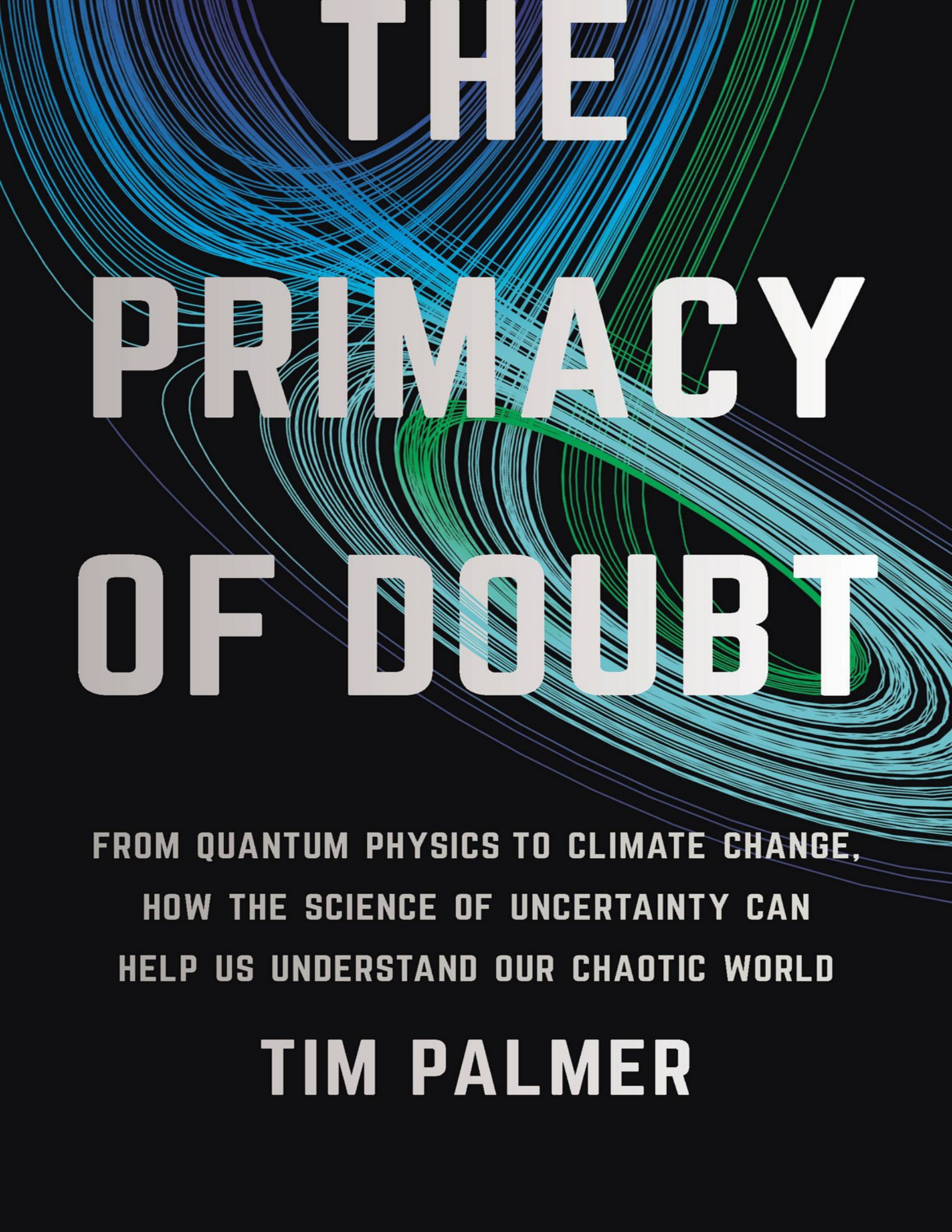




THE PRIMACY OF DOUBT

FROM QUANTUM PHYSICS TO CLIMATE CHANGE,
HOW THE SCIENCE OF UNCERTAINTY CAN
HELP US UNDERSTAND OUR CHAOTIC WORLD

TIM PALMER

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BASIC BOOKS
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Basic Books
Hachette Book Group
1290 Avenue of the Americas, New York, NY 10104
www.basicbooks.com

First Edition: October 2022

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[Figures 2 \(here\)](#), [15 \(here\)](#), and [21 \(here\)](#): vector of people by Olga Tik; background walls image from Freepik.com.

Library of Congress Cataloging-in-Publication Data

Names: Palmer, Tim N., 1952– author.

Title: The primacy of doubt : from quantum physics to climate change, how the science of uncertainty can help us understand our chaotic world / Tim Palmer.

Description: First edition. | New York : Basic Books, 2022. | Includes bibliographical references and index.

Identifiers: LCCN 2022006507 | ISBN 9781541619715 (hardcover) | ISBN 9781541619708 (ebook)

Subjects: LCSH: Weather forecasting. | Uncertainty. | Probabilities. | Quantum theory.

Classification: LCC QC995.4 .P35 2022 | DDC 003/.857—dc23/eng20220726

LC record available at <https://lccn.loc.gov/2022006507>

ISBNs: 9781541619715 (hardcover), 9781541619708 (ebook)

E3-20220901-JV-NF-ORI

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...

To Gill, Sam, Greg and Brendan, for their love,
support, patience and understanding.

In memory of Dennis Sciama, Raymond Hide, Ed
Lorenz and Bob May, who, in their own different
ways, taught and inspired me.



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Our freedom to doubt was born of a struggle against authority in the early days of science. It was a very deep and strong struggle. Permit us to question—to doubt, that's all—not to be sure.

RICHARD FEYNMAN¹

He believed in the primacy of doubt, not as a blemish on our ability to know, but as the essence of knowing.

FROM JAMES GLEICK'S BIOGRAPHY OF RICHARD FEYNMAN²

...

Author's Note

AS A READER, I SUPPOSE IT'S NOT UNUSUAL TO SKIP OVER THE preface to a book and get straight down to the business end of things. At least, that's what I tend to do.

If you do the same, it won't stop you understanding the key theme of this book: how the science of uncertainty can help us make sense of our very uncertain and unpredictable world. But you may wonder, since pretty much everything in life is uncertain, why I focus on the particular, seemingly disparate topics singled out in this book—climate change, economics, quantum physics and cosmology, human creativity and consciousness, to name a few? Why not earthquakes, drug design or cyberattacks? The answer lies in my scientific background.

I studied for a PhD in the 1970s, researching Einstein's theory of general relativity—Einstein had become a hero of mine by the time I was a teenager. I was lucky, the 1970s are considered to have been a golden age for general relativity theory. At the first scientific conference I ever attended, in 1974, I literally heard Stephen Hawking—he hadn't then lost his voice—announce his most important and famous result: that black holes aren't truly black, but radiate quantum particles.

My research supervisor¹, Dennis Sciama, Hawking's PhD supervisor some years earlier, was ecstatic about this result. Sciama was optimistic that we were on the verge of understanding how to unify Einstein's theory of general relativity with quantum mechanics, the theory that successfully describes atoms and elementary particles.

There were problems, however. One problem which frustrated Sciama was that Hawking's calculations were a little arcane. Sciama was convinced that there must be a more physically transparent way of deriving such an important result. He started to look at alternative ideas based on concepts from a field of physics called non-equilibrium thermodynamics. Sciama asked me to look at what is called the principle of maximum entropy production, something I hadn't heard of at the time.

Towards the end of my PhD research [2](#), I had become an expert on black holes and was successful in securing a postdoctoral position in Hawking's group at Cambridge. All seemed set for a career as a theoretical gravitational physicist.

But the closer I came to starting this new position, the less sure I became that this was what I wanted to do. For a start, the work I was doing had absolutely no relevance to the well-being of the person in the street. On top of that, I realised that there were actually only a handful of professional people in the world who were truly interested in the nitty-gritty details of my work.

Perhaps none of this would have mattered if I felt I was going to be the person to unify quantum mechanics and general relativity. This was a holy grail for physics in the 1970s, and indeed it still is today; it remains an unsolved problem. But the more I thought about it, the more I found myself concluding that at a conceptual level these two theories were quite incompatible and weren't going to be unified—at least not by me.

The prevailing view at the time was to ignore these conceptual problems and to press ahead with the mathematics hoping something miraculous would work out in the end (dubbed "Shut up and calculate" by the American physicist David Mermin). However, not everyone felt like this. I began thinking back to that first conference I

attended in 1974. The conference was opened by the renowned British theoretical physicist Chris Isham. In his introductory talk³, Isham remarked,

“It has become unfashionable these days for much notice to be taken of these conceptual problems, most people preferring to work on the more ‘respectable’ technical difficulties. However, in quantum gravity, the conceptual and technical problems frequently go hand in hand and it is possible that by neglecting the former one is rendering irrelevant the latter.”

These words began to resonate. I started to worry about embarking on a career working on things that would ultimately turn out to be irrelevant.

By chance, I met a world-famous meteorologist, Raymond Hide. Hide had interests in both astronomy and meteorology; he was the only person to be president of both the Royal Astronomical Society and the Royal Meteorological Society. I asked him what was new in climate physics. He proceeded to describe a paper by an Australian colleague⁴ showing how properties of Earth’s climate can be derived from the principle of maximum entropy production. It was as if a bolt of lightning had just hit me. Here was some obscure concept in physics whose existence I had only recently become aware of which could seemingly unify everything from the quantum evaporation of black holes to the structure of Earth’s climate system!

At the time, the United Kingdom’s Met Office was looking for scientists. Almost on a whim, I decided to apply, not sure if this was what I really wanted to do. At the interview stage I blathered incoherently about the critical importance of the principle of maximum entropy production for understanding climate. To my great surprise I was offered a job.

But should I turn down an offer to work with one of the world’s top theoretical physicists, and instead move into

the more prosaic world of the scientific civil service, working on a field about which I knew next to nothing? My parents would be disappointed with me if I turned down the job offer from Hawking. I didn't know what to do and walked round in a complete state of indecision for day after day. I decided to shelve the problem for a few days and get back to writing my thesis. And then a week or so later, I walked into my office and it suddenly became crystal clear to me what to do. Before I started to have second thoughts, I wrote to the head of the Department of Applied Mathematics and Theoretical Physics at the University of Cambridge, declining the offer.

What on earth was going on in my brain that suddenly I knew what to do, when earlier I was like Buridan's donkey, unable to move towards either the pile of hay or the pail of water? I became deeply interested in the process by which the brain makes decisions.

For ten years, I didn't think about fundamental physics. I immersed myself in weather and climate science, and was able, with colleagues, to make some exciting breakthroughs. Michael McIntyre and I discovered the world's largest breaking waves in Earth's stratosphere—a process fundamental to understanding why the ozone hole was discovered unexpectedly over the Antarctic. Chris Folland, David Parker and I showed that the decade-long drought in the sub-Saharan Sahel—the drought which motivated the famous Live Aid concert—was caused by fluctuations in tropical Atlantic sea temperatures. Glenn Shutts and I developed a way of representing small-scale atmospheric gravity waves (not the gravitational waves of general relativity that I had previously been researching) in weather forecast models, and showed how this helped airlines improve their fuel efficiency. And James Murphy and I developed the world's first ensemble forecast system, about which I shall be writing in detail. It was during this

time that I met Ed Lorenz, a key protagonist in this book, and became deeply interested in chaos theory.

In fact, in the 1980s, “chaos” was becoming a buzzword around the world thanks to James Gleick’s masterful book *Chaos: Making a New Science*⁵. Working with chaos pioneer Bob May, I began to meet scientists from other fields where chaos theory was relevant, and started thinking about whether ensemble prediction methods could be applied in these other areas. This aroused my interest in ensemble prediction in fields like economics.

My career was developing well and I was happy and felt fulfilled in what I was doing. I had pretty much forgotten about my earlier life in fundamental physics.

But then, sometime in the late 1980s, I was in one of the main bookshops in Oxford. It was 1987, the three hundredth anniversary of Newton’s famous *Principia Mathematica* and a new book celebrating the anniversary, edited by Stephen Hawking, had just appeared. I flicked through the pages. It brought my old world back to me in an instant. The distinguished physicist Roger Penrose had contributed an article in which he mentioned the key experiment—the Bell experiment—which has led physicists to reject Einstein’s views about quantum reality.

And then, some weeks later, completely out of the blue, I had an idea based on chaos theory—a possible way to explain how this key experimental result might after all be consistent with Einstein’s views. I wrote a paper on this⁶ and then tried to forget about it and get back to my post-PhD fields of research. I found I couldn’t. I began to devour papers on the foundations of quantum physics, made easier with the rapidly developing internet, and tried to develop my ideas on quantum physics further. I was being sucked back into the type of physics I thought I had left behind, in the evenings at least.

For this type of work, you simply need pencil and paper. However, to solve the equations that were part of my day job in weather and climate physics, I needed some of the largest supercomputers available. But the more I got to use these computers, the more frustrated I became by how much of a limitation they were on what could be studied. For all their impressive capabilities, being able to perform millions of billions of calculations per second for example, computers nevertheless put severe limitations on the amount of detail that can be coded into a global weather or climate model and on the number of ensemble members that could be run at any time.

Ultimately, the processing power of a supercomputer is limited by the electrical power used to run it. It takes a lot of electrical power—some tens of megawatts—to run a modern supercomputer. I became interested in the possibility that you could reduce the amount of energy a computer uses by turning the voltage down across the transistors⁷. This might allow you to add more chips to your supercomputer for the same amount of input energy, and that would enable you to increase the amount of detail in your model, or the number of ensemble members you could run. But reducing the voltage would make the chips unreliable—the calculations would be corrupted by thermal noise from the uncontrollable jiggling of atoms inside these chips and would no longer be precisely correct. With such noise, computers would no longer be infallible. However, as I will discuss, in chaotic systems noise can be a constructive resource, amplifying signals rather than obscuring them. I proposed ⁸ that we should be developing low-energy, noisy supercomputers for problems like weather and climate prediction. This is starting to happen.

To try to argue my point to colleagues, I began to wonder if there are examples of systems in nature that make use of low-energy noisy computation. I began to

realise the human brain, which needs only 20W to fire its 80 billion neurons, may provide an example. And this got me thinking about what actually it is that makes us the creative species we are.

All these topics have in one way or another fascinated me and have become interwoven in my research career over the years. This is the reason why they appear together in this book.



Acknowledgements

THE DEVELOPMENT OF ENSEMBLE WEATHER AND CLIMATE PREDICTION systems—a concept that lies at the heart of this book—involved the hard work of many dedicated and inspiring colleagues, first at the Met Office and then at the European Centre for Medium-Range Weather Forecasts. I want to start by acknowledging these colleagues. My deep thanks and gratitude to Dave Anderson, Jan Barkmeijer, Roberto Buizza, Philippe Chapelet, Francisco Doblas-Reyes, Dennis Hartmann, Renate Hagedorn, Martin Leutbecher, Jean-Francois Mahfouf, Doug Mansfield, Martin Miller, Franco Molteni, Robert Mureau, James Murphy, Thomas Petroligis, Kamal Puri, David Richardson, Glenn Shutts, Tim Stockdale, Stefano Tibaldi, Joe Tribbia and Antje Weisheimer. In addition, I want to thank Jagadish Shukla, now at George Mason University, who made me realise the potential power of physics-based models for exploiting the predictability of the atmosphere. His work kickstarted my passion for ensemble forecasting. I also want to thank Peter Webster, from the Georgia Institute of Technology, who played such an important role showing how probabilistic ensemble forecasts could be used for disaster preparedness in the developing world. His work is the forerunner of the anticipatory action programmes which are now changing the way disaster relief is enacted by humanitarian organisations.

In my career I have learned from many great scientists. From my undergraduate days at Bristol University, I thank my tutors Michael Berry, Brian Pollard and Ruth Williams.

From my graduate days at Oxford University, I thank my supervisor, Dennis Sciama, and my teacher and my internal doctoral examiner, Roger Penrose. Dennis was the most inspiring speaker I have ever heard. It was impossible to come away from a Sciama lecture without somehow feeling good about the world. And of course Roger, now a Nobel laureate, was something else. When I participated in small discussion meetings in Roger's office, I felt that the Copernican principle was being violated; I was manifestly in a preferred frame of reference at the very centre of the universe. Importantly, Dennis and Roger together taught me not to be afraid to think outside of the box, to think in terms of geometric pictures and simple models, and not to be intimidated by complicated-looking mathematics—things which I believe have served me well throughout my career.

Raymond Hide taught me all I needed to know about geophysical fluid dynamics to make a start in my new career, and I am so grateful to him for this. I was soon lucky to be working with three of the most outstanding scientists in meteorology, Jim Holton, from the University of Washington; Michael McIntyre, from the University of Cambridge; and then Ed Lorenz, from the Massachusetts Institute of Technology. Ed was shy and difficult to engage in conversation, but eloquent on stage when giving a lecture. He reminded me of Dennis's own supervisor, Paul Dirac. Like Penrose, Lorenz was a geometer at heart—by forming simple pictures he finally discovered the geometry of chaos. He gave me great support in developing the ensemble forecast system when others around me were expressing scepticism. Only later did I realise how lucky I was to have interacted with him.

Like Dennis Sciama and Raymond Hide before him, Bob May showed boundless enthusiasm, and from him I learned not only ideas from chaos theory, but also how to communicate these ideas effectively, of which he was a master. Through him I met economists like Mervyn King,

governor of the Bank of England, something I would have hardly imagined possible when I started my career. I was elected to the Royal Society in 2003, when Bob was its president. A highlight of my career was sitting next to him and his wife at the special banquet for the new inductees. I still get goosebumps thinking about this. I am grateful to Bob for all he did for me.

I need to thank the Royal Society in another way. I have been able to pursue the interdisciplinary topics described in this book through an award of a Royal Society Research Professorship (in 2010, the Royal Society's 350th anniversary year). This, I think, is the closest one can come to being given money and being told to get on and do whatever you like! I suppose a position like this could be abused. However, in my case, it allowed me to do research on many of the diverse topics discussed in this book. As such, I am profoundly grateful to the Royal Society for this position. It occurs to me that philanthropic money should be used to fund many more such very light-touch, open-ended research positions. It would be a good way for billionaires to give something back to society and I suspect the bang per buck would be enormous.

I also want to acknowledge the kindness of two of the world's leading philosophers of physics, Harvey Brown and Jeremy Butterfield, in carefully explaining to me the finer points of Bell's inequality in quantum mechanics—discussed in detail in this book—and in welcoming me into their fold as one of their own. The Philosophy of Physics seminars at Oxford University were one of the main ways I managed to get up to speed on the foundations of physics, whilst continuing my day job in meteorology.

And my thanks to Krishna Palem, now at Rice University, for teaching me about low-energy noisy transistors. He, more than anyone, has pioneered the concept of imprecise computers.

I have been lucky that a number of colleagues have given me extremely helpful comments on early drafts of this book, correcting things where they needed correcting. In particular I would like to thank Harvey Brown, Martha Buckley, Mark Cane, Peter Coveney, Clara Deser, Josh Dorrington, Simant Dube, Kerry Emanuel, Doyne Farmer, Charles Godfray, Weisi Guo, Dennis Hartmann, Sabine Hossenfelder, Saleh Kouhen, John Krebs, Beatriz Monge-Sanz, Sebastian Poledna, Juan Sabuco, Chris Shaw, Jagadish Shukla, Nick Stern, Bjorn Stevens, Nick Trefethen, Joe Tribbia and Peter Webster. I thank my editors, Latha Menon (Oxford University Press) and Eric Henney, Emily Andrukaitis and Thomas Kelleher (Basic Books), for extensive comments which improved the book enormously. As ever, remaining errors are my own.

My thanks to my son Brendan Palmer for drawing and preparing many of the figures throughout this book; to my students Josh Dorrington and Milan Klouwer for many of the diagrams showing chaos and turbulence in [Chapters 1](#) to [3](#); to my weather forecast colleagues Martin Leutbecher, Fernando Prates and Cihan Sahin for the ensemble weather forecast figures shown in [Chapter 5](#); and to Paul Lowe and Alex Rattan at the Bank of England for providing the fan chart figure in [Chapter 8](#).

Finally, I need to thank my family for their love, and for their patience and understanding when I would disappear for hours and hours (indeed months and months) in my study: to my wife, Gill; my sons, Sam, Greg and Brendan, who are now all doing so well in their own careers; and to my brother, John, and sister, Rosalyn, for their boundless encouragement and enthusiasm for this project.

Introduction

UNCERTAINTY IS AN ESSENTIAL PART OF THE HUMAN CONDITION. We don't know if we will be run over by a bus in the coming week or win some stupendous prize in the national lottery, should we have bought a ticket. Looking farther into the future, we can't be sure if our investments will be rendered worthless by the next global financial crash, or whether we will succumb to a new pandemic, to the next world war, or indeed to climate change. Perhaps we might wish for supernatural powers to predict the future and relieve us from the stress of living under such uncertainties. But what sort of people would we be if we had these powers? If we knew for certain what was heading our way, would we still be the creative, get-up-and-go species we are?

Apparently, it's not just our lives that are uncertain. According to our most successful theory of physics—quantum mechanics—uncertainty is also an essential part of the life of the elementary particles from which both we and the world around us are made. How would these elementary particles behave if we could excise uncertainty from the basic laws of physics? Would they also end up as indolent good-for-nothings? Would the universe be a very different place if the laws of physics were themselves certain?

These are some of the questions I will try to answer in this book. I will be elevating the notion of uncertainty, or doubt, to a status it is not usually given: not as an “Oh, I suppose we'd better do a risk analysis” afterthought, but as

a matter of primal importance and focus. There are two reasons for doing this. First, there is the practical reason that we are liable to make lousy decisions if we base them on predictions with unreliable estimates of uncertainty. But just as important, at least for me as a scientist, we may be able to understand better the way systems work by focusing on the ways in which they are or can become uncertain. These reasons form the two themes of my book: the science of uncertainty to predict our uncertain world and to understand our uncertain world.

A well-known saying, variously attributed to Danish quantum physicist Niels Bohr and baseball philosopher Yogi Berra, is that prediction is very difficult, especially if it's about the future. However, even tougher than making a prediction is reliably estimating the uncertainty in that prediction. You never hear an astrologer put error bars on their forecast that you'll meet a tall dark stranger. The reason predicting uncertainty is so tough is that the underlying causes of uncertainty can be small and seemingly insignificant. Recall the famous proverb:

For want of a nail the shoe was lost.
For want of a shoe the horse was lost.
For want of a horse the rider was lost.
For want of a rider the message was lost.
For want of a message the battle was lost.
For want of a battle the kingdom was lost.
And all for the want of a horseshoe nail.

But this is only the half of it. Most of the time, kingdoms weren't lost for want of a single horseshoe nail. It's only occasionally that they were. Most of the time, kingdoms ticked along in a reasonably steady and predictable way, even though horseshoes frequently lost nails. The real challenge is to know when the kingdom is at risk of a

missing nail that results in huge repercussions, and when a missing nail is irrelevant and doesn't matter.

Many will have heard of the butterfly effect: the idea that a butterfly flapping or not flapping its wings in the jungle can make the difference between a storm forming or not forming over a remote part of the world. Whether or not this is literally true, our weather forecast models cannot represent all the tiny gusts of air on every part of the planet. Like the lost horseshoe nail, tiny errors in representing small-scale gusts of wind can grow very rapidly to make huge errors in large-scale weather patterns. But most of the time they don't. Most of the time it doesn't matter that we don't know about these tiny gusts of air when we are forecasting the weather for tomorrow or the next day.

But sometimes it does matter.

Michael Fish was a respected weather forecaster on BBC TV. Although his forecasts weren't spot on all the time, by and large they were reliable and the British public trusted what he said—at least until October 16, 1987. Then, overnight, he became an eponym for a failed and discredited forecast system.

In his TV weather forecast on October 15, Fish famously said: "Earlier on today, apparently, a woman¹ rang the BBC and said she heard there was a hurricane on the way. Well, if you're watching, don't worry, there isn't!" Yet in the early hours of the sixteenth, hurricane-strength winds battered the south of England in what was the worst storm to hit the country in over three hundred years. Fifteen million trees were blown down, twenty-two people died and damages exceeded \$3 billion. Of course, if people had known that there was indeed a hurricane on the way, they would have moved cars from under trees and secured boats, cranes and aircraft. They would have cancelled or postponed

inessential trips. But they trusted Fish's assessment that there would be no hurricane.

There was little else on the morning news. The BBC anchorman harangued the weather forecaster on duty:

"Well, you chaps were a fat lot of good last night.... If you can't forecast the worst storms for several centuries three hours before they happen, what are you doing?!"

The nation was incensed and there were calls for Sir John Houghton, the director general of the UK's national weather service—the Met Office—to resign. For all their supercomputers and satellite data, the government agency which provided Michael Fish with the weather forecast had failed woefully in its most fundamental duty: to warn the public of life-threatening weather. Why was this? As we will see, it was the intermittent butterfly effect.

Economists had their Michael Fish moment in 2008 when the global financial markets collapsed without forewarning. So said the Bank of England's chief economist Andy Haldane² in a speech to the UK Institute for Government in 2017. According to Haldane the problem with economic modelling was much worse than with weather modelling: the economic models were "simply unable to function properly when the world was tipped upside down". Why should this be? Is the economy fundamentally more unpredictable than weather and climate, as some leading economists believe ³? Or was the 2008 crash an example of when, like the 1987 storm, the conditions were right for small uncertainties in the economic system to cascade up and create complete loss of predictability at the macroeconomic scale? Does the intermittent butterfly effect apply to the economy as well?

In Part I of this book, I will explain why the butterfly effect is profoundly intermittent—and hence why we can be lulled into a false sense of security about the predictability of systems like the weather and perhaps the economy too.

Indeed, I will discuss how even planets, whose motion is the very epitome of clockwork predictability, might have their own Michael Fish moments. Underpinning this is something I call “the geometry of chaos”: a type of fractal geometry discovered by meteorologist Ed Lorenz, which I rate as important as Einstein’s theories of relativity and Schrödinger and Heisenberg’s theory of quantum mechanics. “The Geometry of Chaos” vied with “The Primacy of Doubt” as the title of this book.

The 1987 storm made it clear that we need to know ahead of time when the butterfly effect is going to be particularly virulent and when it isn’t. In Part II of this book, I discuss practical methods for dealing with uncertainty in a number of different areas where having reliable predictions is critically important for our own personal well-being and that of society more generally. For example, weather services around the world no longer run their model once when they make a weather forecast. Instead, they run it perhaps fifty times in what is called an “ensemble prediction”. The fifty different starting conditions for these forecasts are virtually identical. However, they differ by small “flaps of butterflies’ wings”—uncertain disturbances on the smallest scales resolvable by the models. Not only that, some randomness, or noise, is introduced into the computer-based weather equations to represent the fact that the computer models are also uncertain. When the atmosphere is especially unstable, as it was in October 1987, the members of the ensemble diverge from one another very rapidly. On other occasions, they may only diverge very slowly. By studying how quickly the small differences in the ensemble members grow during the forecast period, forecasters should not be caught out again by an explosively unpredictable situation such as occurred in October 1987.

A consequence of the development of ensemble forecast systems, one that many of us are familiar with on our

weather apps, is that the primary weather forecast variables are now expressed as probabilities, the probability of rain for example. This probability is determined by the fraction of the fifty ensemble members in which it is raining. You may curse these probabilistic forecasts for their imprecision—after all you just want to know if it's going to rain or not. However, probabilistic forecasts now provide an objective basis for disaster relief agencies to act pre-emptively ahead of a possible extreme weather event, rather than simply wait for the event to happen as they did in the past. This is called anticipatory action. And these probabilistic ensemble techniques are vital tools for trying to predict climate change on timescales of decades and longer.

In Part II I also discuss the development of ensemble prediction techniques for pandemic, economic and conflict prediction, areas of enormous societal importance. Indeed, COVID-19 has seen a rapid development of so-called multi-model ensemble methods for predicting hospitalisations and deaths from the pandemic. The Intergovernmental Panel on Climate Change uses a similar multi-model technique to estimate future levels of climate change. Our experience in using these types of ensembles for climate prediction can help assess their strengths and weaknesses for pandemic prediction.

But what about those elementary particles and the uncertain lives that they seem to lead? Einstein thought the idea that the laws of physics are uncertain was nonsense. He believed that the uncertainty in quantum physics was no different, in principle, from uncertainty in forecasting the weather. That is to say, it's not the elementary particles that are uncertain, it's our knowledge of the elementary particles that is uncertain. As I discuss in Part I, most contemporary physicists reject Einstein's view, largely because there are lab experiments which cast doubt on this view.

In Part III, I challenge this consensus view of quantum uncertainty using ideas based around the geometry of chaos. The implications for fundamental physics—currently in crisis given the ongoing failure to understand the nature of the dark universe and a seeming inability to unify Einstein’s theory of gravity with quantum mechanics—are potentially massive.

One of the key results discussed in Part I is that noise can be a constructive resource, and not the nuisance it is often considered to be, when representing complex systems. For example, when modelling a turbulent fluid, it makes sense to use noise to represent eddies and whirls which are too small-scale to be modelled explicitly. In Part III, I claim that, in a similar way, the brain makes use of noise in modelling the world around us and that this is an essential part of what makes us the creative species we are.

But creativity is just one of our defining characteristics. We also have a very visceral sense of having a free will and of being conscious of ourselves and the world around us. But what is free will and consciousness? In Part III, I will suggest that the geometry of chaos can also provide new insights into these age-old conundrums.

In navigating through this book, one can read Part III independently of Part II, and indeed one doesn’t need to absorb all of Part I to read the later chapters. Although I will be discussing mathematically sophisticated ideas throughout the book, I have tried hard to avoid using mathematics in the main text. I have drawn the Lorenz equations, the Navier-Stokes equation and the Schrödinger equation—the three key sets of equations in this book—simply as cartoons. Along with the works of the great artists, these are products of remarkable human creativity. I have written a few equations in the endnotes.

The title of this book is taken from Gleick’s biography of Richard Feynman, one of the greatest physicists of the

twentieth century. But what do we mean by “doubt”? If a friend asserts something and you reply, “Hmm, I have my doubts”, then you are expressing uncertainty about the assertion. Indeed, “to feel uncertain about something” is the first definition in the Oxford Learner’s Dictionary ⁴ for the word “doubt”. However, if you instead reply to your friend, “I really doubt that”, then you are expressing more than uncertainty, you are effectively saying the assertion is unlikely to be correct. In this book, I am using the word “doubt” in the former sense. I do not mean it to imply that something is unlikely to be true. Indeed, as I shall discuss, it is important to neither overestimate nor underestimate uncertainty. Both can, and indeed have, led to poor decisions.

In short, the scope of this book is quite unique. On the one hand we cover the loftiest questions that philosophy has ever addressed and attempt to answer them in novel ways. On the other we’ll describe practical techniques that have transformed the way we go about predicting how our world will evolve over the coming days, years and decades ahead. Some readers, I hope, will be exhilarated by the discussion of long-standing conceptual problems like free will, consciousness and the puzzling nature of quantum physics. Others may be thrilled to see the science of chaos applied for the benefit of society (and some of the poorest parts of society in particular). For yet others, reading this book may help them understand themselves better. Not least they may come to realise that some of our apparent shortcomings are not signs of irrationality or failure, but manifestations of our unique ability to cope with the enormous uncertainties of life. There is, I hope, something for everyone.

PART I

The Science of Uncertainty

The greater the doubt, the greater the awakening.

ATTRIBUTED TO ALBERT EINSTEIN

When I read those words, I get a prickling at the back
of my neck and my hair stands on end. He knew!

Thirty-four years ago, he knew!

IAN STEWART, DOES GOD PLAY DICE? DESCRIBING ED
LORENZ'S DISCOVERY OF THE FRACTAL GEOMETRY OF
CHAOS IN 1963

Three important ideas are discussed in Part I. The first is that a type of geometry—what I call the geometry of chaos—explains why some systems can be stable and predictable for much of the time, and yet their future behaviour becomes completely uncertain on occasion. The second idea applies to systems that are so complex, we will never be able to model them exactly. In such situations, adding noise to a model can be a good way to represent some of the missing complexity. In this way, noise is often a positive constructive resource, and not the nuisance we typically think it to be. I'll give a number of examples of the constructive value of noise. Finally, I discuss uncertainty in quantum mechanics and explain why most physicists believe it is different to the uncertainty which characterises the other chaotic systems discussed in this part of the

book. The ideas developed in this part of the book are applied in Parts II and III.



1

Chaos, Chaos Everywhere

MANY OF US WOULD DESCRIBE OUR LIVES AS CHAOTIC—DISORDERED and confused. Indeed, the science of chaos describes systems whose behaviour is so unpredictable as to appear disordered and confused. Strange, then, that the science of chaos was initially developed by studying what most of us believe to be the epitome of order and predictability: the motion of the planets. We take it for granted that the sun will rise in the east, not only tomorrow but every day for the rest of our lives. And because we can predict with great precision the movement of the Earth about the sun, and the moon about the Earth, we know with confidence the time of high tide and the date of an eclipse, not only days ahead, but for the rest of our lives.

And yet, for all that, the future motion of the planets in the solar system is profoundly unpredictable and hence uncertain.

The story starts with the scientific renaissance, the series of events which marks the emergence of modern science. The renaissance was kickstarted by Nicolaus Copernicus's *De Revolutionibus Orbium Coelestium* published in 1543, and completed with the publication of Isaac Newton's *Principia Mathematica* in 1687.

In his book, Newton uses his famous three laws of motion, together with his law of gravity, to deduce the

formula found empirically by German astronomer Johannes Kepler—that the motion of a planet around the sun is an ellipse with the sun at one focus ¹. To make this prediction, Newton ignored the complications of the real solar system and assumed it comprised just two gravitating bodies: the sun and the planet in question.

There is nothing random or indeterminate about Newton's laws. If you know a planet's position and velocity right now, and you know the forces acting on the planet in the future, you can calculate the position and velocity of the planet at all times in the future. Those of us who studied science in high school may recall the rather tedious homework problems in which one had to calculate the precise distance a projectile would travel if thrown with a given initial speed and angle of elevation. The calculation provides an example of how the future is seemingly determined by applying Newton's laws to a given set of initial conditions. Because of this, Newton's laws are said to be deterministic. In 1814, the French philosopher and mathematician Simon Laplace wrote about a hypothetical daemon² who could exploit Newtonian determinism to predict the future perfectly. Laplace said:

“An intellect which at a certain moment would know all forces that set nature in motion, and all positions of all items of which nature is composed, if this intellect were also vast enough to submit these data to analysis, it would embrace in a single formula the movements of the greatest bodies of the universe and those of the tiniest atom; for such an intellect nothing would be uncertain and the future just like the past would be present before its eyes.”

This reference to “a single formula” is relevant here, because Laplace was one of the many scientists from Newton onwards who tried hard to generalise the formula for an ellipse³ in order to describe the orbits of a hypothetical solar system with three or more

gravitationally bound bodies, e.g., the sun, the Earth and the moon—or the sun, the Earth, the moon and Jupiter. One would input the time into the formula, and the formula would output the positions of the planets at that time.

The problem became known as the n -body gravitational problem. Newton had solved the problem for two bodies (the sun and the Earth for example). Now the race was on to find the formula for $n = 3$ or more. Laplace and his contemporaries failed to find this formula. Finding the formula became a cause célèbre, and to celebrate his sixtieth birthday, King Oscar II of Sweden offered a prize to anyone who could find it.

The problem was finally solved in the late nineteenth century by the French physicist and mathematician Henri Poincaré, cousin of French president Raymond Poincaré. Poincaré's solution took the scientific community by surprise. Whilst all the previous giants of mathematics had assumed from the outset that such a formula exists, Poincaré showed that when $n = 3$ or more, no formula exists.

What does it mean that no formula exists? Using a laptop computer, which of course did not exist in Poincaré's time, we can solve Newton's laws to plot the orbits of the three bodies on the laptop screen, let's say over a million Earth years of simulated time. In principle we—or better, perhaps an artificial intelligence (AI) system—could find a single mathematical formula, more complicated than that of an ellipse, that describes these orbits very accurately.

However, if we extend the computer solution to two million Earth years, then we will find that our formula fails to describe the orbits over the second million years. Our AI system can perhaps find a yet more complicated formula which describes the orbits of the three bodies over two million years. However, this formula will again fail if the simulation is extended to three million years. Indeed, no matter how complex a formula we construct to describe the

motion of the three bodies over any given finite period of time, the formula will eventually fail if the motion is extended to longer periods of time. There is no single formula that can describe the orbits of these bodies over arbitrarily long periods of time. This is what Poincaré found.

A consequence of this is that the orbits of the three bodies never repeat; if they did repeat, then there would exist a formula for the orbits which would apply over arbitrarily long periods of time. We say that the orbits of the three bodies are non-periodic. Poincaré realised that this meant the motion of planets in the solar system is ultimately unpredictable—a property shared with the weather. By studying planetary motion, Poincaré had discovered a phenomenon we now call “chaos”. Because of chaos, there is no formula that Laplace’s daemon can use to see arbitrarily far into the future.

We can get some understanding of this planetary chaos by looking at snapshots from a computer animation for $n=4$ gravitating bodies ([Fig. 1](#))⁴. This example has been set up to illustrate explicitly and dramatically what Poincaré understood could happen from his mathematical analysis. For a limited period of time, the orbits of these four bodies resemble ellipses, and based on data from this limited period, an AI system would likely conclude that the motion would continue indefinitely on these near-elliptical orbits. However, very suddenly and with little warning, the planets decide to spiral off to infinity! The simple near-elliptical formula that describes the orbits of the four bodies in the early period fails completely in the later period.

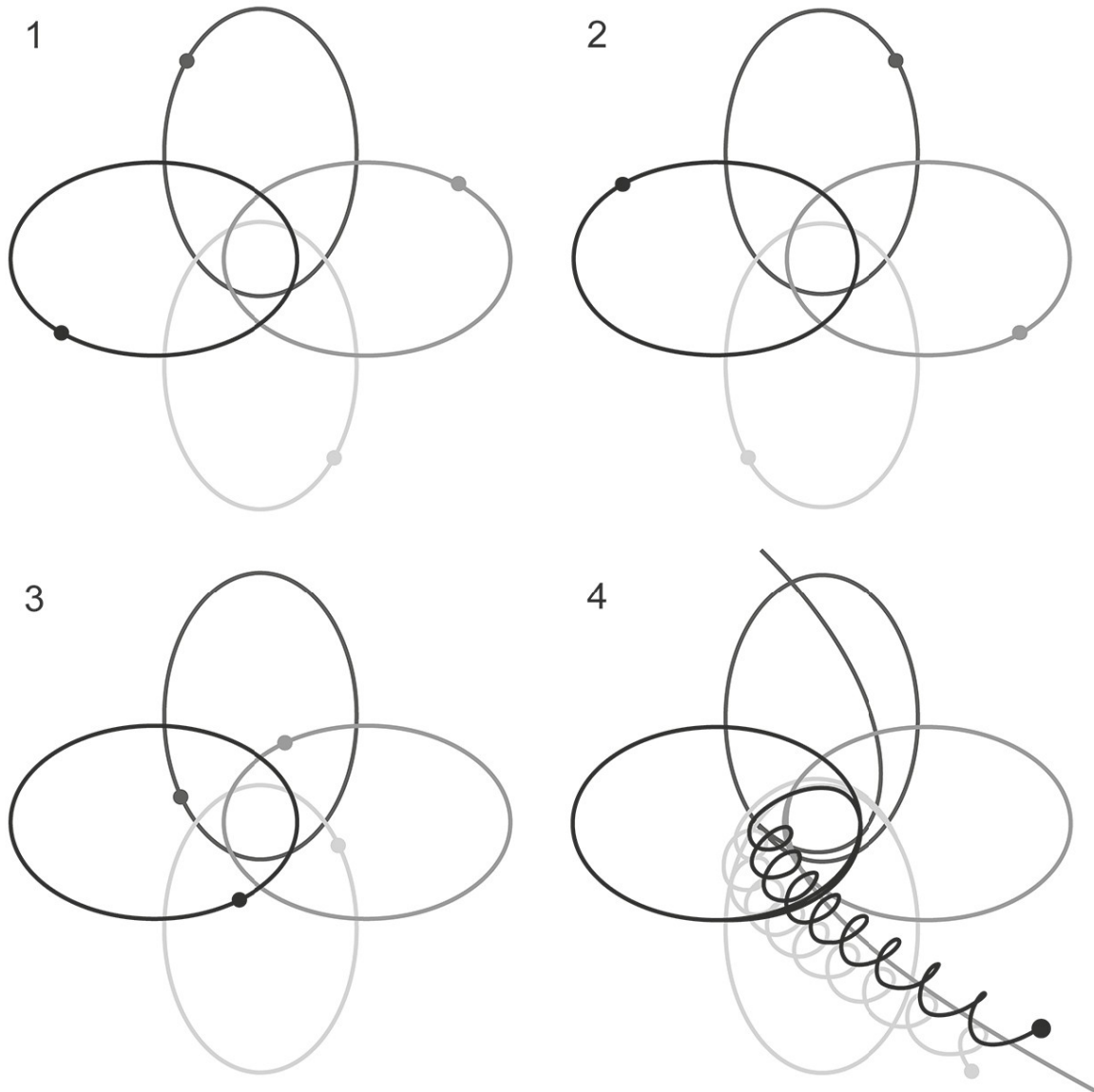


FIG. 1. Snapshots of the orbits of four gravitationally bound objects. For quite a long time, illustrated by snapshots 1-3, the bodies orbit around one another in paths that approximate well to ellipses. However, suddenly, seemingly out of nowhere, the objects spiral out to spatial infinity. The animation of this—see endnote 4—is well worth watching.

[Fig. 1](#) illustrates a feature of chaotic systems which occurs also for weather, and perhaps the economy and many other systems. Like the Michael Fish moment described in the introduction, the system appears

reasonably predictable until, out of the blue, it behaves unpredictably differently.

Could the Earth ever get ejected from the solar system in this way? That, were it to occur, would make global warming, financial crashes and everything else discussed in this book pale into insignificance. How would we find out? The answer is to make a computer model of the solar system and run it into the future.

But can our computer model be relied on to predict such an event? Perhaps, like Michael Fish's weather forecast, the model will tell us not to worry. Perhaps the model will predict that the simulated Earth will continue to orbit periodically around the sun—only for us to find the real Earth getting ejected from the solar system in five years' time.

The way to address this legitimate concern is to understand the key uncertainties in predicting the future orbits of the planets. In this case, the key uncertainty relates to the precise position of the planets. To deal with this, we can run our model hundreds of times with the planets having very slightly different starting positions, consistent with this uncertainty. This is an example of an ensemble forecast, a concept we will meet many times in this book.

We can be relieved. Researchers from Princeton University have run such an ensemble [5](#), and in none of the ensemble members is the Earth ejected from the solar system over the coming billions of years. So we can assign this potential crisis event a probability that is extremely close to (but, strictly, not identical to) zero. The only crisis event of note from these ensembles is that in about 1 percent of the ensemble members, the orbit of Mercury becomes sufficiently eccentric that it collides with Venus some billions of years in the future.

Here is our first example of the application of the ensemble prediction method. It allows us to conclude that it's extremely unlikely that the Earth will be ejected from the solar system. However, such ensemble forecasts indicate that the chance of the Earth being hit by a minor asteroid, one that could easily destroy a large city such as London, is not so completely negligible. This is why we monitor threats from such asteroids as best we can⁶.

« »

AFTER THE DEATH OF Henri Poincaré, a Harvard University mathematician, George Birkhoff, took over the mantle as the world's leading expert on the gravitational n-body problem. In the 1930s, Birkhoff took on a bright student, Ed Lorenz, who was setting out on his postgraduate studies, having studied mathematics at a nearby New England Ivy League college. The mathematical topic that Lorenz worked on under Birkhoff was not the gravitational n-body problem, but was in an area called Riemannian geometry. However, it would seem that some of Birkhoff's thinking rubbed off on the young Lorenz, as, in the early 1960s, Lorenz was to make one of the greatest discoveries of chaos theory, indeed of science in general I will argue. Lorenz discovered the fractal geometry of chaos.

Lorenz's plans to become a mathematician were disrupted by the Second World War, and as a result he had to decide how to best use his mathematical talents for the war effort. Since he had been rather fascinated by the weather as a child, Lorenz signed on for a wartime training course for meteorologists and eventually went out into the Pacific arena as an operational weather forecaster.

After the war, Lorenz had the option to continue his mathematics research at Harvard. However, his wartime

work had rekindled his interest in weather, and he switched fields and worked for a PhD at the Massachusetts Institute of Technology (universally known as MIT) on the new, physics-based weather forecast models that were starting to be developed in the immediate postwar years. Having completed his PhD at MIT and a postdoctoral appointment at the University of California, in 1956 Lorenz was offered a permanent job back at MIT. His duties included leading a group of researchers who were exploring the possibility of predicting weather a month or more into the future.

In those days, little was understood about the predictability of weather more than a day or two into the future. Some of the leading statisticians of the day told Lorenz that long-range weather forecasting was in principle an easy problem to crack. Say you want to forecast the weather a month ahead. Then you simply go to the weather archives and find a weather map that looks close enough to today's weather map—this would be called an analogue. The forecast for a month hence would simply be the weather map in the archive a month following the date of the analogue map.

The method didn't seem to work in practice. However, the statistical experts argued that this was simply because the archives were not large enough, and hence did not contain close-enough analogues. Wait for the archives to grow, they would say, and then you will find that the analogue method will start to work. Lorenz was sceptical about this. For analogue forecasting to work in this way, Lorenz argued, the weather would have to be periodic, repeating itself over and over, as in the movie *Groundhog Day*. Lorenz's intuition was that the weather equations would not support such periodic behaviour. Indeed, this is exactly what Poincaré had shown in the case of the 3-body gravitational problem.

But there are two important differences between the motion of the planets and the evolution of weather. Firstly,

unlike the solar system, with a relatively small number of planets and moons, the atmosphere is a turbulent fluid⁷ with literally trillions upon trillions of whirls or eddies with all sorts of different sizes. If you had asked most of the meteorological experts of the day, they, like Lorenz, would also have been sceptical that the atmosphere ever repeated itself. However, the experts would have said that such non-periodicity arose because the atmosphere is such a complex multiscale system, with numerous interacting whirls from the planetary scale down to the microscale. In other words, conventional wisdom had it that the complexity of the atmosphere was the origin of its unpredictability. The more the weather equations are simplified, the experts of the time would have conjectured, the more the simplified weather equations would produce periodic and hence predictable behaviour.

Lorenz had the idea of trying to use the computer-based weather forecast models he had helped develop as a PhD student to demonstrate his intuition that weather did not repeat itself. However, the very fact that these were complex models posed a practical problem. You might run a model for a hundred years and conclude that the simulated weather did not repeat itself. But perhaps years 101–200 were exact repeats⁸ of years 1–100. Who's to say?

In thinking about this, Lorenz began to wonder whether non-periodicity could be found in simplified systems which could be described with a very small number of equations. Where did he get this idea? Was it from practical experience, running simplified weather models on the computer? Or did he get the idea from the fact that the 3-body gravitational problem was not periodic? In his semi-biographical book⁹, *The Essence of Chaos* Lorenz writes that, despite working with Birkhoff, he was not familiar with Poincaré's work. And so we have to assume it was a

hypothesis he formed from his experience in running weather models.

In any case, Lorenz was right and the meteorological experts were wrong. Lorenz kept simplifying and simplifying until he devised a mathematical model of fluid motion based on just three equations for three variables, conventionally labelled X , Y and Z . This is actually a simpler set of equations than for the $n = 3$ gravitational problem, since each planet in the gravitational problem needs six variables to represent it: three spatial coordinate positions and three components of velocity.

Lorenz's model is so idealised it no longer makes much sense to relate these three variables to specific fluid quantities (though it is possible to make a chaotic waterwheel whose motion is described by the Lorenz equations¹⁰). So, let's just think of X , Y and Z as abstract variables whose values change with time. At a particular instant in time, these variables can be described by three numbers, e.g., $X = 3.327$, $Y = 5.674$, $Z = 0.485$. Lorenz's equations describe how these numbers evolve in time. The equations make use of the mathematical technique called calculus, discovered by Newton and independently by the German mathematician, philosopher and engineer Gottfried Leibniz. Using the calculus, the time rate of change of X , Y and Z can be described in terms of the current values of X , Y and Z and various fixed parameters (see [Fig. 2](#)). Lorenz's equations would have been quite understandable and recognisable to Newton or Leibniz as three so-called nonlinear differential equations.

The word "nonlinear" is of crucial importance here. Nonlinear systems are those whose outputs do not change in direct proportion to their inputs. Assuming you are not a wealthy person, you would be very happy indeed to win \$1 million in the lottery. You would be even happier to win \$2 million, but likely not twice as happy. You would very likely

not be ten times as happy to win \$10 million (substitute billions for millions if you are very wealthy). That's an example of nonlinearity—your happiness (output) does not vary in direct proportion to your lottery winnings (input). In the case of the Lorenz system, if you double the size of the X , Y and Z variables, the time rate of change of the X , Y and Z variables does not double. You, dear reader, are a nonlinear system, just as Lorenz's equations are. Lorenz's system would be predictable if its governing equations were linear.



FIG. 2. THE LORENZ EQUATIONS AS A WORK OF ART.

Solving these equations on a computer of the early 1960s was a challenging task. However, with just three variables, Lorenz could run the equations for long enough periods of time for him to finally realise that the model state was never going to repeat itself. How exactly he came to this realisation will be discussed in the next chapter—it

is one of the great discoveries of twentieth-century science. Published in the *Journal of the Atmospheric Sciences* in 1963, Lorenz's most famous paper announces the discovery of "Deterministic Non-periodic Flow"¹¹. As before, "non-periodic" means non-repeating. Because it was published in such a "niche" journal, it took another decade before the paper became known to the mathematical community at large, and this delayed the explosion of chaos science by a similar period of time.

However, Lorenz's model is not best known for this property of non-periodicity. Instead, it's known for a property which is a consequence of non-periodicity, though Lorenz himself did not fully understand this at first.

Lorenz used to tell the story about how he wanted to repeat a simulation on his computer in order to get the computer to output data more frequently. He took some values which the computer had previously outputted onto printer paper, and restarted the computer initialising the model with these printed values. After he had returned from a coffee break, Lorenz noticed that the repeated values of X , Y and Z bore no relation to the original values that the computer had earlier outputted. An example of what Lorenz saw is shown in [Fig. 3](#). At first, he thought that the computer had become faulty (valves, or "tubes", often failed in these early computers). However, he quickly realised that this wasn't the problem. Rather the problem was related to the fact that the printer did not record the variables at "machine precision" but instead printed only the leading digits of the variables. For example, if the precise value of a variable was $X = 0.506127$, the printer printed only $X = 0.506$. Starting the model with $X = 0.506$ instead of 0.506127 eventually led to big differences in the future values of X , Y and Z .

This "sensitive dependence on initial conditions" eventually became known as the butterfly effect. However,

in some ways this is a misnomer. Butterflies are very small creatures, certainly much smaller than weather systems. The Lorenz model, with only three variables, has no concept of spatial scale. It is too simple to describe weather, never mind butterflies. In particular, the Lorenz model cannot describe how a small-scale error can affect a large-scale weather system. All it can do is describe how a small-amplitude error can grow into a large-amplitude error. We will look at the butterfly effect in terms of small-scale errors growing into large-scale errors in [Chapter 3](#). And we will find something even more mind-boggling than the type of chaos shown here.

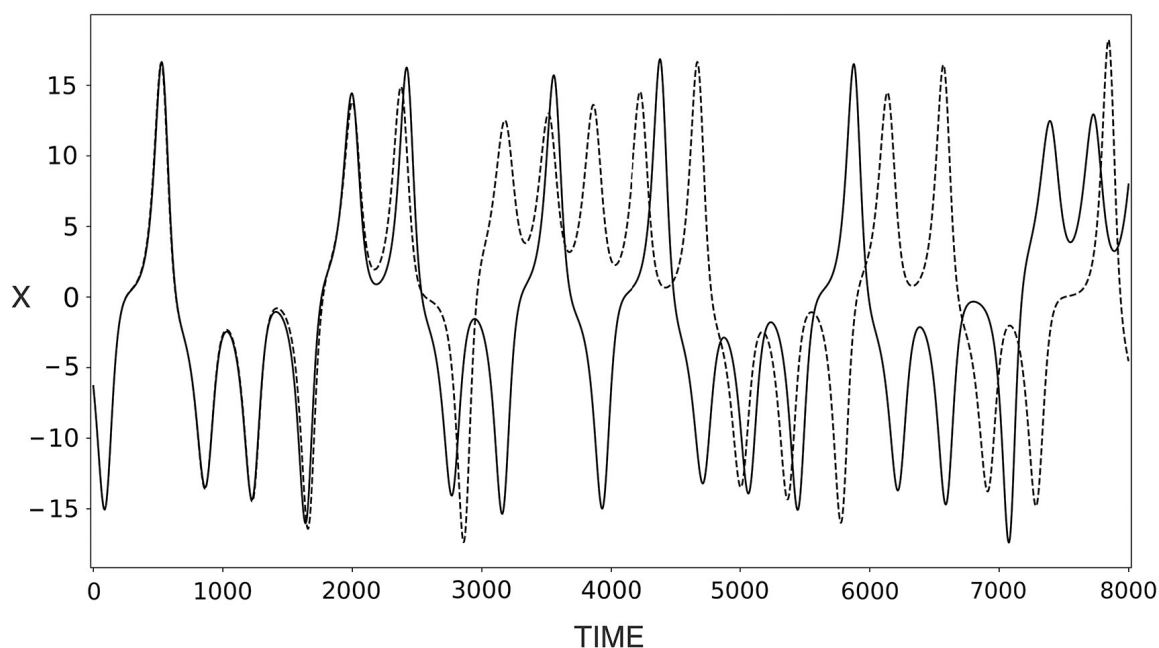


FIG. 3. This is the classic illustration of chaos in the Lorenz model. Two almost but not quite identical initial conditions eventually lead to very different evolutions. This is the way chaos is traditionally described. However, there is something deeper underpinning this behaviour.

In some ways, the illustration of chaos in [Fig. 3](#) seems not so different to that in [Fig. 1](#). In [Fig. 1](#), the position of the planets is unpredictable, whilst in [Fig. 3](#), the more abstract variables in Lorenz's equations are seen to be

unpredictable. We might well ask whether Lorenz's chaos is somehow described by the earlier theory of Poincaré.

The answer is no. There is one very important difference between Lorenz's chaos and Poincaré's chaos. Real-world fluids are viscous. If a fluid's viscosity is large, the fluid is sticky like treacle. Viscosity is a kind of internal friction in the fluid which makes it resistant to small-scale gradients in fluid velocity. In practice, viscosity converts such small-scale changes in fluid velocity into the random motion of molecules and hence into heat. This process is called energy dissipation. All realistic fluids, water or air for example, are viscous to a greater or lesser extent.

Imagine a tank of water at rest that is suddenly stirred vigorously with a paddle. Whilst the paddle is moving, it excites turbulent whirls that spread out from the paddle. However, as soon as one stops stirring, the whirls die out due to the effects of viscosity. The fluid returns to a state of rest but is very slightly warmer than it was, due to the effects of energy dissipation by viscosity. Now, if we videoed this, and ran the video backwards, we would see something completely unrealistic: small eddies growing in the fluid far from the paddle and converging onto the paddle. In this backwards-in-time video, it would seem the paddle's motion is being powered by a slight uniform cooling of the fluid, contradicting the second law of thermodynamics, which says that uniform low-grade heat can't be converted into work in such a way. Because the backwards-in-time motion is unrealistic, we say that the equations that govern the motion of viscous fluids are irreversible in time—the video and hence the fluid equations can realistically be run in only one direction in time.

By contrast the equations that govern the motion of planets are time reversible. If we had a video of the planets moving around the sun and reversed it, the motion would be consistent with the laws of physics. In some sense, it is

only chance that makes the planets rotate the way they do about the sun.

Lorenz felt that it was vital to include this time irreversibility into his equations. He was right. Time irreversibility is crucial for generating the geometry of chaos, a geometry that will form a central part of this book.

The irreversibility we see in fluid motions is much more commonplace than the reversibility in time that we see in planetary motion. For example, when we drop a glass, it smashes into hundreds of pieces. We never see the reverse. A baby grows into an adult. We never see the reverse. We can easily scramble eggs. We can't unscramble them into whites and yolks. We can think of many similar types of examples. What is the origin of such irreversibility? This has perplexed scientists for centuries because Newton's laws of motion, or indeed Schrödinger's equation of quantum mechanics, are themselves time reversible; the same equation describes a system in a movie, irrespective of whether the movie is run forwards or backwards. The way most scientists deal with this problem is to say that the universe must have begun at the time of the Big Bang in an exceptionally ordered state. That is to say, the irreversibility we see arises from the fact that the cosmological initial conditions were special. I will attempt a different explanation, using the geometry of chaos, in [Chapter 11](#).

Now, although the type of chaos studied by Poincaré and Lorenz made use of Newton's calculus, calculus is not itself an essential ingredient of chaos. Before finishing this chapter, I want to give a very simple example of chaos—popularised by physicist-ecologist Bob May and applied to population growth—which doesn't use calculus.

If Adam and Eve had four children, two boys and two girls, and in the next generation, each pair of adult males and females in turn produced four children, then the size of the population for each generation would double (2, 4, 8,

16...). Prior to May, it was assumed that the population would grow until it eventually either came into some kind of steady state consistent with the finite environmental resources available to the population or, at most, would oscillate periodically. As a result, when irregular fluctuations in some species' populations were observed, ecologists assumed this must be because the environmental resources were changing because of external factors—maybe because of irregular fluctuations in the climate, for example.

However, with a very simple equation, May¹² showed that this was not necessarily the case. May's equation¹³ determines the size of a population at the next generation, given the population at this generation. The size of the population at each generation can be computed using a simple calculator—the equation is that simple. In May's equation, there is a free parameter called a . It describes the number of offspring each single adult produces on average, at least in the initial phases of growth. In Adam and Eve's example, $a = 2$. However, once the population becomes large enough, its growth becomes limited by environmental resources. In such circumstances, the population can suddenly crash to small values, before starting to grow again. What May showed was that if a exceeds about 3.57, the evolution of the population over multiple generations is chaotic—the periods of boom and bust start to occur irregularly. This means two things. Like the Lorenz-model variables, the population fluctuates non-periodically without any external forcing in environmental resources. And like in the Lorenz model, it is very difficult to predict these fluctuations—they are sensitive to the starting conditions for the population size.

In this way, chaos theory has had a significant impact in the science of ecology. Indeed, over the years, chaos theory has impacted on almost all branches of science: not only

astronomy, meteorology and ecology, but chemistry, engineering, biology and social science; you name it! But perhaps one field where chaos theory hasn't had so much impact is the foundations of quantum physics¹⁴. The reason is simple: the basic equation which governs the evolution of quantum mechanical systems, the Schrödinger equation, is a linear equation. As such, the governing equation of quantum mechanics doesn't exhibit the type of sensitive dependence on initial conditions that we discussed earlier. It is sometimes argued that because we expect planetary motion, the weather and indeed population dynamics to somehow follow from quantum mechanics (not that anyone has actually shown this), chaos is somehow only an approximate notion and uncertainty in the weather is ultimately due to quantum uncertainty, not chaos. My view is 180° different. I believe chaos actually underpins the dynamical equations of quantum physics; chaos is a property of the planets and the weather, precisely because chaos is a property of the fundamental dynamics of the universe. I will discuss in the next chapter why such a view is in fact consistent with the linearity of the Schrödinger equation.



2

The Geometry of Chaos

ALTHOUGH ISAAC NEWTON WOULD HAVE UNDERSTOOD LORENZ'S equations, not least because these equations are expressed using the calculus which he himself had discovered, Newton would not have had the slightest inkling of the remarkable type of geometry that emerges from them.

For Newton, the word "geometry" was a shorthand for the geometry of Euclid, the Greek mathematician from Alexandria born around 300 BCE. As part of his training to be a natural philosopher, Newton would have studied Euclid's seminal work, *The Elements*. Here Euclid, for example, introduces the cross-sections of a cone. One of these, the ellipse, describes well the motion of a planet around the sun.

The ellipse characterises an important aspect of many of the geometric figures studied by Euclid. Draw an ellipse on a piece of paper and zoom into any portion of it with a magnifying glass. With a powerful enough glass, any small portion will become indistinguishable from a straight line. Similarly, zoom into the surface of an ellipsoid—an egg-shaped two-dimensional figure. Again, with a powerful enough magnifying glass, any small portion of the ellipsoid will be indistinguishable from a flat plane. In Einstein's theory of general relativity, space-time is a curved four-

dimensional space. The curvature of space-time manifests itself by the fact that the internal angles of a triangle do not sum to 180° . However, if you zoom sufficiently into any region of space-time, it looks flat—the internal angles of small-enough triangles sum as close as you like to 180° .

In all these examples, the special properties of the geometric object that make it an ellipse, or an ellipsoid or a solution of Einstein's equations of general relativity, somehow get washed out if you zoom in enough with your magnifying glass. Considered as spaces in their own right, these geometric objects are called "locally Euclidean". By contrast, the properties associated with the geometry produced by Lorenz's equations never get washed out under such a zoom, no matter which part of the geometry you zoom into. This is a concept which would have been alien to Newton and to the generations of mathematicians and scientists brought up studying Euclid's *Elements*. The idea that such a strange geometry could be the product of a simple set of equations based on calculus was also mind-boggling to Lorenz, and it took him a long time to realise what he had actually discovered.

In my view, Lorenz's really great discovery was not so much the butterfly effect—this was, after all, implicit in the work of Poincaré—as the discovery of the fractal geometry of chaos. As we shall see, the consequences of this geometry permeate everything in this book, from modern mathematics, to climate change, to quantum physics, to free will and to consciousness.

« »

IT IS RATHER CONVENIENT that there are only three variables, X , Y and Z , in Lorenz's model of chaos. It means we can represent a point in Lorenz's "state space" (relative

to three right-angled axes labelled X , Y and Z in the same way we would plot a point in ordinary three-dimensional physical space (relative to three right-angled axes conventionally labelled x , y and z in lower case).

The notion of state space is really important in this book. So I want to give a simple example of state space. Suppose you are buying a pair of trousers. The first thing to do is decide what leg length, what waist size and what colour of material to buy. Let's suppose the only choice of colour is a shade of grey. Then a particular pair of trousers can be represented as a point in an abstract space where one direction describes variations in leg length, a second describes variations in waist size and a third the shade of grey. We can call this three-dimensional space "trouser state space" (see [Fig. 4](#)). There is nothing special about the fact that trouser state space is three dimensional. Indeed, if we also had the possibility to choose the type of material used (e.g., wool, polyester or some mixture thereof), then we would need an extra, fourth dimension to additionally account for the choice of type of material, and trouser state space would then be four dimensional (and impossible to illustrate in a single figure in this physical book). In this way, the dimension of state space is equal to the number of independent variables needed to specify the object you are interested in (such as a pair of trousers).

Compared with the theory of physics known as string theory, which posits that the physical space in which we live is not three dimensional but ten or eleven dimensional, there is nothing weird, unusual or incomprehensible about a multidimensional state space. If you get confused about the difference between physical space and state space, just think trousers!

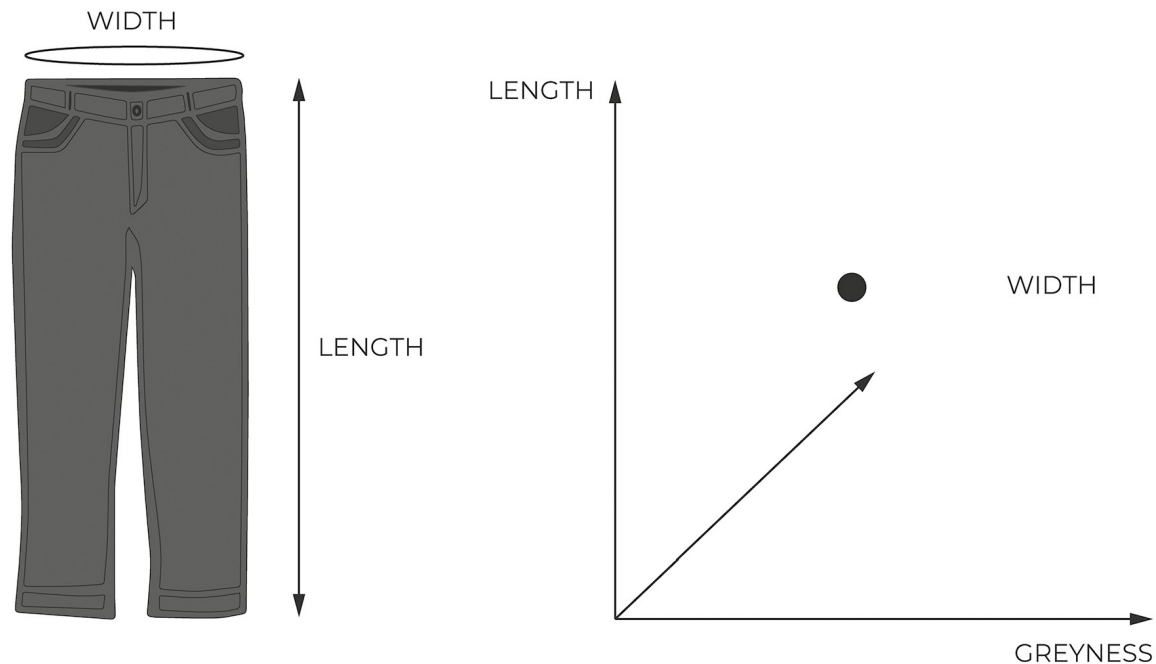


FIG. 4. You want to buy a pair of trousers. You have to know what leg length, waist size and shade of grey to choose. Your choice corresponds to a point in three-dimensional “trouser state space”. If you also had the option of a type of material, then this would correspond to a fourth dimension in trouser state space, making the state space impossible to illustrate but nevertheless easy to understand conceptually.

Let’s think of another example: the state space of 3 gravitationally bound bodies. Each body can be specified by a position and a velocity. In physical 3-dimensional space we need 3 numbers to specify a position and 3 more numbers to specify velocity (the speed of the body relative to the x , y and z axes). This means that the number of independent numbers needed to specify the position and velocity of the 3 bodies is 3 lots of $3 + 3$, which is 18. Hence, the dimension of the state space of the 3 bodies is 18¹. Although it is impossible to visualise an 18-dimensional space, the concept of an 18-dimensional state space is perhaps not especially difficult in principle.

The dimension of the state space of weather is enormous by comparison. If we consider all the whirls that exist in the atmosphere at any time, the state space of weather is

bigger than anything we could ever represent in a computer—not just today’s computers, but ones in the foreseeable future too. The state space of a modern weather prediction model has more than a billion dimensions. That’s a large number of course, and one that a modern supercomputer can handle, but it’s still a tiny fraction of the dimension of the actual state space of weather².

Try to imagine the state space of the whole universe, comprising all the atoms that make all the stars, galaxies and clusters of galaxies. The dimension of such a state space is overwhelmingly, unbelievably large. But as long as it is a finite number, it is in principle no more difficult to comprehend than 4-dimensional trouser state space.

Let’s return to Lorenz’s 3-dimensional state space. Choose a point in this state space. This can be considered as an initial state for Lorenz’s equations. Using a computer, we can solve Lorenz’s equations over a finite period of time. This solution describes a set of points in state space, one for each moment in time. We can represent the overall time evolution of the values of X , Y and Z as a curve in Lorenz’s 3-dimensional state space (see [Fig. 5](#)). We call this curve a state-space trajectory, or simply a “trajectory”.

If Lorenz’s model did have periodic (i.e., repeating) behaviour, and if the state-space trajectory were continued for a long-enough time, it would eventually join up with the starting point, represented by the dot. The states described by such a closed loop would then repeat over and over.

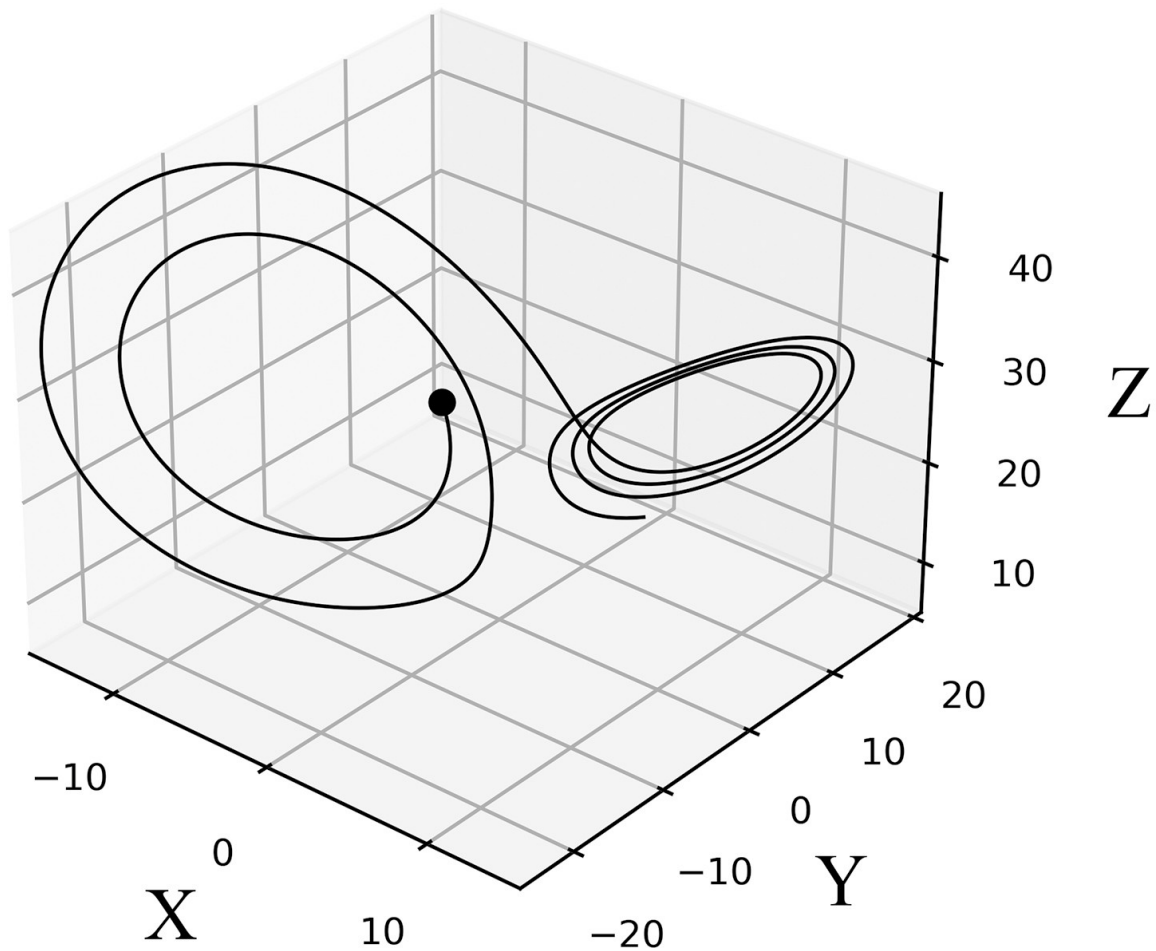


FIG. 5. A short trajectory segment in Lorenz's state space. The initial point is marked with a dot. This can be chosen arbitrarily. The trajectory gives the values of the X , Y and Z variables at future times and is obtained by solving the Lorenz equations on a computer.

[Fig. 6](#) shows a picture of what the trajectories of the Lorenz model look like when you solve the equations over a long period of time. It is now quite a famous diagram. To produce such a picture, it doesn't matter from what state you start. Eventually, the trajectories start following this strange shape with the two lobes. It is as if the trajectories are attracted to this geometric object in state space. As a result, this geometric object is called an "attractor". Curiously, it looks a bit like a butterfly, but that is a coincidence.

The question Lorenz asked himself was this: What is the underlying geometry of this attractor? Is it some kind of Euclidean geometry? One possibility is that, if the computer could continue running Lorenz's model forever, all the tiny gaps that appear in [Fig. 6](#) would be filled in, leaving a solid 3-dimensional object, one that we could perhaps carve from a piece of solid wood.

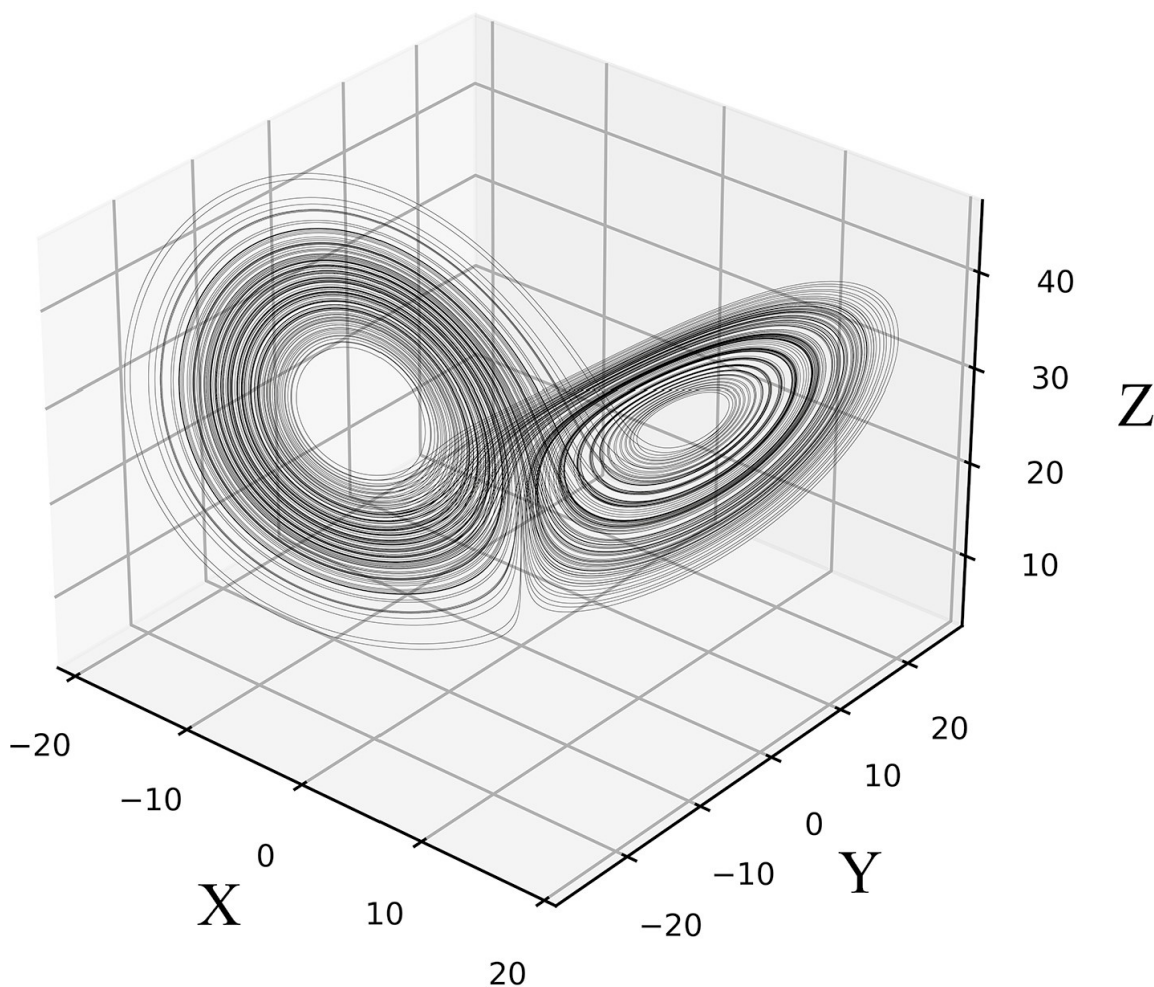


FIG. 6. An approximation of the famous Lorenz attractor as represented by a computer. No matter where you start in state space, the Lorenz equations will take you closer and closer to this attractor. One can say that the attractor is an emergent property of the Lorenz equations. The key question Lorenz asked himself is this: What is the geometry of this attractor? Is it a solid object or a complex folded surface? Lorenz showed it could be neither of these; it had to be a fractal.

However, Lorenz realised this could not be so. This is where the time irreversibility of Lorenz's model is crucial. As we saw in the last chapter, Lorenz included a very simple representation of the irreversible effects of energy dissipation in his model. Hence Lorenz's equations are time irreversible. And this makes these equations fundamentally different to those for planetary motion, studied by Poincaré. Under the effect of the irreversible energy dissipation, a solid 3-dimensional ball of initial conditions in state space would eventually shrink to an object with zero volume, under the action of the Lorenz equations. On the other hand, because of the chaotic unpredictability of the Lorenz equations, the different points of the initial 3-dimensional ball will eventually uniformly cover the whole Lorenz attractor. As such, the volume of the attractor has to be strictly zero.

Now, a 2-dimensional surface has zero volume. So could the attractor be 2-dimensional? For example, instead of carving it from a block of wood, perhaps we could construct the attractor by carefully folding a 2-dimensional sheet, like a piece of origami. Well, Lorenz could rule this out too, by a mathematical theorem proved by Poincaré himself. The Poincaré-Bendixson theorem says that if the attractor were 2-dimensional and had the property of sensitive dependence on initial conditions (which we showed in the last chapter was the case) and lay in some bounded region of state space (which the Lorenz attractor does), then somewhere the state-space trajectories would cross over themselves. However, if the equations are deterministic, then such a crossing point would be impossible: determinism implies that an initial point in state space can evolve in time only into another single point, not into two different points.

Well, perhaps the attractor is just some closed 1-dimensional curve where a trajectory eventually joins up with itself, like a convoluted circle? But if this were the

case, it would imply that the model is periodic, and the computer output from the model suggests otherwise³.

So, what is the geometry of the attractor if it is neither a 3-dimensional volume nor a 2-dimensional sheet nor a 1-dimensional line? Lorenz agonised about this for quite some time. I can easily understand his frustration. He knew he was onto something important, but he couldn't quite put his finger on what it was.

The solution came from the work of the nineteenth-century German mathematician Georg Cantor. Cantor was the founder of a type of mathematics known as set theory. At one level, sets are fairly trivial mathematical things. They simply denote collections of objects. The set $\{1, 2, 3, 4\}$ is the collection of whole numbers between 1 and 4 inclusive. The collection of countries in the world is a set whose elements are individual countries. However, things get mathematically interesting when considering sets which contain unlimited numbers of objects, such as the set of all positive whole numbers, which we can write as $\{1, 2, 3, \dots\}$. Now although this set is infinite, it is, as Cantor was able to show in a precise way, not nearly as big as the set of all numbers, which includes not only whole numbers and fractions like 2.4 and 6.97, but also numbers like $\sqrt{2}$, the square root of 2, and π , the ratio between the circumference of a circle and its diameter. Numbers like 2.4 and 6.97 can be written as fractions (24/10 and 697/100), but $\sqrt{2}$ and π cannot be written in this way and are called "irrational". Such irrational numbers cannot be represented on a computer with a finite number of bits. We call the set of all numbers, which includes such irrational numbers, "real numbers". It is the complete dominance of the irrational numbers over the rational numbers that makes the set of real numbers so much bigger than the set of whole or, indeed, rational numbers.

There is an intimate relationship between sets of real numbers and Euclidean geometry. If you draw a line, say of length 10 cm, then each point on the line can represent a real number, say between 0 and 10. The whole numbers—0, 1, 2,... 10—can be represented by points at a distance 0 cm, 1 cm... 10 cm from one end. The number π corresponds to a point roughly 3.14 cm from one end of the line.

Now, a line is a one-dimensional set. One might imagine that all sets which contain as many points as the set of real numbers are at least one dimensional. However, Cantor constructed a set, and in turn a geometry, in which this intuitive idea is false. Cantor was criticised by his colleagues for his ideas. The great Henri Poincaré said that in the future people would look on Cantor's work as "a disease from which one has recovered"⁴. Ironically, Poincaré had no idea how important Cantor's work would be to understanding the subject that he, Poincaré, had kick-started: chaos theory. Ultimately these criticisms sent Cantor into a depression from which he never fully recovered. Forty years after Cantor's death, Lorenz finally understood the importance of Cantor's ideas. Cantor is now rightly seen as a mathematical pioneer.

Cantor's set, shown in [Fig. 7](#), can be described by a set of steps. These steps are repeated applications, or iterations, of a simple rule. Start with the real-number line between 0 and 1. In the first application, throw away the middle third of this line (technically, one keeps the end points of the two remaining smaller lines). Then, in the second application, throw away the middle thirds of the two smaller lines. Then, in the third application, throw away the middle thirds of the four remaining lines. One can carry on like this indefinitely, removing at each step the middle thirds of the remaining lines. Cantor's set is the set of points that are common to all the iterates. What Cantor showed was that his eponymous set had as many points as

the points of the original line—even though we have thrown away an infinite set of points. In this sense, Cantor’s set is very large (much larger than the set of whole numbers). However, in a different sense, Cantor’s set is also rather small. One way of seeing this is to imagine randomly choosing a point from the original line (between 0 and 1). The probability that this randomly chosen point belongs to Cantor’s set is equal to zero. Mathematicians say that Cantor’s set has “measure zero”.

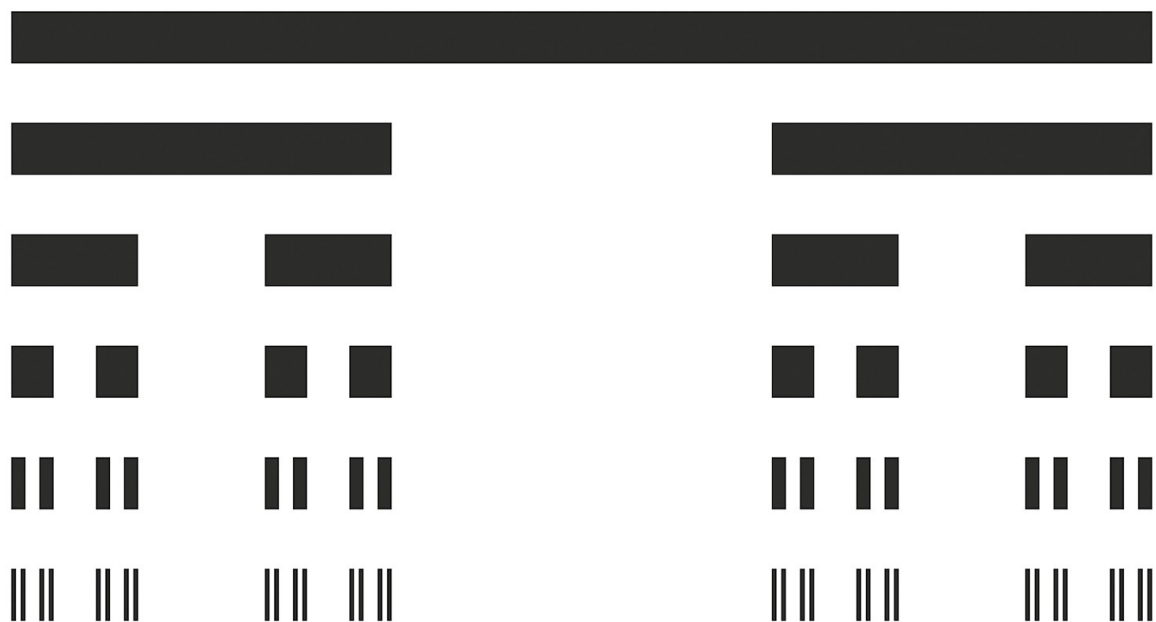


FIG. 7. The Cantor set is created by a series of iterations. Start with the line between 0 and 1 and remove the middle third. Then remove the middle thirds of the remaining two pieces and continue on and on as shown. The Cantor set is the set of points which belong to all the iterates.

The Cantor set (as I shall now call it) reminds me of the time traveller Doctor Who’s spacecraft from the BBC TV science fiction series—apologies to readers who have not encountered this TV programme. From the outside, Doctor Who’s spacecraft looks like a 1950s police telephone box—small and insubstantial. However, from the inside it looks palatial, with rooms everywhere! From the outside, the measure-zero property of the Cantor set makes it appear

tiny. However, it has as many points as the real line and so, looked at internally, it seems enormous. This has big consequences.

We can easily see that the geometry of the Cantor set is different to the Euclidean geometries discussed earlier. No matter how much we zoom into the Cantor set we find the same structure as we started with—small intervals with pieces missing. Whatever the geometry of the Cantor set is, it is not locally Euclidean.

So how can we describe the Cantor set in a way that distinguishes it from the real-number line? It turns out that one way of doing this is through the concept of dimension. Now, a set containing a finite collection of points is a 0-dimensional set. Lines are 1-dimensional sets. Surfaces are 2-dimensional sets, and solid objects are 3-dimensional sets. Space-time in Einstein's theory of relativity is a 4-dimensional set, and, as we have discussed, the unrestricted state space for the gravitational 3-body problem is an 18-dimensional set. It is possible to generalise this notion of dimension in order to describe the Cantor set. With this definition it turns out that the Cantor set has a dimension which lies between 0 and 1—neither 0 nor 1, but approximately 0.631⁵. The type of geometry associated with the Cantor set is known as fractal geometry, so named by the French American mathematician Benoit Mandelbrot⁶. The word “fractal” is in fact short for “fractionally dimensioned”.

The Cantor set lies at the heart of the Lorenz attractor. To explain this, I am going to actually describe the fractal attractor of a set of equations similar to but slightly simpler than the Lorenz equations. These are known as the Rössler equations.

The German biochemist Otto Rössler discovered his set of three equations⁷ as a direct result of Lorenz's work. This is the simplest set of equations which uses calculus to

produce a fractal attractor. The Rössler attractor is shown in [Fig. 8](#) (I have labelled the axes X , Y and Z as before, but now these variables satisfy equations different to the Lorenz equations).

To understand how the Rössler equations generate a fractal attractor, let's start by imagining part of the attractor as a simple 2-dimensional sheet. The action of the Rössler equations is to fold the sheet over as in the left-hand picture of [Fig. 9](#). But we can't join the two ends of the sheet because one end is folded and the other end not. Also, because of dissipation, the total area of the folded end is a little smaller than the original, unfolded end. We could imagine starting with a folded-over sheet, as in the middle panel of [Fig. 9](#). But now the action of the Rössler equations is to create a doubly folded-over sheet. Again, we can't join the two ends together. We could imagine starting with a doubly folded-over sheet, as in the right-hand panel of [Fig. 9](#). But now the action of the Rössler equations is to create a four-times folded-over sheet. Again, we can't join the ends together. We can repeat this over and over and over. In this way, what started as a simple 2-dimensional sheet becomes in fact like a piece of infinitely layered filo pastry! The Rössler attractor is like this filo pastry in the limit where there are an infinite number of folds, in fact where there is a Cantor set's worth of layered filo pastry.

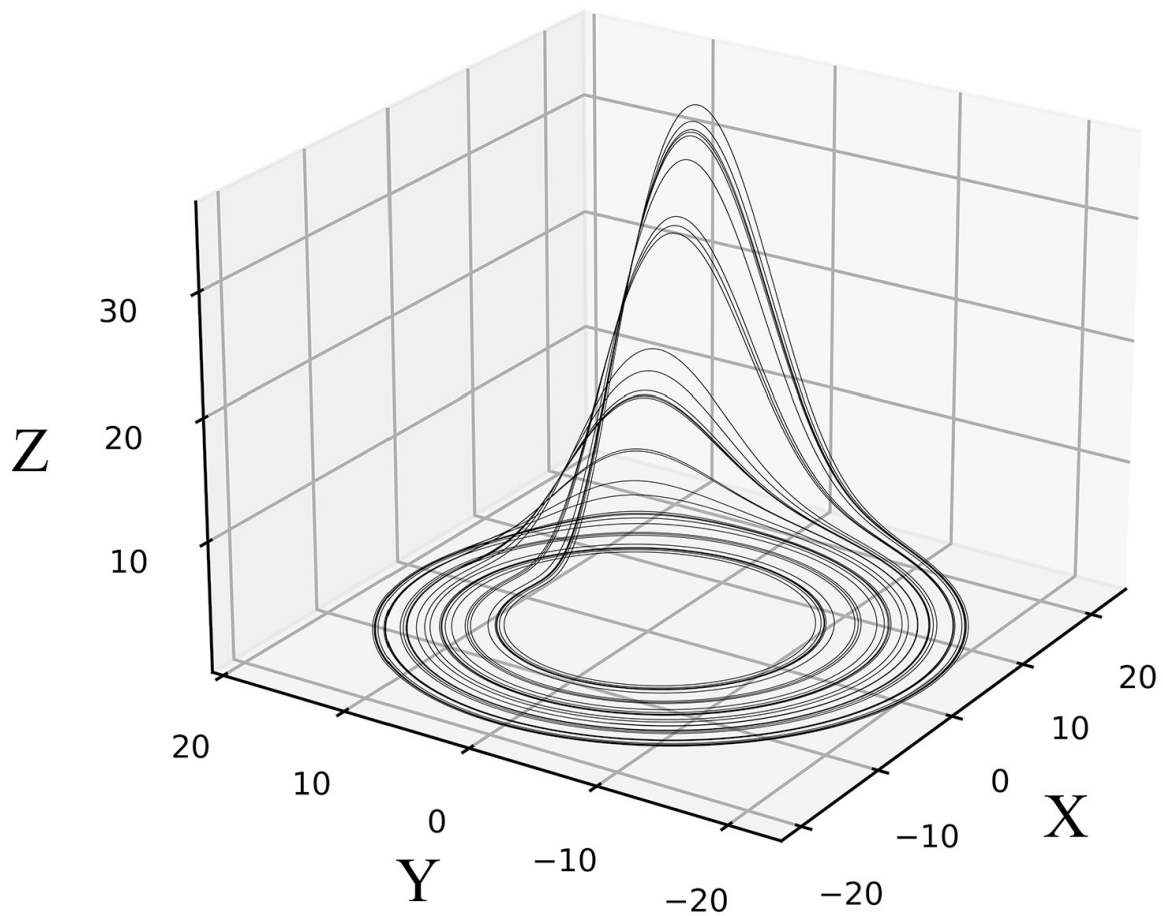


FIG. 8. Like the Lorenz attractor, the Rössler attractor is also a fractal geometry. Its relationship to the Cantor set is a little easier to understand than the Lorenz attractor's.

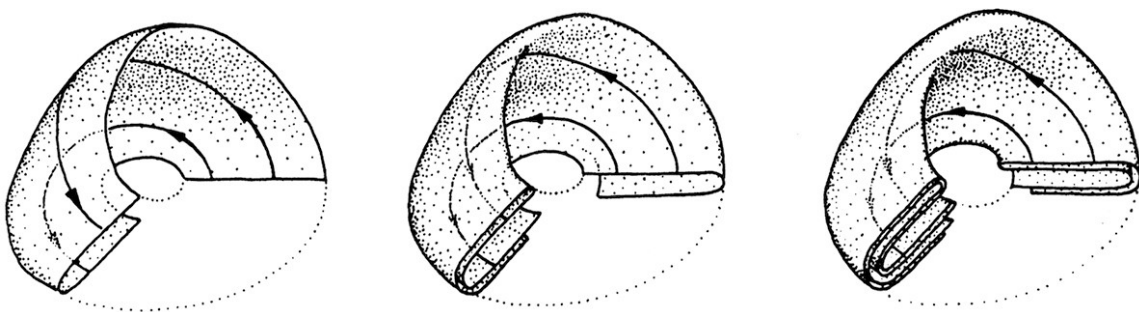


FIG. 9. Constructing the Rössler attractor. From Abraham and Shaw (1984).

Lorenz was the first to have the insight that the Cantor set lies at the heart of his attractor. Here are Lorenz's exact words from his famous 1963 paper:

“We see that each surface is really a pair of surfaces, so that, where they appear to merge, there are really four surfaces. Continuing this process for another circuit, we see that there are really eight surfaces, etc., and we finally conclude that there is an infinite complex of surfaces, each extremely close to one or the other of two merging surfaces.”

Lorenz realised that his attractor, as with the Rössler attractor, can be considered a Cantor set's worth of surfaces on which the state-space trajectories reside. It was this discovery that finally made Lorenz realise that the states of his model would never repeat themselves, that the trajectories would simply be moving from one part of the Cantor set to a different part, without repetition. The fractal dimension of the Lorenz attractor is 2.06 with typical parameter values. That is to say, it is just a little bit more than a 2-dimensional surface. However, the extra bit is absolutely crucial to understanding the non-periodic properties of the Lorenz equations.

It's hard to overestimate the significance of this discovery. Before Lorenz, no one⁸ imagined that a set of three coupled equations based on Newton's calculus could generate such an extraordinary geometry. As mathematician Ian Stewart wrote in 1997⁹ of Lorenz's 1963 paper:

“When I read those words, I get a prickling at the back of my neck and my hair stands on end. He knew! Thirty-four years ago, he knew!... In a mere 12 pages Lorenz anticipated several major ideas of nonlinear dynamics, before it became fashionable, before anyone else had realised that new and baffling phenomena such as chaos existed.”

It actually took another forty years before the Lorenz attractor was shown rigorously to be fractal, i.e., according to the exacting standards of mathematics¹⁰.

LET US RETURN TO the serendipitous experiment that Lorenz performed: starting a run of his model with an approximate value for the starting conditions $X = 0.506$ rather than the model's exact value ($X = 0.506127$). Would we expect some initial error to grow at the same rate, regardless of the starting conditions? [Fig. 10](#) shows that the answer is certainly no. Here is our first application of the geometry of chaos, one that has big consequences, as discussed in Part II of this book.

In [Fig. 10](#) we imagine that we know the initial conditions only approximately. Let us suppose that the actual initial state lies somewhere inside a small ring of points—but we don't know where exactly. [Fig. 10](#) shows how this initial ring evolves on the Lorenz attractor over a fixed period of time, for three different positions of the ring on the attractor.

With the initial ring in the position on the attractor shown in the top left of [Fig. 10](#), there is no growth of uncertainty, even though the ring evolves from one lobe of the attractor to the other. In this case we can be certain that the actual future state will make a transition from the left-hand lobe of the attractor to the right-hand lobe.

With the initial ring in the top right of [Fig. 10](#), uncertainty grows with time. We see that the ring evolves into a distorted banana/boomerang shape. If we wanted to know whether the actual state will make a transition to the right-hand lobe of the attractor, then all we could reasonably say is that there is roughly a 40 percent chance of a transition, and hence a 60 percent chance the true state will stay on the left-hand lobe.

However, when the initial uncertainty ring is in the position on the attractor shown in the bottom of [Fig. 10](#),

then the uncertainty grows explosively. Here we have no idea whether the actual state will lie on the left- or right-hand lobe of the attractor. With a deterministic prediction system, there's a good chance we will forecast the outcome completely incorrectly.

Let's return to the parable of the missing horseshoe nail, as discussed in the introduction. Let's suppose we can model the dynamic state of kingdoms using chaotic dynamics. (Can we? This I discuss later.) Let's imagine the initial uncertainty refers to possible missing horseshoe nails. Let's also imagine that for states on the left-hand lobe of the Lorenz attractor, the king has retained his kingdom, whilst for states on the right-hand lobe of the attractor, the king has lost his kingdom.

Then, for the example on the top left of [Fig. 10](#), the king will certainly lose his kingdom. Here it doesn't really matter whether some horseshoe nails are missing or not, the kingdom is sure to be lost (perhaps the king is totally incompetent). By contrast, in the example at the bottom of [Fig. 10](#), whether the king loses his kingdom or not is critically dependent on the uncertain state of horseshoe nails. It is this parlous state to which the parable refers.

[Fig. 10](#) is an important figure for this book. The fact that the evolution of uncertainty depends so strongly on the position of the initial ring of uncertainty on the attractor is a consequence of the nonlinearity of the underlying equations¹¹. It explains why there are occasions—like the October 1987 storm, and possibly the 2008 financial crash—which are so difficult to predict in a precise manner.



FIG. 10. The initial ring of points represents the uncertainty in our knowledge of the actual initial state. How the ring evolves depends on where it is located on the attractor. In the top left, the ring of uncertainty does not grow at all. In the top right, the ring is distorted into a thin banana/boomerang shape, indicating that it is not certain whether the time evolution of the actual state makes a transition from the left-hand lobe to the right-hand lobe. In the bottom figure, the position of the time evolution of the actual state on the attractor is very uncertain.

However, an ensemble system can warn us when the system is transitioning into a very unstable situation where dramatic things might, but not necessarily will, happen.

A mathematician might ask whether there is a single equation which describes how these banana/boomerang

shapes evolve from simple rings, as shown in [Fig. 10](#). There is indeed. It is called the Liouville equation ¹², named after the nineteenth-century French mathematician Joseph Liouville. It turns out that the Liouville equation is remarkably similar to the Schrödinger equation, the fundamental equation of quantum mechanics.

Indeed, like the Schrödinger equation, the Liouville equation is a linear equation. What does this mean? Earlier, we asserted that it was 100 percent certain that the “true” state lay inside the initial ring of uncertainty. However, suppose instead it was only 80 percent certain that the true state lay inside the initial ring. If this was so, then there would be only an 80 percent probability that the future true state lay inside the evolved banana/boomerang-shaped ring. We can rescale probabilities in this way because the Liouville equation is linear.

In early papers on probabilistic weather prediction, it was suggested¹³ that meteorologists should try to solve the Liouville equation to forecast uncertainty in weather predictions. However, it turned out that this is totally impractical. In fact, even for a simple model based on the Lorenz equations, it is hard to solve the Liouville equation directly. This is because the shape of the probability contours becomes more and more convoluted and distorted the longer the prediction time. It is much easier to estimate probabilities using ensembles.

Let’s look at another feature of the Lorenz attractor. The long time series at the top of [Fig. 11](#) shows the evolution of the X variable in the Lorenz model over a long period of time. The X variable moves irregularly between positive and negative values (corresponding to irregular chaotic oscillations between the left- and right-hand lobes of the Lorenz attractor). These irregular oscillations are completely unpredictable for most of the timescales shown. However, irrespective of the starting condition, overall the

value of the Xvariable is positive as often as it is negative. This is because of a symmetry in the geometry of the Lorenz attractor: the left- and right-hand lobes of the attractor (c.f. [Fig. 6](#)) are of exactly the same shape and size.

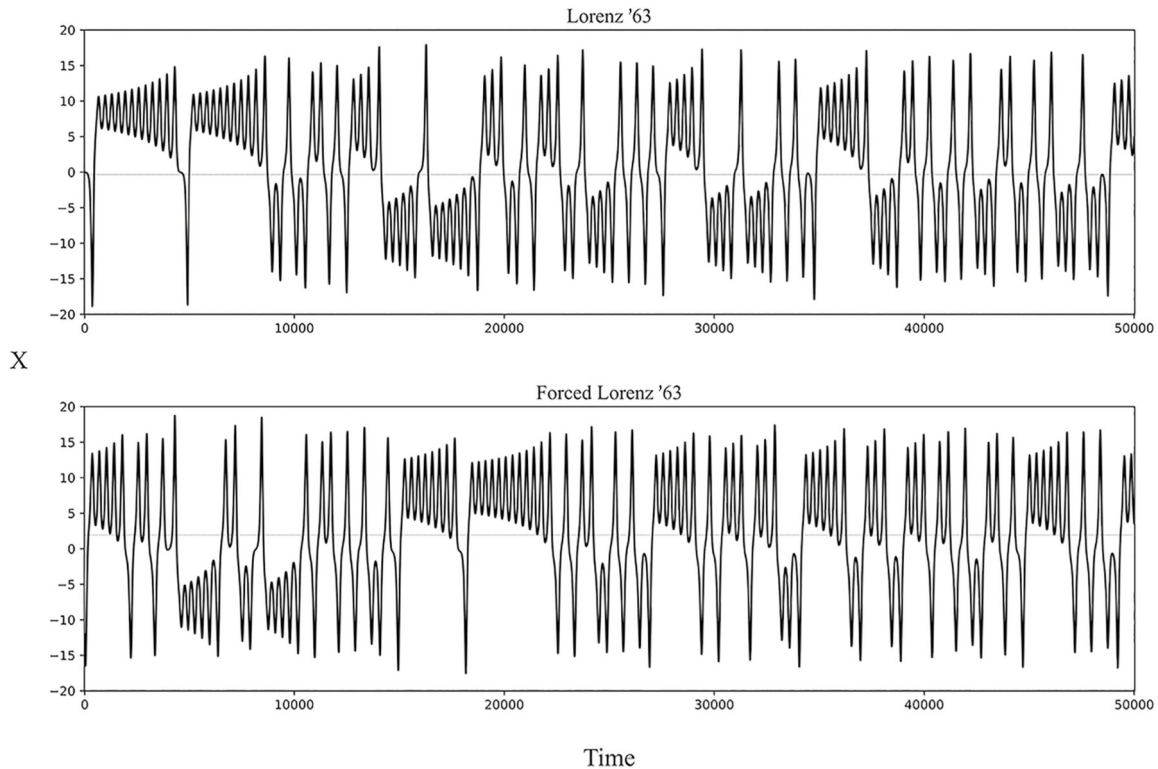


FIG. 11. Top: time series of the Xvariable in Lorenz's original model. Bottom: time series of the Xvariable from the forced Lorenz model, where an extra constant term is added to the equations. The likelihood of being in one of the lobes of the attractor is itself predictable, even if the precise fluctuations in either time series are not.

The long time series at the bottom of [Fig. 11](#) shows the evolution of the Xvariable in a modified version of Lorenz's equations. Here I have added a constant term to the right-hand side of the equations¹⁴. You can think of this extra constant term as representing an external perturbation to the Lorenz system. It is as close as one can get, in a system as simple as the Lorenz system, to a forcing, such as a doubling of the carbon dioxide concentration in the atmosphere.

The irregular oscillations in the lower time series are still unpredictable and depend sensitively on the starting conditions, as before. However, in the lower time series, it is now more likely on average that the X variable has a positive value than a negative value. The external forcing has changed the symmetry of the attractor geometry, making one lobe of the attractor bigger and the other lobe smaller. Let's think of the lobes of the Lorenz attractor as corresponding to opposite types of weather, such as warm weather or cold weather, or wet weather or dry weather. The added forcing to the equations changes in a predictable way the likelihood of having warm or cold weather, or wet or dry weather, even though the detailed way in which the weather evolves between these states is unpredictable. This is an important conceptual point when thinking about climate change.

« »

THERE ARE SOME REMARKABLE and profound connections between the geometry of fractal attractors—the geometry of chaos—and some of the deepest mathematical theorems of twentieth- and twenty-first-century mathematics. Because of this, Lorenz's geometry really does connect Newtonian mathematics with some of the most modern and exciting mathematical developments of recent years. When I learned about these connections, they convinced me of the fundamental importance of fractal geometry, not only for mathematics but also for physics. I leave this latter point for Part III of the book but discuss some of the mathematical connections here. You can skip to the next chapter if this is not to your taste.

Let's go back to the line, the simplest type of Euclidean geometry. As mentioned, if you draw a line on a sheet of

paper, you can think of each point of the line as representing a number, such as 1 , $\sqrt{2}$ or π . There will also be points on the line representing numbers like $1 + \sqrt{2}$, and $\sqrt{2} + \pi$, or even $\sqrt{2} \times \pi$. Indeed, take any two points on the line and convert them to their corresponding real numbers. Now either add or multiply these two real numbers together to get a third real number. Finally, convert this third number back to a point on the line. In this way, we can add or multiply two points on the line and get a third point on the line. We can do arithmetic on the real line using real numbers. This reflects the fact that there is a deep connection between real numbers and Euclidean geometry.

Can we do arithmetic on the points of the Cantor set using real numbers? No! If we take two points on the Cantor set and convert them into real numbers, and then either add or multiply the numbers together, the third number typically won't correspond to a third point on the Cantor set. However, the reason is not because we can't do arithmetic on the Cantor set. Instead, it's because the real numbers are not good numbers to use to describe points on the Cantor set. As we have discussed, fractal geometry is fundamentally different to Euclidean geometry. We should therefore not be surprised that we need a different type of number system to do arithmetic on fractals.

It turns out that there is a number system—called the p -adic numbers—which allows you to do arithmetic on fractals. Here p itself stands for a whole number [15](#). Now, it is probably difficult to imagine that there exist numbers that are not described by real numbers. To get a glimpse at the difference between real numbers and p -adic numbers, a real number such as $\sqrt{2}$ can be written in decimal format as $1.41421356237\dots$ whilst a possible 10-adic number is written instead as... 739620285.643 , where the dots go off to infinity towards the left (corresponding to smaller and

smaller pieces of the associated fractal). It's best to think of a p -adic number simply as a new type of number, better suited to Cantor sets than lines. The rules for adding and multiplying p -adic numbers are very similar to those for adding and multiplying real numbers. For the Cantor set shown in [Fig. 7](#), $p=2$. In this way, each point on a Cantor set can be represented by a 2-adic number [16](#). If you add or multiply two such 2-adic numbers, the sum or the product is a 2-adic number and corresponds to a third point on the Cantor set.

We have to generalise the notion of a Cantor set in order for it to be a geometric representation of 3-adic, or 5-adic, or more generally p -adic numbers. For example, [Fig. 12](#) shows a generalised Cantor set where we start with a 2-dimensional disk and then in the first iteration shrink it down to create 5 smaller disks inside the original disk. We repeat this iteration over and over and the generalised Cantor set comprises the points common to all the iterations. This generalised Cantor set can then be described mathematically by the 5-adic numbers [17](#). As before, if we take two points on this fractal set, convert them into 5-adic numbers, and add or multiply the numbers together, the resulting 5-adic number corresponds to another point on the generalised Cantor set. It's easy to generalise this further to imagine what a p -adic fractal looks like for an arbitrary p

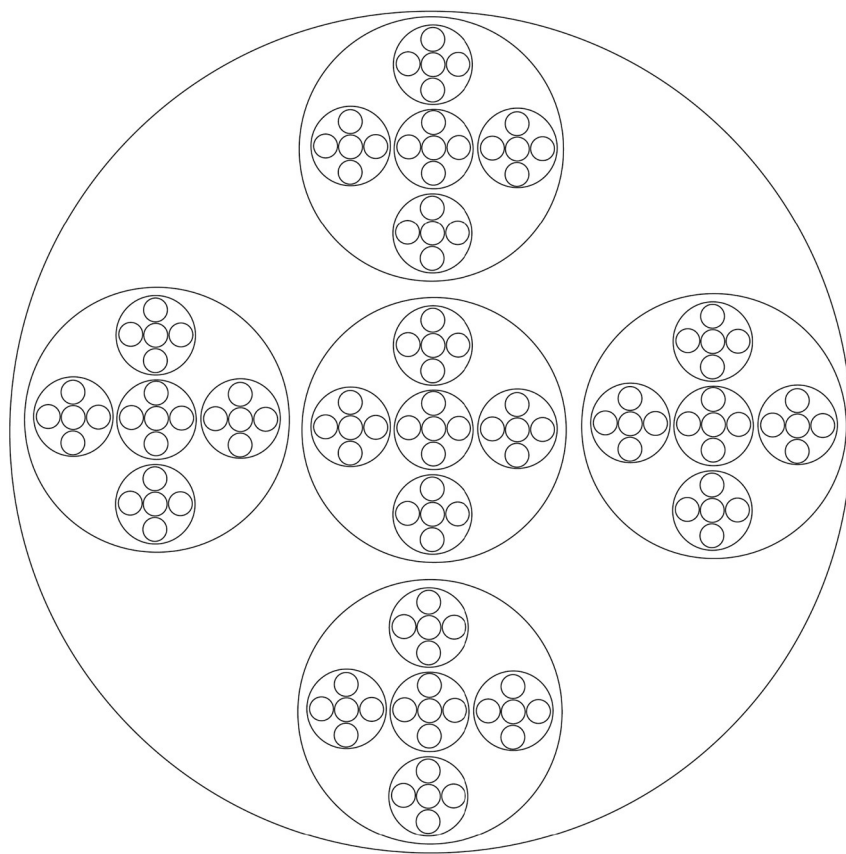


FIG. 12. The figure describes a fractal based on an iteration which takes a 2-dimensional disk and maps it to 5 smaller disks. It generalises the Cantor set with two iterated line segments. Just as the real line is a geometric representation of real numbers, so this fractal disk is a geometric representation of 5-adic numbers. Such numbers play a vital role in modern number theory.

The p -adic numbers are bread and butter for pure mathematicians, and number theorists in particular. A number theorist is someone primarily interested in understanding the properties of the whole numbers $\{1, 2, 3, \dots\}$. Whole numbers are in some sense the “quanta” of all numbers. The greatest mathematician of all time, the nineteenth-century German Carl Friedrich Gauss, described mathematics as the queen of sciences and number theory as the queen of mathematics. His view

would have been that number theory lies at the heart of everything quantitative.

The importance of p -adic numbers was clear to Andrew Wiles, who in 1995 famously proved Fermat's last theorem. Pierre de Fermat was a brilliant French mathematician of the seventeenth century. His so-called last theorem was in fact a conjectured property¹⁸ of the whole numbers that Fermat was himself never able to prove. This conjecture captivated mathematicians over the centuries, and had become, until Wiles, one of the most famous unsolved problems in mathematics.

What is amazing about Wiles's proof is the variety of mathematical ideas and techniques needed to crack Fermat's last theorem. However, one of the really important techniques Wiles used revolves around the theory of p -adic numbers.

Moving into the twenty-first century, one of the greatest contemporary number theorists is the German mathematician Peter Scholze. In 2018 Scholze won the Fields Medal, the most prestigious mathematical prize for mathematicians under the age of forty. The official citation was for "transforming arithmetic algebraic geometry over p -adic fields through his introduction of perfectoid spaces". These perfectoids are examples of fractal geometry. Given what I have written, it is perhaps not surprising that Scholze uses p -adic numbers to describe his perfectoid spaces. In an interview, Scholze said¹⁹:

"Now I find real numbers much, much more confusing than p -adic numbers. I've gotten so used to them that now real numbers feel very strange."

Perhaps one might imagine that there are many different types of geometry in addition to locally Euclidean and fractal geometry. However, an important theorem in number theory, Ostrowski's theorem, says not. If your geometry has an associated distance function, called a

metric, then, according to Ostrowski's theorem, the metric is essentially locally Euclidean or fractal, represented by the real numbers or the p -adic numbers, respectively.

I believe number theory, and p -adic numbers in particular, will play a central role in physics in years to come, reflecting the importance of the geometry of chaos.

A mathematical theorem of the twentieth century rivalling Wiles's proof of Fermat's last theorem in terms of fame is Gödel's incompleteness theorem. Here again there is a link to fractal geometry.

The logician, mathematician and philosopher Kurt Gödel was born in part of the Austro-Hungarian Empire (now in the Czech Republic) in 1906 and did his famous work while in Vienna, before immigrating to the United States and becoming a colleague and friend of Einstein. The key result at the heart of Gödel's theorem is a formula which can be constructed from the rules of any sufficiently powerful logical system. However, it is a strange, self-referential formula. It states that if this formula could be proved, the formula would be false [20](#). We must assume that consistent application of the rules of any logical system will yield only true results. The only way out of the dilemma is to acknowledge that Gödel's formula, whilst true, cannot be proved.

In the 1930s, the British mathematician Alan Turing, who subsequently went on to lead the code-breaking work at Bletchley Park during World War II, realised that Gödel's theorem implies that certain mathematical problems cannot be solved by algorithm: they are "uncomputable". As an example, Turing demonstrated that a universal computer—a concept that Turing defined—would, in principle, be unable to solve a mathematical problem posed by the German mathematician David Hilbert some years earlier. Hilbert's "halting problem" was to construct an algorithm which, on inputting some mathematical

proposition, would halt if the proposition was true. When he posed the halting problem, Hilbert believed that there was no such thing as an unsolvable problem. Turing showed that no algorithm can solve the halting problem.

Roger Penrose has speculated that Gödel/Turing undecidability will underpin the theory of quantum gravity²¹, the as-yet undiscovered theory that synthesises quantum and gravitational physics. Penrose himself won a Nobel Prize for his pioneering work on space-time singularities in Einstein's geometric theory of gravity, general relativity. Indeed, by studying properties of the famous fractal Mandelbrot set, Penrose speculated on the link between undecidability and fractal geometry, and hence between fractals and quantum gravity²².

In 1993, the computer scientist Simant Dube proved the link between undecidability and fractal geometry²³. He showed this by constructing Turing's universal computer using fractal geometry. In this way, undecidable problems such as Hilbert's halting problem can be expressed as problems in fractal geometry. Dube did this using dynamical systems called iterated function systems²⁴, which are even simpler than the Lorenz equations²⁵. An iterated function system can produce fractal figures like the Sierpinski triangle, shown in [Fig. 13](#). This fractal triangle is to the iterated function system what the Lorenz attractor is to Lorenz's three equations. That is to say, no matter what point in the plane one starts from, the iterated function system ends up operating only on the points of the Sierpinski triangle. The triangle is a fractal attractor of the iterated function system.

Dube showed that a problem similar to Hilbert's halting problem was equivalent to asking whether a given line intersects the fractal attractor of the iterated function system. If it could be determined by algorithm whether a given line intersects a fractal, then a point of intersection,

or lack of it, could be used to determine if the corresponding Turing machine would halt (at the state corresponding to the point of intersection). We know that we can't decide by algorithm whether such a Turing machine will halt, ergo we can't decide by algorithm whether the line will intersect the fractal.

The idea that fractal geometry provides a way to understand the mysteries of quantum physics is a discussion point in Part III of this book.

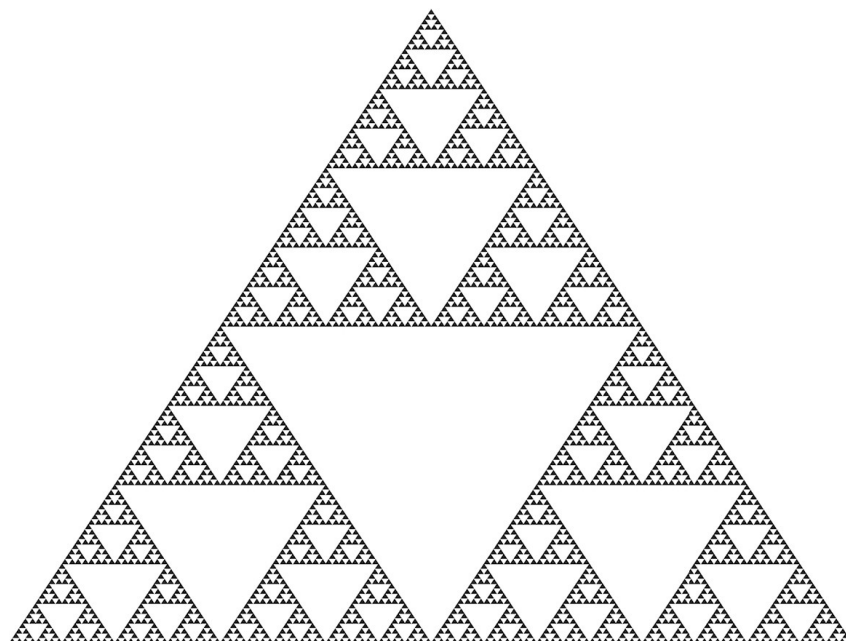


FIG. 13. The Sierpinski triangle, the fractal attractor of an iterated function system.

3

Noisy, Million-Dollar Butterflies

WATCHING WATER FLOW DOWN A WATERFALL CAN BE MESMERISING. The flow at the very top is quite smooth —“laminar”, as it is sometimes called. But as it starts its journey down, the water breaks into a myriad of pieces. We might try to follow the path of one small parcel of fluid, but it breaks into yet smaller pieces which soon become too small to follow individually. Is the science of chaos for such complex systems fundamentally similar or fundamentally different to what Lorenz had found with his three-variable model? We will find that the growth of uncertainty in such complex systems is like chaos on steroids, so much that our deterministic mathematical models seem to fail us. This has important consequences whose implications extend well beyond waterfalls.

« »

THE UBIQUITOUS PHENOMENON OF turbulence has both fascinated and perplexed scientists. Werner Heisenberg, one of the founding fathers of quantum mechanics, famously remarked: “When I meet God, I am going to ask him two questions. Why [Einstein’s theory of] relativity?

And why turbulence? I really believe he will have an answer for the first!"

As we will see, turbulence continues to fox mathematicians today. One of the most important unsolved mathematical problems relates to the question of just how predictable turbulent flow really is. How far down a waterfall could Laplace's daemon predict the motion of an individual small piece of fluid? All the way down? A tenth of the way down? Big prizes (and considerable kudos) await the person who can advance our knowledge of such a problem. As I will discuss in this chapter, the phenomenon of chaos as described in the last two chapters may not be unpredictable enough to describe turbulence adequately.

In the 1940s, the Russian mathematician Andrey Kolmogorov developed a mathematical model of turbulence. In his model, a turbulent fluid is characterised by a spectrum of interacting eddies or whirls, some big, some medium sized, some small. Energy is injected on the largest scale, e.g., by stirring a tank of water with a big paddle or, in the case of the atmosphere, by the sun, which preferentially warms the equatorial regions of the Earth. The energy then cascades¹ down to smaller and smaller scales, where eventually it dissipates by increasing the irregular motion of individual molecules, manifest by a general warming of the fluid. As we have discussed, this dissipation effect is characterised by an internal frictional property of a fluid known as its viscosity. A pithy description of Kolmogorov's cascade of energy was given by the polymath Lewis Fry Richardson (whom we shall meet several times in this book)²:

Big whirls have little whirls
That feed on their velocity,
And little whirls have lesser whirls
And so on to viscosity.

An example of turbulent fluid flow with multiple whirls of different scales is shown in [Fig. 14](#).

The atmosphere itself is a turbulent fluid system. The very largest-scale whirls in the atmosphere are the jet streams³, rivers of fast-moving air in the high atmosphere. Aircraft avoid jet streams when flying from east to west, and try to catch them when flying west to east. The location of the jet streams is not fixed, but varies from day to day, month to month and season to season. The jet streams do not always flow directly from west to east, but can have large meanders towards higher and lower latitudes (perhaps reminiscent of the meandering of a river of water as it wends its way to the sea). These meanders have scales of 10,000 km (6,213 miles) or so. They can cause long spells of unusual weather. In February 2021, a meander in the jet stream caused Texas to experience record-breaking cold weather for around ten days. In July 2021, another meander caused parts of British Columbia to experience record-breaking hot temperatures, reaching almost 50°C (122°F).



FIG. 14. A computer simulation of turbulence, characterised by a spectrum of eddies or whirls, some big, some small. We shall be referring to this simulation again.

The jet streams, in turn, determine the evolution of low-pressure weather systems—or midlatitude cyclones, as they are often called—that routinely bring rain and strong winds to those of us who live in midlatitude regions. These midlatitude cyclones have scales around 1,000 km (620 miles), and so are smaller in spatial scale than the meanders of the jet streams. Inside a cyclone, regions of warm and cold air are separated by features known as fronts, close to which many of the rain-bearing clouds are clustered. Big thunderstorm clouds can have scales between 10 and 100 km (6.2 and 62 miles). Inside individual clouds, the air can be particularly turbulent, which is why flying through clouds can be an unpleasant experience. In extremis, such turbulent eddies can cause a plane to crash. And embedded in one of these sub-cloud eddies are even smaller-scale whirls that may not be

energetic enough to buffet an aircraft but can be detected by small balloons. And so on to viscosity.

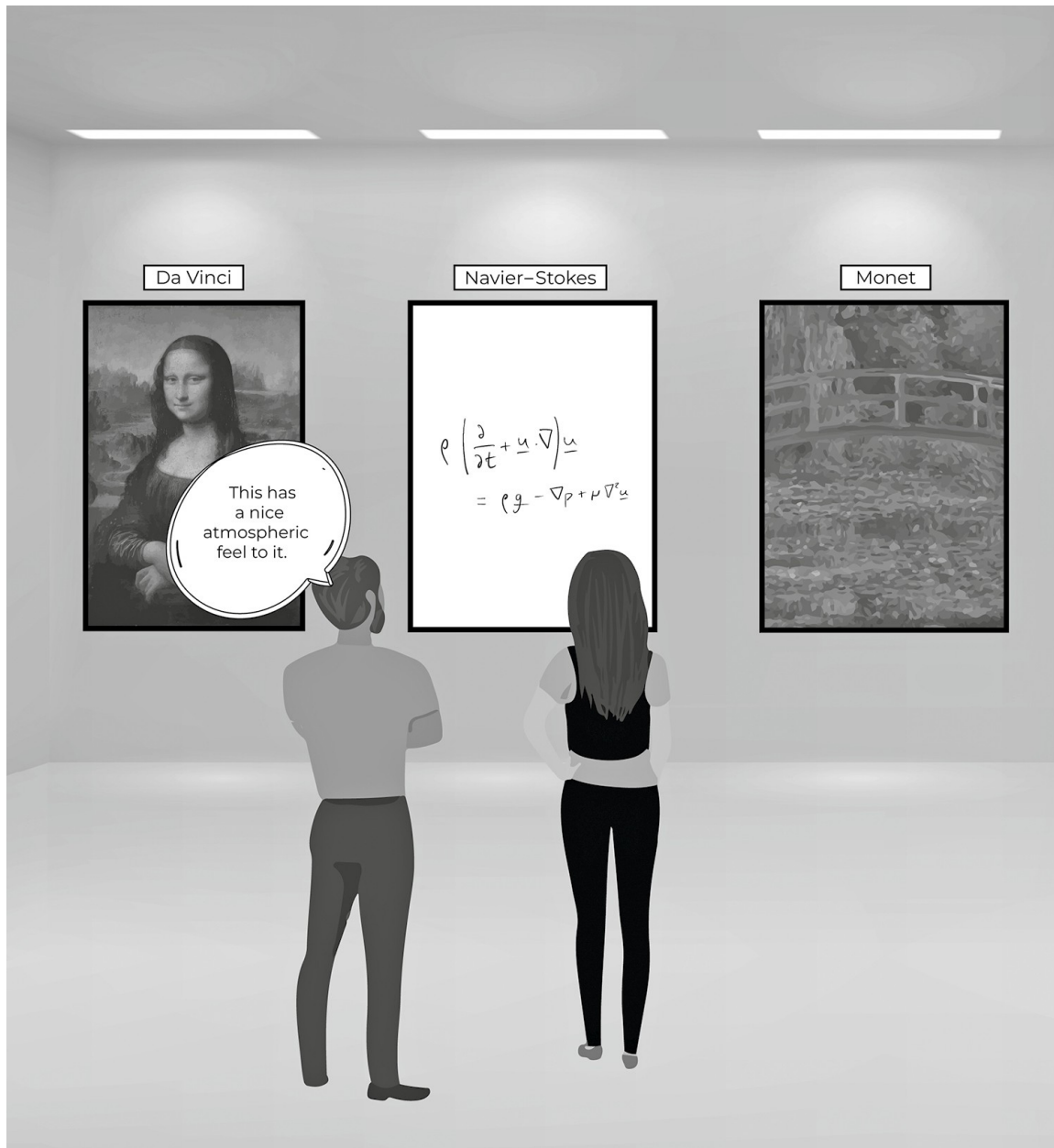


FIG. 15. The Navier-Stokes equation as a work of art.

Fluid motion is described mathematically by an equation discovered independently in the nineteenth century by the French mathematician Claude Navier and the Irish mathematician George Stokes. Based on Newton's laws of

motion, the nonlinear Navier-Stokes equation⁴ describes, with twenty-three symbols, the motion of every whirl in the atmosphere, from the smallest to the largest (see [Fig. 15](#)). Being based on Newton's laws, the Navier-Stokes equation is deterministic; it contains no indeterminate or random terms.

This equation appears to generate a type of unpredictability and uncertainty which is much more radical than the type of chaos we have met so far: not only can Laplace's daemon not predict arbitrarily far into the future, it cannot predict beyond a fixed finite time horizon. This remarkable result was set out in a paper published by Ed Lorenz (yes, him again!) in 1969⁵, six years after his famous paper on the geometry of chaos was published. Here is part of the abstract of Lorenz's 1969 paper:

"It is proposed that certain formally deterministic fluid systems which possess many scales of motion are observationally indistinguishable from indeterministic systems; specifically, that two states of the system differing initially by a small 'observational error' will evolve into two states differing as greatly as randomly chosen states of the system within a finite time interval, which cannot be lengthened by reducing the amplitude of the initial error"

The last clause (my italics) is the killer. In the type of chaos described in the last chapter, the more we can reduce the uncertainty in the initial state, the farther ahead we can predict. Put another way, you tell me how far ahead you want to predict and I will tell you how accurately you need to know the initial conditions. However, according to Lorenz's 1969 paper, in a complex system that contains many scales of motion which interact with each other, you will not be able to predict beyond some fixed predictability horizon no matter how well you know the initial conditions. In this picture, the limit where the uncertainty in the initial conditions is strictly zero is what's called a singular limit.

According to Lorenz, even if the uncertainty was infinitesimally different to zero, we would not be able to predict beyond a finite horizon in time. We will meet singular limits again when we discuss quantum physics.

Lorenz gave a popular account of his 1969 paper at a meeting in 1971 of the American Academy for the Advancement of Science. This 1971 talk has become famous because of the title: “Does the Flap of a Butterfly’s Wings in Brazil Lead to a Tornado in Texas?”; it is the origin of the phrase “the butterfly effect”. What I am going to describe is the original meaning of the phrase “the butterfly effect”⁶. It is more radical than the meaning that is usually attributed to it today, described in [Chapter 1](#).

To solve the Navier-Stokes equation, we will need a computer, indeed a very powerful computer. To make use of the computer, we must first transform the Navier-Stokes equation into a form that the computer can understand. To do this we create a finite mathematical representation of the atmosphere by partitioning it into a number of small but finite volumes which are called “gridboxes” (see [Fig. 16](#)). In this mathematical representation, the atmosphere is assumed to be homogeneous and hence unvarying on scales smaller than the gridbox scale. That is to say, in this finite mathematical representation, all the turbulent scales of motion that are smaller than the gridbox scale are assumed to be unimportant and are therefore removed from the computer’s representation of the Navier-Stokes equation. Hence, the representation of the Navier-Stokes equation that the computer sees is not the original equation, but the mathematical model based on this simplified atmosphere.

The smaller the size of the gridbox, the more accurate the computer’s representation of the Navier-Stokes equation. However, the smaller the gridbox, the more powerful the computer we will need to solve the equation.

Let's suppose that the gridboxes in our model have a horizontal dimension of 100 km⁷, which might describe a weather forecast model from a few decades ago.

To estimate the future weather, we will need a set of initial conditions corresponding to today's weather⁸. These initial conditions—of temperature, pressure, wind velocity, humidity and so on around the world—are determined by observations of today's weather. These come from a combination of remote satellite measurements and in situ instruments on weather balloons.

We will assume that we have plentiful perfectly accurate observations of the atmosphere everywhere around the globe. This doesn't mean that the initial conditions for our weather forecast model are perfect and without error. As mentioned, all of the whirls in the atmosphere on scales smaller than 100 km are necessarily missing from the model's initial state, even though we may have observations of these small-scale whirls. Because of the finite size of the model gridboxes, we are unable to "assimilate"⁹ all the sub-100-km information provided by the observations into the model's initial state.

Now let's suppose we are interested in predicting weather systems only on scales of, say, 1,000 km or more. We might think that, as a result, we don't have to worry that scales smaller than 100 km are missing in the model's initial conditions. However, we would be wrong to think this. Because of the nonlinear nature of turbulence, these initial sub-100-km-scale errors will grow in amplitude and in spatial scale as the forecast proceeds. They will after a few days become large, 1,000-km-scale forecast errors. Eventually, even forecasts of the 10,000-km-scale meanders in the jet stream will be in error because of the missing sub-100-km scales.

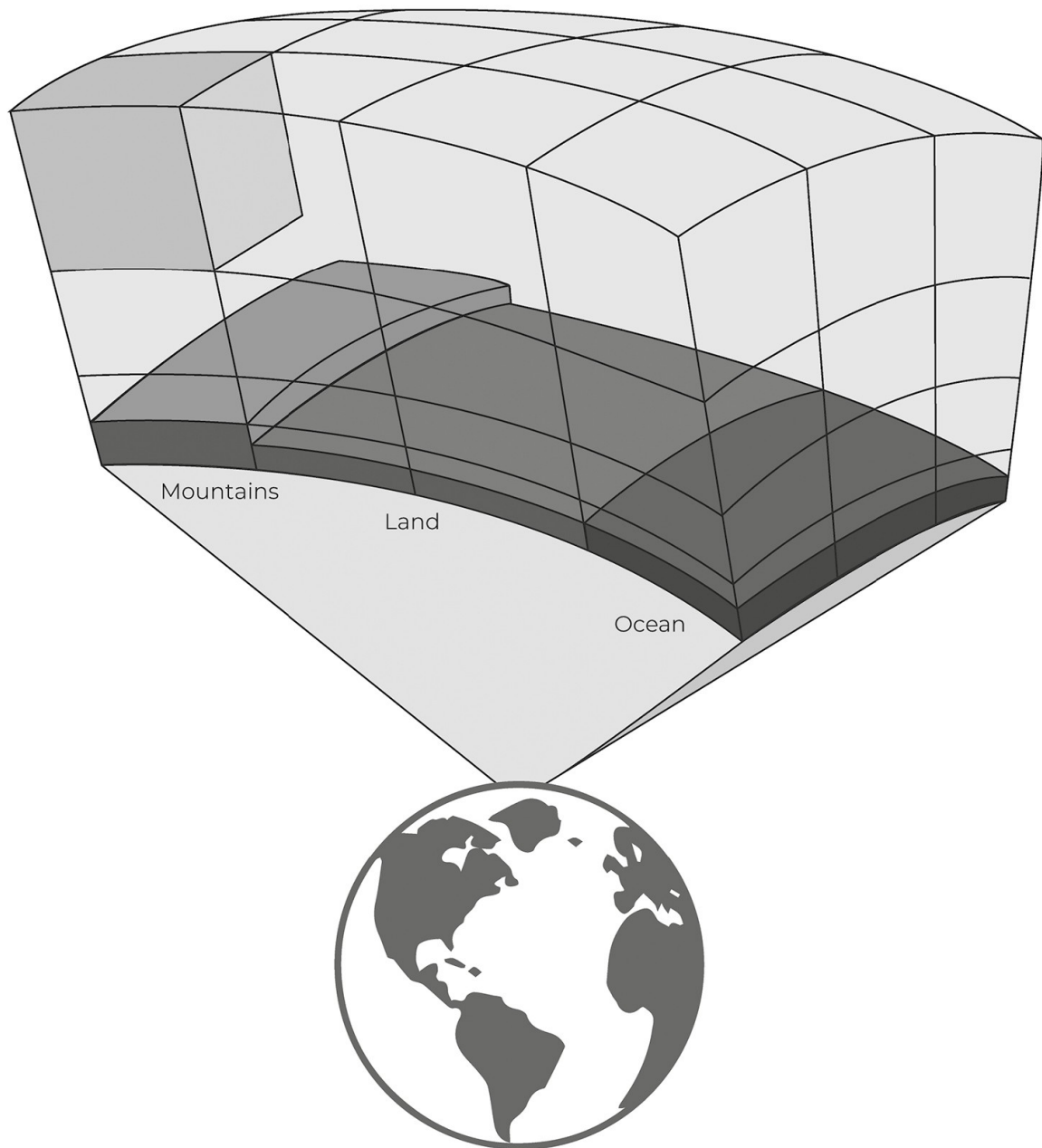


FIG. 16. A schematic of gridboxes in an atmospheric model.

With a 100-km gridbox model, we can, on average, forecast 1,000-km weather patterns to about a week ahead. Not bad, but suppose we really want to predict the day-to-day evolution of the weather three weeks ahead. A way to improve the forecast would be to decrease the size of the gridbox from, say, 100 km to 50 km. This increases the

number of gridboxes in our model, which means we will need a more powerful computer to make the forecasts. OK, suppose we have a new computer to do this. With the new model, errors in the computer model's initial conditions will now be confined to scales smaller than 50 km, rather than 100 km as before. This will reduce the initial-condition error and will allow us to predict the weather for longer than seven days. But how much longer?

Lorenz in his 1969 paper concluded that such a doubling of model resolution would not lead to an extra seven days of skilful forecast time. Using the theory developed by Kolmogorov, he estimated that a doubling of resolution would give us only an extra $7 / 2 = 3.5$ days of prediction time. The fundamental reason for this is that in Kolmogorov's theory, errors grow faster on smaller scales. For example, a small error in the structure of a 1,000-km-scale weather system will get twice as big after a day or two of forecast time, whilst a small error in the structure of a 1-km-scale cloud will get twice as big after only an hour or two.

Continuing in this way, Lorenz argued that each doubling of the resolution of the model will increase our prediction time by only half of what the previous doubling achieved. So, if we decrease the size of a gridbox in our model in a series of steps from 100 km to 50 km, to 25 km to 12.5 km, our prediction time will increase from 7 days to $7(1 + \frac{1}{2}) = 10.5$ days, to $7(1 + \frac{1}{2} + \frac{1}{4}) = 12.25$ days, to $7(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}) = 13.125$ days. This is clearly a law of diminishing returns, since each doubling of model resolution provides a smaller and smaller gain in prediction time¹⁰.

Can you see a problem here? If we keep increasing the resolution of the model again and again and again until the gridboxes become infinitesimally small—in which case, with our perfect observations, the initial error will also become infinitesimally small—we are not going to be able to predict

forever, as we might think we should be able, given the determinism of the Navier-Stokes equation. Instead, we will be able on average to predict for only a maximum of $7(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \dots)$ days ahead, where there are now an infinite number of fractions inside the brackets, each a half of the one before. In this limit, the predictability timescale sums to $7 \times 2 = 14$ days, because $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \dots = 2$. That is to say, even with perfect observations of the weather, we can predict ahead for only 14 days. In this picture, the idea of predicting ahead 3 weeks would be an in-principle impossibility, even for an omniscient being like Laplace's daemon! This is the real meaning behind the butterfly effect.

But is the butterfly effect literally true? Well, the Navier-Stokes equation does not take account of quantum physics, so going to the limit of infinitesimal-scale errors is artificial. A better question would be this: Does the Navier-Stokes equation exhibit a finite predictability horizon? The truth is, we don't know for sure. In his 1969 paper, Lorenz was fully aware that he hadn't rigorously proved that the initial-value problem for the Navier-Stokes equation had a finite predictability horizon; his arguments were plausible but did not come up to the rigorous standard of proof in mathematics. That's why he says in the abstract quoted earlier "it is proposed that", not "it is proved that". Indeed, over fifty years later, we still don't know whether the butterfly effect is literally true or not. Understanding this property of the Navier-Stokes equation remains one of the most important unsolved problems in mathematics: it is one of the Clay Mathematics Institute's Millennium Prize problems, set at the beginning of the twenty-first century^{[11](#)}. This list of problems is a modern update of a set of mathematical problems posed by German mathematician David Hilbert at the beginning of the twentieth century^{[12](#)}. Solving any of the Millennium Prize problems will win you a

million-dollar prize—hence the title of this chapter. The only Millennium Prize problem to have been solved at the time of writing was one posed by Poincaré in the field of topology.

On the other hand, evidence from computer models suggests the butterfly effect is true in practice. We don't have computers powerful enough for us to reduce the size of the gridboxes to the scale of butterflies. However, if we reduce the gridbox size to the size of individual clouds, we do indeed find that forecasts of large-scale weather can be very sensitive to uncertainties in these small-scale cloud structures. Not only that, fourteen days does seem to represent well the limit to which we can predict the detailed motion of weather systems, on average.

This takes us back to the discussion in [Chapter 1](#) as to why weather is unpredictable. Back in the 1950s, the experts would have said it is unpredictable because the atmosphere is turbulent and contains many whirls and eddies of different scales. I mentioned that Lorenz's intuition at the time was that the atmosphere would be unpredictable even if the atmosphere had as few as three degrees of freedom.

Both views are correct. In 1963, Lorenz showed that a model with just three variables could be chaotically unpredictable. However, in 1969 Lorenz showed that the experts were right too—a system with a very large number of variables (big whirls and small whirls) could be much more unpredictable than a system with a very small number of variables. In practice, the latter is actually the key reason why the atmosphere is unpredictable.

For this reason, we must distinguish between what is called low-order chaos (or low-dimensional chaos) and high-order chaos (or high-dimensional chaos). Here the word "dimension" refers to the dimension of state space, i.e., the number of variables needed to describe the system being studied. Recall trouser state space as an example.

The number of variables in a system describable by low-order chaos is sufficiently small that they can all be represented in a computer. By contrast, the number of variables in a system describable by high-order chaos is so large that not all can be represented in a computer. The Lorenz model described in [Chapters 1 and 2](#) is an example of low-order chaos. Turbulence, as described by the Navier-Stokes equation, is an example of high-order chaos.

However, we are now on the horns of a dilemma. On the one hand, we must limit the number of whirls in order to be able to solve the Navier-Stokes equation. Essentially, we must reduce the dimension of weather state space to something a powerful supercomputer can manage, a billion or so. However, in so doing, the model has lost all knowledge of the butterflies (i.e., tiny whirls) in the atmosphere. As we have seen, these butterflies can and indeed will cause the weather forecast to go completely off the rails after some finite time. A model without these butterflies could produce detailed forecasts that may in reality be complete nonsense. And if that's the case, the forecasts will lead us up the garden path in predicting things that may never happen, or, in Michael Fish's case, in not predicting things that do happen.

We solve this problem by representing the unresolved flapping butterflies by random numbers. That is to say, we deliberately make our models noisy¹³. A fancy word for randomness is "stochasticity". In this way, modern weather forecast models (and, increasingly, climate models—see [Chapter 6](#)) are stochastic in nature. If we run an ensemble of forecasts using our stochastic weather model, even when each ensemble member has absolutely identical initial conditions, the individual members of the ensemble will still spread apart from each other, because each ensemble member has different realisations of the random noise in the model equations. Put another way, to represent a high-

order chaotic system on a computer, we first truncate the system so it is a low-order chaotic system and then add noise to represent the scales of motion (the tiny whirls) that have been excised by the truncation.

The consequences of using noise to model high-order chaotic systems extend well beyond weather prediction, as we will see.

« »

IF WE SAY THAT noise is created by random numbers, then this raises the question: What are random numbers? That's a difficult question to answer. It's easier to say what randomness isn't. The sequence of numbers (0, 1, 2, 3, 4, 5...) is not random because there is an obvious pattern to the sequence: the next number is one more than the last. So, certainly, a sequence of random numbers should not exhibit any pattern like this.

In computer science, we create random numbers using "pseudorandom-number generators", or PRNGs for short. PRNGs are in fact deterministic pieces of computer code that produce sequences of numbers without any obvious patterns. Now, this might seem like a contradiction in terms: How can you generate randomness from determinism? Indeed, it is a contradiction, and the contradiction has consequences. A computationally cheap PRNG will spit out numbers that may superficially look random, but when analysed in more detail will reveal patterns. Worst of all, the numbers from a PRNG will eventually start to repeat themselves, a sure-fire guarantee that the PRNG isn't truly random. The cure is to increase the complexity of the computer code for the PRNG. However, if we keep increasing the complexity of the PRNG, it will eventually be as computationally expensive to

run as representing the butterflies that we truncated away in the first place! We therefore have to make a judgement. We need a PRNG that is sufficiently complex to emulate randomness well enough, but still computationally cheap compared with simulating the processes the randomness represents.

However, there may be a fascinating alternative to PRNGs in future-generation computers: using noise created directly by the computer hardware. For example, as mentioned in the introduction, if we turn the voltage down across a transistor in a computer, it will start to become rather “noisy” in the way it operates. The transistor will become sensitive to the seemingly random motion of atoms and molecules associated with their thermal energy. Indeed, as discussed, if the transistors are small enough, then the noise will start to become directly associated with quantum physics (for example, through a process called quantum tunnelling). In this way, the ultimate source of such hardware-based noise is quantum noise, believed to be the purest form of randomness possible in the universe.

In this way, it may be possible to generate almost perfectly random noise for truncating our high-order chaotic models, not at some small computational cost as would be the case with a PRNG, but at negative computational cost. That is to say, by turning the voltages down, we can run the computer using less energy than we did when the computer was operating deterministically. I hope this is the way some future-generation supercomputers will work¹⁴—many of their chips being low-energy indeterministic processors.

But hang on. Why are we trying to generate noise? We normally think of noise as a nuisance, something to be eliminated or at least minimised. In signal detection theory we often speak about a “signal to noise” ratio, and try hard

to maximise this ratio by boosting the signal or reducing the noise as much as possible.

However, in chaotic systems, noise can be a positive and useful resource and not one we should be trying to eliminate or minimise. This is perhaps a paradoxical message, but one that lies at the heart of the nonlinear philosophy that underpins this book. This is such an important point that I want to give some examples of when noise can play an important role, not in making a signal harder to see—the intuitive picture we have of noise—but in making it easier to see. In all the examples next, nonlinearity plays a crucial role.

Look at the percentage figures on the left of [Fig. 17](#). The pixels in the figures are made from different shades of grey (where solid black would correspond to 100 percent and plain white would correspond to 0 percent).

Let's suppose we want to transmit the greyness of one of the figures, but we can transmit only one bit of information per pixel: 0 (white) or 1 (black). For the percentages in the middle of [Fig. 17](#), the lighter shades of grey are simply truncated to 0, and the darker shades truncated to 1. We have clearly lost a lot of information. In fact, we now can't see half of the figures, because they have been truncated to the same whiteness as that of the page.

However, there is an alternative way to truncate a shade of grey to 1 bit, and that is to use noise. The results are almost magical—see the right-hand figures in [Fig. 17](#). As an example, consider a pixel element in the “30 percent” figure. Instead of deterministically truncating to 0, we first randomly choose a fraction between 0 and 1 (e.g., using a PRNG) assuming all fractions are equally likely. If the randomly chosen number is greater than 0.3, then we truncate the shading of the pixel to 0 (white). If the randomly chosen number is less than (or equal to) 0.3, then we truncate the value of the pixel to 1 (black). There is a 70 percent chance that the randomly chosen number will be

greater than 0.3. Hence it is more likely than not that the pixel will be truncated to white. By contrast, consider a pixel element in the “70 percent” figure. We now truncate to white if the randomly chosen number is greater than 0.7. It is more likely than not that the pixel will be truncated to black.

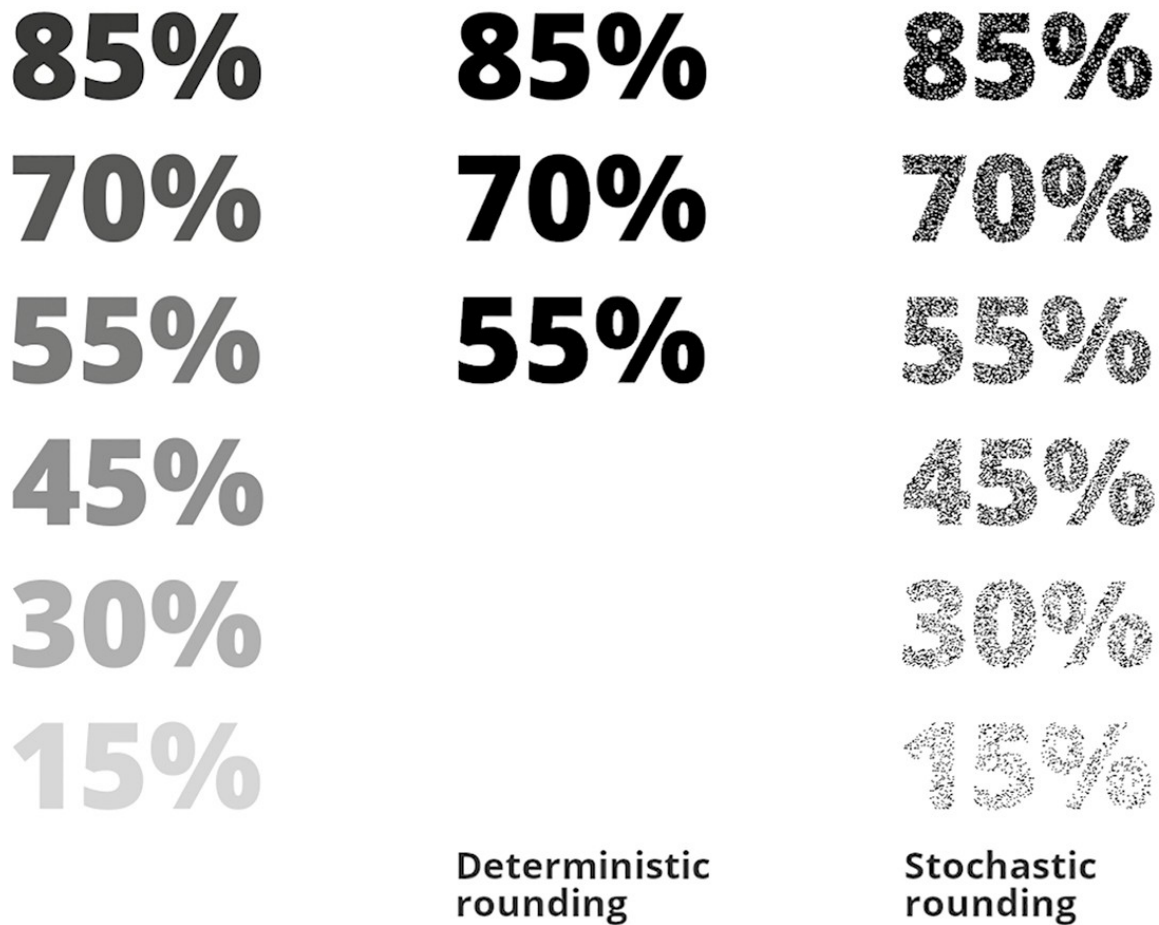


FIG. 17. Stochastic rounding. On the left are percentage figures shown in different shades of grey. In the middle, the shade of grey is rounded to one bit (black or white) in a systematic way. On the right, the shade of grey is also rounded to one bit (black or white) but in a more stochastic way, i.e., using random noise. There is clearly more information on the right-hand side than in the middle.

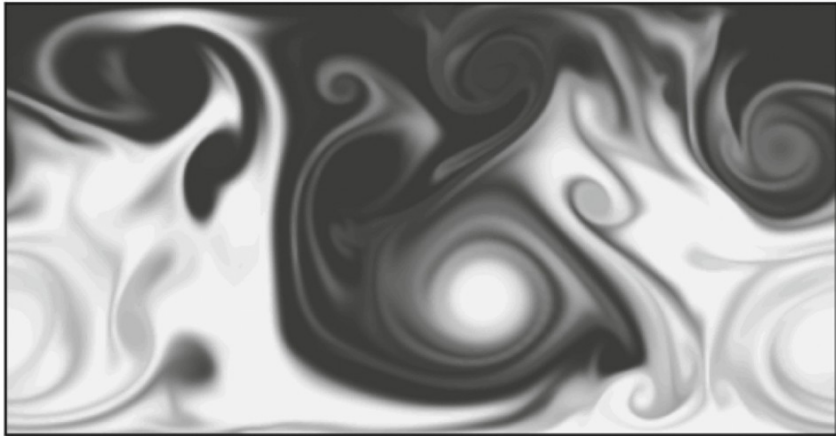
If we use a similar procedure, performing this “stochastic rounding” to all the pixels and choosing independent random numbers for each pixel, we end up with something looking like the right-hand side of [Fig. 17.](#)

Hey presto, the missing percentages have reappeared! The noise has amplified the signal!

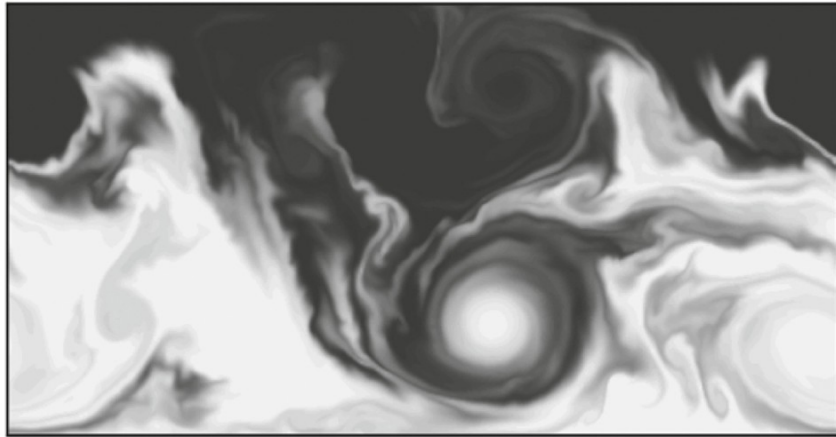
Stochastic rounding is beginning to play an important role in weather modelling¹⁵. [Fig. 18](#) is a simulation of turbulence, like that in [Fig. 14](#), but where each fluid variable is represented by 16 bits instead of the default numerical “precision” in scientific computation, which is 64 bits. There is a good reason to try to use as few bits as possible when doing a computation. Most of the energy of a modern supercomputer is used transporting bits of information inside the computer, e.g., from memory to a processing unit. If you can condense the information into packages of 16 bits rather than 64 bits, then you will save about three-quarters of the energy otherwise used for moving data. The problem is that the 16-bit computation is definitely not as accurate as the 64-bit computation. It looks like we have lost information in the 16-bit computation. However, when applying stochastic rounding to the 16-bit representations of variables, as shown in the bottom panel in [Fig. 18](#), the result is almost identical to the original 64-bit computation.

I strongly believe that future electronic computers must develop this type of stochastic rounding capability in hardware¹⁶. With several billion variables in a weather model, tens of billions of bits are being transported unnecessarily inside the computer when each variable is represented with 64 bits. This is energetically inefficient and constrains the speed at which the computer can do calculations.

Float64



Float16



Float16 + Stochastic rounding

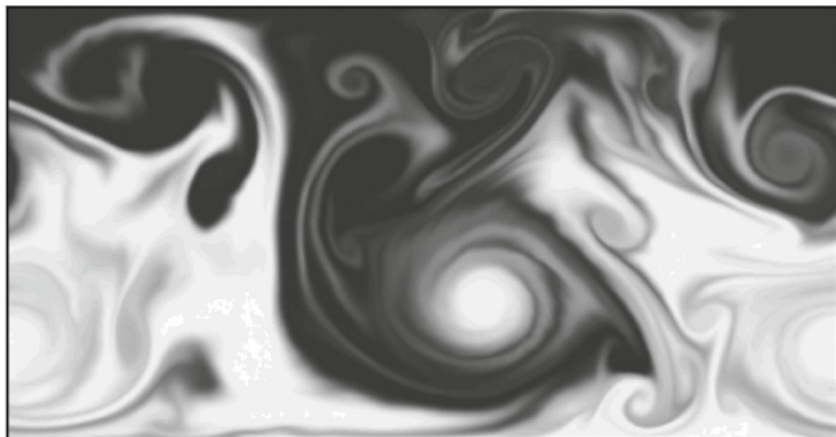


FIG. 18. Top: a copy of the simulation of the fluid turbulence shown in [Fig. 14](#). Here, the individual fluid

variables in the simulation are represented by 64-bit numbers. This representation is the default in scientific computation but uses a lot of computer resources. Middle: where the number of binary digits representing the individual fluid variables in the simulation has been reduced from 64 to 16. The turbulent whirls in the simulation are degraded in realism compared with the top figure. Bottom: the same simulation but where the stochastic rounding technique has been used when truncating to 16 bits. Now the simulation is virtually identical to the top figure. Adding noise has made the simulation more accurate. From Paxton et al. (2022).

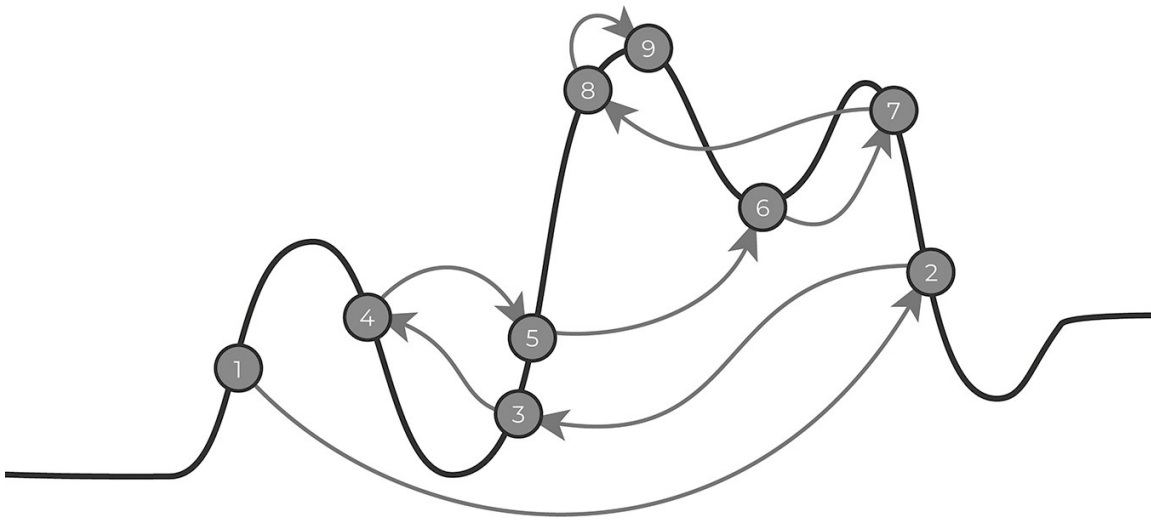


FIG. 19. Simulated annealing is an algorithm which uses noise to find the highest point of a complicated set of peaks and troughs. Here the algorithm is shown progressing to the highest peak in nine steps.

These are not the only examples where noise can be shown to be a valuable resource.

Suppose we want to find the highest point of some complicated curve, such as the one shown in [Fig. 19](#). A deterministic algorithm for finding this highest point would typically start at some arbitrarily chosen initial point on the curve and then simply move one step to the left or one step to the right depending on which direction is up. However, such an algorithm will get stuck at a local peak. Once at this local peak, both directions are “down”, and the

algorithm will wrongly think it has reached the highest point and stop.

There is, however, an algorithm (called “simulated annealing”) which uses random numbers to find the highest point. Let’s imagine the algorithm has been running for a while and has reached a point on the curve which is not yet the highest point. Let’s see what happens in the next step of the algorithm.

First of all, the algorithm consults a PRNG, which randomly suggests a new point on the graph for the algorithm to consider. The algorithm first looks to see if the suggested point is higher up the curve than the old point. If it is higher, then the algorithm definitely accepts the PRNG’s suggestion and we move on to the next step of the algorithm. If, on the other hand, the new point is lower than the original point, the algorithm doesn’t automatically reject the PRNG’s suggestion. However, as the algorithm proceeds, it becomes less and less likely that a point which is substantially lower than the current point will be accepted. That is to say, in the early stages, the algorithm won’t be too fussy about accepting a substantially lower point, but later on it becomes more discriminating about accepting such a degradation. This latter feature is reminiscent of annealing metals, hence the name of the algorithm—when a metal is hot, in the early phases the metal is quite pliable; however, as the metal cools, it becomes more rigid.

[Fig. 19](#) shows the simulated annealing algorithm at work. The steps from 1 to 9 correspond to the steps suggested by the PRNG and accepted by the algorithm. The rejected steps are not shown. The step from 1 to 2 is accepted because 2 is higher than 1. The step from 2 to 3 is accepted because, although 3 is lower than 2, we are in the early phases of the algorithm; as mentioned, there is a fair probability that lower steps will be accepted early on. The step from 4 to 5 is accepted because 5 is only very slightly

lower than 4. The likelihood of actually finding the highest point of the graph depends on the length of time the algorithm is run. However, for a given length of time, the algorithm appears very efficient at estimating the highest point compared with more deterministic algorithms.

In the final example, we add noise (not too weak, not too strong) to the chaotic Lorenz model. The result, shown in [Fig. 20](#), is startling. Instead of washing out the structure of the two lobes of the Lorenz model, the noise exaggerates these lobes. The system typically spends longer in one lobe before making an unpredictable transition to the other lobe. This is a rather profound example of the way in which nonlinearity can partially stabilise a nonlinear system, making it more predictable.

In conclusion, for nonlinear systems, noise can be your friend. It's not necessarily the nuisance we often think it to be. Perhaps nature makes use of this fact much closer to home than in weather models. I will be discussing this in Part III of the book.

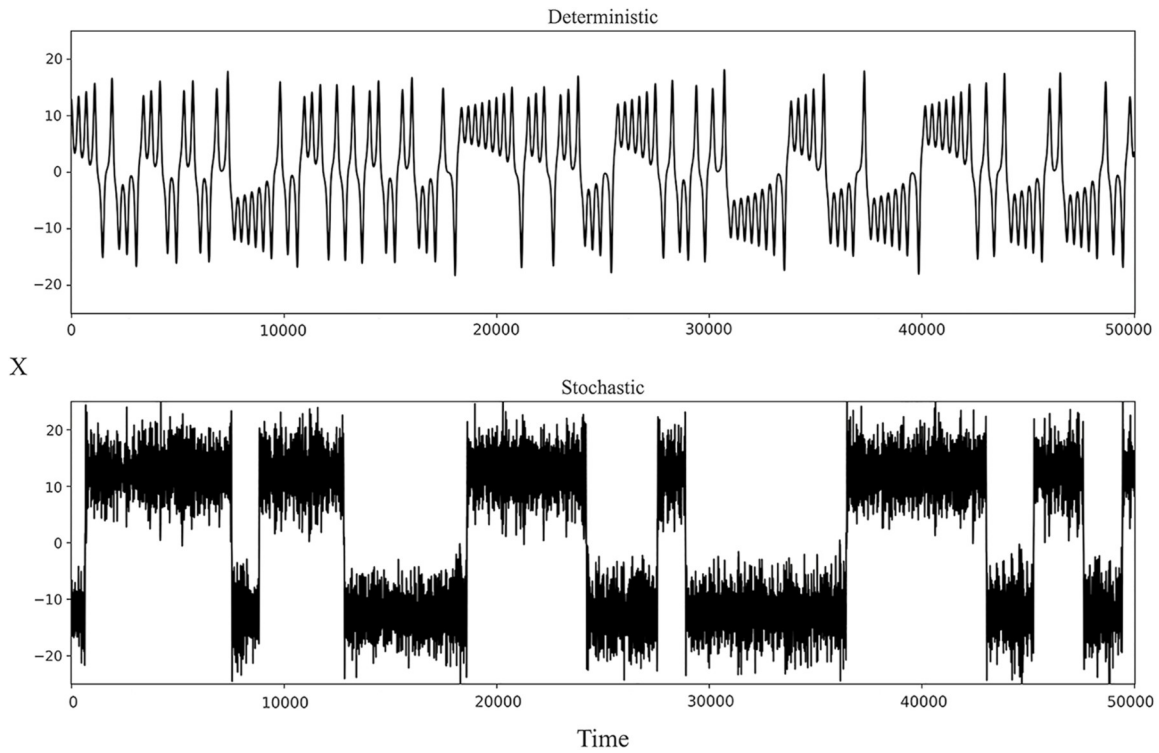


FIG. 20. Top: a time series of the X -variable of the standard Lorenz model. Bottom: a time series of the X -variable of the Lorenz model where noise has been added to the equations. The presence of noise makes the “regime” structure of the model much more pronounced, with the system spending longer in one of the lobes^{[17](#)}. The transitions between the positive and negative X regimes remain completely unpredictable.

4

Quantum Uncertainty

Reality Lost?

THE BUTTERFLY EFFECT IS PART OF WHAT IS CALLED “CLASSICAL” physics. Classical physics describes everything that predates quantum mechanics, as formulated in the mid-1920s by Niels Bohr, Werner Heisenberg, Erwin Schrödinger and others ([Fig. 21](#)). Hence, Einstein’s theory of relativity is considered “classical” even though it was formulated in the twentieth century. Because Lorenz’s equations of chaos are ultimately based on Newton’s laws of motion, these equations are also described as “classical”. Should we describe the algorithmically undecidable fractal geometry of chaotic systems as “classical”? I would say not—the discovery of this geometry postdates quantum mechanics. But we are jumping the gun.

Uncertainty lies at the heart of quantum physics. Indeed, it is elevated to a principle: Heisenberg’s uncertainty principle. According to the uncertainty principle, there are pairs of properties of a quantum particle, say its position and its momentum, and if you know one of these properties perfectly, then you know nothing about the other property. There’s the old (and not very good) joke about the traffic cop pulling Heisenberg over and asking whether he knows

how fast he was going. “No,” replies Heisenberg, “but I know where I am.” “Well, you were doing exactly one hundred miles per hour,” says the cop. “Oh great,” says Heisenberg, “now I’m completely lost!”

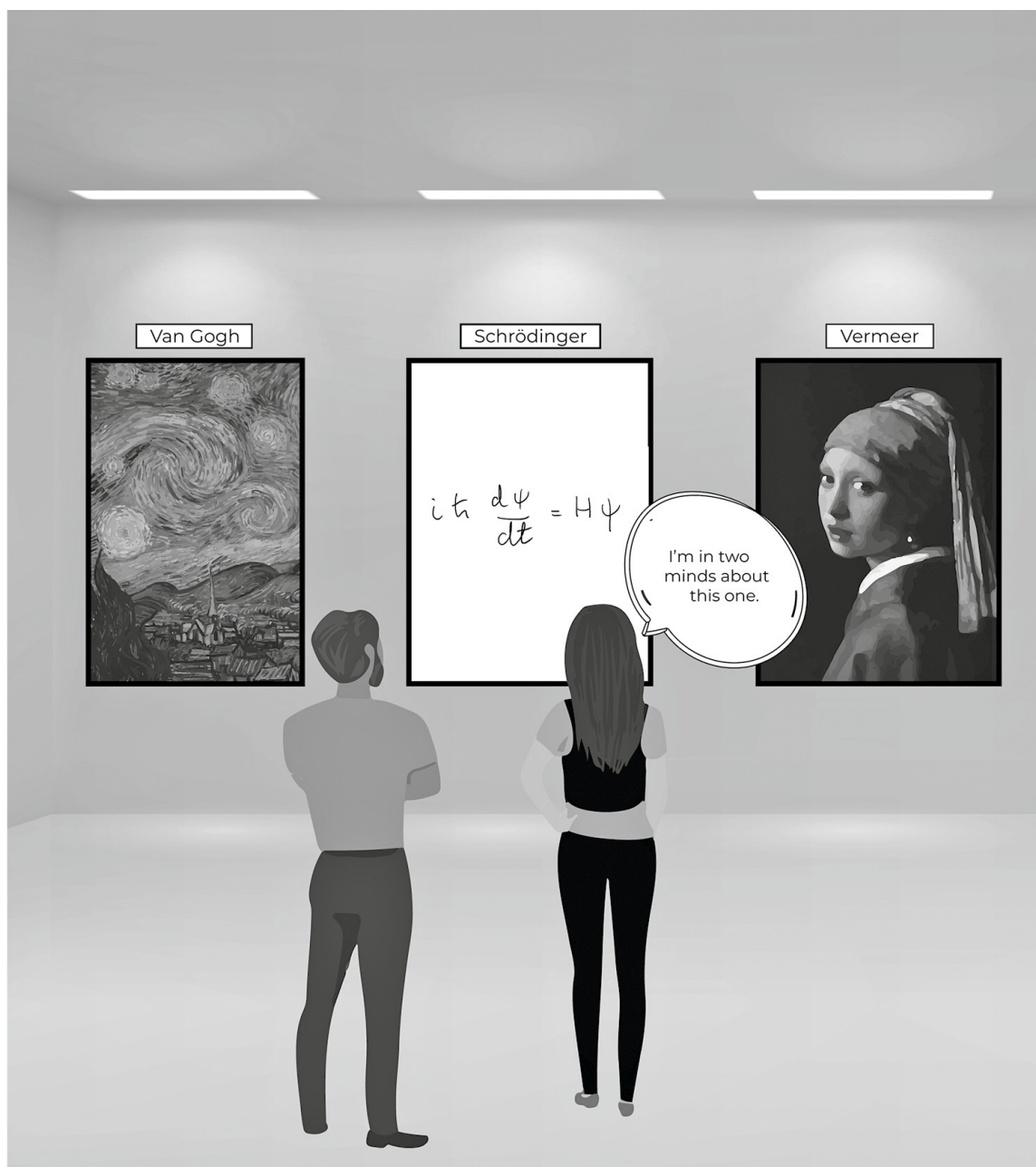


FIG. 21. The Schrödinger equation lies at the heart of quantum mechanics. Like Newton’s laws of motion, it makes use of Newton and Leibniz’s calculus. Unlike Newton’s laws, it describes a picture of physical reality which physicists and philosophers argue about to the present day.

According to Heisenberg's uncertainty principle, if the precise position of the particle is mathematically determined, its precise momentum is mathematically undetermined. What on earth does this mean? I think it's fair to say no one really understands what it means. However, it has sparked a debate that originated with the birth of quantum mechanics, and continues to the present day. Is quantum uncertainty "epistemic" or "ontological" in nature? We can define these words as follows. If it is epistemic, then quantum uncertainty merely describes our uncertainty about an otherwise precisely defined system. If it is ontological, then quantum uncertainty describes some inherent property of a quantum system, independent of whether we are trying to study it or not. Most physicists believe that quantum uncertainty is ontological for reasons I will discuss in this chapter. Since what we call reality is comprised of myriads of elementary particles, this means that the very concept of reality—which most of us take for granted—is itself uncertain.

The debate about the nature of quantum uncertainty goes back a long way. In his 1930 book, *The Physical Principles of the Quantum Theory* Werner Heisenberg tried to explain his eponymous principle by imagining locating a particle (we'll call it the "object-particle") by looking through a microscope. He argued that the smaller the wavelength of the light used by the microscope, the better the position of the object-particle can be determined. In fact, modern microscopes use electrons with wavelengths up to 100,000 times smaller than visible light to see small objects with ultra-high resolution. However, the smaller this wavelength, the more energetic are the photons (or electrons) and hence the more they will recoil unpredictably with the object-particle being measured, making it increasingly difficult to determine the momentum of the object-particle.

This seems like a nice explanation of the uncertainty principle, except it isn't right, at least according to Niels Bohr, Heisenberg's mentor. Bohr argued that variables like position and momentum are inherently "complementary"—an experiment to measure one necessarily excludes the possibility of measuring the other. Heisenberg's explanation suggests that the quantum uncertainty in the momentum of the particle describes our uncertainty, necessarily arising when we try to measure the quantum properties of the object-particle by shining light, or other particles, on it. Bohr complained that quantum uncertainty is more profound than this, and that it somehow refers to the particle's own uncertainty.

Bohr came to this point of view because of so-called wave-particle duality: the notion that to explain quantum reality we must sometimes use the language of waves, whilst other times we must use the language of particles. A good example is the so-called double-slit experiment, as shown in [Fig. 22](#). In the nineteenth century, the British polymath Thomas Young demonstrated the wavelike nature of light by passing light waves through a screen with two closely spaced slits. At the other side of the screen, the waves spread out from the two individual slits and interfered with each other. In some regions the intensity of the light was increased by this interference effect; in other regions the intensity was decreased. In the twentieth century we were able to pass single photons through the same double-slit screen. Providing you don't try to observe which slit a photon goes through, the individual photons tend to be detected in places where the light waves interfered positively to create regions of enhanced brightness, and tend not to be detected in regions where the waves interfered negatively to create regions of darkness. On the other hand, if you try to observe which slit a photon goes through, then the interference pattern disappears and the photons behave like small classical

particles. In this way, we can describe the quantum double-slit experiment using both wave and particle descriptions of light. But neither description alone is sufficient. Bohr characterised these wave and particle descriptions as being complementary to each other. Why quantum physics should be like this is a mystery¹.

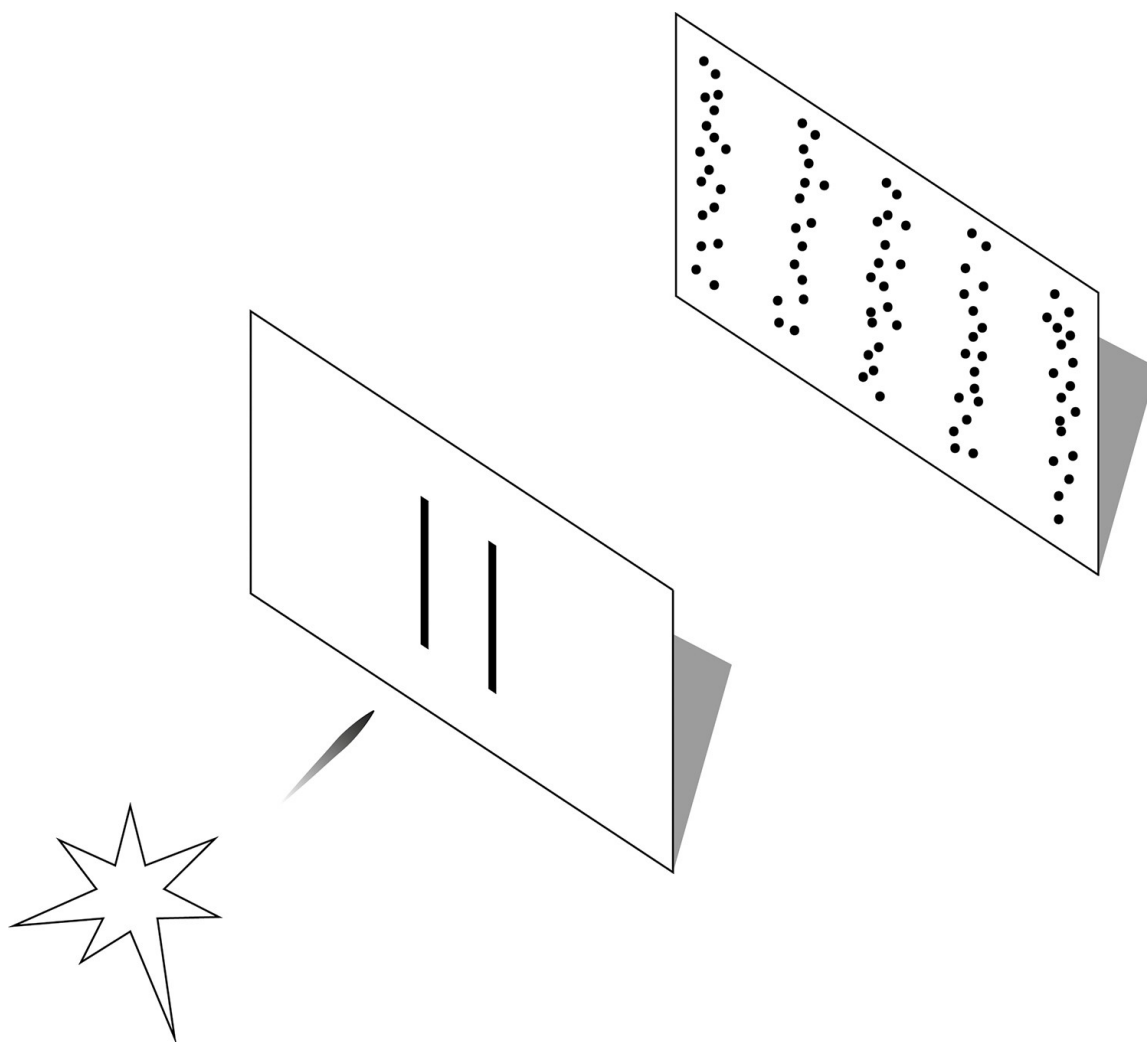


FIG. 22. Individual photons passing through two slits produce a wavelike interference pattern on a screen behind the screen—unless we try to observe which slit a photon goes through. How this happens remains a deep mystery and is illustrative of wave-particle duality in quantum physics.

Einstein famously battled with Bohr about this issue. He believed that the standard interpretation of quantum

mechanics, in which the laws of physics contain some inherent uncertainties, made no sense. At the famous 1927 Solvay Conference in Brussels, Einstein posed a simple thought experiment to challenge Bohr and the other proponents of quantum mechanics. It illustrates his two major objections to quantum mechanics. The first is its inherent indeterminism, made famous by Einstein's phrase "Does God play dice?" The second refers to what he called "spooky action at a distance". Einstein's thought experiment still poses a profound challenge for today's advocates of quantum mechanics.

Consider a single photon emitted by a light source ([Fig. 23](#)). The particle passes through a small hole in a screen and is sent to a point on a hemisphere coated with phosphorescent material. As suggested in [Fig. 23](#), it seems reasonable to imagine that the photon travels to the hemisphere in some particular direction, which we don't know, but the photon does, until the photon hits the point q on the hemisphere. Then both we and the photon know where the photon is, and hence which path it took from the source. Surely that's the sensible way to imagine what's going on.

But no! According to quantum mechanics, before it excites the photoluminescent material and reveals itself, the photon is described by a mathematical quantity called a "wavefunction". The wavefunction is represented by the Greek letter ψ in the art gallery in [Fig. 21](#). The wavefunction describes mathematically the wavelike features of quantum systems. It spreads out in all directions from the hole in the screen like a wave propagating out from a source. In popular books, the wavefunction is sometimes called a "probability wave". But that's a dumbed-down description which does not do justice to the incomprehensibility of quantum mechanics. According to quantum mechanics, the wavefunction does not assert that there is (say) a 40 percent chance of the

photon being here and a 60 percent chance of it being there. Rather, the wavefunction describes a photon which is somehow simultaneously here and there, with the degree of “hereness” and “thereness” being weighted by quantities that resemble² probabilities (like 40 percent and 60 percent) but aren’t. This is usually called a “superposition”. For example, in the quantum two-slit experiment, we don’t say that there is a 50-50 chance of a photon going through one slit or the other; instead the wavefunction appears to require us to say that the photon went, equally, through both slits at once. Confused? Join the club. But that’s only the half of it!

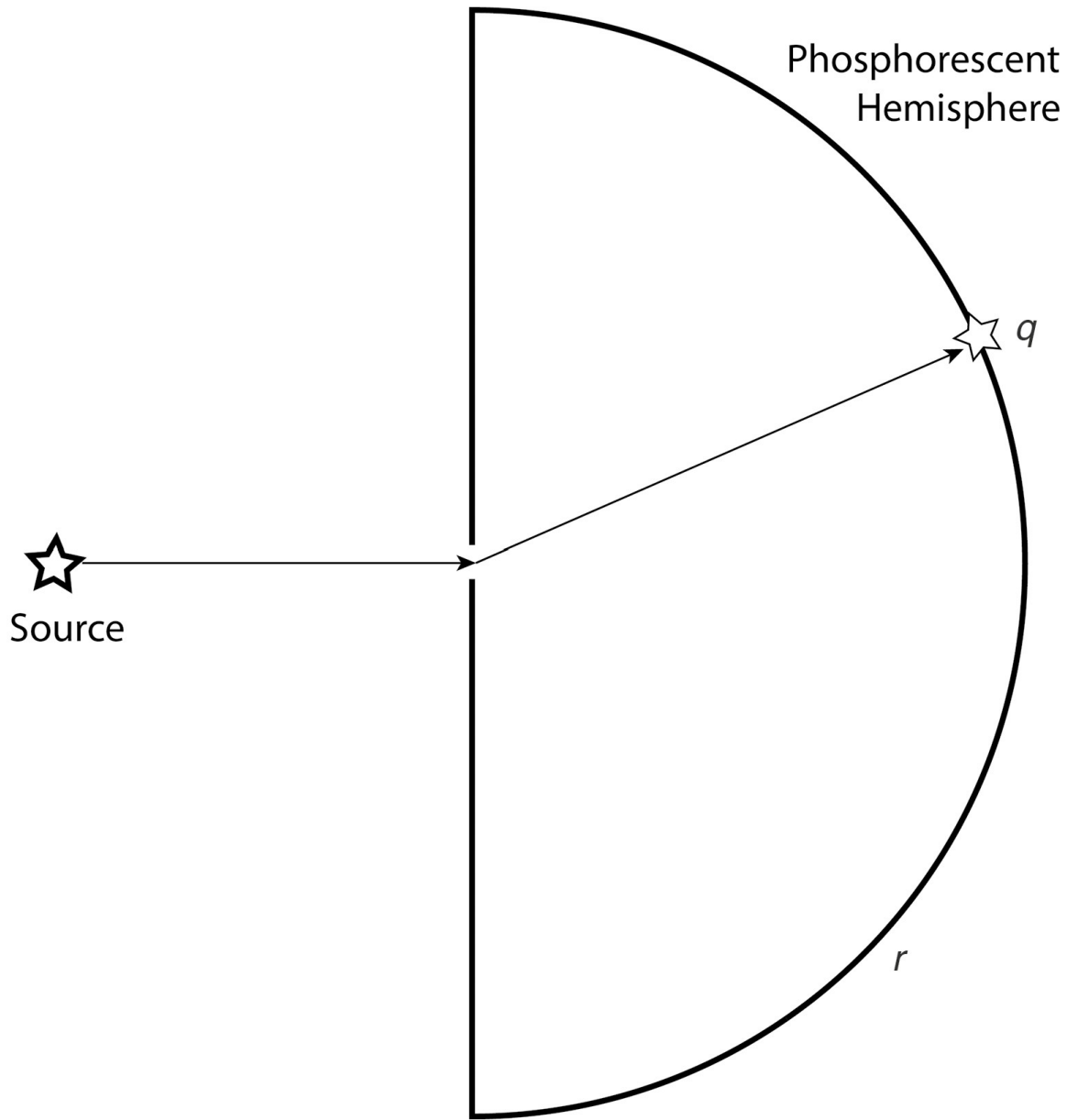


FIG. 23. An example Einstein raised at the Solvay Conference in 1927. A photon hits a point on a hemisphere coated with phosphorescent material. According to quantum mechanics, the wavefunction associated with the particle collapses randomly at q to reveal a particle. At all other points, such as r , the wavefunction collapses to not reveal a particle. How the wavefunction achieves this without “spooky action at a distance” is a mystery.

When the photon hits the sphere and causes the photoluminescent material to glow at q then suddenly the photon exists only at q and nowhere else. The

superposition no longer applies. In standard quantum mechanics, this process is described by the “collapse of the wavefunction” and is associated with what is called “measurement”: the process by which we get to know about quantum information. During collapse, the photon suddenly (and mysteriously) goes from being everywhere at once, describable by a wavefunction, to existing only at q . Put another way, the wavefunction mysteriously goes from looking like a wave to looking like a particle. According to standard quantum mechanics, the position of the point q where the wavefunction collapses to reveal a definite particle, is inherently random.

Maybe we might be prepared to accept such randomness in the laws of physics. But before so doing, we should be aware that this does not resolve the conceptual dilemma associated with this thought experiment. If the place q where the wavefunction collapses to reveal the particle, is random, so too are the other places on the sphere where the wavefunction collapses to reveal no particle. This means that all these random processes on the sphere cannot be operating independently of one another; if the processes were operating independently, then there is a chance we would see more than one flash on the photoluminescent screen. But we don't. So the random processes on the hemisphere have to be correlated with one another. How could this correlation arise in practice? Is it that, when the wavefunction randomly collapses at q to reveal a photon, quantum information is somehow sent to all the other points on the sphere telling the wavefunction at these other points not to collapse to reveal a photon, i.e., to instead collapse to reveal no photon? If so, how quickly does this apparent information propagate? Well, it would have to be instantaneous. Otherwise, there would be a chance that at the precise instant that the wavefunction at q collapses to reveal a photon, at some other point r on the sphere the wavefunction would collapse to produce a

second photon before the information from q arrives telling the wavefunction at r not to collapse in this way. Again, this can't happen because we know that the source produced only a single photon—and indeed we see only a single photon. And so, even if the hemisphere were light years in radius, the apparent quantum information telling the wavefunction at r not to collapse to reveal a photon would have to have been propagated instantaneously from q .

If quantum uncertainty was epistemic, i.e., it was our uncertainty, then this process would not be conceptually problematic. The instant we know that the particle is at q then we also know it cannot be anywhere else. If we hear that we have won the lottery and that the lottery has only one winner, then we know instantly that no one else has won the top prize. But if the wavefunction really does describe the evolving nature of quantum reality at different places in space-time—and that is the standard way of thinking—then this instantaneous propagation of quantum information poses severe conceptual problems. Indeed, if you think this just sounds completely crazy—then you are in very good company; this is what Einstein thought too.

Newton's theory of gravity (which is, on Earth at least, a very accurate theory of physics) also shows this type of instantaneous propagation, called "action at a distance". Imagine in a distant part of the universe, two neutron stars colliding with each other. How long does it take the information about the changing gravitational field of the two stars to arrive here on Earth? According to Newton's theory, it takes no time at all; the information propagates instantaneously. Newton himself realised this couldn't be right, but it took Einstein to solve the problem. In general relativity theory, gravitational information propagates at the speed of light, in the form of gravitational waves³.

Having found action-at-a-distance again appearing in quantum mechanics, Einstein concluded that there must be

something wrong with the theory, just as there was something wrong with Newton's theory of gravity. Einstein's thought example is important because many physicists think that quantum action-at-a-distance—or nonlocality, as it is now called—is a feature of having two or more interacting particles⁴. However, this is not the case. If we can understand what is happening with a single quantum particle, we will likely be able to understand what happens with more than one.

Here is what Einstein said about quantum mechanics⁵:

“The attempt to conceive the quantum-theoretical description as the complete description of the individual systems leads to unnatural theoretical interpretations, which become immediately unnecessary if one accepts the interpretation that the description refers to ensembles of systems and not to individual systems” (emphasis mine).

The question of whether it is possible to interpret the quantum wavefunction as a description of an ensemble of systems is central to the debate about whether quantum uncertainty is epistemic or ontological. According to Einstein, there exists a deeper theory of physics in which the photon does indeed just go from the source to $\mathbf{q}$. From the point of view of this deeper theory, the wavefunction simply approximates a probabilistic description of an ensemble of possible but hypothetical worlds, in one of which the photon travels to $\mathbf{q}$ in another it travels to $\mathbf{r}$ and in yet other worlds the photon travels to other points on the sphere. Until the photon hits the sphere, we humans cannot be sure which of the ensemble members corresponds to reality. In Einstein's picture, the photon itself knows where it is and to which point on the sphere it is heading. Reality is something definite, it corresponds to one of the ensemble members, and the photon knows which ensemble member corresponds to reality. In this interpretation, uncertainty in the particle's movement is fundamentally no different to

uncertainty in how the weather will evolve: we don't know how the weather will evolve, even though the weather itself (or, at least, the set of equations describing the weather) does. According to Einstein's ensemble interpretation, quantum uncertainty is all ours (epistemic uncertainty), not the particle's (ontological uncertainty).

So, there you are—straight from the horse's mouth—ensembles are the solution and quantum uncertainty is epistemic. Now you see why quantum physics fits so easily into a book focussing on ensemble forecasting for weather and climate prediction.

If Einstein was right, that quantum uncertainty reflects merely our uncertainty, then as far as the particles are concerned, their properties and behaviour must be determined by precise rules and laws. It may be that the precise forms of these rules are hidden from us, but they are not hidden from the particles themselves. A theory which assumes such a set of rules is called a "hidden-variable theory".

The first attempt at such a model was made by one of the quantum pioneers, the French physicist Louis de Broglie in the 1920s. Einstein himself had a go at developing a hidden-variable model, but abandoned the attempt when he found it was inconsistent with his theory of relativity—more about that later. The physicist best known for developing hidden-variable models of quantum physics (and in so doing, rediscovering the earlier work of de Broglie) is the American physicist David Bohm. Bohm was one of the leading theoretical physicists of the 1950s. But, having earlier written one of the standard textbooks on quantum mechanics, he became disenchanted with the theory. Bohm suffered badly at the hands of the McCarthy witch hunts of the time and was forced out of the country, eventually settling in England. However, he also suffered at the hands of his physics colleagues, who considered him as

no better than an amateur crank for questioning quantum mechanics.

As I will describe, physicists typically do not believe quantum physics can be described by hidden-variable theories. As a result, Einstein is almost universally considered to have been misguided in his ensemble interpretation of quantum mechanics. To understand why, we need to probe a little more deeply into the world of quantum physics. To do this, I will introduce a quantum experiment using what is called a Stern-Gerlach device, or SG for short, named after German physicists Otto Stern and Walther Gerlach. I'll be using SG devices for two different (but ultimately related) purposes ahead.

Again, imagine a source of particles. Instead of spitting out photons, the source spits out electrons⁶. An electron has a property usually referred to as “spin”. The spin can be measured by passing the electron through a magnet whose north and south poles are aligned in a direction chosen by the experimenter. When it passes through the magnet, the electron will be deflected either upwards or downwards relative to this chosen direction. If the electron is deflected up, we say it is “spin up” relative to the chosen direction. If it is deflected down, we say that the electron is “spin down”. In the following, SG_z denotes an SG device whose magnet is oriented in the z direction (as in [Fig. 24](#)), and SG_x denotes an SG device whose magnet is oriented at right angles in the x direction. We will suppose that the electrons are moving in the y direction—the third orthogonal direction in physical space.

We can link two or more of these SG devices together, as in [Fig. 25](#). These are called “sequential SG experiments”. Let's first analyse the double-SG experiment in [Fig. 25a](#). The first SG device in [Fig. 25a](#) is an SG_z. We take the “spin up” output from this device and feed it into a second SG device oriented in the orthogonal x direction—an SG_x.

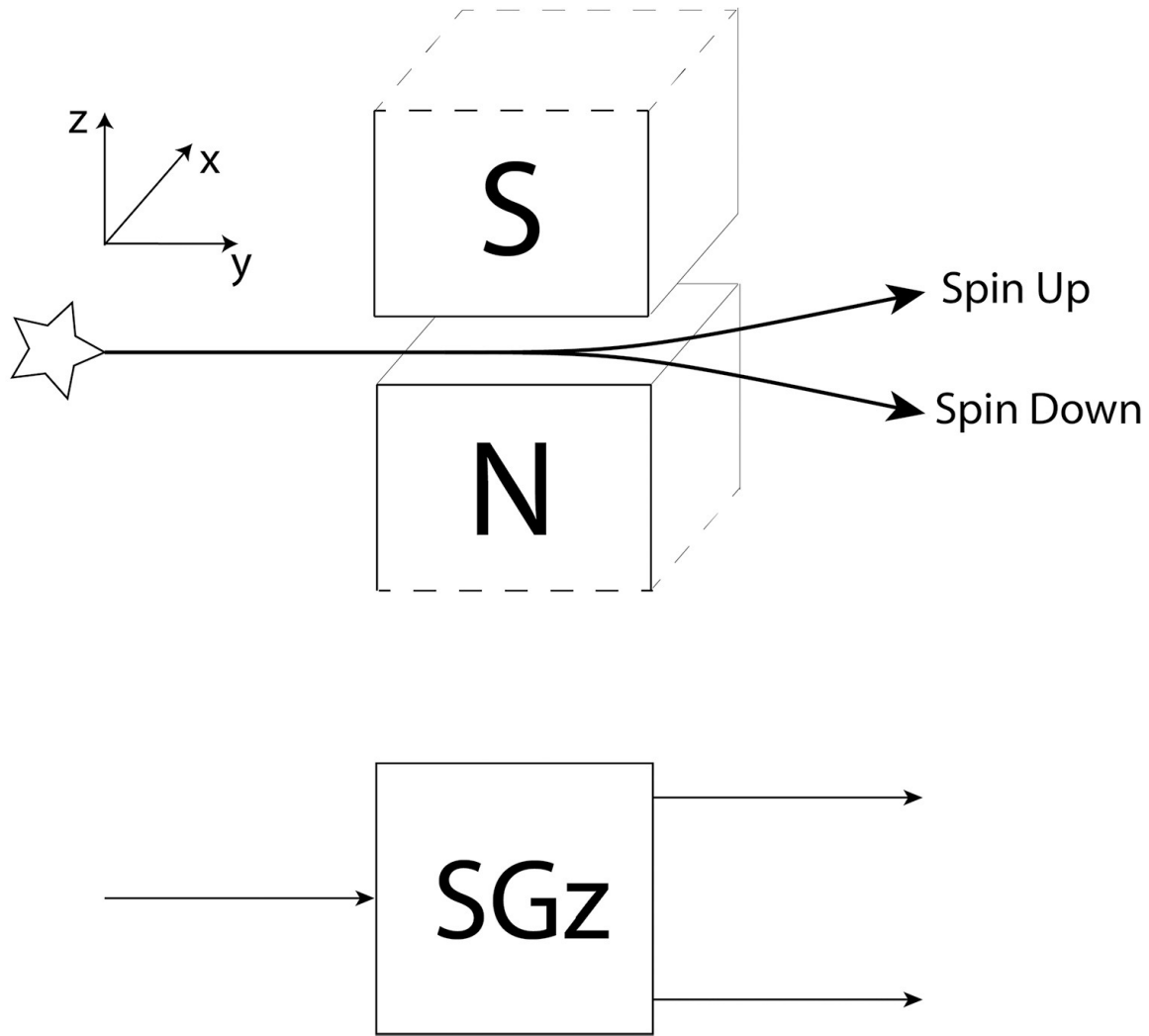


FIG. 24. Top: the Stern-Gerlach device. Here a beam of electrons passes through a magnet (with north and south poles showing) oriented in a certain direction, z and is split into two beams, traditionally referred to as “spin up” and “spin down”. Bottom: the same Stern-Gerlach device represented more schematically.

According to quantum mechanics, it is completely uncertain whether an electron will be deflected to the spin-up or the spin-down output channels of the second SGx device. All we can say is that there is a 50 percent chance of an electron being output to the spin-up output channel and a 50 percent chance of it being output to the spin-down output channel.

A hidden-variable theory of quantum physics can easily explain this behaviour. In such a theory, it is predetermined how each particle will react to being sent through magnets aligned in different directions. This implies a hypothetical “look-up table” which says, for example, that a particular particle will have an up spin when passed through a magnet oriented in the z direction, and a down spin when passed through a magnet oriented in the x direction. [Table 1](#), which has two rows and twelve columns, is an example of such a look-up table, where each column corresponds to a different electron. The first row (+ for “spin up”, – for “spin down”) shows the spin of the twelve electrons when measured in the z direction. The second row shows the spin of the twelve electrons when measured in the x direction, as implied by the hidden-variable theory. The table accounts for the results shown in [Fig. 25a](#).

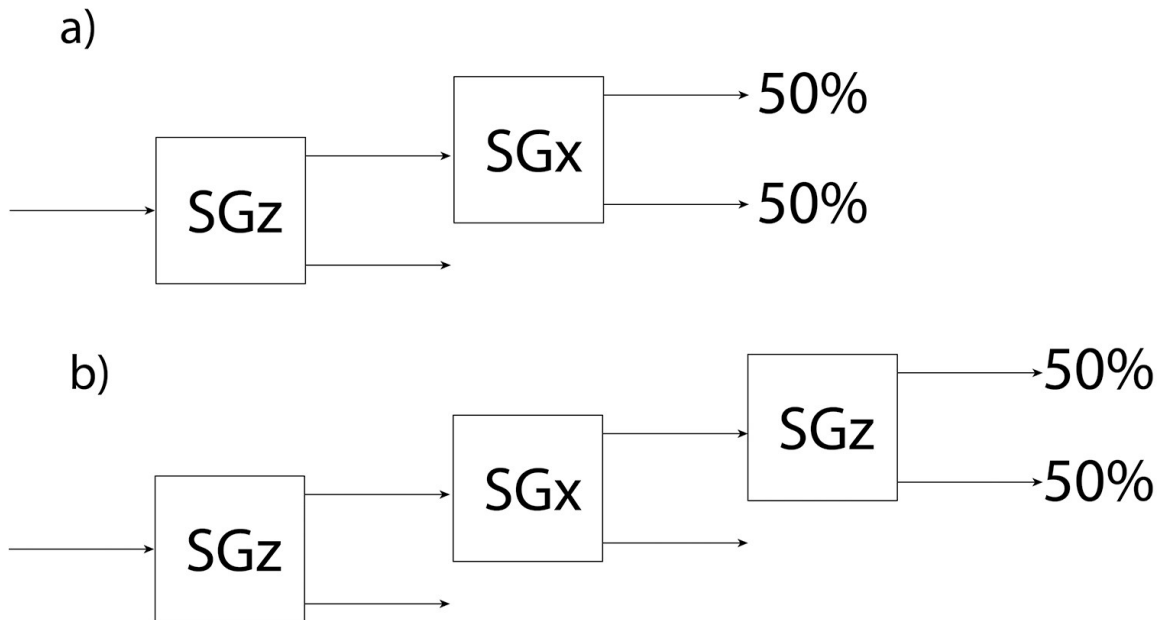


FIG. 25. Two sequential SG experiments performed on an ensemble of particles: a) SG z —SG x and b) SG z —SG x —SG z . The fraction of particles in each of the two output channels of the final device is shown. The second of these is not straightforward to explain: Since the first SG z filters out all spin-down particles, how come they reemerge from the second SG z ?

Although the hidden-variable model is able to describe the results of SG experiments with two sequential SG devices, problems appear if we couple three SG devices together. In [Fig. 25b](#) we have added a third SG device, oriented again in the z direction—an SG z . What would we expect to see coming out from this second SG z ? Well, since the first SG z device has effectively filtered out all the $-$ particles, i.e., particles which are spin down in the z direction, we would surely expect to see only spin-up particles emerging from the second SG z . But that’s not what we see. In reality, half the particles coming out of the second SG z are spin up, and half are spin down. So where on earth have these spin-down electrons come from, if the first SG z supposedly filtered them all out beforehand?

z	+	-	+	+	-	+	-	-	-	+	-	+
x	+	+	-	-	-	+	+	+	-	+	-	-

TABLE 1. The twelve columns refer to twelve different electrons, showing part of a hypothetical “look-up” table giving the spins of these twelve particles for two different possible orientations of the SG magnet. The table is a consequence of assuming that quantum physics can be explained by a deterministic hidden-variable theory. For each particle, the first row gives the result of a possible SG measurement in the z direction. The second row represents the result of a possible SG measurement in the x direction. The table is consistent with the probabilities shown in [Fig. 25a](#). In particular, taking all columns which are $+$ in the first row, half are $+$ in the second row and half are $-$.

Explaining the triple-SG experiment is problematic for a hidden-variable model, though it’s possible to fudge some sort of explanation⁷. However, there is another experiment, which can also be described using SG devices, that is generally considered to provide the killer blow to hidden-variable theories and, with it, Einstein’s ensemble interpretation of quantum physics. This is the so-called Bell experiment. I am going to describe the Bell experiment next and return to it in [Chapter 11](#). In this experiment, the

notion of nonlocality rears its ugly head in a way which seems even more unavoidable than in Einstein's thought experiment shown in [Fig. 23](#). By contrast, the triple-SG experiment shown in [Fig. 25b](#) seems to have nothing to do with nonlocality. Despite this, in [Chapter 11](#), I will try to explain how both the Bell and triple-SG experiments can be understood using ensembles in exactly the same way

The Bell experiment makes use of the phenomenon of "entanglement", a word coined by Schrödinger to describe interacting quantum systems. The importance of entanglement for testing hidden-variable theories experimentally was realised in the 1960s by Irish physicist John Bell⁸, whose day job was as a high-energy physicist at CERN⁹. Bell showed that under seemingly unexceptionable assumptions, hidden-variable theories that do not have action-at-a-distance must satisfy a certain statistical inequality that is violated in quantum mechanics. At the time Bell published his paper, no one had tested whether the inequality was violated experimentally. However, there was very little doubt that if quantum mechanics predicted that the inequality was violated, then it would indeed be violated in reality.

To understand Bell's inequality, we construct another table like [Table 1](#), made of +s and -s. This time, however, we have three rows instead of two (see [Table 2](#)). As with [Table 1](#), I have used twelve columns by way of illustration, though in principle we can have as many columns as we like. There is nothing special about the way I have filled the boxes of [Table 2](#) with +s and -s. You can choose your own entries if you want. In fact, for the time being, just imagine the table as something without any particular connection to physics.

Now count the number of columns where there is a + in the first row and a + in the second row, and call this A . From [Table 2](#), $A = 2$ (corresponding to the fourth and

eighth columns). Now count the number of columns where there is a $-$ in the second row and a $+$ in the third row, and call this B . From [Table 2](#), $B = 6$. Finally, count the number of columns where there is a $+$ in the first row and a $+$ in the third row, and call this C . From [Table 2](#), $C = 5$. In this example, $A + B = 8$ is clearly bigger than $C = 5$.

As Bell showed in 1964 [10](#), $A + B$ will always be bigger than or equal to C , no matter how we fill in the entries to this table with $+$ s and $-$ s. This result is known as Bell's inequality. I give a proof of Bell's inequality using elementary mathematics in the endnotes [11](#).

1	+	-	-	+	+	+	-	+	+	+	+	+
2	-	+	-	+	-	-	-	+	-	-	-	-
3	-	+	-	-	+	-	+	-	+	+	+	+

TABLE 2. We fill $+$ s and $-$ s into a table with three rows and as many columns as we like (here twelve). The statistics of these $+$ s and $-$ s necessarily satisfy a mathematical inequality, known as Bell's inequality, described in the text.

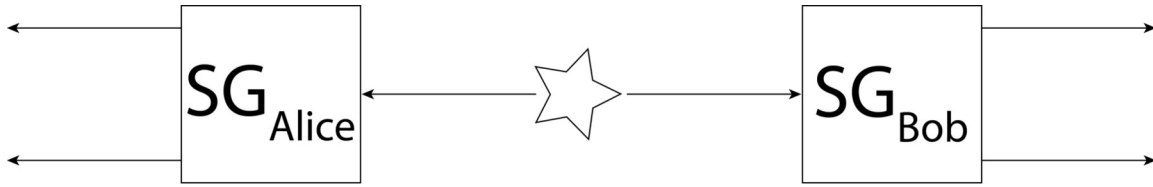


FIG. 26. The famous Bell experiment, using pairs of entangled particles moving in opposite directions. In this experiment, Alice's and Bob's SG devices can be oriented arbitrarily. However, if they are oriented in the same direction and Alice measures her particle to be spin up, Bob will measure his particle to be spin down. In practice, the experiment is typically performed with photons and polarisers instead of electrons and SG devices.

Bell applied his eponymous inequality to a situation where a source emits pairs of entangled particles, say electrons¹², moving in opposite directions, as shown in [Fig. 26](#). Here the particles are sent to two SG devices which measure the particles' spins. Each SG device is oriented by the experimenter. Traditionally, "Alice" sets the left-hand SG device and hence measures the spin of the left-moving particle, and "Bob" sets the right-hand SG device and measures the spin of the right-moving particle. As before, we will denote spin up by a + symbol and spin down by a - symbol. But now we imagine that each SG device can be oriented in any direction Alice or Bob chooses and not just the z and x directions.

A crucially important property of such entangled particle pairs is that if Alice's and Bob's SG devices are oriented in the same direction, and if one device outputs a +, then the other device, measuring the spin of the other entangled particle, will output a -, or vice versa. That itself is not a surprising fact and can easily be explained with hidden variables¹³. It relates to the fact that the combined spin of the two particles leaving the source must always be zero.

If the particles' spins are determined by hidden variables, then we can imagine Alice's and Bob's two sets of particles each having a look-up table. An example of such

a look-up table was in fact shown in [Table 2](#). We can imagine it's a look-up table for twelve of Alice's particles, where the "1", "2" and "3" that label the rows of the table correspond to three possible orientations for Alice's SG device. Bob's look-up table for the twelve particles that are entangled with Alice's, and for the same three orientations, would be the same as Alice's look-up table except that each + in Alice's look-up table would be a - in Bob's, and each - a +. In this way, the mathematical inequality—that $A + B$ is greater than or equal to C —becomes an intrinsic property of this type of hidden-variable theory of quantum physics.

The question that Bell asked himself was this: Can we test experimentally the hidden-variable model prediction that $A + B$ is greater than or equal to C , in Alice's (or Bob's) look-up table? It turns out that there is a way, using real-world entangled particle pairs. For example, as mentioned, the quantity A describes the number of columns where there is a + in the first row and a + in the second row of Alice's look-up table. We can estimate A experimentally by measuring the spins of an ensemble of entangled particles, where for each pair, Alice measures her particle in the 1 direction and Bob measures his in the 2 direction. We now use the "zero-spin" property to infer that if Bob's particle was measured to be a +, then Alice's particle, if it had been measured in the 2 direction, would necessarily have given a - outcome, and vice versa. In this way we can estimate the quantity A in Alice's look-up table from an experiment on an ensemble of entangled particle pairs using measurements that Alice and Bob actually make. We can similarly estimate B and C on separate ensembles of entangled particle pairs.

So which theory is correct in reality: the conceptually simple deterministic ensemble-based hidden-variable model, where $A + B$ is necessarily greater than or equal to C , or quantum theory, where the inequality can be violated? The exact experiment that is performed is a little more

complex than the one suggested and tests a variation of Bell's inequality known as the CHSH [14](#) inequality. However, the answer is not good for hidden variables, and hence for Einstein's ensemble interpretation of quantum physics. Bell's inequality is violated in practice [15](#), exactly as predicted by quantum mechanics.

So, what is going wrong with our hidden-variable model in failing to predict the experimental violation of Bell's inequality? There's something wrong with this assumption that Alice and Bob's particles each have their own separate look-up tables. But what is wrong?

One conclusion is that the concept of the look-up table doesn't work because there's no such thing as definite reality in quantum physics. That's what most physicists believe (even though it's really hard to know what this means physically), and that's the reason why most physicists believe Einstein was misguided to think that uncertainty in quantum physics can be explained by ensembles.

It's important to understand that this is not the same as saying that quantum physics contains some randomness. As discussed in Einstein's thought experiment, one needs also to posit some kind of action-at-a-distance to ensure that the randomness is properly correlated over space. It's the same with Bell's inequality. One can posit that the hidden-variable model contains some stochastic terms which make the spin outcomes defined in only a probabilistic sense. However, such a stochastic hidden-variable model is still constrained by Bell's inequality (essentially because the quantities A , B and C are statistical quantities defined from the statistics of the look-up tables). Randomness alone does not explain the experimental violation of Bell's inequality.

A different way to invalidate the existence of separate look-up tables for Alice's and Bob's particles would be to assume that reality is definite, but that the spins of Alice's

electrons are determined not only by the orientation of Alice's SG device but also by the orientation of Bob's SG device. This would mean giving up on the idea that Alice's and Bob's particles have their own separate look-up tables, as in [Table 2](#). This is what physicists typically mean by the word "nonlocality". But it is a truly horrible idea, in my view. If Bob made up his mind how to set his SG device only just as his electron was arriving, his decision would have to somehow be communicated instantaneously to Alice. It's the same old action-at-a-distance idea which we saw earlier and which Einstein—rightly in my view—hated so much.

On the other hand, nonlocality is a subtle matter. As with Einstein's thought experiment, we cannot use nonlocality to send classical information instantaneously from Alice to Bob. For example, Bob cannot communicate the result of some horse race superluminally to Alice using these entangled particles. If he could do this, then the hidden-variable theory would explicitly violate a basic property of causality in relativity theory: that causes always precede effects. If this idea is violated, then Alice and Bob could make certain money at the bookies.

Because it's not possible to signal superluminally with these types of experiments, most physicists take the view that understanding what is going on in Bell's experiment is not something to lose sleep over, e.g., when doing low-temperature experiments or quantum computing.

But for fundamental physics I believe this is something we should lose sleep over. Even if quantum nonlocality does not allow superluminal signalling of classical information, this superluminal communication of quantum information goes against the spirit of relativity theory. Perhaps our failure to understand quantum nonlocality is the primary reason why we are still struggling to unify quantum mechanics and general relativity theory into a single unified theory of physics—as the Penrose quote at the beginning of Part III of this book suggests.

So, does the experimental violation of Bell's inequality require us to either abandon our intuitive concept of reality or accept spooky action-at-a-distance? If pressed hard, most physicists will admit that there is likely something deeper going on, which we have yet to understand. In [Chapter 11](#), I will suggest there may be another way to explain both Bell's experiment of [Fig. 26](#) and the sequential SG experiment of [Fig. 25b](#). The explanation is consistent with Penrose's speculation that Gödel/Turing undecidability may play a key role in any future theory of quantum gravity. In particular, the explanation requires us to exploit the geometry of chaos. If the explanation is correct—without clear-cut experimental tests I cannot be certain—then it will be possible to describe the mysteries of quantum physics using Einstein's ensemble interpretation. In particular, God will not play dice, there will be no action at a distance, quantum uncertainty will be epistemic and reality something definite. In claiming this, I am not proposing a return to pre-quantum classical physics: as mentioned at the beginning of the chapter, we should not think of the algorithmically undecidable geometry of chaos as classical.

However, before developing these ideas, we have more practical matters to discuss.

PART II

Predicting Our Chaotic World

I will give you a talisman. Whenever you are in doubt, or when the self becomes too much with you, apply the following test. Recall the face of the poorest and the weakest man whom you may have seen, and ask yourself if the step you contemplate is going to be of any use to him. Will he gain anything by it? Will it restore him to a control over his own life and destiny?

ONE OF THE LAST NOTES LEFT BY MAHATMA GANDHI¹

The guidelines, which May described as “the most important thing I did”, set out three principles for providing scientific advice to government: fully acknowledge uncertainties (sometimes the answer is “we don’t know”); seek a wide range of views (it is rare that the scientific and other communities totally agree on the evidence); be totally transparent about the process and outcome of advice. These principles are simple, but as salient today as when they were first published.

FROM THE ROYAL SOCIETY’S BIOGRAPHICAL MEMOIR OF LORD ROBERT MAY, PIONEER OF CHAOS THEORY AND UK GOVERNMENT CHIEF SCIENTIFIC ADVISER²

We apply the ideas discussed in Part I to develop practical tools for predicting complex, inherently uncertain systems.

The key idea here is ensemble prediction: running our models multiple times while varying uncertain initial conditions and model equations. When the spread of a reliable ensemble system is small, we can make fairly precise forecasts with confidence. By contrast, when the spread is large, we can make forecasts only using the language of probability. In this way, the geometry of chaos is manifest in the variable spread of the ensemble. Ensemble prediction methods are applied to weather and climate (where the techniques are reasonably mature and well validated), and to disease, economics and conflict (where techniques are still in development). Since we are dealing with extremely complex systems, the best models used for making ensemble predictions are, consistent with ideas discussed in Part I, inherently noisy. We show that the value for a forecast user of a reliable noisy ensemble prediction system can greatly exceed that of a more precise but less reliable best-guess deterministic forecast.



5

The Two Roads to Monte Carlo

THE PREVIOUS FOUR CHAPTERS HAVE BEEN QUITE INTENSE, developing concepts which underpin our understanding of uncertainty in the world around us. It's now time to apply this understanding to matters of practical concern—to efforts to predict our chaotic world.

I need to issue a caveat. I am not an expert in some of the topics discussed in this part of the book, notably disease, economic and conflict prediction. Nevertheless, I have taken the liberty of expressing my views about what I see as the state of play in these areas, based on my experience in the areas of predictive science where I do have some in-depth technical knowledge. I expect some experts will be rather exasperated with my meddling in this way. I can fully understand this feeling as I myself have been at the butt end of criticism from those expressing their views to me about the inadequacy of climate science. In truth, it is good to have multiple views from different fields about a topic, and in my experience progress is often made by taking existing ideas from one field and applying them to another.

THE EMERGENCE OF MODERN science, sparked by the scientific renaissance, seemed to have passed by the dark art of weather prediction. The rather desperate state of affairs coming into the twentieth century was well described in Thomas Hardy's novel *The Mayor of Casterbridge* published in 1886. The novel's protagonist, a hay and grain merchant, has fallen on hard times. In an attempt to regain some of his lost wealth, he decides to consult a "weather prophet" about the harvest weather. The self-proclaimed prophet tells him:

"By the sun, moon and stars, by the clouds, the winds, the trees, and grass, the candle-flame and swallows, the smell of the herbs; likewise by the cats' eyes, the ravens, the leeches, the spiders, and the dungmixin, the last fortnight in August will be—rain and tempest."

Unfortunately for the grain merchant, the forecast turns out wrong, and he is ruined.

In fact, just a few years before Hardy's book was published, the first attempts to make weather prediction more scientific had begun. This development was spurred by a tragic event. On October 25, 1859, the steam clipper the *Royal Charter* was on the last leg of its sixty-day voyage from Melbourne to Liverpool, having crossed the Pacific and navigated the infamous Cape Horn. There were four hundred passengers aboard, many bringing the fortunes they had made in Australia back home to Britain. As they passed Anglesey, an exceptionally violent storm developed. The captain of the *Royal Charter* could not control her and she was thrown against the rocks of an Anglesey headland. Only forty-one passengers survived; none were women or children. Victorian Britain was traumatised.

As others were wringing their hands, one man thought he could do something to prevent such a loss happening again. Robert Fitzroy was a naval officer who had made a name for himself captaining the ship, the *HMS Beagle* that carried Charles Darwin on his epic voyage to the

Galápagos, inspiring him to the theory of evolution by natural selection that would revolutionise biology.

Four years before the Royal Charter disaster, Fitzroy had been put in charge of the Meteorological Office. In those days, the Meteorological Office did not make forecasts. Indeed, the word “forecast” did not exist—Fitzroy would later invent the word. But Fitzroy had a brilliant idea. Using the electric telegraph invented a few years earlier, he realised that if all of the coastal stations taking weather readings could relay their information to a central office, then Fitzroy, with his expert meteorological knowledge, could analyse the information to create a spatial picture of the current weather patterns, and with that picture make a prediction of the expected weather for the coming days.

The first storm warning was made on February 6, 1861. It was a success and The Times congratulated Fitzroy for his pioneering work. For the next few years, things generally went well. The number of shipwrecks dropped and the media and public alike were impressed by Fitzroy’s forecasts. However, a number of inaccurate predictions began to be made. As a result, questions were asked about the expense of Fitzroy’s telegraphic network. And finally, some of his scientific colleagues began complaining that there was no scientific basis to the method Fitzroy used to make his predictions. By 1865 Fitzroy’s health was beginning to fail him and he ran out of energy and enthusiasm. He sank into one of the black moods he would suffer from time to time. On Sunday, April 30, 1865, just before he was due to go to church, he pulled out a razor and cut his throat.

Fitzroy had correctly identified one key aspect of how to make skilful weather forecasts: the need for plentiful observations, not just at one location but across a geographical domain. In modern parlance, he realised the importance of accurate “initial conditions”. However, knowing the initial conditions was only part of the solution.

AROUND THE SAME TIME, in a part of the British Empire far from the stormy coasts of Britain, meteorology was taking another step forward.

As winter turns to spring, the sun melts much of the snow that has fallen over the Asian land mass, the Tibetan plateau in particular. When the snows have largely melted, the land warms and surface temperatures begin to rise. To the south, the Indian Ocean also feels the changing energy of the sun, but here the sun's energy evaporates moisture from the sea surface and the sea warms less than does the land¹. As a result, a strong surface temperature gradient develops between the Indian Ocean and the land mass to the north. The gradient is the reverse of what one normally finds, where low latitudes are warmer than higher latitudes. In turn, a vast current of moist air starts to flow north over the Indian subcontinent—a sea breeze of massive proportions. As the air is forced up over the Western Ghats mountains of India, the air cools and releases its moisture through torrents of rain. The summer monsoon has arrived. Many give thanks as the stultifying heat and humidity of the early summer are finally broken. Under the plentiful rainfall, the crops grow, and the food sustains the inhabitants of the Indian subcontinent for another year.

And yet, the chaos in the climate system ensures that the strength of the monsoon is not the same year in, year out. Some years the monsoon fails altogether, bringing drought and misery to the population.

In the 1870s, there were a series of monsoon failures. Exacerbated by British imperial policy, the drought led to famine and death. The British authorities looked to their scientists to see if these monsoon failures could be

predicted. In 1882, the India Meteorological Department, established a few years earlier and under the control of its first director, Henry Blanford, began to make the first monsoon forecasts for India. Unlike a weather forecast for a few days ahead, this was an attempt at a long-range seasonal prediction for rainfall over all of India averaged over the summer months of June through September, and made before the summer monsoon actually broke.

Blanford was convinced that the key to successful prediction of the monsoon lay in estimating the amount of snowfall that fell over the Tibetan Plateau during the preceding winter: the less snow that falls during the winter, the less there is to melt in the spring, and the more the sun can warm the plateau. From this, the larger will be the temperature difference between the continental land mass and the more constant ocean temperatures to the south, and the stronger will be the Asian summer monsoon.

Blanford used observations of the amount of snow on the Himalayan mountain peaks as an indicator of more general snow cover over the Tibetan Plateau to the north. Like Robert Fitzroy's early predictions back in Great Britain, Blanford's first predictions were reasonably successful. However, after a series of forecast failures, it became apparent that there must be other things going on.

Things took a big step forward when in 1904 Gilbert Walker became the third director of the India Meteorological Department.

Walker attempted to improve the long-range forecasting techniques of his predecessors by correlating monsoon strength with observations of atmospheric surface pressure from weather stations around the world. He discovered that the strength of the Indian monsoon tended to be correlated with the difference in surface-pressure observations taken at Tahiti and Darwin, Australia, some months earlier. A barometer effectively measures the mass of air in a column above the instrument. Systematic

variations in mass between one location and another are an indication of some large-scale circulation in the atmosphere. From this, Walker deduced that the strength of the Indian monsoon was correlated with, and could be predicted by, indicators of some large-scale atmospheric circulation over the tropical Pacific and Indian Ocean basins—a circulation we now call the Walker circulation.

Walker referred to the variations in the pressure difference between Tahiti and Darwin as the “Southern Oscillation”—in contrast with another pressure oscillation he had discovered involving stations in the North Atlantic region, which he called the “North Atlantic Oscillation”. But what caused these so-called oscillations? Walker had no idea.

From Fitzroy to Walker, Victorian meteorologists had identified that observations of the weather are vital to making good forecasts. But by themselves, they aren’t enough.

« »

AROUND THE TURN OF the twentieth century, two meteorologists, the American Cleveland Abbe and the Norwegian Vilhelm Bjerknes, who incidentally studied under Poincaré, independently suggested that we could make much better forecasts if we put the laws of physics to work on these observations. Specifically, they suggested that we should try to solve the Navier-Stokes equation (amongst others) as discussed in [Chapter 3](#). They proposed that weather forecasting should be seen as a scientific initial-value problem: we use the laws of physics to determine the future state of the atmosphere, given the initial conditions from today.

The first person generally credited with trying to make a forecast using these ideas was Lewis Fry Richardson [2](#). Richardson was born in 1881 into a Quaker family, and this instilled in him a deep belief in pacifism that strongly shaped his life. He graduated from Cambridge in 1903, and worked at a number of places including, just before the outbreak of World War I, the Meteorological Office (now called the Met Office). When war broke out, he refused to fight, resigned from the Met Office and instead served in the Friends' Ambulance Unit near the front. In free time during this period, he performed his pioneering weather-forecast calculations.

In practice, Richardson wasn't making a forecast at all, because the computations of how the weather would evolve over a few hours took him many months of hard work. However, the method he developed to solve the Navier-Stokes equation is still, broadly speaking, the way it is done today. Richardson's method was described in [Chapter 3](#), and is based on dividing the atmosphere up into a number of gridboxes, inside which one assumes the atmosphere is uniform and unchanging. After weeks of tedious computations, Richardson finally reached the endpoint of his forecast calculation: an estimate of the change in surface pressure over six hours for one of the gridboxes. It was a disaster. His estimate was over one hundred times bigger than occurred in reality.

The error Richardson introduced, which led to such a disastrously wrong forecast, was in fact a subtle one. The problem wasn't the way the forecast calculations were made. Rather it lay in how the initial conditions were created from the limited observations available. Relatively recently, Irish meteorologist Peter Lynch developed an algorithm to improve the way in which the available observations could be used to create initial conditions for the model gridboxes. After applying this algorithm, Lynch

showed that Richardson's retrospective forecast became remarkably accurate after all³.

The more serious problem with Richardson's method was that the calculations were proceeding much more slowly than the weather was itself advancing. Hence in practice the method was useless. Richardson dreamt of a future weather forecast factory where thousands of human "computers" toiled away doing mindless arithmetic calculations at breakneck speed, in an attempt to beat the weather to it. This vision was remarkably prescient if instead of a human computer, we think of an individual processing unit inside a massively parallel electronic supercomputer.

It took the end of World War II to realise Richardson's dream, thanks indeed to the development of the first electronic digital computers. The mathematician and physicist John von Neumann, based at Princeton University, put together a team of meteorologists led by Jule Charney. Since these models are based on laws of physics such as the Navier-Stokes equation, I will henceforth call them "physics based", to distinguish them from statistical-empirical models (sometimes called data-driven models) of the type that Fitzroy, Blanford and Walker developed. To run their physics-based models, Charney's team used the first fully programmable electronic digital computer, known as ENIAC, which stood for "electronic numerical integrator and computer".

Weather forecasting wasn't ENIAC's main use. Although it was originally conceived as a tool to help in the computation of artillery look-up tables, one of ENIAC's main uses was for hydrogen bomb computations⁴. Here von Neumann and his colleague Stanislaw Ulam were major users of the computer.

These bomb calculations turned out to be hard because the hydrogen bomb is in part a classical device, describable

using classical laws of physics, and partly a quantum device—after all, the basic release of energy is due to the fusion of nuclei of the hydrogen atom. As we saw in [Chapter 4](#), quantum mechanics deals in abstract quantities, which under the right circumstances can be interpreted as probabilities.

Ulam was only too aware how hard it was to make calculations when some of the basic variables were probabilistic and others not. When recuperating from an illness, he began playing a game of solitaire, which he lost many more times than he won. He began to think about how one might calculate from first principles the probability of winning this game. Like his H-bomb calculations, this was a difficult mathematical problem. However, in the case of solitaire there was an alternative—you could simply play the game enough times and count the fraction of times that you win. That gives you the probability you are looking for.

This was a eureka moment for Ulam. What he realised was that instead of treating the diffusion of neutrons in terms of probabilities, he could use pseudorandom-number generators (PRNGs) to create an ensemble of possible deterministic paths for the neutrons, get ENIAC to do the computations separately for each of these deterministic paths, and then finally average the results from all these computations. This was so much simpler than dealing with probabilities explicitly. Essentially the idea was to let the computer do the hard work—ENIAC wouldn't mind repeating, over and over, calculations for how the neutrons might diffuse.

This new method was considered so important that Ulam and von Neumann decided they needed to give it some kind of secret name, and the code name “Monte Carlo” emerged, based on the casino in Monaco where Ulam's uncle had gambled.

Here is a simple example of the Monte Carlo method at work. Suppose you want to estimate the value of $\pi / 4$ without using a calculator. The area of a circle is $(\pi / 4) \times D^2$ where D is its diameter. The Monte Carlo way to estimate π would be to take a sheet of paper and draw on it a square (of area D^2) which just contains the circle (see [Fig. 27](#)). You then randomly scatter a number of small seeds on the sheet of paper. It is the random scattering of small seeds that is the repeated operation mentioned earlier. Now count the total number N_1 , of seeds that lie inside the square. Of these count the number, N_2 , that also lie within the circle. If indeed the seeds are randomly distributed inside the square, then the ratio N_2 / N_1 will get closer and closer to $\pi / 4$ the greater the number of seeds that are scattered on the sheet of paper.

The circumstances of Ulam's eureka moment are similar to those of many other eminent scientists'. The moments occurred when they were relaxing—playing a game of solitaire in Ulam's case. Why do eureka moments seem to occur during moments of relaxation? I will have some things to say about this later in the book.

The methods of Ulam and von Neumann would have been known to Chuck Leith, an American physicist and mathematician who also worked on the hydrogen bomb after the war. However, in the 1950s, Leith decided to change direction and work in the newly developing field of climate modelling. With his knowledge of Ulam and von Neumann's work, Leith wrote a paper in 1974 suggesting that the Monte Carlo method could also be applied to weather forecasting. Leith proposed that instead of running a weather forecast model once, you run it several times from slightly different initial conditions, each initial condition being consistent with the slightly imperfect observations you have at the initial time. Like the Monte

Carlo simulations for the bomb, you then average all the forecasts together.

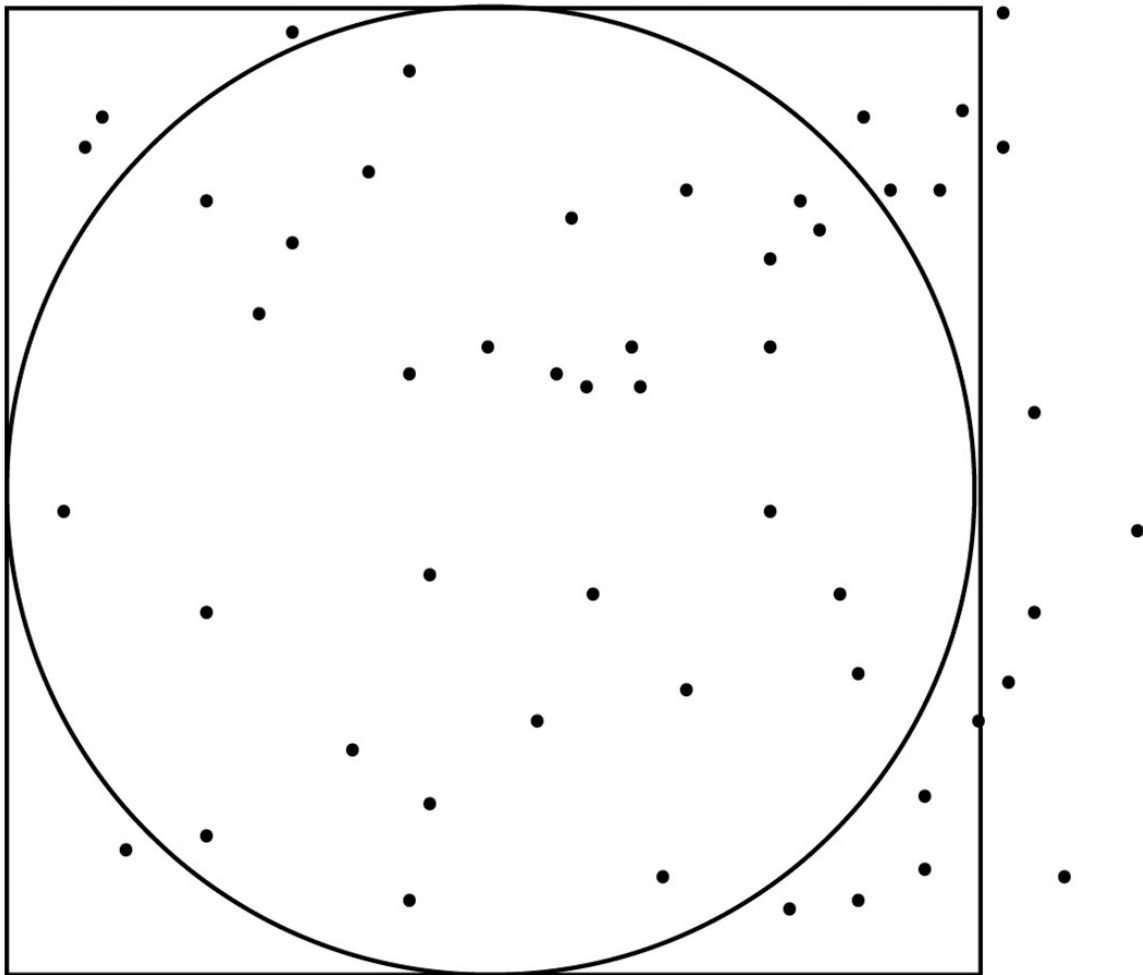


FIG. 27. An illustration of the Monte Carlo method. Here we randomly scatter small seeds on a sheet of paper on which a square and the largest circle which fits inside the square have been drawn. The ratio of the total number of seeds lying inside the circle to those lying inside the square provides an estimate of the (irrational) number $\pi / 4$.

Suppose a particular low-pressure system is especially unpredictable. Then whilst one member of a Monte Carlo forecast might predict a drop in the pressure system, another member might predict an increase in pressure in the same spot. By averaging the two forecasts together, the averaged prediction would be for no change in pressure at all, neither positive nor negative. In this way the averaged

Monte Carlo forecast would filter out the unpredictable parts of the weather, leaving only the predictable part: the part common to all Monte Carlo members. Leith estimated that in order to get the most benefit from such a Monte Carlo forecast, you would only need to average together about ten forecasts.

The idea that averaging together many individually uncertain forecasts can lead to improved predictions wasn't itself new. In 1906, British statistician Francis Galton observed a phenomenon now referred to as the "wisdom of the crowd". At a country fair, eight hundred people took part in a contest to guess the weight of an ox. Galton noted that the average estimate from the eight hundred uncertain guesses was within 1 percent of the true weight. In another example, in the 1980s, finance professor Jack Treynor famously asked his fifty-six students to guess the number of jellybeans in a jar. The actual number was 850. Only one student did better than the average guess of 871, an error of 2 percent.

If only von Neumann and Ulam had talked more to Charney and his team at Princeton, Monte Carlo weather prediction could have first happened on ENIAC. But the Monte Carlo methods were secret and so they didn't talk. It would take another forty years for this synthesis to occur.

Instead, the early weather forecast models attempted to forecast tomorrow's weather based on a single deterministic run of the forecast model. It would take several decades before the operational weather forecasters came to rely fully on these forecast models and completely abandon their own empirical rules of thumb for weather prediction.

As these computer forecasts using physics-based models started to become more and more successful, meteorologists began to wonder how far ahead they could forecast the day-to-day evolution of weather. Certainly, the 1963 paper of Ed Lorenz, discussed in [Chapter 1](#), had made

it clear one could not predict indefinitely far ahead. However, by the end of the 1960s, there was a general feeling, backed up by some research, that the day-to-day evolution of weather should be predictable to about two weeks ahead, consistent with the analysis I described in [Chapter 3](#). Two weeks became known as the “limit of deterministic predictability”.

In 1975, the countries of Europe got together to pool their human and computing resources and create the European Centre for Medium-Range Weather Forecasts (ECMWF). The primary purpose of this new organisation, located in Reading in southern England, was to realise a forecast system that could predict the detailed evolution of weather up to the two-week limit of deterministic predictability.

In the early days of ENIAC, when forecasts were made only one day ahead and for a specific region of the world—the United States—it was not necessary to model weather in the tropics or in the southern hemisphere. Hence the early physics-based models had regional domains with lateral boundaries. However, when forecasting two weeks ahead, the weather anywhere on the globe can affect the weather anywhere else. It became clear that to realise the goal of forecasting to two weeks ahead, physics-based models with a global domain had to be developed.

In setting up ECMWF, the European countries rightly realised that there was no point in each member state developing its own global model, and so the effort focussed on developing a single global forecast system which would benefit all the citizens of Europe equally. From the day it started producing forecasts, ECMWF became de facto the most skilful centre in the world at so-called medium-range weather prediction. This forecast system now benefits all the citizens of the world, especially those at risk of the most extreme forms of weather.

IN THE 1960S, WILHELM Bjerknes's son, Jacob, started to ponder Gilbert Walker's finding that the strength of the Indian monsoon could be forecast knowing the Southern Oscillation index (the surface pressure difference between Tahiti and Darwin, Australia). Bjerknes junior had emigrated in 1940 to the US, where he founded the Department of Meteorology at the University of California, Los Angeles. There, he asked himself a question: What makes the Southern Oscillation vary from year to year? He began to realise that the answer depended on an oceanic phenomenon known for many centuries to Peruvian fishermen.

Under normal circumstances, sea temperatures along the equatorial Pacific Ocean are relatively warm in the west and relatively cold in the east. This is a consequence of the trade winds that blow from east to west across the equatorial Pacific Ocean. The winds blowing over the ocean near the South American coast cause cold water to upwell from the deeper parts of the ocean. However, the fishermen knew that in some years this normal state of affairs is completely disrupted—again the chaotic variability of climate at work. When these disruptions occur, the trade winds and the cold upwelling in the eastern Pacific weaken and the water warms. This is bad news for the fishermen, since the upwelled waters carry nutrients such as nitrates and phosphates, the product of the decomposition of sinking organic matter. So when the upwelling ceases, the source of nutrients is shut off and the fish either die or move elsewhere. When these disruptions occurred, they tended to be first noticed around Christmastime and so were referred to as El Niño events—"El Niño" meaning "the Christ Child"⁵.

Jacob Bjerknes realised that these occasional disruptions to sea temperatures in the equatorial eastern Pacific are intimately related to variations in Walker's Southern Oscillation. Bjerknes postulated a dynamical process involving feedbacks between the tropical Pacific Ocean and the atmosphere above it to explain both the El Niño event and the Southern Oscillation. Thanks to Bjerknes's work, we now inexorably link El Niño to the Southern Oscillation through the joint acronym ENSO (El Niño/Southern Oscillation). Indeed, in part due to Bjerknes, we now think of the atmosphere and oceans as components of a single, dynamically coupled entity.

In the 1970s, it began to be realised that when an ENSO event occurs, the effects not only influence the Indian monsoon, they can also be broadcast around the globe through what are called atmospheric teleconnections⁶. Indeed, it turns out that through these teleconnections, El Niño can also influence Gilbert Walker's North Atlantic Oscillation and hence the weather of Europe⁷.

In this way, the interaction between the atmosphere and the underlying ocean (and to a lesser extent land) provides a source of predictability for the atmosphere on timescales much longer than the limit of deterministic weather predictability of two weeks.

Does the possible existence of such long-range predictability conflict with the butterfly effect discussed in [Chapter 3](#)? No. ENSO changes the statistical characteristics of weather during a season. For example, knowing ENSO, we can make a prediction of the likely number of Atlantic hurricanes in a season, and their typical intensity and track, even though the intensity and track of any single Atlantic hurricane may be rather unpredictable more than a few days ahead.

By the end of the 1970s, a kind of meteorological schism had arisen. Complex physics-based models had been

developed to forecast the weather within the two-week limit of deterministic predictability, whilst simple statistical-empirical models were used to predict weather on month-to-seasonal timescales. The forecasts from the physics-based models were precise and deterministic (“it will be sunny” or “there will be no hurricane”), whilst the forecasts from the statistical-empirical models were based on probabilities and hence uncertain in nature.

The two groups of meteorologists developing these different forecast systems hardly talked to each other—their methodologies had next to nothing in common.

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I JOINED THE UK’S Met Office in the 1970s having completed a PhD in general relativity theory. After spending a few years working on the nonlinear dynamics of the upper region of the atmosphere known as the stratosphere⁸, in 1982 I was posted to the branch making long-range weather forecasts. At the time, I wasn’t super keen about this posting, but in the scientific civil service you do not always get to choose. As it turned out, it was probably the best thing that happened to me in my scientific career.

Pioneering work on monthly prediction using physics-based models of the atmosphere had begun to be performed by Indian American meteorologist Jagadish Shukla, from NASA, and Japanese American meteorologist Kiku Miyakoda, from Princeton. In Shukla’s studies, the observed slowly varying sea-surface temperatures provided what are called “lower boundary conditions” for the atmospheric models, and these boundary conditions generated predictable atmospheric circulations inside the

models on the monthly timescales⁹. These studies made me realise that physics-based models could play a valuable and powerful role in long-range weather forecasting, alongside the more traditional statistical-empirical models. My colleagues Chris Folland and David Parker had developed statistical-empirical models for monthly-averaged weather prediction over the UK. These statistical models predicted probabilities of different types of time-averaged atmospheric circulation patterns, and the forecasts were sold to utility companies—water, electricity and gas—to help their planning for the weeks and months ahead.

We began wondering, Could the physics-based models be used to strengthen the skill of these statistical-empirical models for long-range predictions over the UK? Maybe—but in order to blend the statistical-empirical model approach with the physics-based model approach, the latter would need to be probabilistic somehow.

It was clear that some kind of Monte Carlo method of the type Leith had been advocating some years earlier was needed. But instead of averaging the forecasts together as Leith had suggested, we would need to estimate forecast probabilities of different types of weather patterns from the members of the ensemble. In this way, in November 1985, my colleague James Murphy and I implemented the world's first ensemble weather forecast system based on the latest generation of physics-based models¹⁰. It was a very crude system by present-day standards.

I recall thinking at the time: Why are we making these ensemble forecasts only for the monthly timescales? Why don't we also run these ensemble forecasts for predictions on shorter timescales, i.e., within the two-week limit of deterministic predictability? I joined ECMWF in 1986 with the aim of trying to do something about this. I wanted to take the ensemble system that James Murphy and I had

recently developed and implement something similar within the two-week forecast range, the focus of work at ECMWF.

This idea met some resistance both from my research colleagues and from the weather forecasters who made forecasts for the media. They saw the two-week limit of deterministic predictability as a natural dividing line between deterministic prediction and probabilistic prediction. My colleagues' view was that when predicting weather on timescales shorter than two weeks, you just make a single best-guess forecast from your single best-guess initial state using your single best-guess model. Hence, these colleagues argued that when new computer resources became available—this was the era when computer speed was increasing exponentially with time—they should be focussed on improving the best-guess model and best-guess initial state, and not on making multiple ensemble forecasts to estimate forecast uncertainty. In addition, the forecasters didn't like the idea of ensembles because they felt that by explicitly estimating uncertainty, these predictions were somehow undermining their professionalism. Their job, they felt, was to tell it like it was—and not how it might or might not be.

I did not agree. Because the atmosphere is nonlinear, its predictability will vary from day to day. This variation in predictability can be seen in simple chaotic models—see [Fig. 10](#). On many days, the atmosphere might indeed be predictable enough so that deterministic predictions would be quite accurate for a long time ahead (like the top left of [Fig. 10](#)). However, there will be days when the atmosphere is sufficiently chaotic that the deterministic predictions could become inaccurate even after a few days of forecast time (like the bottom of [Fig. 10](#)). And yet we were releasing these deterministic forecasts to the public every day, without any idea whether the atmosphere is in a predictable state or in an especially chaotic state. If the latter, then the best-guess forecasts could be worse than

useless if people used them to make important decisions. What is it, after all, that distinguishes science from pseudoscience? Surely one key feature is an ability to handle uncertainty, to estimate error bars. Weather forecasting inside the limit of deterministic predictability had no reliable means of estimating these error bars.

However, to estimate these error bars reliably, ensemble systems would need to have many more members than Leith's estimate of ten. This method really was going to take a lot of computer time. My deterministic colleagues were not happy.

The 1987 storm, discussed in the introduction, was something of a godsend. We ran our ensemble system retrospectively on this storm. In the computer simulation, we initialised the ensemble just over two days before the storm hit the UK. The fifty ensemble members started with virtually, but not quite, identical initial conditions—flaps of butterflies' wings if you like. Just two days later the fifty ensemble members had diverged more than I would have believed possible (see [Fig. 28](#)). Some ensemble members showed intense storms. Others showed rather benign weather. The atmosphere was in an exceptionally unpredictable and chaotic state during this period and the diversity of possible solutions was remarkably large. This was the equivalent in the real world of the explosive growth of uncertainty shown at the bottom in [Fig. 10](#) for the Lorenz system.

What can one do with these fifty forecasts? The last thing you would want to do is average the forecasts together in the way Leith was suggesting. If you did that, the storm solutions would just get smoothed away and there would be no chance of predicting the possibility of hurricane-force winds.

Instead, one can simply count the number of ensemble members with exceptional storms and treat this as a frequentist-based estimate of the probability for an

exceptional storm. [Fig. 29](#) is a map showing just this. It is a forecast of the probability of hurricane-force winds. From [Fig. 29](#) you can see that such winds are predicted with a probability of around 30 percent or so over southern England. Given that “in Hertford, Hereford and Hampshire, hurricanes hardly ever happen [11](#)”, a 30 percent probability is enormously large compared with one’s prior expectations of hurricanes over southern England. At the very least, knowing such a probability, it might be prudent to park your expensive new car in the garage and not under a big oak tree.

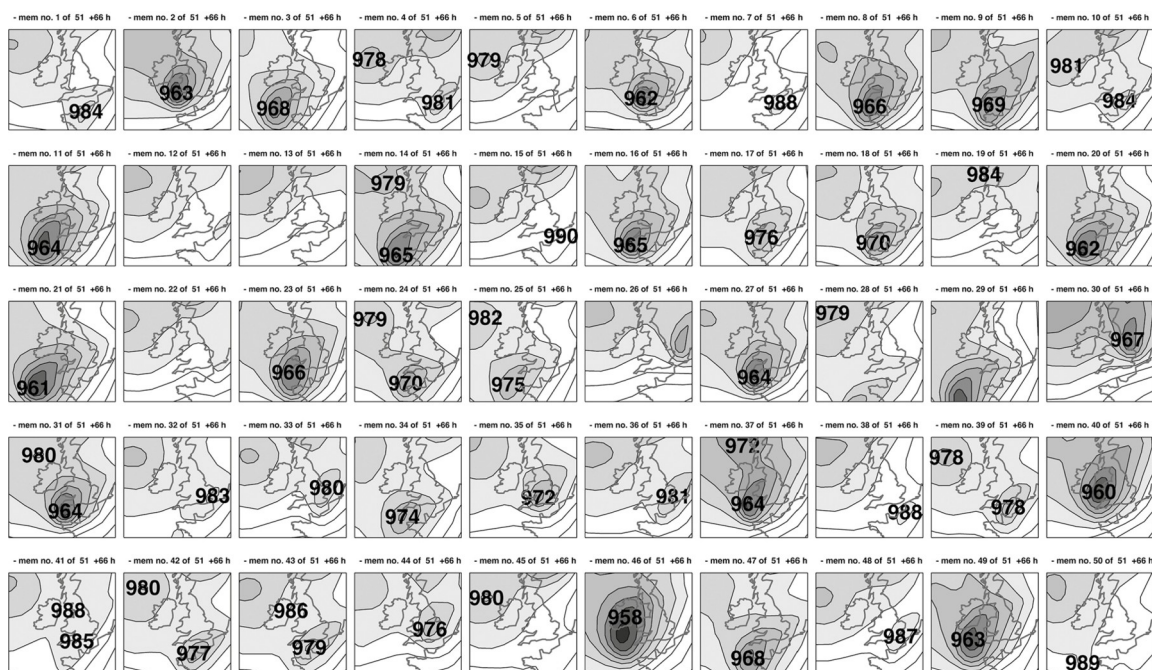


FIG. 28. An ensemble forecast for the morning of October 16, 1987, from initial conditions from midday on October 13. Illustrated are fifty “stamp maps” of forecast surface pressure. The uncertainties in the initial conditions and in the model equations have led to enormous divergence in possible forecast states for the sixteenth. Some ensemble members have severe storms, others benign high-pressure weather.

Tuesday 13 October 1987 12UTC ECMWF EPS Perturbed Forecast t+(66-72) VT: Friday 16 October 1987 12UTC
Surface: **wind gust at 10m (Exp: ep2b) - Ensemble member number 1 of 51

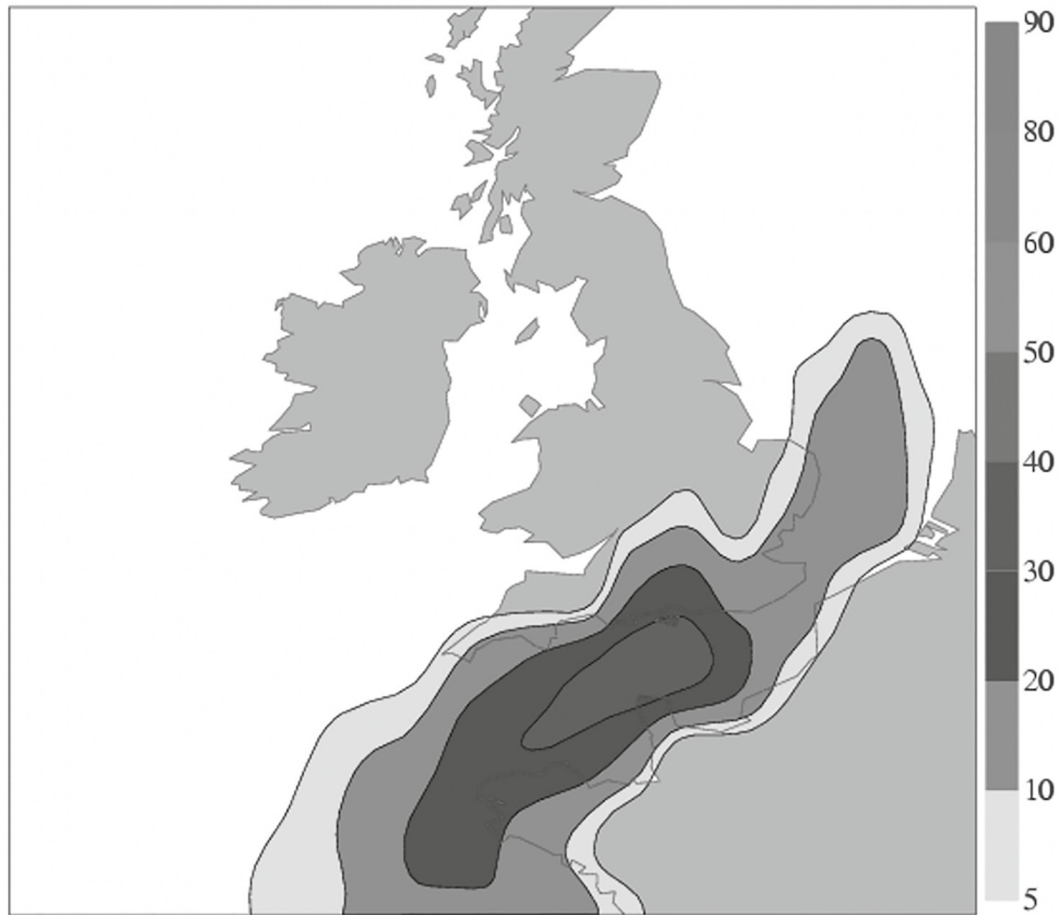


FIG. 29. A map showing the probability of hurricane-strength winds on October 16, 1987, based on the ensemble forecast shown in [Fig. 28](#).

If ever there was a slam-dunk example of the need for ensemble forecasts, this was it. The arguments and discussions were over. The ECMWF ensemble forecast system^{[12](#)} went operational late in 1992. At the same time, colleagues Eugenia Kalnay and Zoltan Toth implemented a similar scheme at the US National Weather Service. Over the coming years, all operational weather forecast centres around the world would implement ensemble prediction systems of one sort or another.

Of course, an ensemble system like this would be useless if it predicted hurricanes over England all the time. [Fig. 30](#) illustrates a set of ensemble “stamp maps” for a rather typical situation (a three-day ensemble forecast at the time of writing). Compared with [Fig. 28](#), the fifty pressure maps look fairly similar to one another and suggest a rather high probability of a cool northerly air stream across parts of central and western Europe. There is no chance of a hurricane over England.

Of course, it's not just midlatitude storms that benefit from ensembles. [Fig. 31](#) shows weeklong ensemble forecast tracks for three tropical-cyclone/hurricane forecasts. The top shows the fifty forecast tracks for Cyclone Sidr, which hit Bangladesh in 2007. This is an example where the path of the storm was fairly predictable. The Bangladesh authorities sent out clear warnings of the storm and few lives were lost. The middle shows the forecast tracks of the infamous Hurricane Katrina five days before it hit New Orleans. Here the path of the storm was much less predictable. At this range the most likely path was actually up over Florida. However, there was a second, weak maximum in the probability of predicted tracks over the Gulf of Mexico and on to New Orleans. This path became more and more likely as landfall time approached. Katrina is an example of an only partially predictable event. The bottom shows an ensemble forecast for Hurricane Nadine in the Atlantic in 2012. In this case, the direction Nadine would take was completely uncertain.

The widely familiar probability forecasts on weather apps today give ensemble-based probabilistic predictions right down to postcode or zip code scales. To do this, some additional “downscaling” software is applied to the physics-based model output. This is an area where AI techniques are becoming more and more important.

But what does a probability forecast mean? It's worth pausing to think about this. After all, it's easy to make a

probability forecast, and it seems you can't go wrong. Just say, "There is an eighty percent chance of it raining tomorrow". If it rains you say, "Well, I did say it would very likely rain!" and if it doesn't rain you say, "Well, I did say it might not rain!" So, if we are putting a lot of computer resources into estimating these probabilities, we'd better be sure we know what a forecast of 80 percent probability actually means and whether our forecast system captures this properly.

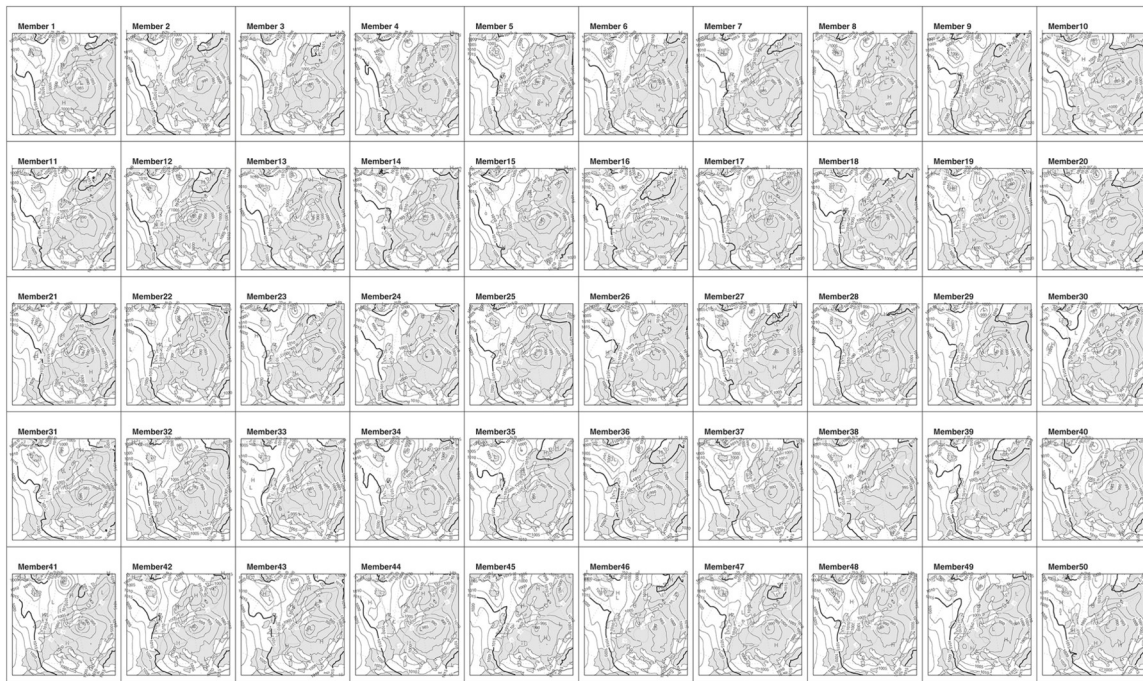
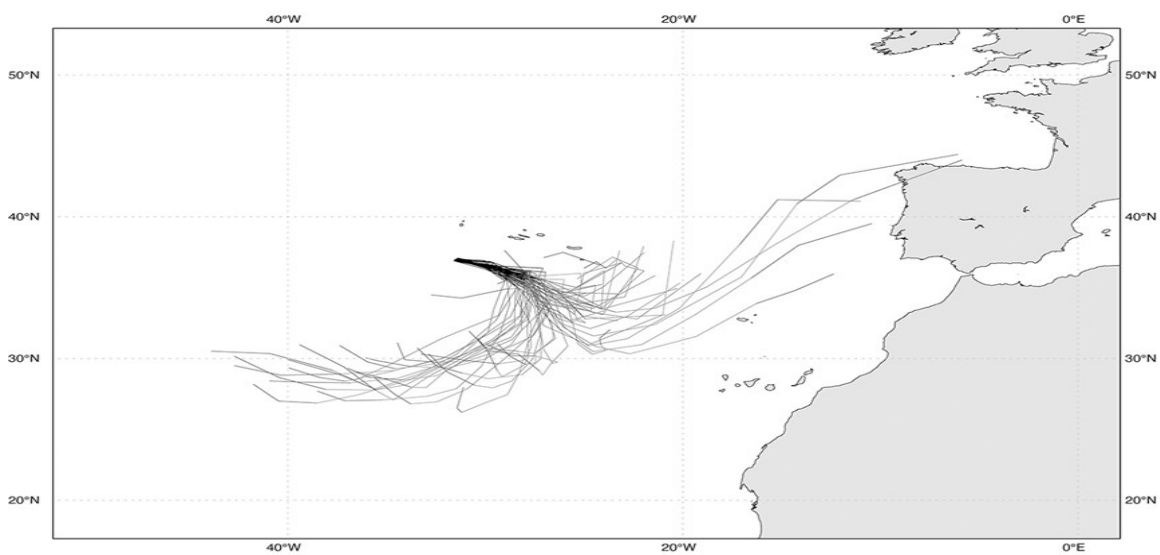
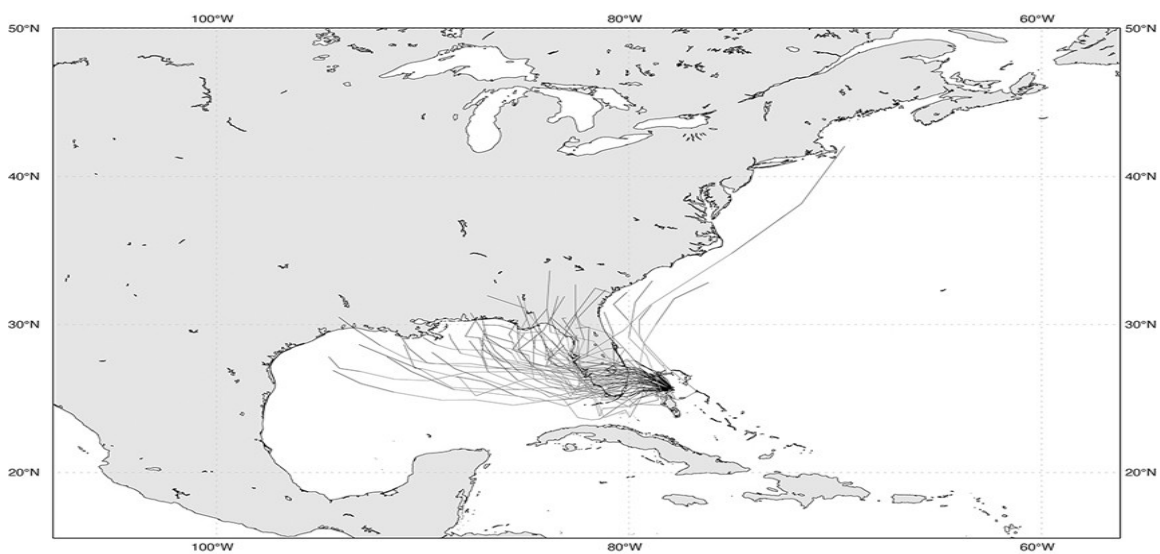
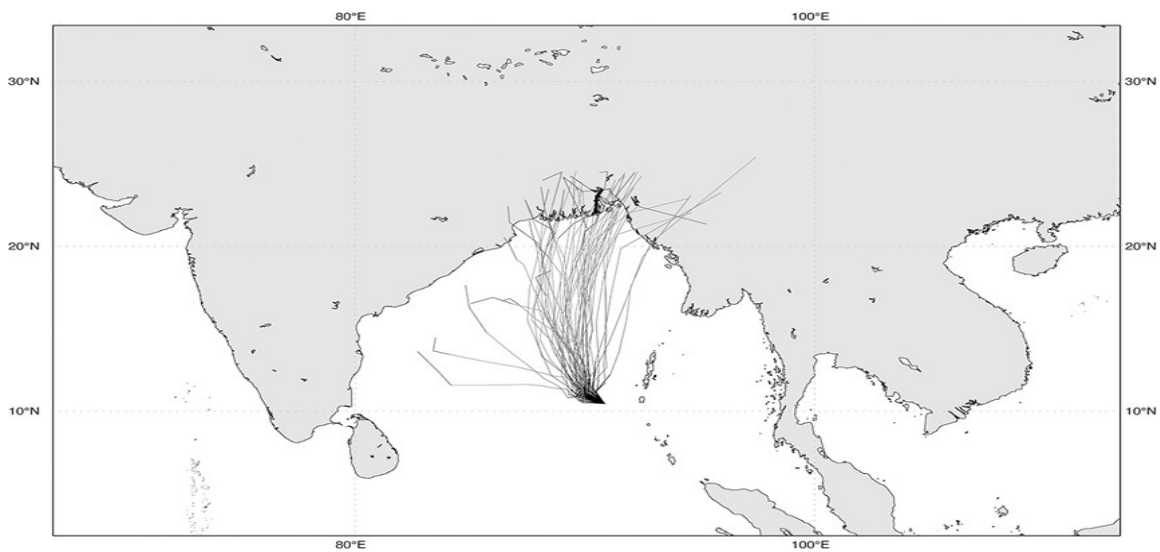


FIG. 30. An ensemble of surface pressure “stamp maps” over Europe for a three-day weather forecast starting November 26, 2021. Compared with [Fig. 28](#), the ensemble forecasts have not diverged much. This is a more typical situation in weather prediction.



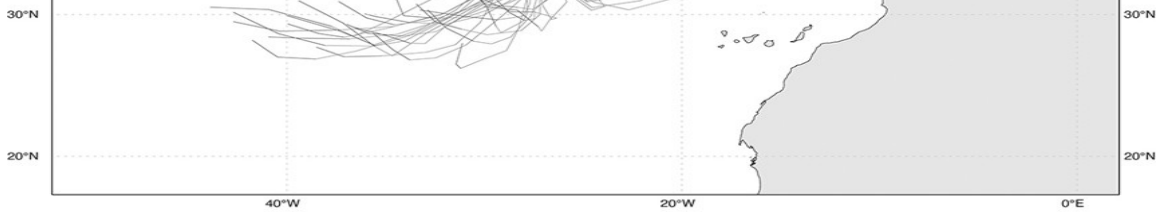


FIG. 31. Ensembles of week-long forecast tropical-cyclone/hurricane tracks, made using the ECMWF ensemble forecast system.

Top: Sidr—fairly predictable.

Middle: Katrina—fairly unpredictable.

Bottom: Nadine—totally unpredictable!

Suppose my weather app says that the probability of rain between 6 and 7 p.m. over my town on Tuesday is 80 percent. This doesn't mean that it will rain over 80 percent of my town on Tuesday between 6 and 7 p.m. Neither does it mean that it will rain for 80 percent of the time between 6 and 7 p.m. over my town on Tuesday. And finally, it doesn't mean that only 80 percent of weather forecast providers think it will rain over my town between 6 and 7 p.m. on Tuesday. So what does it mean?

Let's suppose that we have access to all the ensemble weather forecasts that have been made over the last year (from a particular weather provider, like ECMWF). That's 2 ensemble forecasts per day for 365 days, making 730 ensemble forecasts altogether. Now let's extract from all these ensemble forecasts the ones that predicted rain between 6 and 7 p.m. over my town with 80 percent probability. That is to say, with an ensemble forecast comprising 50 individual forecast members, we are looking for ensembles where 40 out of the 50 ensemble members (i.e., 80 percent of them) predicted rain over my town between 6 and 7 p.m.

Let's suppose there were 35 days over the last year when the forecast probability of rain was 80 percent. Now, if these forecast probabilities are reliable, we should expect that in reality it actually rained on 80 percent of these 35 forecast dates; that is to say, it actually rained on 28 occasions. If, for example, we found it rained only on 3 of the dates when an 80 percent probability had been forecast, it is likely [13](#) that our ensemble forecast system is

not reliable and we would need to go back to the research and development drawing board to try to find out why we are not representing forecast uncertainties properly in our ensemble system.

More generally and more succinctly, if the ensemble system produces reliable forecast probabilities, then, from all occasions that a weather event is predicted with a probability of p percent, the weather event should actually occur on p percent of these occasions. What we are doing here is taking a probability from the ensemble forecast and comparing it with a frequency of occurrence from the real world. The first time I referred to “ p percent”, it was as a forecast probability. The second time it was as a frequency of occurrence in the real world. It is this link between estimated probabilities of occurrence and actual frequencies of occurrence that makes ensemble forecasting useful.

It turns out that developing an ensemble forecast system whose probabilities are reliable in this way (i.e., precisely calibrated with frequencies of occurrence) is very challenging, especially for the most extreme types of weather. Merely adding random perturbations to the initial conditions doesn't lead to reliable ensembles at all: with such perturbations, the ensemble members do not typically diverge from one another nearly as rapidly as they should. As a result, such ensembles underestimate forecast uncertainty, and the corresponding forecast probabilities are overconfident. Decisions on when to evacuate a city based on unreliable probabilities of hurricane-force winds could be disastrous. In order to forecast uncertainty reliably, you really have to understand the nature of the uncertainties you are dealing with—it's the primacy of doubt again.

Trying to represent the butterfly effect properly in an operational ensemble weather prediction system took many years of hard work, not least because the error

characteristics of many processes relevant in determining the forecast initial state are difficult to quantify (e.g., how exactly information in observations is lost when assimilating these observations into truncated models of the atmosphere). To compensate for such “uncertainty about uncertainty”, we implemented techniques from what is called generalised instability theory. Essentially the idea is to focus initial ensemble perturbations along state-space directions in which the atmospheric circulations are most unstable. Such perturbations are known as “singular vectors”. Without such singular-vector perturbations, the ensemble members tend to be too tightly clustered and the ensemble is overconfident and generates unreliable probabilities. We will return to these singular vectors when discussing ensemble conflict prediction, because the concept has relevance more generally. Because of these special types of perturbations, operational ensemble forecast systems are not the simple Monte Carlo forecast systems originally envisaged by Leith, and so instead we now use the more general phrase “ensemble prediction” to describe them.

From the time of implementation in 1992 to the time of writing, the fifty-member ECMWF ensemble has been run in parallel with a more traditional single-member, best-guess deterministic forecast. The latter is made with a model whose spatial resolution is around twice that of the ensemble model. With its higher resolution model, the deterministic forecast provides more detail than the ensemble model. However, there is no way of knowing whether this extra detail is trustworthy or not. As I write these words, ECMWF has just announced that in 2023 the horizontal resolution of the ensemble prediction system will be increased to 9 km (5.6 miles), the same as the deterministic forecast model. Essentially, in 2023, the ensemble system will be ECMWF’s only forecast system. This event will mark the end of a very long road of research

and development to introduce ensemble prediction into weather forecasting.

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TO FINISH THE STORY, let's briefly return to the month-to-seasonal forecast timescale—beyond the so-called two-week limit of deterministic predictability. A major advance in our ability to predict ENSO using physics-based models of the coupled ocean-atmosphere system came in the 1980s. This is an area where scientists in the United States led the way, notably Mark Cane and Steve Zebiak, from Columbia University, who developed a physics-based model of the tropical Pacific Ocean system and atmosphere above, which encoded the basic coupled ocean-atmosphere dynamical processes that Jacob Bjerknes had uncovered back in the 1960s. Based on Cane and Zebiak's work, it seemed ENSO was predictable over a year ahead [14](#). I still recall with great excitement the time I first heard Cane talk about his model at a meeting at the Royal Society in London in 1986. At the end of the talk, he showed his prediction of the 1986/87 ENSO event. It happened just as predicted. Like with Fitzroy and Blanford before him, this might have been beginner's luck. But in this case, it wasn't.

Nevertheless, none of this could be done reliably with single deterministic forecasts, and ensemble prediction is now a vital part of any seasonal prediction system. Such predictions can be used to warn of the risk of prolonged droughts in parts of the world, or of exceptional hurricane seasons, or of the likelihood of cold and snowy winters. Indeed, with many operational centres now producing seasonal forecasts, and with so much of the forecast uncertainty arising from shortcomings in the resolution of these ocean-atmosphere models (as I will discuss in

[Chapter 6](#)), it makes sense to combine different model outputs from different forecast institutes into a so-called multi-model ensemble system. I myself led an international project to develop one of the first such multi-model seasonal ensemble forecasts¹⁵. We used it to probabilistically predict crop yields in Europe and outbreaks of epidemic malaria in Africa, two of the many applications of seasonal predictions.

This multi-model ensemble technique is also used by the Intergovernmental Panel on Climate Change for their projections of climate change ([Chapter 6](#)). And multi-model ensembles have also been used to predict the spread of COVID-19 ([Chapter 7](#)).

6

Climate Change

Catastrophe or Just Lukewarm?

CLIMATE CHANGE HAS VARIOUSLY BEEN DESCRIBED AS A CATASTROPHE in waiting or a lukewarm perturbation to our climate system of no great significance. That's quite a difference of opinion. Which is scientifically correct? How should we view climate change from a perspective consistent with the "primacy of doubt"?

Let's start by reviewing two opposing sides of the argument. On the one hand:

We are in a climate crisis. We have reached this state by burning fossil fuels such as coal, oil and gas. This started in earnest with the Industrial Revolution and continues to the present day. Over this time, we have released back into the atmosphere hundreds of billions of tons of carbon, most of which had been stored in the ground for hundreds of millions of years. In this way, we have suddenly increased the concentration of carbon dioxide in the atmosphere to levels not seen for millions of years.

These concentrations will continue to increase into the future unless we stop burning fossil fuels. Because carbon dioxide is a greenhouse gas, our carbon emissions will warm the atmosphere, leading to catastrophic changes to

our climate, including deadly heatwaves, storms of unprecedented intensity, and metres of sea-level rise caused by ocean warming and the disintegration of ice sheets.

And the carbon we release into the atmosphere will stay there for hundreds if not thousands of years. We must stop our emissions now.

On the other hand:

Such alarmist warnings are completely overblown. Carbon dioxide is what is called a “trace gas” in the atmosphere. Its concentration is measured in parts per million. Since the start of the Industrial Revolution, the concentration has increased by just one part in ten thousand. That’s a tiny amount. Indeed, if you calculate the amount of warming due to a doubling of carbon dioxide concentration, it is barely more than 1°C (1.8°F). Not only is this not a catastrophe, it’s barely noticeable. Temperatures change by more than 1°C from one day to the next.

And of course, Ed Lorenz’s great insight was to realise that a chaotic climate never repeats itself. On this basis, a chaotic system like climate is always changing and hence we can never be sure that some unusual warm spell in the atmosphere is not just the effect of internal variability, even if the warm spell is record breaking. In any case, climate-change prediction is just a long-range weather forecast, and, again, we know from chaos theory that such long-range predictions are impossible. Hence, these forecasts of deadly heatwaves and substantial sea-level rise are completely bogus.

On top of this, the climate models are flawed and unreliable. If we look at the predictions of climate change made thirty years ago, the models predicted too much warming. Indeed, when has a climate model ever made a prediction that has actually turned out to be correct?

And in any case, this increase in carbon dioxide concentration is something we should be happy about—having those extra molecules of carbon dioxide in the atmosphere will help plants grow more strongly, thereby greening the planet.

There—I have summarised both sides of the argument, hopefully in an accurate and disinterested way. Perhaps the arguments on both sides sound plausible.

Before diving into the science of climate change, I need to find some words that describe these two points of view. Sometimes the phrase “climate sceptic” is used to describe the second point of view. However, all scientists, including climate scientists, are inherently sceptical people. Bob May, whose work on chaos and ecology was described in [Chapter 1](#), characterised science as “organised scepticism”. Another phrase that is sometimes used is “climate denier”. However, the overtones of Holocaust denial make this phrase offensive to many people. In an attempt to be neutral, I propose to call the first a “climate maximalist” position and the second the “climate minimalist” position¹. An important question is this: Where, on a spectrum in which these positions lie at the extremes, does climate science lie? Near one end, halfway along?

The maximalists’ certainty that we are headed for disaster seems inconsistent with the “primacy of doubt” mantra of this book. Perhaps, therefore, we should heed the minimalists who emphasise the need to be cautious and who stress uncertainties in our understanding of the climate system.

However, as stressed in the previous chapter, estimates of uncertainty are useful for making decisions—e.g., whether to cut carbon emissions or not—only if they are reliable, i.e., neither overconfident nor underconfident. Exaggerating uncertainty can be just as damaging as underestimating it. If Michael Fish had warned of a

possible hurricane hitting southern England every evening he appeared on TV, no one would have paid attention on the day when the hurricane did actually hit.

In their influential book *Merchants of Doubt*², Naomi Oreskes and Erik Conway describe how a group of scientists and scientific advisers exploited the inevitable uncertainties in research to obscure scientific knowledge on a variety of issues from tobacco smoke to global warming. Oreskes and Conway's book appears to give *The Primacy of Doubt* a bad name! However, the authors focus on the deliberate inflation of uncertainty for ideological or commercial reasons. Clearly, we should be just as wary of inflation of uncertainty as of attempts to make predictions more certain than can be justified.

In short, in trying to estimate what climate change holds in store for us, it is vital that our estimates of uncertainty are reliable. How do we achieve this? We typically estimate uncertainty using ensemble techniques of the type described in the previous chapter for weather prediction. Indeed, it's simply not possible to describe climate change in an objective, scientific way without explaining the three roles of ensembles: for estimating the uncertain feedback effects in climate science; for estimating the impact of climate policy on climate change; and for separating the effects of natural chaotic variability from human-induced effects.

From the first way of using ensembles, we can address how big an impact our emissions of greenhouse gases are having on global temperatures. With the second set of ensembles, we can assess whether mitigating actions can be effective or not. And from the third set of ensembles, we can not only assess the extent to which observed changes in weather and climate are natural ("climate is always changing"), we can try to attribute specific weather events to climate change, at least in a probabilistic way.

But let's start at the beginning. What is a greenhouse gas? We say that carbon dioxide is a greenhouse gas because it is relatively transparent to the predominantly visible and ultraviolet photons emitted from the sun, but is relatively opaque to the less energetic infrared photons emitted back to space by the cooler Earth³. Imagine looking down on Earth from space, wearing infrared goggles⁴. Then instead of seeing Earth's surface, the blue of the ocean, for example, we will see a fuzz of photons emitted from the atmosphere a few kilometres above the Earth's surface. These are the photons emitted by the greenhouse gases in the atmosphere. We would not be able to make out the Earth's surface at all from this fog of infrared photons.

Suppose we could suddenly double the amount of carbon dioxide in the atmosphere. Then the Earth's surface would become even more invisible. The greenhouse blanket is now stronger than before. Carbon dioxide is what is called a "well-mixed gas", which means that if we double the amount of carbon dioxide in the atmosphere, its concentration will double everywhere in the atmosphere. As a result, the photons that we see with our infrared goggles will on average have been emitted from greenhouse gas molecules a little higher up in the atmosphere—the original layer we were seeing with our goggles is now partially obscured by extra greenhouse molecules above this layer. But higher up in this part of the atmosphere, the air is a little cooler. This means that the energy of the infrared photons that are emitted out to space is less than it was before. The Earth is now in an energetic imbalance: the sun is giving the Earth more energy than the Earth is re-radiating back to space. The only way to restore this balance is for the Earth to warm. It does this at the level where the photons are radiating energy back to space, and, as a consequence, it warms at

the surface too⁵. Directly from this effect, a doubling of carbon dioxide from pre-industrial times will warm the surface of the Earth by a little more than 1°C—perhaps not something to make a big deal of. Here the minimalists are correct.

However, carbon dioxide is not the only greenhouse gas in the atmosphere. Indeed, it is not even the strongest greenhouse gas, molecule for molecule. As discovered by the Irish physicist John Tyndall in the nineteenth century, the most important greenhouse gas in the atmosphere is actually water vapour—the gaseous form of water. When the sky is blue and cloudless, there is still plenty of water vapour in the air. This water vapour is largely transparent to visible light, which is why we don't see it. However, it is opaque to light at infrared wavelengths. Hence, like carbon dioxide, it can trap the outgoing energy from the Earth.

We often think of the important role that liquid water plays in sustaining life on Earth. However, Tyndall realised that, as a greenhouse gas, the gaseous water vapour plays a vital role too. In a wonderfully eloquent turn of phrase, Tyndall wrote,

“This aqueous vapour is a blanket more necessary to the vegetable life of England than clothing is to man. Remove for a single summer night the aqueous vapour from the air that overspreads this country, and you would assuredly destroy every plant capable of being destroyed by a freezing temperature. The warmth of our fields and gardens would pour itself unrequited into space, and the sun would rise upon an island held fast in the iron grip of frost.”

With this in mind, you may well wonder why we are so bothered by an increase in carbon dioxide concentration. Surely, we should instead be worried about the effect we might be having on the amount of water vapour in the air.

The answer is that we are worried about the effect we are having on the amount of water vapour in the air. But we aren't worried so much by any direct emissions of water vapour as by the knock-on effect on water vapour from our emissions of carbon dioxide. Specifically, if the global atmosphere warms a little from our emissions of carbon dioxide, this warming will be amplified by a knock-on increase in the amount of water vapour in the air. The process is perhaps more familiar in reverse. During cold autumn nights, the warm, moist air of the day cools and the water vapour condenses into small droplets we call fog (or a low-level cloud). The cooler nighttime air contains fewer water vapour molecules than the warmer daytime air. And then, as the next day breaks, the air warms again and the opaque water droplets slowly re-evaporate back into optically transparent water vapour.

The amplifying knock-on effect is an example of what is called a positive feedback process. We emit carbon dioxide into the air. The air warms globally a small amount. Because of evaporation (e.g., from the oceans and land surface), the warmer air becomes more moist. The greenhouse effect from this extra water vapour increases the warming of the air from carbon dioxide alone. The direct warming due to a doubling of carbon dioxide is a little over 1°C. However, if we add this water vapour feedback, the warming doubles to just over 2°C (3.6°F). If we also take into account the fact that reflective ice and snow cover on the Earth's surface start to disappear as the Earth gets warmer, so that more of the sun's energy is absorbed at the surface, the warming increases to about 2.5°C (4.5°F). Now climate change starts to become something to worry about.

We understand these feedback processes reasonably well. However, there is another feedback process associated with water that we understand rather poorly. This is the cloud feedback process. Clouds occur when the

water vapour in moist air condenses into small water droplets or small ice crystals. Compared with water vapour, very little of the water in the atmosphere is in the form of clouds⁶ (less than one hundredth). However, clouds have a big effect on climate. They punch well above their weight, so to speak.

There are two key questions: How will clouds adjust to the increasing levels of carbon dioxide, and will this lead to an amplification of the warming or an off-setting cooling effect? These are especially difficult questions to answer. This is partly because, depending on its type, a cloud might warm or cool the Earth's surface. For example, a layer of cloud just above the Earth's surface will predominantly scatter photons from the sun back to space. Earth's surface will never see these photons, and so the surface will be cooler than it would be if the layer of cloud weren't there. However, a thin, wispy cirrus layer of cloud, mainly comprising ice crystals high in the atmosphere, would have an opposite effect. These clouds let sunshine through. However, like carbon dioxide in the atmosphere, they trap infrared photons emitted from the atmosphere below. The Earth's surface is warmer as a result.

How do clouds respond to, say, a doubling of carbon dioxide? If low-level clouds increase and high-level clouds decrease, then clouds will likely save us from catastrophe⁷. They will offset the warming due to the increase in carbon dioxide. In this scenario, clouds will feed back negatively on climate change. It won't take us all the way back to the minimalist description of climate change, but perhaps to somewhere in the middle ground between minimalism and maximalism. By contrast, if it's the other way around—low-level clouds decrease and high-level clouds increase—then clouds will exacerbate the warming due to the increase in carbon dioxide. In this scenario, clouds provide a positive feedback on climate change. Another positive feedback,

compounding the water vapour feedback, would be very bad news for humanity. In this case we will be facing climate maximalism.

The question of whether clouds act as a positive or negative feedback on climate change can't at present be answered unambiguously: indeed, I would say it is the biggest unsolved problem in physical climate-change science. There are two reasons why it is a difficult problem.

Firstly, the physics of the small droplets and ice crystals in a cloud—what is called cloud microphysics—is itself rather complicated. For example, the balance between ice crystals and water droplets determines how reflective a cloud is to incoming solar photons, and this reflectivity is crucial in determining the effect of the cloud on the temperature at the surface. Now, it isn't the case that cloud droplets simply freeze into ice crystals once the temperature of the cloud falls below 0°C (32°F). In particular, cloud droplets frequently comprise supercooled liquid water at temperatures well below 0°C. The conversion of supercooled water droplets into ice crystals depends not only on temperature but also on impurities (natural and manmade) in the air—what are called “aerosols”. Indeed, these aerosols are important for determining how much water vapour in the cloud turns into water droplets and ice crystals in the first place. In addition, once the ice crystals do freeze, their precise shape and size can be important in determining things like cloud reflectivity. Hence, trying to estimate how individual clouds will change as the atmosphere warms depends on quite complicated cloud microphysical processes.

The second problem is that clouds also depend on the larger-scale environment in which they exist. There are many types of clouds in the atmosphere, and assessing how each of these changes in response to increasing levels of carbon dioxide is a complicated business. One type which is believed to be important is what is called “marine

stratocumulus". Over the Pacific Ocean, for example, the extent of these types of clouds depends on the strength of the Walker circulation, the large-scale atmospheric circulation associated with Gilbert Walker's Southern Oscillation, as described in [Chapter 5](#). The marine stratocumulus clouds broadly occur in the sinking branch of the Walker circulation, over the tropical eastern Pacific [8](#). By contrast, deep thunderstorm clouds form in the ascending branch of the Walker circulation, over the tropical western Pacific, bringing rain to these regions. The conversion of water vapour to liquid water in these thunderstorm clouds plays a crucial role in determining the strength of the Walker circulation. And so if we want to know what will happen to the marine stratocumulus clouds under climate change, it's not enough to simply study the microphysics of the stratocumulus clouds in the eastern Pacific; it is also important to understand what will happen to the thunderstorm clouds in the remote western Pacific [9](#). And this is just one small part of the cloud feedback jigsaw. It's all consistent with the complicated picture of high-order chaos described in [Chapter 3](#). It is why climate-change science is unquestionably complicated and why clouds are the most complicated and uncertain part of it all.

At present, there is no observational evidence that global cloud cover is systematically changing. On the other hand, neither is there evidence that it is not changing. The record we have of high-quality global cloud observations is relatively short, and the changes we are looking for are small compared with natural year-to-year variations in cloud cover. The problem of detecting changes in clouds is compounded by the fact that individual satellite instruments do not produce data over multidecadal timescales and the precise calibration of one instrument relative to its successor or predecessor is delicate and difficult. However, if we do one day start to see systematic

changes, and they are in the direction of reducing global low-level cloud cover, and/or increasing high-level cloud cover, indicating a positive cloud feedback, this will be a bleak day indeed for humanity. Here the word “positive” does not denote something good.

Precisely because observations of global cloud cover have yet to change in a systematic way, it is impossible to estimate how much warming we can expect over the rest of this century by simply looking back at the amount of warming that has happened over the past fifty years or so and extrapolating into the future. These cloud feedbacks are highly nonlinear processes.

So, we are faced with a problem. We cannot model clouds using the proper laws of physics (i.e., using the Navier-Stokes equation of [Chapter 3](#)) because the gridboxes in current global climate models are too coarse in spatial scale, in turn because the computers climate scientists use aren't powerful enough. Instead, clouds have to be represented by simplified formulae known as parametrisations. These parametrisations are, by their nature, approximate and uncertain. The real problem is that we are trying to compress the complex fractal-like multiscale structure of clouds into a highly simplified formulaic representation. Just as Poincaré found there is no simple formula to describe the motion of planets, so climate scientists realise there is no simple formula to describe the structure of clouds. As discussed in [Chapter 3](#), a way to incorporate the inherent uncertainty in cloud and other small-scale turbulent processes is through stochastic noise. Such noisy parametrisations are beginning to be introduced into the current generation of climate models.

The first physics-based climate model was developed in the 1950s by Norman Phillips, a meteorologist at the Institute for Advanced Study in Princeton. Phillips showed that many of the properties of climate, such as the jet streams and cyclonic weather disturbances, could emerge

naturally from the Navier-Stokes equation and other primitive equations of classical physics. However, Phillips's model was crude. The model knew nothing about the continentality of the Earth or the seasons, and had no representation of clouds or rain. In addition, the model had very coarse-scale gridboxes.

In 1958, one of the pioneers of climate science, Nobel laureate Syukuro Manabe^{[10](#)} (known to his colleagues as "Suki"), moved from the University of Tokyo, where he had just completed his PhD, to the Geophysical Fluid Dynamics Laboratory at Princeton University, then being led by Jo Smagorinsky, another of the early ENIAC pioneers. Together Smagorinsky and Manabe began to develop the first comprehensive global climate models with realistic geography and seasonal cycles and with hydrological cycles and hence (parametrised) clouds.

Whilst these models were important for understanding the emergence of Earth's climate from the laws of physics, Manabe realised that they could also be useful in estimating the impact of human emissions of greenhouse gases, a matter which was already of concern in the 1960s in scientific and environmental circles. In the mid-1970s, Suki Manabe and his colleague Richard Wetherald^{[11](#)} performed the first climate-change experiment with the three-dimensional climate model they had developed. In this model they estimated the effect on climate of doubling the carbon dioxide concentration from pre-industrial values, as discussed earlier. [Fig. 32](#) shows the temperature response of the model to such a doubling, as a function of latitude and height.

Let's take a careful look at this early physics-based climate model simulation of climate change. You can see two clear predictions, over and above the prediction that the surface of the Earth will warm due to the doubling of carbon dioxide concentration. The first is that in the

stratosphere, the region of the atmosphere above about 15 km (9.3 miles), the doubling of carbon dioxide will actually cool the air. The second is that the surface warming will be particularly marked in the Arctic.

The stratosphere is a region of the atmosphere where temperatures increase with height due to the absorption of sunlight by stratospheric ozone, a special molecule made of three oxygen atoms. The absorption of ultraviolet light by stratospheric ozone is important in preventing us from getting skin cancer. So why, then, does the stratosphere cool when the carbon dioxide concentration increases? The first point to make is that carbon dioxide is a well-mixed gas. So, when we emit carbon dioxide into the atmosphere, it mixes throughout the atmosphere, including the stratosphere, fairly quickly. The second point is that the temperature in the stratosphere arises from a balance between a warming from the absorption of solar radiation by ozone and a cooling from the emission back to space of infrared radiation by carbon dioxide. If the amount of carbon dioxide in the stratosphere increases, then the emission to space dominates over the ozone absorption, and the stratosphere cools. As the stratosphere cools, it emits less infrared radiation to space, and in this way the stratosphere gets back into equilibrium at a reduced temperature.

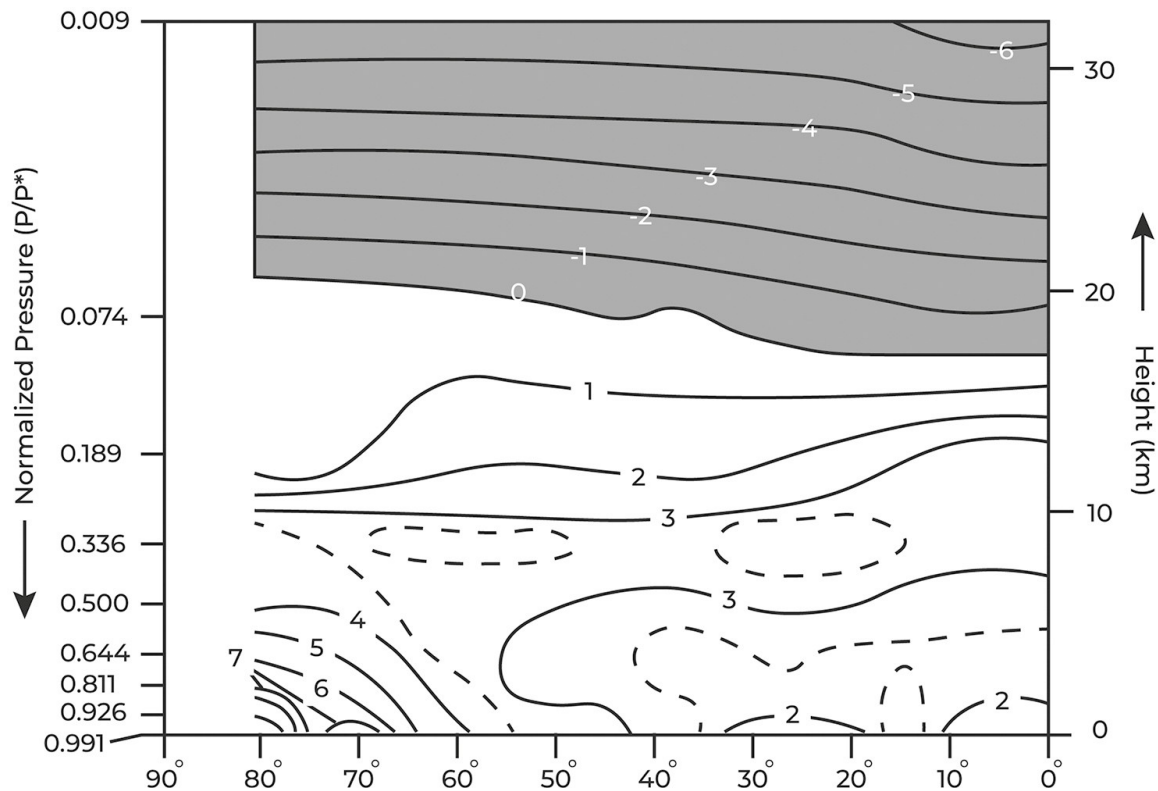


FIG. 32. A simulation of climate change from Manabe and Wetherald's seminal 1975 study. It shows the simulated change in temperature, averaged around a line of constant latitude, as a function of latitude and height in the northern hemisphere, with the North Pole on the left and the equator on the right. Note the cooling in the stratosphere (at a height of about 15 km [9.3 miles]) and the "hot spot" warming over the surface Arctic. Both stratospheric cooling and enhanced Arctic warming have occurred in reality. Published in 1975 by the American Meteorological Society.

This differential heating effect—a warming of the surface and a cooling in the stratosphere—would not be expected if the warming of the Earth's surface was due to an increase in energy from the sun. In that case, the stratosphere should warm. Observations show that the Earth's stratosphere is indeed cooling as the Earth's surface warms. This is a clear prediction of Manabe and Wetherald's model that has been verified by observations.

The second prediction from Manabe and Wetherald's study, the enhanced warming in the Arctic, has also come

to pass—large parts of the Arctic are now ice-free in the summer. This is in part (there are other reasons) due to a positive feedback effect associated with water. The sea ice over the Arctic reflects sunlight straight back to space. However, as the Arctic warms, the sea ice melts and exposes darker water underneath, which absorbs these solar photons, which then adds further to the warming of the Earth.

After the success of Manabe and Wetherald's climate-change papers, many institutes around the world began to develop their own climate models. The number has continued to grow to the present day and is approaching one hundred. What was technically innovative has become relatively straightforward for individual climate institutes. These models are all based on the same principles: a “dynamical core” where the Navier-Stokes equation (amongst others) is solved up to some imposed spatial resolution, and a set of simplified parametrisation formulae for unresolved subgrid scale processes like clouds. The models contain representations not only of the atmosphere but also of the oceans, the land and the cryosphere (the regions of Earth covered in ice), and more recently the biosphere.

This provides a natural “ensemble of opportunity” to study climate change. It is an example of the so-called multi-model ensemble mentioned in [Chapter 5](#). Each model differs from the others in the precise computational techniques used to solve the Navier-Stokes equation, and, more importantly, in the parametrisation formulae for unresolved processes. Every few years, the World Climate Research Programme, partly under UN auspices, oversees the production of a set of highly coordinated multi-model ensembles to study climate change. Results from these multi-model ensembles are fed into the Intergovernmental Panel on Climate Change (IPCC) assessment reports.

Let's use these ensembles of opportunity to estimate how much global warming would arise if we were to double the amount of carbon dioxide in the atmosphere—from roughly 300 parts per million (pre-industrial values) to 600 parts per million (which we may reach later this century) [12](#). In [Fig. 33](#) I have shown these estimates in the form of a histogram combining together output [13](#) from climate models which contributed to either the fifth or the sixth IPCC assessment report. It can be seen that the most likely warming is around 3°C (5.4°F) though a number of models predict increases of more than 5°C (9°F). No models predict the Earth will warm less than about 2°C (3.6°F). The Earth's surface has already warmed by over a degree since pre-industrial times.

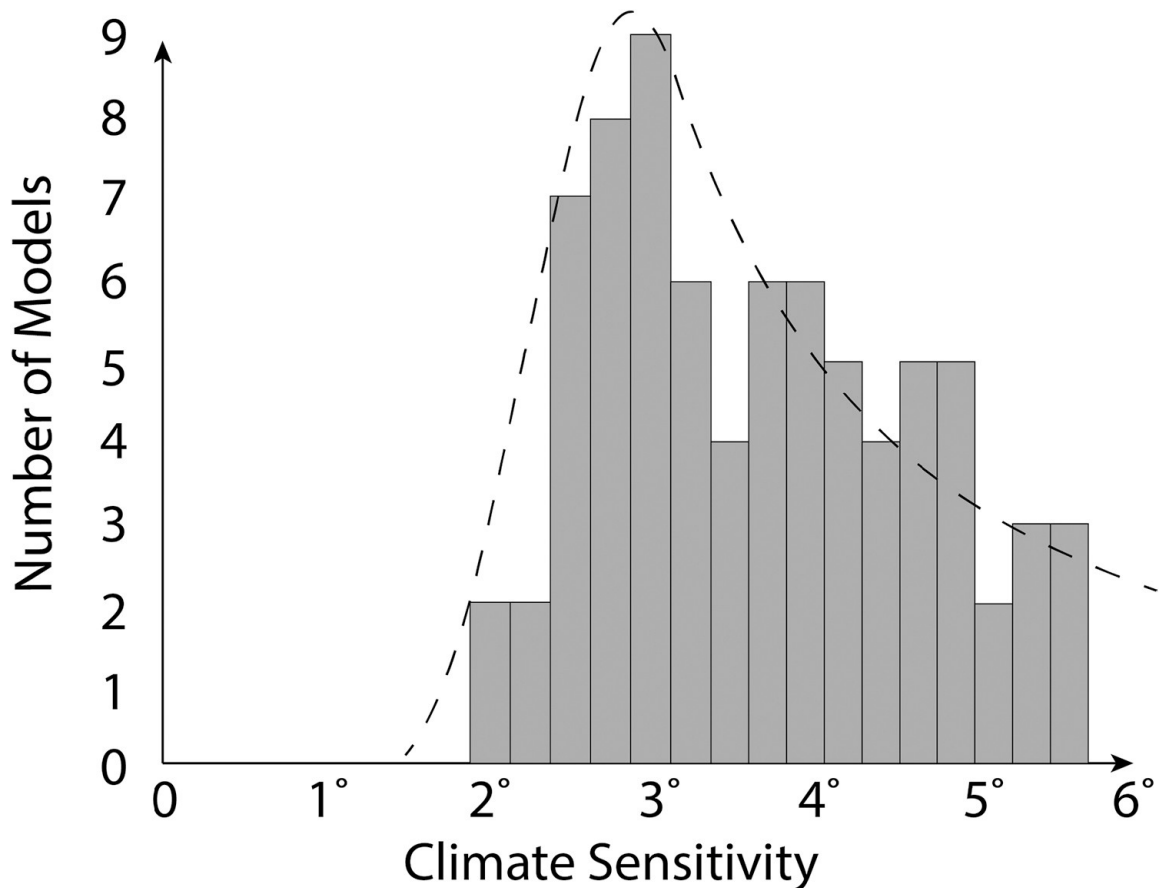


FIG. 33. A histogram of estimates of climate sensitivity—the global warming that arises from a doubling of carbon dioxide concentration—from models which contributed to either the fifth or sixth IPCC assessment report (Coupled Model Intercomparison Project Phase 5 [CMIP5] or CMIP6). The dashed line is a probabilistic estimate of such warming. As expected from theory, it is skewed towards large values of warming.

Motivated by expectations from theory [14](#), I have fitted a simple curve to this histogram. The higher the curve, the more likely the corresponding increase in global mean surface temperature shown on the horizontal axis. The distribution has a characteristic shape that statisticians call “skewed”, which means the distribution is not symmetric about its peak, i.e., about its most likely value. It has a long, fat^{[15](#)} tail going out to large warming values^{[16](#)}. By contrast, it goes to zero for global warming values less than one degree. We will see other examples of skewed

distribution when we study pandemics and financial crashes. Skewed distributions with long, fat tails are indicative of an underlying nonlinear system¹⁷.

One consequence of such a skewed distribution is that the warming we can expect to see will be greater than the most likely warming predicted by a single model. To illustrate this, suppose we have 10 hypothetical models which differ by the value of some uncertain parameters, and suppose the 10 predictions from the 10 models are described by the highly skewed sequence of numbers {1; 1; 1; 1; 1; 1; 1; 1; 1; 100,000}. If each set of parameter values is equally likely, the most likely single-model prediction is the value 1. However, if we average the predictions over the 10 models, then the average or “expected” prediction is close to 10,000—much larger than the most likely single-model prediction. Based purely on the ensemble data in [Fig. 33](#), the most likely warming from a single-model prediction is between 2.5° and 3°C, whilst the expected warming from the ensemble is about 3.6°C (6.5°F). If this is the warming we can expect, it is the level of warming we should be prepared for.

The skewed shape of the climate sensitivity curve can be largely understood from the feedbacks discussed earlier ¹⁸. We have already mentioned that if climate change were associated only with an increase in carbon dioxide, then we would expect the Earth’s surface to warm by a little over 1°C. As mentioned, with water vapour feedback this increases to just over 2°C. The fact that we have this rather broad, skewed distribution is largely due to uncertainty in the cloud feedback effect ¹⁹. Whilst a few models have negative cloud feedback effects, most climate models predict positive cloud feedbacks but with different strengths.

Although climate-model experiments in which carbon dioxide levels are doubled (such as shown in [Fig. 33](#)) are

useful in quantifying the possible changes in weather due to climate change, they are in one particular way too idealised to be useful for policy-making. These experiments don't tell us when we will reach a given level of climate change. If our emissions reduce substantially, but not to zero, it could take many hundreds of years before carbon dioxide concentrations in the atmosphere will double. Politicians tend to lose interest about things too far in the future.

This brings us to the second important use of ensembles in climate prediction. If we are to estimate the future climate for, say, the year 2100, we need to know how our emissions of carbon dioxide will change with time over the twenty-first century. This of course depends on how we humans respond to the model simulations of climate change. Climate models are like economic models in this respect—just as output from economic models can affect the economy, so output from climate models can affect the climate. Earth's climate could turn out very differently if, on the one hand, we ignore the simulations or, on the other hand, we heed the simulations. In principle one could imagine incorporating these different human responses into a single coupled human-climate ensemble—in which humanity ignores the predictions in some ensemble members and cuts emissions in others.

However, this may not be especially useful for policy-making. Policymakers need to know how climate will change both without regulatory policy and with regulatory policy. From such information they can assess the value of such regulatory policy.

To do that, separate ensembles are run for different emissions “scenarios”, as they are called. For example, one emissions scenario could be considered a worst-case situation in which we burn fossil fuels as rapidly as we can²⁰. We appear to have moved away from this scenario

because we are not burning coal as much as we might be. In another emissions scenario, we try to decarbonise at a leisurely pace. In a third emissions scenario we not only decarbonise rapidly, we find ways of sucking carbon dioxide back out of the air—e.g., by planting more trees or by more technological means—through what are called negative emissions.

In this way, climate scientists tend to talk not about climate predictions but climate “projections”. A projection is a prediction conditioned on an assumed scenario for carbon emissions (and other related processes). We create an ensemble of climate projection for each emission scenario.

Being aware of this dependence on scenario is particularly important when examining the accuracy of climate projections made in past decades. For example, one of the best-known historical sets of projections was made in 1988 by one of the leading climate scientists: Jim Hansen from NASA. Hansen presented to a US congressional committee results based on the NASA climate model, run using three different emissions scenarios. In retrospect we can see that his Scenario A overestimated the emissions that have actually occurred. This does not mean that Hansen’s model was wrong with this projection—Hansen could not have been sure how our emissions would evolve with time. Indeed, his somewhat less aggressive Scenario B has turned out to be a more accurate representation of our emissions. Some climate minimalists use Hansen’s Scenario A projection to suggest that the models overpredict global warming. However, in terms of Scenario B, the observed and simulated changes in global temperature are quite close. Indeed, looking more broadly at those climate projections whose emissions scenarios were close to what actually happened, estimates of increases in global mean

surface temperature were found to be broadly consistent with observations²¹.

So far, we have focussed on global mean temperature. Scientists describe climate change in terms of increases in global mean surface temperature, not because it's a societally important variable but because the chaotic internal variability of climate affects global mean temperatures much less than more regional temperatures. The temperature over London can change by 1°C from one hour to the next. However, the temperature averaged over the whole surface of the Earth is much more stable. Changing it by 1°C is a big deal.

Using global mean surface temperature to describe climate change comes with a major downside—it conveys no real sense of what climate change means to us humans. We, after all, are affected by local weather, not by global mean temperatures. Perhaps the best book I know that gives a plausible sense of what a warmer world would be like at the regional scale is Mark Lynas's *Our Final Warning: Six Degrees of Climate Emergency*

What I want to do now is fix on a specific level of 4°C (7.2°F) of warming. According to [Fig. 33](#), it's a little more than the expected warming given a doubling of carbon dioxide but certainly one that is plausible in the future. This is what Lynas says about a 4°C warmer world:

“At four degrees a whole new process has begun to unfold, making substantial areas of the planet biologically uninhabitable for humans. Whilst the increased mortality risk in today's heatwaves tends to be among older, very young and other vulnerable people, temperatures at four degrees of global heating are high enough to reach a critical threshold that will kill anyone, however fit and healthy, because of the laws of thermodynamics.”

Here Lynas is referring to the fact that the human body loses excess heat (e.g., when air temperature exceeds body

temperature) by sweating, i.e., by evaporative cooling. However, once temperature and humidity pass a certain threshold, we are unable to lose heat by sweating, and, without going into an air-conditioned building, death will result. It is no good sitting in the shade for respite. The critical threshold occurs when a thermometer whose bulb is encased in a wet piece of cloth—the temperature of a so-called wet-bulb thermometer—reaches 35°C (95°F).

We have not yet experienced such wet-bulb temperatures, though in 2015 the wet-bulb temperature in Bandar Mahshahr in Iran reached 34.6°C (94°F)²². Lynas recounts how, in a 4°C warmer world, such deadly heatwaves will become commonplace in large parts of the Middle East and South Asia, including large parts of India and China. The consequence would likely be mass migration either polewards or to higher altitudes.

But this won't necessarily bring complete relief because, according to the models, subtropical regions like the Mediterranean, whilst not as hot as the Middle East, will have turned into deserts as the rain-bearing storms themselves shift polewards²³. The desertification of large parts of the world will impact the crops we grow for food, and simultaneous drought in the main breadbaskets of the world will not be uncommon.

On top of that, the more humid conditions will make extreme storms more intense. The latent heat released when water vapour condenses into liquid water plays an important role in determining the intensity of such storms. More water vapour in the air allows for greater latent heat release. With sea-level rise (caused by a combination of thermal expansion of water and disintegrating ice sheets), the storm surges associated with some of these extreme storms will be devastating for many coastal cities.

From Lynas's description, a 4°C or warmer world describes a vision of hell. I cannot imagine anyone would

want to live in such a world, even if it were possible to find ways of adapting to it in part.

However, we should not have to take Lynas's word for it. Since climate models are very similar in structure to weather forecast models—historically, they evolved from the weather models of the 1950s—they not only simulate changes in global temperature, they also simulate changes in regional weather patterns.

Indeed, because of this, we can in principle use climate models to study whether some actual observed weather event, let's say a heatwave in Canada or a flooding event in Germany, is somehow caused by climate change. This is a contentious issue, as these days climate maximalists tend to associate any observed extreme weather event with climate change, whilst minimalists dismiss them as part of the natural chaotic variability of climate.

This brings us to the third use of ensembles. Two separate ensembles of runs of physics-based climate models are made. In the first ensemble, the climate model is run for, say, one hundred years with an atmospheric carbon dioxide concentration that is fixed at values from just before the beginning of the Industrial Revolution. In the second ensemble, the carbon dioxide concentration is doubled. Let's suppose we are interested in whether climate change will affect the probability of a heatwave over California. We simply count the number of times such heatwaves occur in each of the two ensembles. If the model is realistic enough, Californian heatwaves will spontaneously occur in the first ensemble since they are part of the natural variability of climate. Comparing the two ensembles' heatwave frequencies will give us an estimate of the change in likelihood of a Californian heatwave due to our carbon emissions [24](#). The technique of quantitatively attributing real-world weather events to climate change was pioneered by my Oxford colleague

Myles Allen²⁵ in 2003, and the application of his ideas features strongly in the most recent IPCC assessment report.

It's certainly true that because a chaotic system never repeats itself, we can never be 100 percent certain that some real-life weather event was caused by our carbon emissions. It's true but pretty much irrelevant. What is relevant is knowing that, under climate change, the likelihood of this type of event has changed from 1 in 1,000 years to, say, 1 in 10 years. From such a statistic, we can decide whether it's worth trying to mitigate the effects of such weather events, for example by changing forestry management practices or by building better flood defences.

This methodology works if the weather events we study are not too extreme. However, there is a problem if the weather events are particularly extreme. If we try to perform attribution studies on some of the recent extreme events (e.g., from 2021, the extreme summer temperatures in British Columbia or the extreme rainfall in the Eifel region of Europe, Henan Province in China or New York), we find that current-generation models are not capable of simulating events of this intensity. This is simply because the climate model gridboxes aren't small enough to allow the model equations to generate the intensity of rainfall or temperature or wind associated with such extremes. The probability of such events in both pre-industrial times and in the twenty-first century is 0, according to current-generation models. Hence, according to the recipe I've described, the effect of climate change is equal to $0 / 0$, which is an undefined number—it could literally be anything. And yet it is these extreme events in which we are most interested. This very much suggests that if we try to use current climate models to estimate the nature of weather extremes in a 4°C warmer world, we will likely underestimate them.

The conclusion is not that climate change isn't affecting the likelihood of such extreme events. Rather, it means we need to put more resources into climate modelling²⁶ to decrease the size of model gridboxes enough to allow climate models to simulate such extremes. I return to this point further on.

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MAXIMALIST OR MINIMALIST: WHICH is scientifically correct?

Let's start by pointing out a few scientifically incorrect statements on both sides. Clearly what is wrong with the maximalist position (as stated earlier) is the assertion that climate change will be catastrophic. This is not supported by the science. Because of uncertainties in climate feedbacks, most importantly with clouds, the science is not that certain.

However, the minimalist position (as stated earlier) is scientifically incorrect in a number of ways.

For example, one cannot dismiss century-long climate projections just because a deterministic weather forecast goes wrong after a few days. Climate-change prediction is not an initial-value problem of the type suggested by Cleveland Abbe and Vilhelm Bjerknes, as we saw in [Chapter 5](#). In weather forecasting, we try to estimate where we will be on the climate attractor as accurately as possible at some point in the future. In climate forecasting, by contrast, we try to estimate how an external forcing (e.g., coming from our carbon emissions) changes the shape of the climate attractor as a whole. [Fig. 11](#) from Lorenz's model illustrates the point well: we can predict a change in the shape of the attractor with much more

confidence than we can predict where the state will be on the attractor at any time.

A second factually incorrect point in the minimalist argument as stated is that models from thirty years ago predicted too much warming and that these climate models made no other verifiable predictions. As mentioned, given the correct emissions scenario, the early models did predict the correct amount of warming. In addition, the earliest climate models correctly predicted the cooling of the stratosphere and the fact that the Arctic would be especially prone to warming.

Then there are statements in the minimalist position that are true but misleading. It is true that we can never be sure that some record-breaking event is not part of natural climate variability. However, as I have explained, this is a red herring. What we want to know is whether such types of events are becoming more likely, and by how much, as a result of climate change.

Another example of a misleading statement is that increases in carbon dioxide will help to green the planet. Such beneficial impacts will occur only as long as the weather doesn't become uncondusive to plant growth. It's no good having that extra plant material if there is too little or too much rain for it to grow or it is so hot that it wilts. Perhaps there is an analogy with addictive recreational drugs. Initially they seem beneficial to your sense of well-being, but ultimately, they can destroy you.

Insofar as the science of climate change is concerned with estimating the probabilities of different levels of climate change, it is unscientific to assert that we are heading towards climate maximalism or climate minimalism, or indeed any specific point in between.

It may appear from what I have written that I am more sympathetic with the climate maximalist position than the climate minimalist position. However, there is one element of the climate minimalist statement that I cannot

completely dismiss: the possibility that the climate models are, in part, unreliable. To be clear, I would find it very hard to believe that the probabilistic projections of global warming are completely unreliable, for the simple reason that these estimates are entirely consistent with basic physics in a moist atmosphere. On top of which, it is hard to think of any external forcing (like solar variability), or an internal dynamical mechanism, that would generate the observed changes in global temperatures in both the lower and upper atmospheres on a multidecadal timescale.

Nevertheless, there are good reasons to suppose, for example, that we are not estimating the cloud feedback term as accurately as we might. For example, no cloud parametrisation scheme can properly represent what is called the “mesoscale” structure of clouds: structures that are larger than individual clouds but smaller than the grid spacing of the models. For example, holes can appear in sheets of stratus cloud, like in a regular honeycomb²⁷. Such structures can’t be properly represented in current climate models because the gridboxes—typically 100 km (62 miles) or more in horizontal length—aren’t small enough. Another type of mesoscale structure which can’t be represented in climate models is associated with the way thunderstorm clouds sometimes aggregate together to form superstorms. When they do aggregate, the surrounding air is clear and relatively dry. Such structures change the radiative balance of the Earth. If the mesoscale structure of clouds evolves under climate change, this will contribute to the cloud feedback term.

Indeed, because of such inherent shortcomings, current-generation climate models depart systematically from observations in quantities like regional seasonally averaged rainfall. Indeed, these systematic departures (indicating a bias of the models compared with observations) are somewhat larger than the climate-change signals we are

asking the models to simulate ²⁸. In a nonlinear system like climate, this is an indication of unreliability on the regional scale.

We can imagine a climate version of the “Turing test” for AI. In the original Turing test—the imitation game²⁹—we ask the AI system questions and, based on the answers, try to tell whether we are talking to a human or a computer. Current AI systems cannot pass the Turing test. Similarly, current climate models cannot pass the climate Turing test; looking at output from these models, we can tell that it is coming from a computer model rather than from the real world³⁰.

Despite these shortcomings, we cannot simply ignore climate models. These models are our only tool to try to understand and estimate the future. It’s not like other areas of science where we can simply do a laboratory experiment. There are no lab experiments that can emulate climate change.

Climate models are central in determining the best strategy on carbon emissions. One particularly important issue is whether we will pass “tipping points”—changes to ice sheets, rain forests or ocean circulations which cannot be undone by negative carbon emissions at a later time. We can plant as many trees as we like, but if they are planted after one or more of these tipping points have been passed, they will be ineffective in returning us to current conditions. Current climate models are not able to assess reliably whether such tipping points will occur, as the models’ resolution is too coarse.

Output from climate models is also vital for assessing societal vulnerability to climate change on the regional scale, and so for determining the investments needed to make society resilient to changes in climate. This is particularly important in the developing world where extremes of weather are likely to become especially

extreme (e.g., wet-bulb temperature exceeding 35°C). Prioritisation of possible adaptation strategies will be crucial for such countries: Is preparing for drought and heatwaves more important than preparing for exceptional storms and deluges of rain, or vice versa? Again, this requires high-resolution simulations of climate.

Climate models are also vital if we are ever to consider a so-called Plan B in which we try to actively control the climate, e.g., by spraying aerosols (or their precursors) into the stratosphere to reflect sunlight back to space³¹. The danger with such alternatives is that they may have unexpected consequences, such as diverting the monsoons or shutting off the moisture supply to the rainforests. Reliable climate models are the only tools we have to quantify these risks.

It is for all these reasons that I am not happy with the status quo in climate modelling—i.e., with the multi-model ensembles of relatively low-resolution models, as used in the IPCC reports. They have done their job of warning the world of the dangers of climate change, but they are not well suited for estimating climate change at the regional level, particularly for estimating changes in the likelihood of extreme events. The problem is that individual institutes often lack the necessary human and computational resources. They simply can't afford to run their models at a higher resolution. As a result, we do not have the capability to produce high-resolution ensemble projections of climate.

My own view is that we need a new framework for developing next-generation climate models if we are to reduce some of these key uncertainties about climate change. Instead of a large number of subcritical institutional efforts, we need to pool resources into a "CERN for climate change"³², CERN being the international institute which found the Higgs boson. In this way, scientists should work more collaboratively to develop

next-generation, high-resolution climate models for the benefit of society worldwide. There is a relevant African proverb: if you want to go fast, go alone—if you want to go far, go together. Perhaps philanthropy can play a role in supporting such an institute. The cost per year need not be much more than that of launching a few craft into space.

With such an institute—or better, a federated hub of institutes around the world—we can envisage applying ensemble prediction in a fourth sense, one which connects with its use in weather prediction as described in [Chapter 5](#). As we discussed, it doesn't matter that weather is chaotic and unpredictable when making projections of future climate. As such we do not have to try hard to find precise initial conditions for the projections, as we would in weather forecasting. On the other hand, there is some evidence that decadal-timescale variability in the oceans may be partially predictable. For example, as mentioned right at the beginning of this book, decadal-timescale drought in the Sahel region of Africa may be linked to ocean circulations in the Atlantic. There is evidence that these circulations may be inherently predictable, that is to say, can be predicted given an ensemble of forecasts in which the ocean circulations are accurately initialised from observations. This raises the prospect of making regional climate projections over multidecadal-timescales using ensembles which are both carefully initialised with observations and in which scenarios for our carbon emissions are also included³³. Such an endeavour would need high-resolution climate ensembles. It would be an ideal goal for an international federated institute with pooled human and computing resources.

Here I am not arguing for a single, monolithic climate model. However, I am arguing for a rationalisation of effort around a few modelling systems (perhaps one or two per continent) which individually have sufficient resolution to

pass the climate Turing test. I hope this is something both climate minimalists and climate maximalists will support. The better we can represent our climate system with the primitive laws of physics, the more confidence we can have in the output of these models.

Let's summarise. Taking a specific position—minimalist, maximalist or indeed any specific point in between—is simply inconsistent with the science. The key message of this chapter is that one's attitude to climate change, like to weather prediction, should be framed in terms of risk: Is the risk of undesirable changes to climate high enough to warrant taking precautionary action now? Consider the possibility of the Earth spiralling off to infinity, as discussed in [Chapter 1](#). If this were to occur, it would wipe out humanity. We could take precautions of a sort by diverting massive resources into populating Mars (assuming Mars won't itself be ejected at the same time). However, the risk is so small—recall that not one single member of the ensemble of solar system forecasts predicted this would happen—that it is clearly not worth taking such precautions. However, as we have seen, the risk of a hellish 4°C or warmer world is far from negligible—it is only a few tenths of a degree warmer than what we would expect from a doubling of carbon dioxide concentration.

If we think the risk is worth the cost, then manifestly the uncertainties discussed in this chapter should not deter us from taking precautionary action. That is, uncertainty should not be a cause for inaction. However, given the importance of the problem, if we do decide to act, we must surely continue to fund the research that will help reduce uncertainties, or will at least increase confidence in the reliability of our estimates of uncertainty. Whether the risk really is worth the cost is something I will address in [Chapter 10](#).

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7

Pandemics

COVID-19 HAS KILLED MILLIONS AROUND THE WORLD. AND YET, IF we can leave aside the misery it has brought, the disease provides a fascinating example of the role of ensemble prediction in practice. COVID-19 also illustrates an extraordinary number of parallels with the climate-change problem. What can we learn from this?

Early in the days of the pandemic, governments around the world faced difficult decisions about which most had no prior experience. Indeed, although some, like the UK, had made plans for a major outbreak in disease and illness, the plans were not really appropriate for COVID-19, where many in the population were asymptomatic carriers. In any case, the crucial question was whether to impose strict lockdowns to try to minimise the spread of the virus, and hence minimise the risk that the countries' health services would collapse under the weight of COVID infections, but with the consequence that the countries' economies would suffer badly. The alternative was to simply let economies continue as unimpeded as possible, suffer the health consequences and hope for some eventual herd immunity. Many policies were based on the former approach, though some governments took a more laissez-faire attitude.

This strikingly paralleled strategies on tackling climate change: try to cut emissions and get to net-zero as quickly

as possible, or do little that might threaten economic growth. As far as I could tell, those who advocated minimal restriction to the economy during the pandemic tended to also advocate minimal restrictions on the use of fossil fuels.

In the UK at least, the politicians repeatedly said that we must “follow the science” in dealing with the pandemic. However, as with climate change, science does not itself advocate a particular policy. If you want to reduce the risk of dangerous levels of climate change, then, according to climate science, you should reduce carbon emissions. Climate-change science does not say that you must reduce emissions—that involves making value judgements. Similarly, if you want to reduce the risk that health services get overrun, then, according to COVID-prediction science, you should restrict the way people interact with one another. The science does not say that you therefore must restrict such interactions. That too involves value judgements.

To invoke measures that minimise the risk of hospitals being overrun, governments need ways of predicting the number of people needing hospitalisation as a function of different possible policy measures. The predictions will therefore be contingent on these policy measures. This is exactly the same as occurs in estimating climate change. Because of this, the COVID-modelling community uses the same terminology as the climate-modelling community: predictions contingent on particular policies are called “projections”. This is important when trying to assess the accuracy of a particular projection. It’s no good comparing real-world data with a projection of hospital admissions that assumes a policy of normal, unrestricted societal interaction if in the real world the government implemented a policy to restrict interaction. This is like comparing the observed global warming rate with climate projections based on bigger carbon emissions than occurred in reality.

The core of typical COVID prediction models is quite simple. The population at any time is divided into three classes: individuals who have not yet been in contact with the virus but are susceptible to infection; individuals who have the virus and can infect others; and individuals who have been infected and have either recovered or died, but in either case cannot infect others. This is often referred to as an SIR (susceptible/infected/removed) model.

The development of effective vaccines does not invalidate the SIR model. Now vaccinated individuals are less susceptible to infection (and are much less likely to need hospitalisation and much less likely to die). This effectively increases the number of “removed” individuals in the SIR model.

The difficulty in predicting the spread of the disease is less to do with uncertainty in classifying the stages of infection, and more to do with uncertainty in the way we humans interact with each other, and hence uncertainty into which of these three classes an individual will belong. At one extreme, one can imagine a hypothetical population of individuals which is fully homogeneous in the sense that each individual has the same chance of interacting with, and hence of passing the infection to, any other individual. This is described by the R_0 parameter: R_0 describes how many people each infected person will infect on average. With $R_0 = 2$, on average 1 infected person will infect 2 people, these 2 people will infect 4 people, these 4 people will infect 8 and so on. This is called exponential growth. More generally, if R_0 is greater than 1, the disease grows. If R_0 is less than 1, the virus will eventually die out on its own.

Clearly no population is homogeneous in this way. We each live in our own regions, our own cities, towns and villages. Living in the south of England, I am much less likely to come into contact with someone living in Scotland

than with someone living in, say, London. We can try to account for this by estimating the R_0 parameter for different regions. On the other hand, if this heterogeneity extends down to the smallest scales—as well it might—then R_0 ceases to be a simple parameter, and instead becomes a highly dynamic variable which fluctuates wildly from one grouping of people to another, and from time to time.

The most obvious manifestation of the heterogeneity of society is the potential existence of “superspreaders”—individuals who infect many more people than would be implied by a single R_0 parameter—or “superspreader events”, where many more people would get infected than would be expected from a more homogeneous society. An important question is whether overall infection rates can be significantly influenced by a small number of superspreader events. To answer this, the assumption of homogeneity must be replaced with one which is not overly complex and intractable, and yet takes account of the way in which individuals interact as a function of age, geographic location and other factors.

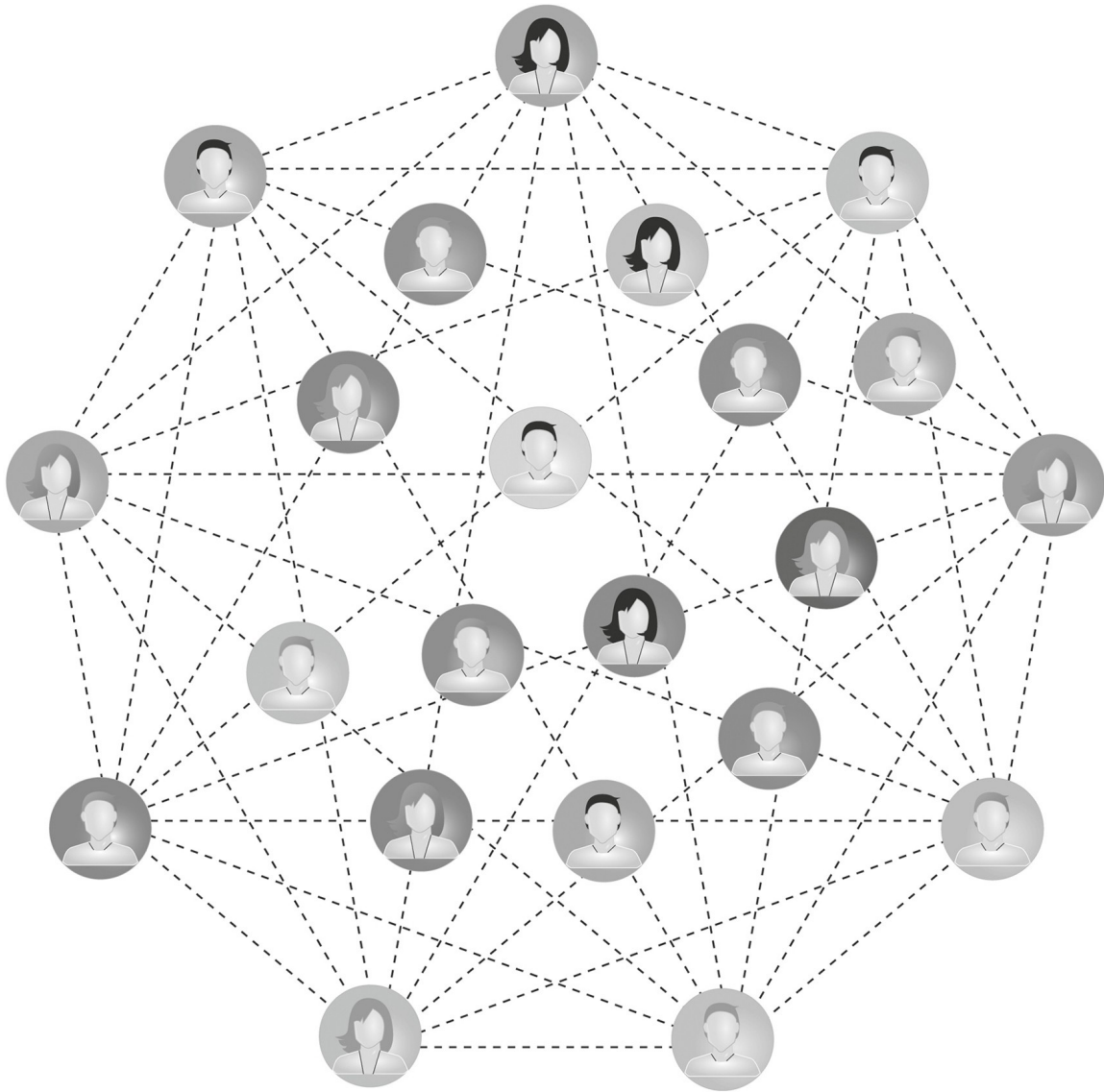


FIG. 34. An example of a network of people. Individuals correspond to nodes of the network, and they interact if they are joined or linked together.

One way to model the more heterogeneous aspects of society is through what are known as networks¹. From a mathematical point of view, a network is easy to describe: it is a set of points, called “nodes”, and lines joining some of these points, called “links”. A simple example of a network is shown in [Fig. 34](#). Here nodes reflect individuals, and they are joined by links when the individuals interact.

For example, imagine a cocktail party where initially no one knows anyone. The nodes of the associated network would correspond to the individuals at the party. If two people talk to each other during the party, we draw a link between the two corresponding nodes. Since it is fairly unpredictable who will talk to whom, we can model the interactions between individuals in a stochastic way, i.e., using random numbers. That is to say, we assign a probability p that these individuals will interact. If all the individuals are introverts, then p will be quite small; if they are all extroverts, p will be larger. Nodes are then joined randomly, consistent with the probability p .

This is sometimes called a random network. One of the interesting properties of a random network is that it exhibits what is sometimes called the “small-world” property: if you choose any two individuals on Earth, you will find that one of them knows someone, who knows someone, who knows someone, who knows someone, who knows the other person, i.e., that any two people are linked by at most six intermediate acquaintances. For many years after network theory was developed, it was thought that such random networks described societal and many other types of interaction quite accurately.

In random networks of the type I’ve described, it is very unlikely that nodes with a large number of links will exist if p is small. For example, with 100 nodes and with $p = 0.02$, then we would expect each node to have about 2 links emanating from it on average. The likelihood that a node has, say, 8 links emanating from it is virtually zero. This is because of the assumption that if the probability that individual A interacts with individual B is equal to p then the probability that individual A will interact with both B and C is equal to $p \times p = p^2$. If p is a small number (e.g., 0.02), then p^2 is a very small number (0.0004). Because of

this, the number of nodes with k links to other nodes decreases exponentially the larger is k . Just as exponential growth represents very rapid growth, so exponential decay represents very rapid decay.

However, in the real world, this assumption frequently turns out to be incorrect. Imagine that at our cocktail party we had invited a few celebrities. Then, likely, we would find that many people would try to interact with those particular individuals. In terms of the corresponding network, there would be many more links emanating from the celebrity nodes compared with the non-celebrity nodes. If we analyse the World Wide Web, where the URLs (websites) are represented by nodes, then we will find numerous nodes with a small number of links coexisting with a few nodes with a very large number of links—we call these “hubs”. If we look at an air-traffic network, we find a few hubs coexisting with many nodes of small degree. These hubs correspond to superspreaders in COVID transmission.

In networks with hubs, the number of nodes with k links does not decrease exponentially as k increases. Instead, the probability that a node has k links is described by what is called a “power-law” distribution, where the probability decays with k much more slowly than exponential decay.

Such power-law distributions also describe how energy, in the spectrum of whirls in a turbulent fluid, varies with the size of the whirl. Indeed, fractal geometries can also be described by such power-law distributions. The mathematician Benoit Mandelbrot—the one who coined the word “fractal”—drew attention to these connections in his book *The (Mis)behaviour of Markets*². He made the point that extreme events, like financial crashes, which might superficially be assessed as extremely unlikely, are not unlikely at all in models with this type of power-law behaviour. The existence of power-law behaviour is telling

us about a type of nonlinearity which underpins multiscale systems, which is common to the spread of disease, the economy, the weather and maybe even the whole universe.

Using power-law networks of ten thousand nodes, where nodes represent individuals who are either “susceptible”, “infected” or “recovered”, it was indeed found that the best strategy for curbing the spread of the infection was to reduce the extent to which the superspreader hubs spread the infection³. Here the spread of the disease when the number of links emanating from a hub is reduced is compared with a strategy in which the number of links emanating from randomly chosen nodes is cut.

Whilst such network models are useful in drawing attention to the importance of heterogeneity of the population, they are still too idealised to be useful in actually estimating infection and hence hospitalisation numbers. For a country like the UK, one would need to increase the number of nodes from ten thousand to about 67 million if one were to capture the population using one of these networks (and model the way in which all such individuals interact with each other). And that would not include the infections coming from people travelling to the UK from abroad. Although this is not completely inconceivable—as I discuss later on—with present supercomputers it is not feasible.

Instead, state-of-the-art epidemiological models—such as the Imperial College COVIDSim model⁴—work in a somewhat different way. COVIDSim is an important model as its projections provided one of the key pieces of information that led the UK government to order a nationwide lockdown in March 2020 to try to contain the virus. In COVIDSim, the country is tiled with gridsquares, in much the same way a weather or climate model tiles the Earth with gridboxes. Each of the gridsquares contains information about the population within the gridsquare—

for example, population density, age structure and household structure, along with information about schools, universities and workplaces. This information is represented in the form of parameters—some 940 of them in total—numbers which are fixed and don't change with time during a model forecast.

A key question for any government is whether the projections from a model like COVIDSim are reliable. Can the projections be trusted to give a plausible picture of the future? Clearly this depends on the accuracy of the parameter values. It turns out that there is real uncertainty in about sixty of the parameter values and the model output is especially sensitive to the value of about nineteen of these uncertain parameters.

Is it important to know how uncertain the values of these sensitive parameters are? Suppose we wanted to know only the expected numbers of hospitalisations and deaths from COVID. Then we might argue that all we need to do is run the model with the most likely values of the parameters. As such, the uncertainty in these parameter values wouldn't be judged to be particularly important.

However, this argument is valid only if the system we are dealing with is fundamentally linear. The emergence of power-law network scaling suggests that we are not dealing with linear systems. Here we can get some guidance from climate-change projections. In [Chapter 6](#) we saw that if we vary the uncertain subgrid parametrisations in a climate model, the global warming caused by a doubling of carbon dioxide appears to have a rather asymmetric, skewed probability distribution with a long, fat tail going out to large warming values (and no corresponding tail at low warming values). For such a distribution, the expected warming—the average over an ensemble of models with different parametrisations—is larger than the warming obtained by setting the parameter

values at their most likely values. The same situation occurs here.

In 2021, a study⁵ was published of ensemble projections with COVIDSim in which these uncertain parameters were varied according to plausible estimates of their uncertainty. These estimates were obtained by what is called “expert solicitation”, which means asking the experts how much they felt a parameter value could be in error: 10 percent, 20 percent, 50 percent? Ensembles were then run in which the parameter values were randomly perturbed by the experts’ estimates of parameter uncertainty. One of the key results from the study is shown in [Fig. 35](#). It illustrates probability distributions of projected COVID deaths for two different policy scenarios.

Interestingly, the ensemble-based probability distribution of projected deaths, particularly for the first policy scenario, is remarkably similar to the probability distribution of global warming due to a doubling of carbon dioxide—a skewed distribution with a long, fat tail going out to large numbers of deaths.

As with global warming, this means that the expected number of deaths cannot be reliably estimated using a model simulation run with the most likely values of the uncertain parameters. This is shown explicitly in [Fig. 35](#). The expected number of deaths is higher than the number predicted using the most likely values of the parameters. In the case of the second, more laissez-faire policy, the estimated cumulative number of deaths at day three hundred using the most likely value of the parameters is around 25,000. However, the expected number of deaths based on the full ensemble with perturbed parameters is significantly larger than this, around 40,000.

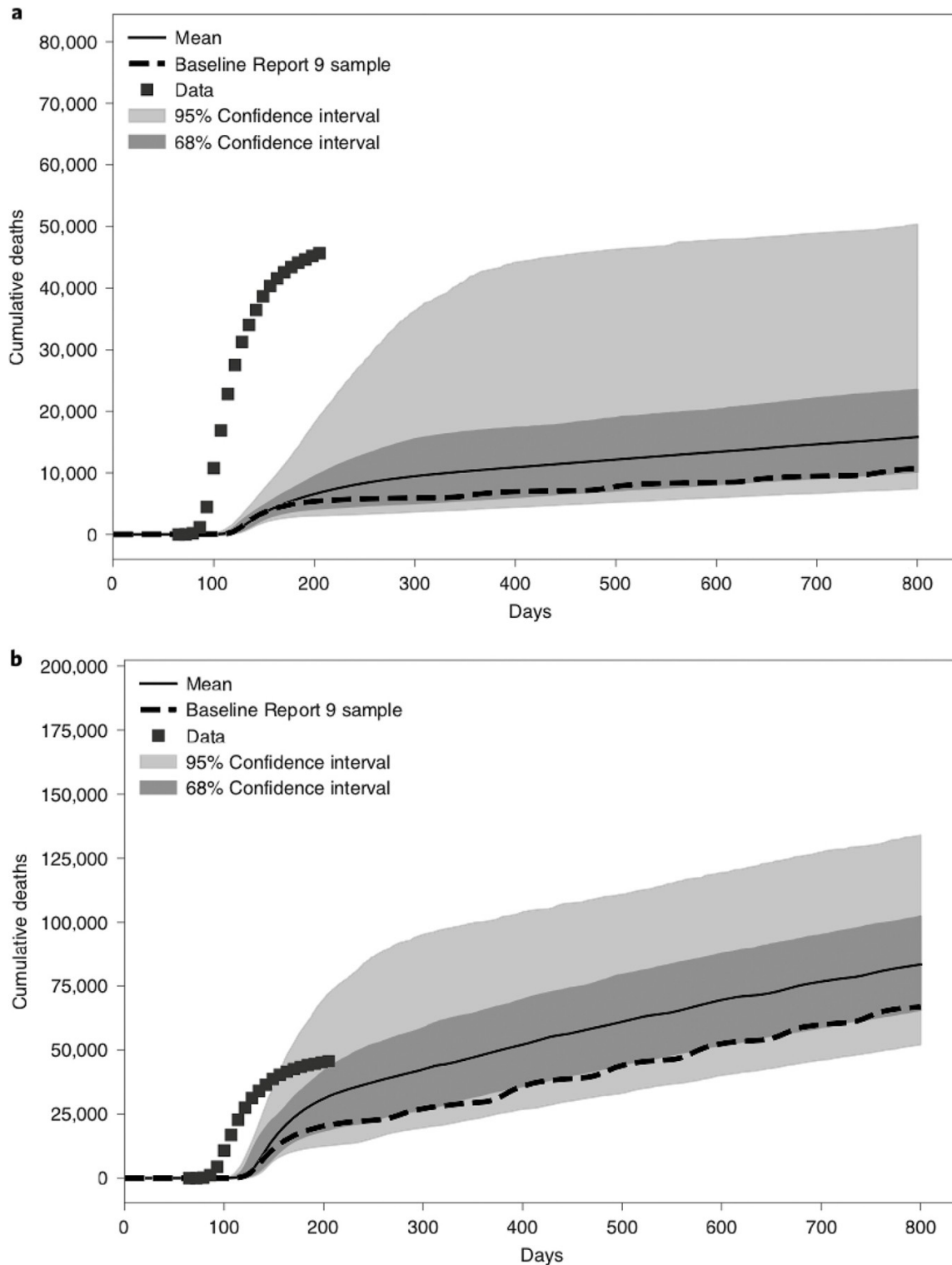


FIG. 35. (Facing Page) The grey shading denotes estimates of uncertainty in COVID deaths based on ensembles of Imperial College COVIDSim model projections. The darker the shading, the more likely is the estimated number of deaths. The difference between the top and lower illustrations

reflects different policy assumptions (e.g., whether social gatherings are allowed). The thin solid lines show the average of the ensemble distributions. The dashed lines show the estimated number of COVID deaths projected by a model with its parameters fixed at their most likely values. The former is higher than the latter, providing an important reason to make an ensemble of projections. The squares show real data. The fact that reality does not lie within the ensemble estimates of uncertainty suggests that the ensemble does not fully capture the true uncertainty in these projections (especially initial infections). From Edeling et al. (2021).

Estimates of the expected numbers of deaths and hospitalisations may not be the most relevant information for policy decisions. Having a health system that essentially collapses because it is overwhelmed with new patients is something a government may not wish to risk at all. As such, plausible worst-case outcomes can be just as important, if not more so, as expected outcomes. This is similar to the situation in weather forecasting—the averaged forecast can hide the possibility of some catastrophic weather event. Hence, for example, if we take, as an estimate of a plausible worst-case scenario, the value given by the distribution at the edge of the heavier shading (the region where, according to the ensemble, there is a two-thirds probability of occurrence), then the number of deaths rises to 60,000, well over double the prediction with best-guess parameter values.

However, [Fig. 35](#) also illustrates a problem. As discussed, an ensemble prediction is useful only if the probabilities it generates are reliable. For the ensemble predictions I have helped develop in weather and climate science, far and away the largest part of research time was spent on developing initial-condition and model perturbations which would produce reliable probabilities. We can see in [Fig. 35](#) that in the first hundred days or so, the actual numbers of deaths rose much more sharply than the ensemble suggested could occur. This means that the ensemble probabilities, in this case, weren't reliable.

Clearly some other sources of uncertainty are important. In this case, the missing source of uncertainty is associated with the date the disease started to grow—in reality it must have been growing sooner than was thought possible. That is to say, the projections used unduly certain initial conditions. This result is consistent with weather prediction, where uncertainty in the forecast initial conditions is typically more important than uncertainty in the values of model parameters.

However, there is a second type of uncertainty in these projections which is not well represented by simple parameter perturbations. For example, COVIDSim is based on a tiling of the country. Are the assumptions of homogeneity within a tiled region accurate? How small should the tiles be for this assumption to be correct? Is it ever correct? More generally, there may be many assumptions in the models where uncertainty goes far beyond simple parameter uncertainty. Such uncertainties are often referred to as “structural model uncertainties”.

This is perhaps a good time to dust off the much-repeated saying from Donald Rumsfeld, US secretary of defense at the time of the invasion of Iraq. Speaking about reports of the lack of evidence for weapons of mass destruction, he famously said in 2002⁶:

“Reports that say that something hasn’t happened are always interesting to me, because as we know, there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns—the ones we don’t know we don’t know. And if one looks throughout the history of our country and other free countries, it is the latter category that tends to be the difficult ones.”

In the context of modelling, and this could be COVID modelling or weather and climate modelling, the known

knowns could refer to a subset of parameter values which are known to great certainty. In a weather or climate model, this could be the value of the gravitational constant or the amount of energy the Earth receives from the sun. In COVIDSim, this could be some of the demographic data. Continuing, the known unknowns are the parameters where we can quantify the range of uncertainty in the parameter values. In weather and climate models, the values of some of the parameters relating to the microphysical properties of clouds are uncertain, but the range of uncertainty can be quantified.

In this context, Rumsfeld's unknown unknowns refer to structural model uncertainties—to uncertainties in the way in which the computational equations are formulated in the first place. Here it is not easy to quantify the range of uncertainty of a parameter value, because the process in question may not be representable by a fixed parameter value at all. In a weather or climate model, the structural form of the parametrisation formulae used to represent unresolved processes such as clouds are profoundly uncertain; there is no good reason to suppose that one can represent clouds accurately with such formulae. Hence, some of the parameters in such a parametrisation are ontologically rather than epistemically uncertain, to borrow from the philosophical language used in [Chapter 4](#). Similarly, in COVIDSim, the basic assumptions around the way in which humans interact with one another, and indeed respond to governmental restrictions, are themselves ontologically uncertain.

If we are to aspire to truly reliable predictions and projections, we have to find some way of handling these structural model uncertainties. However, they can be some of the hardest uncertainties to represent and quantify. There's an old North English saying: "There's nowt so queer as folk!" For sure, there's nowt as uncertain as folk. In [Chapter 3](#), I discussed the representation of unresolved

scales in models of fluid turbulence using random noise. Such noise can in principle also provide a representation of some of the uncertainty associated with structural model error—even with the uncertainty of folk. Now whilst models like COVIDSim do have some inherent stochasticity in their equations, the associated noise processes do not contribute much to the ensemble spread⁷. As such, the use of stochasticity as a means to represent inherent structural uncertainty in these COVID models is likely being underdone. One argument I heard in this regard is that if we were to use stochasticity to represent structural model uncertainty, then the spread from the ensembles would be unusably huge. But of course, that is the whole point of ensembles! When ensemble spread is truly huge, as it was for the retrospective ensembles for the weather for October 1987, then the ensembles tell you, on this occasion at least, not to trust any of the outcomes and to be wary of the possibility of extreme outcomes. The worst thing, under such circumstances, is to act on the basis of unreliably confident, underspread ensembles.

In [Chapter 5](#) I discussed another approach to the problem of structural model error: the multi-model ensemble. For seasonal-timescale climate predictions of ENSO, the multi-model ensemble was more skilful than were single-model ensembles. On a very pragmatic level, the rather difficult issue of the ontological uncertainty of structural model error was addressed by constructing ensembles using multiple models developed in different institutes. In some ways, this is no different to the “wisdom of the crowd” concept, where here the “crowd” is not the views of members of the public, but the views of differently constructed climate models developed in different institutes.

A limited form of multi-model ensemble was used by the UK government’s Scientific Advisory Group for

Emergencies when making COVID projections⁸. However, the multi-model ensemble approach has been developed more extensively in the United States, where there are many more individual groups developing COVID models. Here, the US Centers for Disease Control and Prevention (CDC) partnered with an academic research lab at the University of Massachusetts Amherst to create the COVID-19 Forecast Hub, launched in April 2020. Predictions focus on the one-to-two-month timescale, which can inform operational decisions about allocation of healthcare supplies, staffing needs or such matters as school closures. Essentially, any team in the US was permitted to submit a forecast model to the multi-model ensemble as long as data was provided in a specified format and a full description of the methods used to generate a forecast was given. According to a paper describing this technique⁹, some twenty-three models had met the eligibility criteria and were used in the evaluation of the multi-model ensemble. Separate forecasts were submitted for all fifty states in the US, and a national forecast for the whole of the US.

The results were entirely in line with what we found for seasonal climate prediction. Although for any one particular forecast it was possible to find a model that performed better than the ensemble, the ensemble performed better on average, both in terms of an ensemble-averaged prediction and as a probability forecast. Clearly, consistency in skill is important when making decisions, indicating that the multi-model ensemble was preferable to any individual model forecast.

Since the US COVID-19 Forecast Hub went live, a similar multi-model ensemble system has been implemented in Europe. There is perhaps one further step to go: a global multi-model ensemble forecast system for disease prediction, to which all nations of the Earth can contribute

their models in a coordinated way. There is a precedent for this.

In the mid-1990s, the World Climate Research Programme—an international body of climate scientists sponsored by the World Meteorological Organisation and the International Oceanographic Commission (both United Nations bodies) as well as the International Science Council began to organise a set of coordinated multi-model ensemble experiments that would feed into the Intergovernmental Panel on Climate Change assessment reports. The project which organised these coordinated experiments became known as CMIP—the Coupled Model Intercomparison Project. In the first multi-model CMIP experiment, climate modelling groups from around the world ran experiments with their models with a prescribed 1 percent increase in carbon dioxide concentration per year. In the latest CMIP experiment, the prescribed increases in carbon dioxide are based on different socioeconomic scenarios, ranging from what is sometimes called “business as usual”, to net-zero by 2050. There are now some thirty different modelling groups contributing to the CMIP multi-model ensemble.

It seems to me that there is a need for global collaboration like this for disease prediction in general—not just for COVID but for other diseases such as malaria, dengue, influenza and Ebola. Perhaps the World Health Organization could take the lead.

But there are downsides to the multi-model ensemble. The “wisdom of the crowd” works only if estimates from members of the crowd really are independent. For example, if there is a rumour going round that Fred guessed five hundred jellybeans, and that Fred is quite a smart guy, then the possibility that other people’s guesses are influenced by Fred’s estimate can completely negate the benefit of the ensemble approach to guessing the number of jellybeans in the jar.

In practice, individual modelling groups are influenced by other groups' assumptions, and so the models are typically not independent of one another. They are at best quasi-independent. This is certainly true in climate modelling, and I would find it hard to believe it is not also true in disease modelling. The types of model representations and parametrisations that different institutes use can be strongly influenced by one dominant player in the group. That is to say, in a multi-model ensemble, the models may be less independent of one another, and hence less useful, than an outsider might imagine them to be.

There is another problem. Each modelling group is typically not composed of a team of one hundred researchers, say. It may be only a half a dozen. Indeed, the team may comprise just one person. The question then arises—Which is a better use of resources: funding the development of a large number of quasi-independent models, or focussing human and computing resources to produce a much smaller number of (necessarily stochastic) models? In terms of climate, I believe the latter is a preferable route. I am not expert enough in disease modelling to know which strategy is better. However, it strikes me as an important topic for discussion.

There is one aspect of these COVID ensembles that hasn't been discussed. If we study, say, the one-to-two-month multi-model ensembles across a range of initial conditions, is there evidence that the spread of the ensembles varies from one initial condition to another (as illustrated in [Fig. 10](#))? If so, can the spread of the ensemble be used to predict the likely skill of, say, the ensemble-average forecast? For example, if the spread of an ensemble is especially small, we would expect each ensemble member to be close to the truth. By contrast, when the spread is especially large, many of the ensemble members will turn out not to be skilful at all. This is only

now starting to be studied. Results will have to await a new edition of this book.

At the time of writing, COVID-19 isn't over. The highly transmissible Omicron variant has spread exponentially. The primary uncertainty about Omicron when it first appeared was its virulence—how many hospitalisations and deaths would it cause? Fortunately, it seems not to be as virulent as some of the previous variants. In the UK, some politicians to the right of centre have complained that scientists have somehow exaggerated the Omicron predictions, giving politicians only worst-case scenarios¹⁰. Of course, this is not so. As we have discussed, the job of the scientists is to provide information about the whole range of possible future hospitalisations and deaths, including the expected as well as plausible worst-case and best-case scenarios. In the earlier phases of development, when little is known, the predictions are necessarily broad. As with climate change, it's up to the politicians to decide on policy implications given such forecast information—further lockdowns, lighter measures to try to restrict social interactions or a complete relaxation of such measures. Of course, it isn't easy to decide when predictions are highly uncertain. However, making politicians' lives easier by giving them forecasts which are more precise, but less reliable, is not the right way to go—though more precise forecasts would provide politicians with a scapegoat (the scientists) if policies turn out badly.

In the meantime it seems we need a new generation of ensemble systems which, coupled to these SIR models, can probabilistically predict the spread of COVID virus mutations. Some first steps in this direction have already been taken with models trained using the many millions of genome sequences of COVID viruses (mutations of the SARS-CoV-2 virus to be more specific). In one paper ¹¹, the authors develop a model for predicting virus mutations that

will spread across the population for different phases of the pandemic. The key information for making such predictions is an analysis of the recent patterns of spread of a particular mutation/variant, and whether particular mutations appear in different viral subtypes (a criterion called “convergence” in biology). Based on observations outside the training data, the authors show that the model has predictive power out to about four months.

Ultimately, one wonders whether it might be possible to predict the spread of virulent virus mutations from first principles, i.e., directly from the basic genomic structure of the virus. However, there are so many potential mutations to consider that predicting viral evolution in a general way would be completely intractable with present-day computational technology. On the other hand, this is the sort of problem that might conceivably be solvable with some future-generation quantum computing technology, discussed later in the book.



8

Financial Crashes

ACCORDING TO ANDY HALDANE IN 2017, THEN THE BANK OF England's chief economist, economists had their "Michael Fish moment" when the global financial markets collapsed without warning in 2008 ¹. The economic models were simply unable to function properly when "the world was tipped upside down".

When I read Haldane's comments, his reference to Michael Fish brought to mind something that I had wondered from time to time: whether the same ensemble techniques that have transformed weather and climate prediction could also play a role in economic prediction. If it was possible to create a reliable ensemble prediction system, would the 2008 crash turn out to have been something inevitable, insensitive to the precise economic starting conditions and economic model equations, or would it look like the actual Michael Fish storm, something highly unpredictable and only forecastable in a probabilistic sense? Even if the latter, a probabilistic ensemble prediction would still be useful. Just as forecasting the October 1987 storm in a probabilistic way is enough to make simple planning decisions, so too presumably if an imminent global financial crash is predictable only probabilistically.

However, responses from economists about the possibility of creating such an ensemble forecast system were mostly disappointing. There were a number of different reactions, all of which were broadly negative in one way or another. The first response was that in economics, more than in weather and climate prediction, there is an element of uncertainty that is itself unknowable. This is often called “radical uncertainty”². It’s related to the type of ontological uncertainty discussed in [Chapter 7](#). For example, no one could have predicted the 9/11 attack on the Twin Towers, and yet this had a major impact on the state of the world, including its economies, resonating to the present day. The view was that such radical uncertainty would make the prediction of financial crashes impossible.

Another point raised by the economists was that whilst weather follows along the “tramlines” provided by the Navier-Stokes equation, there is no known equation which governs the state of the economy. Economists are in the dark, so the argument went, about the equations which govern financial shocks and crises.

Then there was the “self-referential” argument. A weather forecast doesn’t change the weather, whilst an economic forecast can change the economy. Hence the two systems are completely different from each other and not much can be learned comparing one with the other. Finally, I met economists who, although somewhat sympathetic to what I was suggesting, said that economists had not yet developed techniques to assimilate economic data into models with the same mathematical sophistication that meteorologists use to assimilate weather observations into their models.

Dispirited, I let the idea drop. But in planning this book, I decided to revisit these issues. Perhaps things had moved on. Alas, writing to some economists I knew, I got pretty much the same responses as the first time around. Are they

right? Is the idea of creating some kind of ensemble prediction system for predicting economic shocks a complete non-starter? Of course, as an outsider in this field, it would be arrogant of me to claim that the experts are categorically wrong. However, I was not convinced by their objections. For example, radical ontological uncertainty does certainly play a role in weather and climate prediction. Indeed, I might make a probabilistic seasonal forecast for six months from now, only for it to be completely undone by an unpredicted volcano whose aerosols spread around the atmosphere and dim the sun's radiation. In principle, one could include a volcano in one or two ensemble members. However, that is not what we do. We just accept that, occasionally, even probabilistic forecasts can go wrong as a result of the radical uncertainty of a volcano. I don't think that makes the probabilistic El Niño forecasts any the less useful.

However, as discussed in [Chapter 3](#), we do include a certain amount of radical uncertainty in our weather forecast models by introducing noise into the truncated version of the Navier-Stokes equation. It really isn't the case that the Navier-Stokes equation provides hard-and-fast tramlines on which the forecasts evolve. Now it may be that to model the radical uncertainties that exist when modelling the economy, the amplitude of the stochastic noise would have to be much larger than it is when modelling the weather. If realistic amounts of noise mean that the ensembles are always so highly spread that no useful signal can be extracted, then so be it. However, as with forecasts of weather, it may be that sometimes the spread isn't that large, meaning that the state of the economy is able to absorb the inherent radical uncertainties that exist. On other occasions, the ensembles may tell us that the state of the economy is like the kingdom whose existence is so fragile that it can be compromised by a missing horseshoe nail.

Regarding the self-referential issue, although it is the case that weather forecasts don't change the weather, it certainly is the case that climate forecasts can change the climate, as discussed in [Chapters 6 and 7](#). For many years, climate scientists have known that they are not really making predictions of climate change but projections; they are "merely" estimating how the climate will respond to certain emissions scenarios for greenhouse gases. These projections can be verified once the emissions scenarios are known. In this sense, the difference between climate and economic projections need not necessarily be fundamental.

Economists do in fact make efforts to present their projections, such as the inflation rate or the gross domestic product, with estimates of uncertainty. For example, the Bank of England's projections are in the form of what are known as "fan charts"³. These essentially give error bars on predictions and were inspired by discussions with meteorologists back in the 1990s. They are called fan charts because the uncertainties get larger, as one would expect, for longer forecast times. However, these fan charts are not themselves based on ensembles. Instead, they are based on a statistical analysis of errors from past forecasts. However, there are times when subjective judgements are added to widen or narrow the fan charts from one forecast to the next, based on whether the outlook is deemed to be more or less uncertain.

This was the key question for me. As a nonlinear system, the predictability of the economy should itself be variable, perhaps similar to what was shown in [Fig. 10](#) for the Lorenz model. Instead of thinking of the two lobes of the Lorenz system as, say, fair weather and foul weather, let's imagine them as representing a vibrant economy and a depressed economy. [Fig. 10](#) showed that there were situations where the transition from one lobe to the other

was quite predictable, and hence rather inevitable, and yet others where it was very unpredictable, and hence not at all inevitable. Is there a way to characterise the predictability of the economic system under capitalism as it moves between boom and bust in this way?

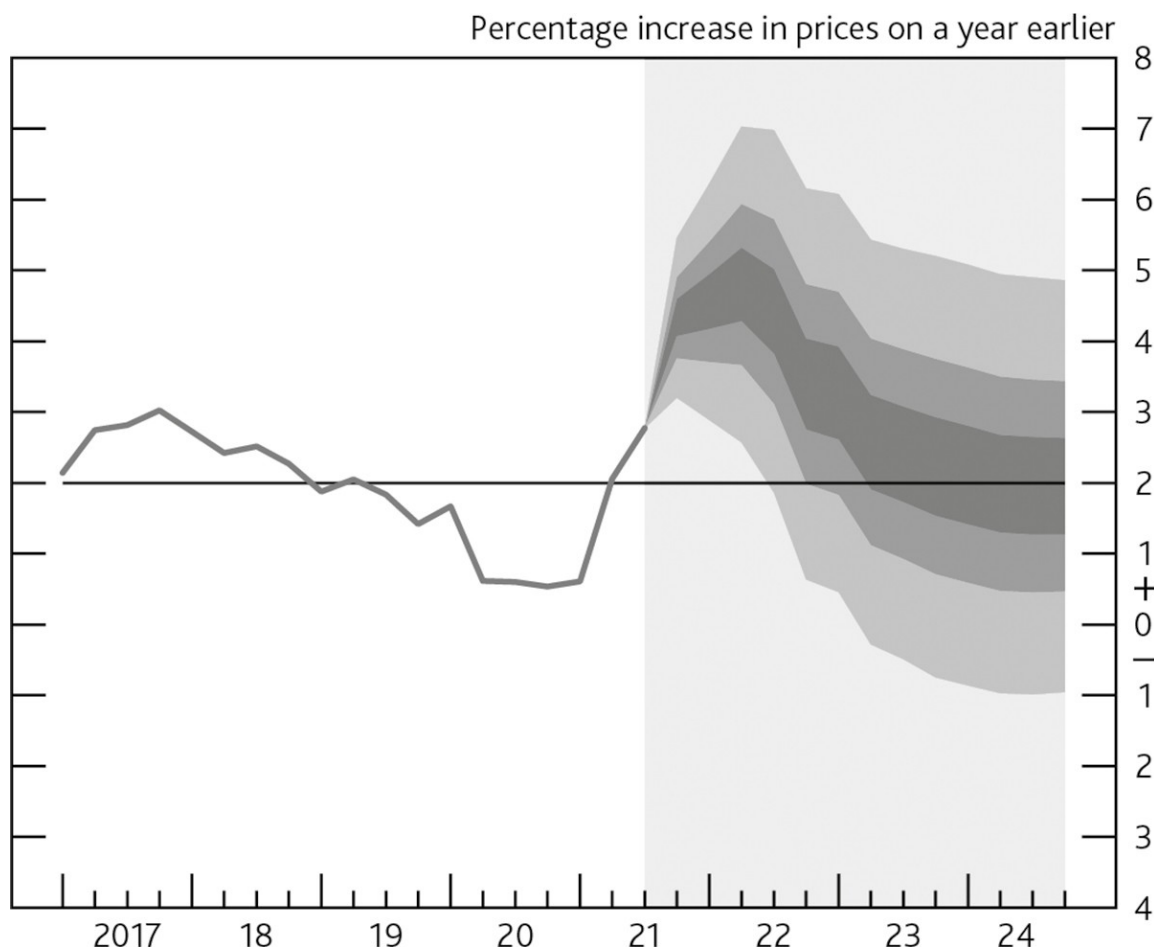


FIG. 36. A Bank of England “fan chart” predicting the rate of inflation (from November 2019). The shaded regions denote the likelihood that the true rate of inflation lies within certain values. The heavier the shading, the more likely lies the true rate of inflation. However, these fan charts are based not on ensemble forecasts but on a statistical analysis of past error, together with subjective judgements about future uncertainty. From the Bank of England Monetary Policy Report, November 2021. With permission from the Bank of England.

To try to quantify such predictability, we need good models of the economy. What is a “good” economic model? Based on discussions with economists, the authors of a

paper entitled “The Seven Properties of Good [Economic] Models⁴” identify goodness in terms of parsimony, tractability, conceptual insightfulness, generalisability, falsifiability, empirical consistency and predictive precision. The authors then go on to comment that only the first four are well accepted by economists, the last three not so.

This struck me as very odd. If I were to ask a group of my meteorologist colleagues to characterise the principal difference between a good weather forecast model and a bad weather forecast model, the answer would hardly be surprising: a good weather forecast model makes accurate weather forecasts and a bad one makes inaccurate weather forecasts. There may be some discussion about how one might measure the accuracy or inaccuracy of weather forecasts (e.g., how much emphasis one should give to forecasting extreme weather situations rather than common or garden situations) but that is a matter of detail.

In other words, the properties of falsifiability, empirical consistency and predictive precision, which are at the bottom of these economists’ list, would be at the top of the list for meteorologists, and vice versa. Indeed, I have never heard a meteorologist use the word “parsimony” as a desirable trait for a weather forecast model. This does not mean they put processes into their models that are irrelevant, like the weather on Jupiter. However, there is no conscious effort to pare things down to the bare minimum—rather the philosophy is to add as much detail into the model as the computer resources allow.

However, there are circumstances where parsimony, tractability and conceptual insightfulness are important attributes of meteorological models. In their research, meteorologists develop hierarchies of models. At one end of the hierarchy, there are the models which are good at forecasting the weather. Because these models are so complex, with millions upon millions of lines of code, and

billions of individual variables, they are not at all tractable and are difficult to generalise, the things that economists hold dear. However, at the other end of the hierarchy are the simple conceptual models, like Lorenz's prototype model of chaos. The conceptual models are better for understanding why the weather is the way it is, but they are typically useless at forecasting it.

It was a simple conceptual model of the atmosphere that drew me into a career in meteorology. It was based on the assumption that the behaviour of weather could be understood using the so-called principle of maximum entropy production, an idea drawn from the physics of thermodynamic systems (like steam engines). This principle can't be rigorously shown to apply to large-scale weather systems. But it is an interesting concept, and it's certainly interesting to explore what properties of the weather can be derived from it. From this perspective, midlatitude weather systems can be thought of as nature's way of transporting heat from hot tropical latitudes to colder polar latitudes as efficiently as possible on a rotating planet. Such a model predicts how the temperature of the Earth, as well as more complex variables like mean cloud cover and poleward fluxes of heat, varies between the equator and the poles on average.

These simple models certainly are parsimonious, in the sense that they are stripped to the bone of irrelevant complexity, and can sometimes be solved with pencil and paper, or at worst with a laptop computer. They are certainly tractable and conceptually insightful. During periods when the weather seems relentlessly wet and windy, it is somehow reassuring to know that the atmosphere is being a ruthlessly efficient thermodynamic machine—if it weren't like this, then lower latitude regions would become unbearably hot, and higher latitude regions unbearably cold. So, the rainy, stormy weather we frequently experience here in the UK is at least making life

more equable for others on the planet, and that provides some modicum of comfort when the weather is especially bleak!

On the other hand, it is important to understand the limitations of these parsimonious conceptual weather models. They are useless for actually predicting weather. And they are utterly useless in predicting the sorts of extreme weather that occurred over southern England in October 1987. I am not aware of any way that the principle of maximum entropy production could have been used to predict the October storm.

What conclusion can we draw from this? Could it be that the economists are trying to have their cake and eat it too? They can have highly parsimonious models, good for understanding how the economy works in general terms, or they can have predictive models, good for actually forecasting the economy. However, perhaps one size does not fit all.

So, what are these economic models and why did Haldane paint such a bleak picture of their ability to predict the 2008 crash? Well, it seems there are two traditional types of models.

The first are so-called econometric models. An econometric model is really the same type of statistical-empirical analogue model which Ed Lorenz's statistical colleagues said could be used to predict the weather weeks or months ahead. That is to say, an econometric model predicts the future based on patterns it finds in past data. Essentially, we try to find an analogue of the current state of the economy and base our prediction on what happened in the months and seasons following the analogue state. These days, one might use sophisticated AI to do the forecasting, but the underpinning methodology is essentially the same.

Lorenz had shown that this method doesn't work for a nonlinear system like the weather. As such, perhaps it's not

surprising that it doesn't work well for a nonlinear system like the economy, especially during periods of extreme volatility.

The second traditional type of economic model is based on what is called neoclassical economics. The fundamental idea in neoclassical economics is the notion of an equilibrium arising from an agent maximisation problem. A simple example arises in determining the price of something like a car. The more expensively a car is priced, the more profit the seller can potentially make. However, the more expensive the car, the less likely a potential buyer will want to buy it. An "equilibrium" is reached when the seller's and buyer's desires (to make as much profit as possible and to buy as cheaply as possible) are jointly optimised.

The value of something is encoded into a concept called a utility function. This originated in the theory of utilitarianism by nineteenth-century moral philosophers Jeremy Bentham and John Stuart Mill as a way of measuring pleasure or happiness. In this way, the utility function can be used to discriminate quantitatively between different possible choices. For example, I typically get more pleasure eating chocolate ice cream than, say, strawberry-flavoured, and would in principle pay (a little) more for a chocolate ice cream than another flavour.

Neoclassical economists treat agents as idealised rational beings who make decisions to maximise some relevant utility function. In the neoclassical framework, an agent is really an idealised representative of a set of individuals or businesses. As the idealised agent behaves, so, it is assumed, the set of individuals represented by the agent behaves. However, we have seen (e.g., in both climate and pandemic prediction) that in a nonlinear system, this assumption can break down. The prediction made by a "representative" prediction model with its most likely set of model parameters can differ greatly from the

expected prediction when we vary the model parameters according to their uncertainties.

Why do economies crash, as they sometimes do? The standard neoclassical view is that it is because of factors which are external to this world of utility maximisation. This can be in the form of a shock—some kind of noisy input from the outside which would make the system non-deterministic—or some “externality”, an outside factor that the capitalist system of price setting does not take into account. For example, environmental pollution or degradation is traditionally considered an externality in neoclassical economics (despite the fact that it is us, the agents, who cause it in the first place). One of the mantras of environmental economics is the need to internalise the externalities—the price of goods should reflect their damage to the environment, e.g., in shipping them from one part of the world to another.

The more I thought about it, the more the “maximisation of utility” philosophy of neoclassical economics sounded like the “maximisation of entropy production” used in meteorology to understand why weather behaves the way it does. In both cases, we are taking a principle which has some relevance under certain specific circumstances, and applying it at a scale where it no longer rigorously holds. For example, the principle of maximum entropy production holds for systems which are close to being in what’s called thermodynamic equilibrium. However, the climate system is not like this—the energy the atmosphere receives from the sun every day means it is far from thermodynamic equilibrium. In a similar way, the idea that we can model the economy based on the actions of representative agents who are maximising some specific utility function may not be a good assumption for a highly heterogeneous nonlinear system in which the individuals may be acting according to some individually tailored optimisation principles. In both cases we can form some understanding of the systems of

interest—the weather and the economy, respectively—from these heuristic optimisation principles, but equally it is important to understand their shortcomings when making actual predictions of systems far from equilibrium.

Pursuing this analogy, and given that we do not use the principle of maximum entropy production to forecast the weather, is there an alternative to the principle of utility maximisation by representative agents to forecast the economy? In [Chapter 7](#) I discussed how we can model the spread of disease using networks in which nodes represent individuals. Could such a framework also be relevant in economic modelling? Indeed, it seems it can. In 2009, the year after the crash, economists Doyne Farmer and Duncan Foley published an influential paper in *Nature* explaining why neoclassical economic models were fatally flawed when it came to predicting the crash. In the paper they claimed that so-called agent-based models will stand a much better chance of predicting the next big crash than neoclassical models ever could.

I knew Farmer's name when I first read this paper, as he features prominently in James Gleick's masterful 1988 book,⁵ *Chaos: Making a New Science*, the book that brought Ed Lorenz's work to the public's attention. In Gleick's book, Farmer was portrayed as one of the new generation of "complexity theorists"—scientists steeped in the concepts and methods of chaos theory. In his youth, Farmer perfected a way of beating the casinos at roulette by embedding a small computer in the sole of his shoe. The computer was fed with data on the position of the roulette ball as it went around the wheel, in the form of taps of the foot. From this, the computer would calculate where the roulette ball would finally land. Farmer had great success until the casinos discovered what was going on and banned him from entering again! Farmer went on to found Prediction Company and made a small fortune. He now

uses his expertise in complexity theory to understand macroeconomic cycles and crashes; he is director of complexity economics at the Institute for New Economic Thinking at the University of Oxford, where he is now a close colleague and friend of mine.

Farmer has remarked on the sociology of mathematical sophistication in economics⁶. He makes the point that if your model isn't based on the standard mantra—about representative agents maximising their utility functions—all wrapped up in elegant mathematics, then you won't get your paper published in one of the top economics journals. If you are a young academic seeking tenure, you will be at a disadvantage if your paper deviates from this mantra. When I first read this, I have to confess, I wondered whether there was an element of hyperbole in this statement, or whether, if it was once true, it is no longer so. But it seems there is some truth in Farmer's views today, as we shall see.

In an agent-based model, an agent represents an actual economic unit. It could be an actual individual, or it could be an actual firm or a bank. In an agent-based model, one tries to represent as many of these microeconomic agents as possible—thousands, millions (or potentially billions, which we'll see ahead). Agents are perceived as acting as a unit from the outside. They have an ability to act and to react to external stimuli, and they can interact with each other. A key question is what sort of macroeconomic behaviour is emergent when studying such large numbers of microeconomic agents. An example of emergent behaviour—though nothing directly to do with economics—can be seen in studying the flow of traffic on a road⁷. Suppose we think of each car or truck as an agent who merely responds to the agent in front. When the density of vehicles is small, the traffic flows well. However, as the density increases, we start to see the familiar shock waves

as the traffic becomes congested. These shock waves are examples of macro-scale emergent behaviour.

In fact, a relatively simple agent-based model may be able to describe some of the essentials of the 2008 crash [8](#). In this model, there are three types of agents. The first are described as “noise traders”. They more or less trade stocks and shares at random. However, they believe that these shares have some intrinsic fundamental value. The second type of agent is a hedge fund. It holds stocks which the hedge fund managers believe to be underpriced. Otherwise, they hold cash. The third type of agent is the bank that can lend money to the hedge fund, allowing them to buy more stock.

Under normal circumstances, the presence of hedge funds dampens volatility in the value of the stocks, since hedge fund trading should move the price of a stock towards its intrinsic worth. However, there is a problem. To contain their risk, the banks will limit the amount of money that they will lend to a hedge fund. In the jargon of finance, the banks cap leverage at a predetermined maximum value, determined by the wealth of the hedge fund. A problem then arises if a stock’s value falls when a hedge fund is fully leveraged. This can take the amount borrowed beyond the predetermined cap. The hedge fund will have to sell off its stock to pay off part of the loan, to bring it back to the predetermined maximum value. But selling off the stock in this way can cause the value of the stock to fall further. In extremis, this can lead to a death spiral that it is impossible to get out of, leading to disaster for both the hedge fund and potentially the bank too. Of course, this is a very simple example of an agent-based model. However, it can relatively easily be extended, making it more realistic. For example, the banks can themselves borrow money and be limited by leverage caps.

Well, we seem to be getting somewhere at last. Is it possible to perform a retrospective agent-based model ensemble forecast of the 2008 financial crash, much like the retrospective ensemble forecast we had performed of the October 1987 storm? Would it show exceptional spread, like the bottom example in [Fig. 10](#)? Or would the crash be shown to have been very predictable, like the top left example in [Fig. 10](#)—an inevitable consequence of the exceptionally leveraged positions that many individuals, hedge funds and banks had taken over subprime mortgages?

I approached several agent-based modellers around the world and was again disappointed by the responses. It seemed my idea wasn't intrinsically silly, it was just that the agent-based models weren't sufficiently sophisticated to answer such questions. However, unexpectedly, I was emailed by a postdoctoral researcher, Juan Sabuco, at my own university, Oxford, who said that he was using an agent-based model and would like to develop it to have ensemble capability. Would I like to work with him to explore this idea? And so, I began a small piece of research with Juan.

In Sabuco's highly idealised agent-based model, we can think of the agents, of which there can be as many as 100,000, as divided into two: firms and unemployed workers. The agents move randomly in a two-dimensional space that simulates a virtual job market. Every "time step" (which in this model has no specific linkage to real time), the workers move around looking for a job opportunity. Firms also move around in this marketplace looking for new workers. When a firm hires workers, these workers are removed from the unemployed pool of workers. Now firms lose staff for any number of reasons, and so for a particular firm to be viable, it must be able to attract new labour. If it doesn't attract new labour over a predefined number of time steps, it will go out of business. In addition, it is

assumed that unemployed workers appear every time step, either due to workers being fired or because of new workers entering the market. These new workers are placed randomly in the marketplace, but close to the other unemployed workers already present in the simulation. It is also assumed that new firms are created in the market every time step. We also assume that the newly created firms are placed in regions close to other existing firms⁹.

[Fig. 37](#) shows a long run of Sabuco's agent-based model over a quarter of a million time steps. Shown on the vertical axis is the employment rate of the workers, with 100 percent representing full employment. The agent-based model shows considerable internal variability in this employment rate with a number of "crashes" in employment rate. Economists would call such internal variations "endogenous", to contrast with externally forced variations, which would be called "exogenous". We will call this the "truth" run, i.e., we will assume this is what actually happens in Sabuco's agent-based model world. The question is whether we can predict the endogenous fluctuations with ensemble forecasts.

Like weather prediction models and COVID models, Sabuco's agent-based model has a number of free parameters, some of which are represented using stochastic formulae (hence involving random numbers). It is straightforward to generate large ensembles by using different draws from the pseudorandom-number generators. [Fig. 38](#) shows some results from these ensembles. The top left figure shows an economically healthy situation where, according to the truth run, employment rates are stable and stay high throughout the forecast period. Some ensemble members predict drops in employment, but the bulk of the ensemble suggests that this will be a period of stability. The top right figure shows a situation where, later in the forecast period, the truth run

does show a dip in employment. Some of the ensemble members predict this (i.e., the ensemble predicts the dip with some non-zero probability), but most do not—at this range, the dip in employment is not really predictable, according to the ensemble. The bottom two figures show situations where there is an employment crash relatively early on in the forecast, where one would hope for some predictability. One crash (bottom left) appears very predictable according to the ensemble, whilst the other (bottom right) appears less so. If one tried to make a single best-guess deterministic forecast for this last situation, it seems quite likely that the forecast would underestimate the true drop in employment (and indeed predict a faster recovery than actually occurred). In this case the ensemble provides an alert of the chance of a deeper and longer-lasting crash. If this were to occur, then with the aid of the ensemble, the crash would not come as a complete surprise.

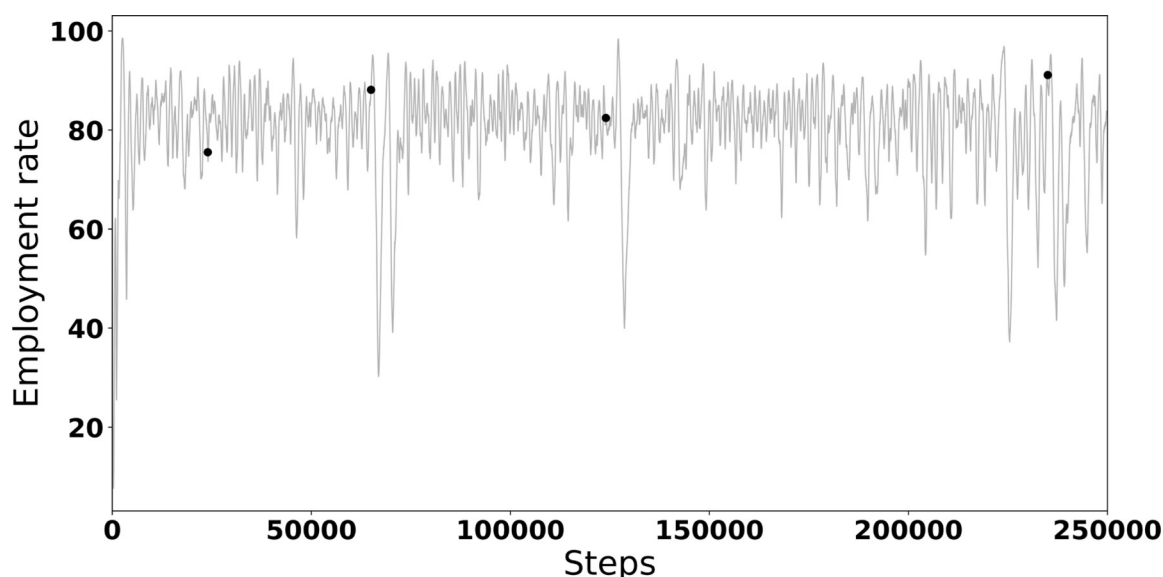


FIG. 37. Output from a long truth run of Sabuco’s agent-based model. In the next figure, ensemble forecasts are made from the four points marked in the figure.

This was all quite encouraging. However, we were looking at very idealised situations.

And then, when I had all but given up on real agent-based model forecasts, Doyne Farmer pointed me to a new paper in which a group of economists led by Sebastian Poledna from Austria had used an agent-based model to make retrospective predictions of the Austrian economy [10](#). They had developed a simple data assimilation system for inputting the observed economic data into their model. There is nothing special about Austria except that two of the authors' home institutes are based there. The authors say that their model could be applied to make predictions for the UK or the US, for example.

A pure mathematician looking at the equations in Poledna's paper would not find the mathematics very endearing. It's engineering mathematics. It's the sort of mathematics used in comprehensive climate models. But precisely because of this, I personally found myself drawn to the paper. It seemed to me to be exactly the type of inelegant mathematics you need to represent the messy realities of the real-world economy. In Poledna's predictions, the agent-based model is run five hundred times in ensemble mode, varying uncertain aspects of the initial conditions and sampling the pseudorandom-number generators that drove the stochastic agent-to-agent interactions in the model.

The proof of the pudding is in the eating, and the authors put their model to the test with some retrospective "out of sample" predictions. This means that the predictions were for years which weren't used to set the values of parameters. The authors report that the ensemble-mean forecasts are better than those from both econometric models and the neoclassical equilibrium models.

I was excited reading this paper. I emailed Poledna and asked if he had looked at retrospective forecasts for 2008. He answered that he had literally just produced some results for 2008, but hadn't started to write up the results. This was too good to be true! I had to speak to him. We arranged to chat over the internet.

[Fig. 39](#) shows the spread in retrospective forecasts of Eurozone gross domestic product (GDP) growth in two five-hundred-member ensembles from Poledna's model. The first is for a normal year. The second is for the financial crash year of 2008. The most likely predictions don't really show any differences between the two years. However, the spread of the ensemble in 2008 is substantially larger and the tail of the distribution extends to values twice as negative as from the normal year. Like the ensemble predictions of climate change and of COVID, the ensemble distribution is skewed towards the dangerous end. Nonlinearity is at work again.

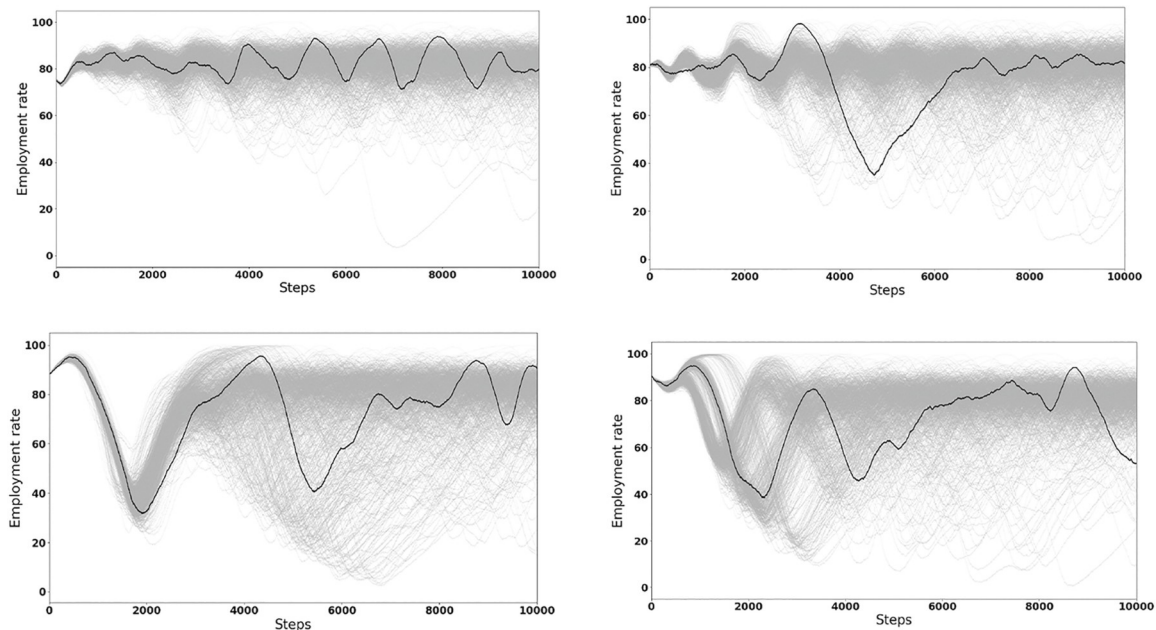


FIG. 38. Ensemble “forecasts” from four different initial states chosen from the truth run in [Fig. 37](#). The darker line itself shows the truth run.

Impressive as the results were, the ensemble for 2008 did not actually capture the observed downturn for one quarter, which was around -0.03 . I suggested to Poledna that it was because the ensemble was not large enough; perhaps he should try running five-thousand-member ensembles instead of five-hundred-member ensembles. However, his explanation was more plausible. He did not feed into his model the economic downturn from the world outside the Eurozone. He did not do this because he wanted to see what would happen from purely endogenous dynamical processes.

But this raises an extremely important question. Since the world is, of course, globally connected, we really need to extend these agent-based models to cover the whole world. How many individual agents can an agent-based model handle? Indeed, Poledna has been involved in the development of an agent-based economic model with 331 million individual agents¹¹, implemented and run efficiently on a supercomputer. Is that the limit? Certainly not. As discussed, a modern weather forecast model has several billion variables, about the same number as there are people on the planet. It is therefore perhaps not completely ridiculous to imagine an agent-based model with one agent per human on the planet. I look forward to seeing some ensemble results for 2008 when the agent-based model is truly global in extent.

Sadly, Poledna confirmed Farmer's conclusions concerning barriers to progress in economics. He believed it would be impossible to get his paper considered by the top economics journals. Indeed, he said that traditional economics colleagues had largely ignored his paper. He was not at all optimistic that his work in this area would help him get a tenure-track position in a top economics department.

Just as ensemble weather prediction means we should never again get these Michael Fish weather-forecast moments, so perhaps once these ensemble-based agent-based prediction models and their associated data assimilation schemes have been developed on the global scale, we will no longer have the Andy Haldane/Michael Fish economic-forecast moments either¹².

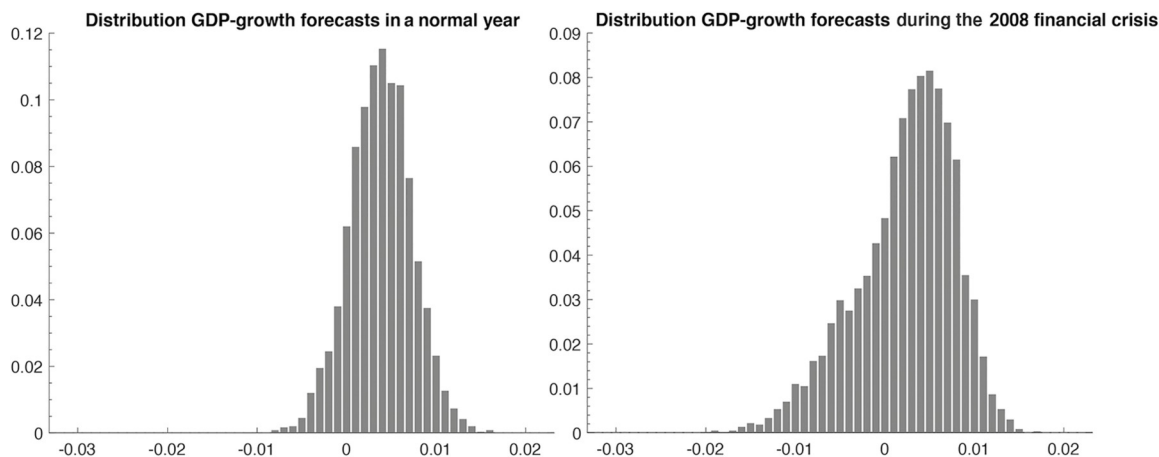


FIG. 39. Histograms of predicted quarterly growth in Eurozone gross domestic product for a “normal” year and for 2008, containing the financial crash.

Or is this wishful thinking? As I complete this book, inflationary pressures are increasing strongly and there is a danger of a global economic recession. Was this predictable a year ago? These economic impacts can be linked in part to the fact that Russian forces have been shelling Ukrainian cities, something most of us would hardly have believed a year ago. These actions have led the West to impose significant sanctions on Russia, pushing up (already high) energy prices and depressing stock markets worldwide. Hence the answer to the question above depends on whether conflicts are predictable. We will look at this in the next chapter.

Indeed, such events make it clear that the development of ensemble economic systems based on agent-based models are really just the tip of the iceberg. We need to

develop ensemble systems in which human and physical systems (such as the climate system) are fully integrated on a global scale. Welcome to the digital twin of planet Earth, another topic for the next chapter.



9

Deadly Conflict and the Digital Ensemble of Spaceship Earth

WHO'S TO SAY WHY GENERAL OSKAR POTIOREK, AUSTRO-HUNGARIAN governor of Bosnia and Herzegovina, didn't tell the driver that there had been a change to the planned route for the archduke's car. There had been an assassination attempt on Franz Ferdinand earlier in the day, and changing the route seemed like a sensible precautionary step. Perhaps Potiorek was about to tell the driver, but was momentarily distracted by a butterfly passing in front of him—we'll never know.

But in any case, he didn't tell the driver, and he and the car's occupants, including Potiorek, Archduke Franz Ferdinand and his wife, turned right near the Latin Bridge in Sarajevo. Realising his mistake, Potiorek finally shouted to the driver to stop and reverse. In so doing, the car stalled. Gavrilo Princip, still a teenager, thought he had missed his chance to assassinate the archduke earlier that day. And then, to his amazement, the stalled car was right in front of him. Princip shot the archduke dead.

If only the butterfly had not passed by, the archduke might not have been killed. And perhaps the First World War would not have happened. And perhaps the Second World War would not have happened either. That is, if it's permissible to argue in this counterfactual way.

This is a real-world example of the parable of the missing horseshoe nail. Because of a flapping butterfly (perhaps), the driver was not told to change his route. Because the driver was not told to change his route, the archduke was assassinated. Because the archduke was assassinated, Austria-Hungary declared war on Serbia. Because of this, Russia, an ally of Serbia, mobilised its army against Austria-Hungary. Because of this, Germany declared support for Austria-Hungary and mobilised against Russia and its ally France. Because of this, and as a pre-emptive strike, Germany attacked France through neutral Belgian territory. Because Britain had a treaty to defend Belgium, it went to war against Germany and Austria-Hungary. Because of this, the United States in 1917 joined the war, and Germany and Austria-Hungary (and the Ottoman Empire, which had joined the war on the Austro-Hungarian side) lost their empires—all for the want of a horseshoe nail. Indeed, Britain came out of the war so much poorer that this marked the beginning of the end for its empire too.

However, as many historians have written over the years, the fact that the countries of Europe were bound by these various treaties and alliances, and that military spending was generally on the increase, meant that the transition to war was in fact quite predictable. Perhaps if the archduke had not been assassinated, some other relatively minor incident could have triggered a similar sequence of events. Like with weather, pandemics and the economy, it seems that the dynamics of conflict are similarly nonlinear, varying from periods when its evolution is stable to small perturbations, to periods when its evolution is highly unstable.

Can we think about world wars in terms of the geometry of chaos as shown in [Fig. 10](#)? Let's now imagine that the left-hand lobe of the Lorenz attractor corresponds to states of the world where the nations are at peace with one

another, and the right-hand lobe corresponds to states of the world where the nations are at war. Is World War I a situation like the top left—inevitable—or the bottom—highly unpredictable? Does it make sense to imagine ensemble forecasting being applied to conflict prediction?

If ensembles are to be part of any serious attempt to predict conflict in the world, one needs to ask first whether conflict prediction is amenable to the methods of mathematical modelling. This question fascinated Lewis Fry Richardson, whose seminal contributions to numerical weather prediction were described in [Chapter 5](#). Richardson felt strongly that war could be predicted using mathematical equations. Coming from a Quaker family, he felt that trying to develop a mathematical theory of conflict could be his most important contribution to humanity.

We left Richardson in [Chapter 5](#) as an ambulance driver at the front in World War I. He had resigned his position at the Meteorological Office, and during rest periods he worked not only on performing the first weather forecast by numerical methods, but also on developing what he called “the mathematical psychology of war”.

After the war, Richardson rejoined the Meteorological Office. However, his pacifist principles caused him to resign again, on grounds of conscience, when the office joined the Air Ministry in 1920. Instead, he focussed on the two things that fascinated him, turbulence and the causes of war. To help with the latter he enrolled in psychology courses.

However, Richardson realised that to make real progress, he needed data that would measure things like a country’s preparedness for war—e.g., how much it spends on armaments, its war weariness and its degree of internationality (which could be gauged from the extent of its international trading). And he grouped conflicts using the same logarithmic (Richter) scale that we use to group earthquake events. In this way, gang conflict and world

wars are merely points on a continuous spectrum of deadly conflict.

When he did this, Richardson found a phenomenon that we have commented on in the context of weather, turbulence, networks, economics, pandemics and perhaps even the universe as a whole—power-law structure. As we noted in earlier chapters, power-law behaviour is an indication of some underlying nonlinearity. Here was an indication that some rationality and nonlinear order could be brought to the seemingly irrational world of conflict.

Based on his experience with weather, Richardson sought a system of differential equations (equations using calculus which included rates of change of key quantities with time) similar in many respects to the types of equations Lorenz sought years later. For example, he posited that the rate of change of a nation's armament build-up is directly proportional to the number of arms its rival has and also to the grievances felt towards the rival, and inversely proportional to the number of arms it already has itself.

One of the most curious spin-offs from his work came when he tried to include in his equations a term which described the propensity of two countries to go to war as a result of having a common border. Richardson reasoned that the longer the length of this common border, the greater the propensity. In trying to find data that would support his hypothesis, he found that the books he consulted gave wildly different estimates of the length of borders, e.g., between Spain and Portugal or between the Netherlands and Belgium.

Richardson eventually realised that the estimated length of a border depends strongly on the resolution of the measuring system used to determine its length. Imagine trying to measure the length of the coastline of Britain with a 200-km ruler. It gives us some sort of estimate. But now cut the ruler in half and measure with a finer-scale 100-km

ruler. The estimate increases. Cut the ruler in half again and remeasure—the estimate increases yet again. We cut again and again and the estimate increases and increases. Richardson imagined a mathematical coastline—indistinguishable from Britain’s coastline on any perceivable scale—whose length was literally infinite.

What is going on here is a generalisation of the fractal process we discussed in [Chapter 2](#). There we discussed the Cantor set, an infinite set of points characterised by a fractional dimension between 0 and 1. Richardson had found—in the idealised coastline of Britain—an example of an infinite set of points characterised by a fractional dimension between 1 and 2. A simpler example of such a fractal is the Koch snowflake, illustrated in [Fig. 40](#). We start with an equilateral triangle and iterate by replacing the middle third of each side with two sides of a corresponding equilateral triangle.

Richardson’s work on fractal coastlines and borders inspired the mathematician Benoit Mandelbrot. Recall that it was Mandelbrot who coined the word “fractal”, essentially meaning some fractionally dimensioned object, and in so doing discovered one of the most intriguing fractals known: the eponymous Mandelbrot set.

Lewis Fry Richardson died in 1953. His book *The Statistics of Deadly Quarrels* was published posthumously in 1960. In some ways, he reminds me of Alan Turing. Whilst Richardson is best known for his work on turbulence and weather prediction, his later work became the basis of much that followed in conflict prediction. Similarly, whilst Turing is best known for his work on computing and cryptography, his later work forms the basis of many studies in mathematical biology. Interestingly, the mathematics behind Richardson’s work on conflict and Turing’s work on biology are very similar. Both involve the mathematics of nonlinear equations based on calculus—the equations on which the geometry of chaos is based. I am

not aware that Turing and Richardson ever met. If they had, I expect they would have got on like a proverbial house on fire.

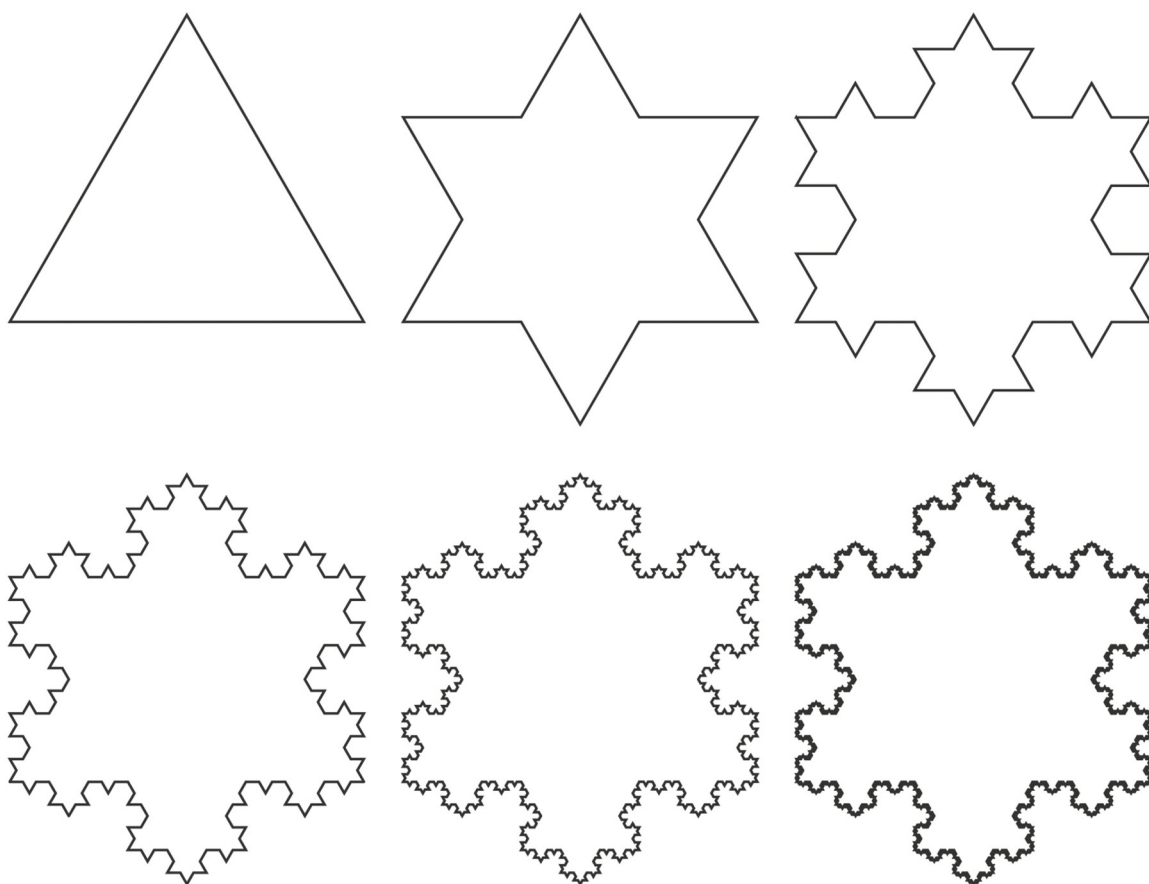


FIG. 40. The Koch snowflake at various levels of iteration. The length of the edge of the snowflake increases with iteration number. Lewis Fry Richardson found that a similar phenomenon made the length of nation borders uncertain—complicating his theory of deadly conflict, but making him aware of fractal geometry.

Today, appropriately perhaps, some of the important work on conflict prediction has been taking place at the Alan Turing Institute, the UK institute for data science and artificial intelligence (perhaps they need a Richardson wing for their institute). Their work on deadly conflicts started in 2015 when team leader Weisi Guo began studying a map of where fighters of the so-called Islamic State were operating. He found that there was some overlap with

places along the old Silk Road. He started to notice that many of the areas prone to repeat violence were along ancient trade routes more generally. He realised that regions which are geographical bottlenecks, i.e., routes travellers have no choice but to use, are inherently unstable and prone to violence.

Guo likened this to a chessboard, whose central four squares are contested by almost every significant piece at some point. Guo created a human connection network and found the geographical equivalents of those key chess squares for various locations around the world.

With colleagues at the Turing Institute, Guo created a prediction system based on such networks. It is called GUARD, which stands for “global urban analytics for resilient defence.” In this model, states of war and peace are represented by what are called “potential wells”, a little like the two lobes of the Lorenz attractor as described earlier in this chapter. In between these are states of instability, which Guo translates as meaning that countries can’t be on the brink of war for long. Running the model in ensemble mode, in order to try to estimate uncertainties in its predictions, will I hope be a focus of future work at the Turing Institute.

A project that has made progress with ensemble conflict predictions is called ViEWS¹: Violence Early-Warning System (it seems to be a fact of modern-day science that all projects have to have acronyms). The project is run from the Department of Peace and Conflict Research at the University of Uppsala in Sweden. Currently, ViEWS provides early warnings for three forms of political violence: armed conflict involving states and rebel groups, armed conflict between non-state actors, and violence against civilians. The outputs from ViEWS are probabilistic assessments of the risk and likely severity of the three types of conflict events, at three different spatial scales:

national, subnational and individual actors. Like GUARD, the prediction model makes use of key data that affects the likelihood of conflict in a given place: a history of conflict, political institutions, timing of elections, economic development, natural resources, demography and geographical proximity to conflict. However, the ViEWS researchers have developed a kind of multi-model ensemble from which to make probabilistic forecasts. On the ViEWS website they have analysed the skill of their forecasts from the period of 2011 to 2013. One thing is completely in line with everything written so far: the skill scores for the ensemble forecasts are much improved compared with any of the single-model forecasts. Having said that, I haven't yet seen a convincing account of whether the spread from the ViEWS ensemble is a good predictor of the skill of the ensemble-averaged predictions. As with other applications of ensemble prediction, this will provide a good indication of whether the project's ensemble predictions are reliable.

One of the consequences of regional conflicts is the displacement of local populations who then seek refugee sanctuaries. Derek Groen from Brunel University has developed a network-based model which predicts conflict migration based on expected local conflicts. His predictions are fed to the Save the Children charity. The network nodes mark potential conflict regions, refugee camps and other settlements.

In Groen's model, a conflict can be generated randomly at a node, and the model then predicts the resulting movement of the displaced population along a simplified travel network. In this way, the numbers arriving at the refugee camps can be estimated. Alternatively, a conflict can be specified deterministically and numbers of displaced people arriving at the different camps estimated as a consequence of such a conflict. The difference between these two uses of the model is similar to the "prediction vs. projection" distinction discussed in the context of climate

and pandemics. Groen makes these predictions using an ensemble of estimates, where uncertain parameters are varied.

As mentioned in [Chapter 8](#), at the time of completing this book, Russia invaded Ukraine. On timescales of months this was predictable, at least in a probabilistic sense, simply because Russia had begun amassing troops on the Russia-Ukraine border a few months prior to invasion. But more than that, Richardson's conditions for conflict were met: a long common border, perceived grievances that the Russian president felt about Ukraine after the demise of the Soviet Union, and a general build-up of Russian armaments in recent years. Weisi Guo confirmed to me that conflict models had identified Ukraine as a region of concern for a number of years.

In thinking about the development of ensemble systems for conflict prediction, perhaps some ideas can be borrowed from the way in which operational weather prediction ensembles are initialized. In particular, as described in [Chapter 5](#), the initial perturbations of a weather ensemble deliberately target the dominant instabilities in the atmospheric circulations, using mathematical quantities known as singular vectors. These are perturbations in state-space directions which can be expected to have a large effect on the forecast. Without these initial singular-vector perturbations, the ensemble members would be too tightly clustered and the predictions too confident. An analysis of the corresponding "singular vectors" of the global geopolitical system would surely identify Vladimir Putin's actions and decisions (rather than mine, say) as being critical to global peace and security. That is to say, in developing a global ensemble conflict model, uncertainty in what people like me—and billions like me—think and do can be represented by general stochastic noise. But Putin's actions and decisions represent geopolitical singular vectors, and need to be considered

explicitly. Any ensemble system which purports to predict conflict probabilities reliably must surely focus on such geopolitical singular-vector perturbations.

« »

WHAT HAVE WE LEARNED from the different contexts discussed in [Chapters 5](#) through 9? First, for complex nonlinear systems, ensemble prediction is vital if we are to make reliable predictions which include estimates of uncertainty. Such ensemble prediction systems can help avoid the Michael Fish moments when a prediction of some disastrous event fails because the prediction is especially sensitive to the flap of a butterfly's wings, or a missing horseshoe nail. These ensemble forecasts provide quantitative estimates of the probability of such events. As discussed in the next chapter, prediction probabilities provide the basis for making rational decisions about whether to take anticipatory action to mitigate the worst consequences of these possible disasters.

On the other hand, we have treated the various topics in this part of the book as if they were completely independent of each other. Clearly, they are not. For example, one of the biggest drivers of forced migration and conflict in future years is likely to be climate change. If heatwaves exceed the level at which humans can maintain their body temperature at safe limits, people may have no alternative other than to move polewards. In migrating polewards, the migrants are likely to pass one or more of the hotspots identified by Guo as places where conflicts will emerge and develop.

So how do we go about bringing all these elements together in a more unified way? My answer to this question is to first step back in time to April 1970.

One of the most famous phrases uttered during the Apollo space programme was from the commander of Apollo 13². “Houston”, Jim Lovell said to ground control, “we have a problem.”

This isn’t exactly what he said, but that doesn’t matter. Essentially, he reported, in a matter-of-fact way, that he had heard a thud, apparently linked to a computer message of a “Main B Bus Undervolt”. What did matter at the time was that, unknown to both the astronauts and ground control, an explosion in the spacecraft’s oxygen tanks had critically damaged their main engine and left the oxygen supplies leaking into space. With the explosion, the Apollo 13 service module had become a dangerously crippled spacecraft moving farther from Mother Earth with each passing hour. On their current trajectory, if the astronauts were unable to perform any further engine burns, they would stay on an elliptical orbit for the best part of eternity, never coming closer to the Earth than 70,000 km (43,500 miles). The key question was whether the spacecraft was fatally crippled. How was mission control going to bring the astronauts home, knowing that trying possible solutions could well make matters worse?

Fortunately, NASA had created a number of simulators of the Apollo spacecraft. These had been used to train astronauts and mission controllers ahead of the space missions. As Gene Kranz, the chief flight director for Apollo 13, said, “These simulators were some of the most complex technologies of the entire space programme: the only real things in the simulation training were the crew, cockpit and the mission control consoles. Everything else was make-believe, created by a bunch of computers, lots of formulas and skilled technicians.”

These simulators had already proved their worth for the Apollo 11 mission, which landed the first men on the moon. In the final simulation of the Apollo 11 mission, the

guidance computer issued a “1201 alarm code”, which had not been seen before. Thinking this meant that the computers were somehow getting overloaded, the mission controllers called for an abort. However, having later consulted the team at MIT that had written the computer software code, they realised that a “1201 alarm code” was only a warning and did not indicate a critical error. If it were not for that final simulation, it is likely that the landing of the Apollo 11 Eagle, during which 1201 and 1202 alarms occurred in the minutes before touchdown, would similarly have been aborted.

In the case of Apollo 13, the simulators proved vital to the safe return of the crew. The back-up crews had to develop and practise the three manual engine burns—the computers had not been programmed to do these burns because it was assumed that the lunar module would not still be attached on the return journey. Perhaps one of the most complex problems occurred after the astronauts had powered the command module down to save power and used the lunar module as a life raft. Now they had to power the command module up again so that they could re-enter Earth’s atmosphere. The simulator teams worked around the clock to define a power-up sequence using the tiny amounts of energy left in the batteries. This involved operating hundreds of switches, in precisely the right order. Any mistake could instantly deplete the remaining electrical energy and thus be fatal.

What’s the point of this story? Well, like Apollo 13, we can think of planet Earth as a damaged spacecraft hurtling through the emptiness of space. Unlike Apollo 13 with its three astronauts, planet Earth holds many billions of human astronauts. Unlike Apollo 13, there is no external Houston for us to call for advice.

Planet Earth has not been damaged because of some design flaw, as was the case for Apollo 13; it has been damaged by its astronauts. As Partha Dasgupta’s influential

report³ to the UK government put it, we have failed to manage our global portfolio of national assets sustainably. Between 1992 and 2014, produced capital per person doubled. However, the stock of natural capital per person declined by nearly 40 percent. It is as if the Apollo crew squandered the few precious joules of energy in the spacecraft's batteries needed for re-entry so that they could play music and watch videos to make their journey back to Earth more comfortable. The problem with climate change is one manifestation of this more general problem.

And yet at the same time, we have been developing weapons of unimaginable destructive power. We astronauts have the power to annihilate ourselves almost overnight, if we should so wish. Those that need reminding might wish, as I did just a year or two ago, to read Nevil Shute's *On the Beach* written in 1957. It's a grim reminder of what could be in store for us if we reach another point of instability. Indeed, could climate change trigger such an instability? Let's indulge in some speculation about sometime mid-century.

We'll suppose international climate-change talks fail and carbon emissions keep rising. Attempts to cut emissions in the developed world are half-hearted and developing-world countries say they won't seriously contemplate cutting emissions until the standards of living of people in their countries have reached levels comparable with those in the richer countries. Temperatures keep rising, but none more so than in the western half of the United States, where wildfires become a yearly fact of life. Temperatures exceeding 50°C (122°F) are commonplace around the world, including high-latitude regions, where such temperatures were previously unimaginable. Lake Mead has dried up and the Hoover Dam no longer produces electricity for large parts of the year. Wheat yields plummet, not just one year in ten, but pretty much every

year. European countries are suffering similar problems with flooding events, which also destroy crops.

A joint grouping of European and United States ministers comes to a conclusion: something has to be done. They resort to Plan B. In this plan, military aircraft from these countries fly continuously around the clock spraying sulphuric acid vapour into the lower stratosphere. From this, sulphate aerosols are formed which reflect sunlight back to space. The atmosphere now has a haze of aerosols which, it is hoped, will offset the effects of global warming and cool the atmosphere down again. They justify this action on the basis it will help humanity as a whole.

However, the impact of such “geoengineering” on climate isn’t as simple as it might seem at first sight. As discussed in [Chapter 6](#), the global warming problem is caused because we are trapping electromagnetic radiation in the infrared part of the spectrum. Sulphate aerosols in the stratosphere increase the reflection of sunlight in the visible parts of the spectrum. The one is not precisely offsetting the other. What could be the consequences of this? This is an area where estimates from current-generation models are unreliable.

So, back to the story. After several years of spraying, both Russia and India find that the atmospheric circulation patterns over their countries have changed in such a way that rain-bearing weather systems no longer track across the main agricultural regions. The monsoons start to fail, and as a result the crops fail. India and Russia blame these failures on the US/Europe policy of spraying aerosols into the stratosphere and tell these countries to stop spraying immediately.

There is a big international study which tries to determine whether the Russian and Indian crop failures are caused by the stratospheric spraying. The results are inconclusive because the models’ resolution is too coarse to give clear-cut results (the “CERN for climate change”

advocated for in [Chapter 6](#) hasn't been built because the nations of the world could not agree to pool resources on this project). The US and Europe claim that climate models show that the monsoon failures are just part of the natural variability of climate and not caused by the spraying. India and Russia claim that the model results are unreliable and cite as evidence the fact that they typically fail to simulate the occurrence of extreme climatic and weather events with anything like the frequency observed.

Eventually, Russia and India tell the US and Europe that if they do not stop spraying, they will shoot down the planes doing the spraying. The US responds that this would constitute an act of war. Russia and India respond by saying that the destruction of their food supplies is already an act of war.

The first plane is shot down, leading to massive economic sanctions by the US. A second plane is shot down, leading to airfields in India being bombed. The US and Europe issue a final ultimatum: desist or face the very serious consequences. A third plane is shot down, and the consequences, both immediate and knock-on, ensue.

Within a short time, a radioactive cloud spreads across the world, cooling the planet by several degrees, reducing the population of Earth to one-tenth of what it was, and making climate change, as we currently understand it, a thing of the past.

Oh Lord! How can we prevent such a situation developing in the first place?

Like Apollo 13, we need credible simulators with which these problems can be played out and addressed at the international level before they spiral out of control.

The simulator system that Apollo 13 had would now be called a "digital twin". Even though not all aspects of the Apollo simulators were digital, they performed the tasks that today we would expect a digital twin to perform, notably of providing information about a physical system

that for various reasons can't be directly studied by human intervention. As with Apollo 13, it is too risky to try these Plan B options directly. But we need the tools with which we can reliably simulate the consequences of a Plan B.

Here we need not so much a digital twin as a “digital ensemble twin”. Clearly the climate models discussed in [Chapter 6](#) would be a critical part of a digital ensemble twin of planet Earth. However, they must be integrated with the other models discussed in this book: economic, health and conflict models, as well as agronomic models, hydrological models, population growth models and so on. In discussing these impact models, the unrealistic, equilibrium-based models are no good at all. We need to move towards more agent-based models in which individuals are represented and connected through networks of interactions.

Could we extend the concept of an agent-based model of global society with our 8 billion inhabitants? It's not a crazy idea. As we have discussed, a weather forecast model has some billions of degrees of freedom, so doubling this to incorporate degrees of freedom of individuals is not completely out of the question. Of course, individual agents will have to be treated as having some inherent stochasticity. But as I have explained, we have to do this anyway for weather variables. AI will play a central role in such an endeavor.

Of course, such a digital ensemble twin would not only be able to tackle the socioeconomic problems of climate geoengineering, it should be able to provide credible estimates of future migration, future conflict, future health risks, future food supplies, the future health of the oceans and so on.

Clearly, such a project is not one that could or, I would argue, should be developed at a national level. Its output would have to have the imprimatur of the international community, likely the United Nations. It would be run using

literally the most powerful supercomputers in the world. We will very soon be in the era of the exascale computer, which will be able to perform a billion billion calculations in each second of operation. Completely dedicated exascale computing would be needed to even contemplate building a digital ensemble of planet Earth, and perhaps it won't really be a serious option until we have zettascale computers, capable of performing one billion trillion computations a second. Perhaps we will have these in the 2030s. Perhaps by then we will in part use quantum computers, or photonic processors as well as silicon processors, and high-performance low-power imprecise chips that produce noise in hardware.

This would be an extraordinarily interdisciplinary exercise; perhaps a more interdisciplinary exercise than the scientific community has ever contemplated. And since handling uncertainty underpins all sciences, the project to build a digital ensemble of global society would bring together the various approaches to representing uncertainty that have developed in the different subdisciplines. This project really would recognise the primacy of doubt.

A first step in this direction is the European Union's Destination Earth (DestinE) project⁴. Evolving from an earlier EU flagship project proposal, ExtremeEarth, that some colleagues and I developed a few years ago, Destination Earth will see not only the construction of extremely high-resolution climate models, but also a range of socioeconomic-impact models to which the global climate models will be coupled. DestinE is the first step towards building a digital ensemble of global society. Sadly, the UK having left the EU⁵, UK scientists will not be contributing to DestinE.

10

Decisions! Decisions!

AVERAGING ONE HUNDRED INDEPENDENT GUESSES OF THE NUMBER of jellybeans in a jar usually provides a better estimate of the actual number than does any one guess. More generally, ensemble-averaged forecasts are typically better than best-guess deterministic forecasts. However, when we have to make decisions, simply knowing the expected outcome is often not good enough. We need to know the likelihood of some plausible worst-case scenario. If the consequences of a plausible worst-case scenario are truly dire, we should try to avoid such a scenario at any reasonable cost. For example, COVID lockdown policies were imposed, at great economic cost, to avoid a plausible worst-case scenario in which hospitals were overwhelmed and unable to cope with the number of patients needing treatment. But how plausible is “plausible”, and what do we mean by any “reasonable” cost? In this chapter we will look at the use of ensembles for decision-making. This is an area where really exciting new approaches to disaster relief management are being developed and deployed.

A FEW YEARS AGO, before weather apps told us the probability of rain, a friend of mine called. He was having a garden party in ten days' time and had an option to rent a tent (what in British English would be called a "marquee"). However, he had to let the owner of the tent know by lunchtime that day whether he wanted it or not. So, he called me and asked: Is it going to rain in his back garden next Saturday between 2 p.m. and 6 p.m.?

I told him that I would look at the latest forecast but it would be in the form of a probability. I could hear the groan at the other end of the phone line.

"How on earth are probabilities going to help me?" he complained.

"Well, who's coming to your party?"

"What's that got to do with it?" he replied.

"Suppose the Queen is coming. You would not want to risk the Queen getting wet, would you? If the Queen got wet, your chances of that knighthood would go right down the drain—along with the rainwater. So, even if the probability of rain was as low as five percent, say, I guess you would likely still rent the tent. Is the Queen coming?"

"No, of course not!"

"Well, what about the town mayor?"

I sensed he was losing patience. "Look," I continued, "if the town mayor is coming, you would not want him to get wet either. However, you probably wouldn't be quite as bothered about him getting wet as you would the Queen getting wet. Let's say that if the town mayor was coming, perhaps you would rent the tent if the probability of rain exceeded twenty percent, say. Is the town mayor coming?"

"No."

"So, who is the most important person coming to the party?"

He thought about it for a while and replied, "The mother-in-law."

“And how much does it matter if she gets wet? That is to say, what is the smallest forecast probability that your mother-in-law will get wet above which you would decide to rent the tent? If you don’t care whether she gets wet or not, the threshold is one hundred percent. If she’s as important as the Queen, the threshold is five percent. I’m guessing your threshold probability is somewhere between these two.”

Again, he thought about it for a few seconds and said, “About fifty percent.”

“Good, then we have made a decision. I’ll take a look at the forecast. If the probability of rain exceeds fifty percent, then you hire the tent. If it does not exceed fifty percent, then you don’t. OK?”

I looked at the latest forecast. The probability was only around 30 percent. He didn’t rent the tent/marquee. Fortunately, it didn’t rain and everyone was happy.

What’s the essential idea from this story? It’s that probabilities aren’t some wishy-washy things that make forecasts imprecise and difficult to use. In fact, these probabilities help you make better decisions—providing the probabilities are reliable.

Let’s generalise the story. Imagine some weather event which I will call E . E could represent the occurrence of rain or perhaps freezing temperatures or perhaps storm-force winds. However it is defined, E either occurs or doesn’t occur. In the story about the tent, E occurs if it rains.

If E occurs, then Jim, as we will call him, will incur a financial loss, L , if he has taken no measure to protect himself. However, we will assume L can be avoided altogether if Jim takes some kind of anticipatory action. But this anticipatory action will cost him an amount, C . L and C have values in real monetary units such as dollars or euros, but the ratio C/L is just a dimensionless fraction which we can assume lies between 0 and 1 (if it were greater than 1, then there will be no point taking anticipatory action, since

it would cost more than the avoidable loss). We call C/L Jim's "cost-loss ratio".

If C/L is small enough, i.e., the cost of anticipatory action is cheap enough relative to the loss, Jim might just as well take anticipatory action regardless of what the weather is likely to be. Conversely, if C/L is close enough to 1, then Jim might just as well never take anticipatory action, and just suffer the loss when the bad weather hits. But if C/L lies between these two extremes, then it's worthwhile for Jim to consult a weather forecast to decide when to take anticipatory action.

If Jim has access to only an old-fashioned deterministic weather forecast, then the decision process is straightforward: take action when the forecast predicts that E will occur; otherwise don't take action. This is all well and good, except, as we have seen, these old-fashioned deterministic forecasts can be unreliable. If E denotes the occurrence of hurricane-force gusts in southern England for the morning of October 16, 1987, then on the basis of the deterministic forecast, on that day no one will have taken anticipatory action (moved cars, secured boats and aircraft, cancelled travel plans). Deterministic forecasts have limited value because they are unreliable.

However, if Jim has access to probabilistic forecasts based on a reliable ensemble prediction system, then he has a much more valuable strategy for deciding when to take anticipatory action. Suppose an ensemble forecast system predicts E with probability p . Then the best decision Jim can make is to take action when p exceeds his cost-loss ratio, C/L . For example, in the case where L is twice the cost of C , then Jim should take action when E is forecast with a probability of at least 0.5, or 50 percent. On the other hand, if L is twenty times the cost of C , then it makes sense for Jim to take precautionary action when E is forecast with a probability of as low as 0.05, or 5 percent.

[Fig. 41](#) shows an estimate of the value of a modern weather forecast system for decisions based on whether it will rain four days into the future. The horizontal axis shows all possible values of C / L between 0 and 1. The vertical axis shows how valuable the forecast system is for deciding when to take anticipatory action. A value of 0 means that the forecast system has no value (you can make decisions that are just as valuable knowing only the climatological likelihood that it will rain in your region). By contrast, a hypothetical perfect oracle of truth has a value of 1; nothing can be better than that. The solid line shows the value of decisions using a deterministic forecast system (i.e., one that predicts that it either will rain or won't rain with 100 percent likelihood). It can be seen that the forecast system has value only for a limited range of cost-loss ratios C / L . Outside this range, it is useless. The dashed line shows the value of an ensemble forecast system. It has value for almost the entire range of cost-loss ratios. Not only that, for any cost-loss ratio where there is value, the value is greater than for the deterministic system¹.

This is all very well, but does it really matter if the Queen gets wet? In any case, she will have an army of staff who, with umbrellas and the like, ensure she never actually gets wet. Indeed. So, let's instead think about a farmer in Bangladesh. We'll call him Ahmadul. In the days before computer-based weather forecasts, individual tropical cyclones would kill hundreds of thousands of people like Ahmadul; as many as half a million individuals on one occasion².

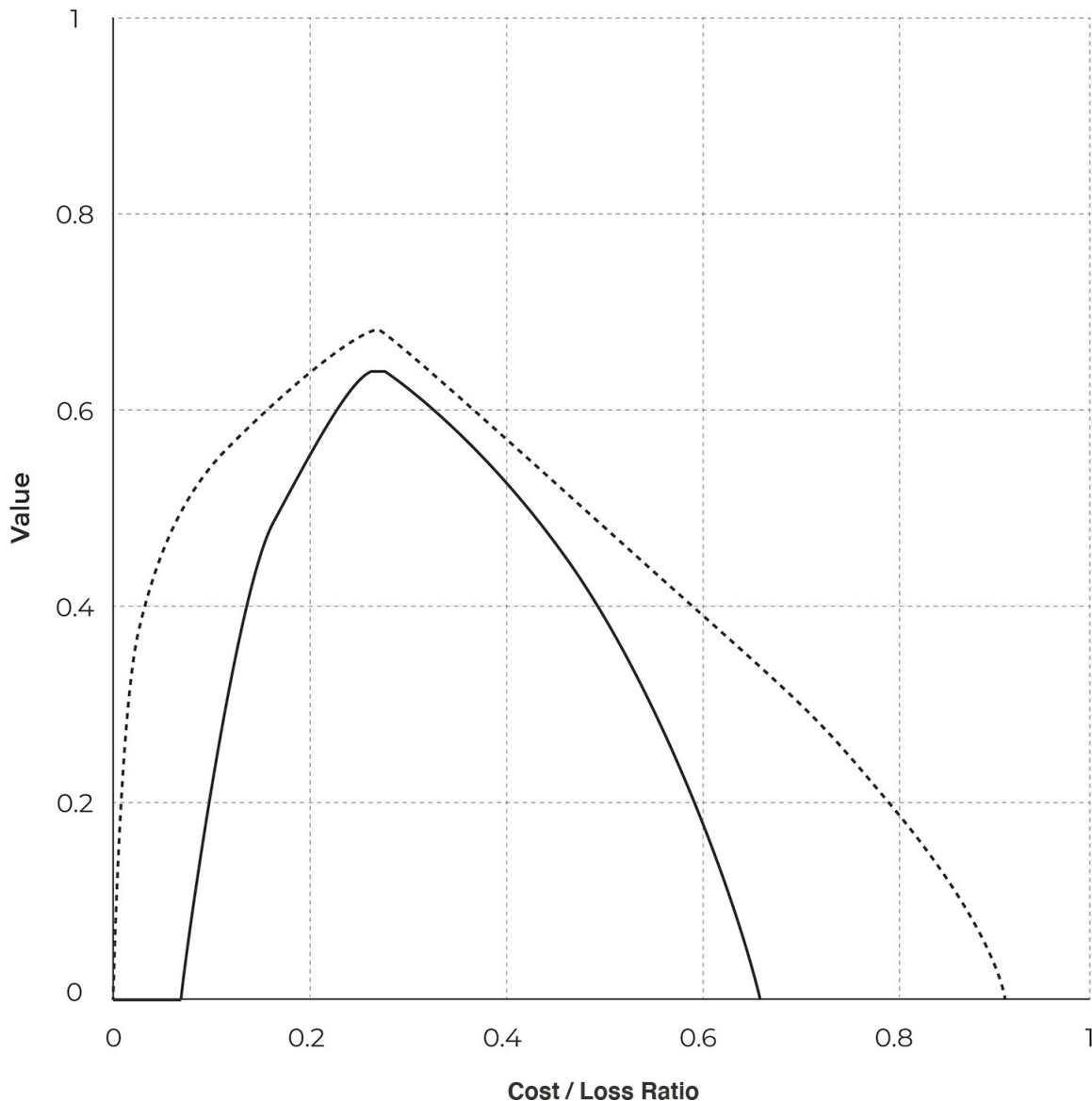


FIG. 41. The potential economic value of operational forecasts from the European Centre for Medium-Range Weather Forecasts for the European/North African region over the period October–December 2020 for the event of rainfall over a six-hour period, four days into the forecast. The lines show the value of two different forecast systems for decision-making, where losses associated with the occurrence of rain can be mitigated at cost C . The horizontal axis denotes the user's C/L . The vertical axis shows the value of the forecast, where 0 is useless and 1 is perfection. The solid line is for a state-of-the-art high-resolution deterministic forecast system. The dashed line is for a lower-resolution ensemble forecast system.

Because modern weather forecasts are so much more skilful, far fewer people get killed by extreme weather

events today than in decades past. However, that doesn't mean that we are making the best use we can of these forecasts. Disaster relief and humanitarian aid agencies have tended to react to extreme weather events only after they have hit. It can take many days, perhaps a week or more sometimes, for emergency food, water, shelter and medicines to reach a stricken region. And of course, once an extreme weather event has hit, reaching affected communities is that much harder.

It would be so much better if these agencies were able to be more proactive—to use the forecasts to target aid, allowing action to be taken before the event hits. The problem is that aid agencies aren't flush with cash. Because the old-fashioned deterministic forecasts weren't reliable, if they acted every time such forecasts predicted an extreme event, valuable resources would be spent on weather events that never happened.

Because of ensemble forecasting, one can be much more discerning about when to become proactive. Targeting aid before the event hits is called “anticipatory action” by humanitarian and disaster relief agencies. Like my friend deciding whether to rent a tent based on a probability threshold, aid agencies predetermine a probability “trigger”, based on cost/loss estimates, above which it makes good sense to take anticipatory action.

The story starts over thirty years ago. It's not just tropical cyclones that Ahmadul has to worry about. His life's investments are bound up in cattle. Like many in the country, Ahmadul lives in a low-lying area and the land on which his cattle graze is prone to flooding from the nearby Brahmaputra River. The flooding need not be caused by local heavy rainfall; indeed, the rain may have fallen many hundreds of kilometres upstream. But if a major flood occurs without warning, Ahmadul's cattle, and thereby his life savings, will be lost.

My colleague Professor Peter Webster from the Georgia Institute of Technology has had an illustrious research career. He is one of the world's experts on the theory of tropical meteorology³. He led a number of observational field campaigns to improve our scientific understanding of the climate of the tropics. However, in 1992, Webster spent a sabbatical year working with me, and during this period he completely changed his research direction. He became a convert to the recently developed ensemble-based forecasts and decided to spend the next years of his career showing how this new way of doing weather forecasting could help people in some of the poorest parts of the world.

Webster began by consulting with local officials in regions in Bangladesh affected by the Ganges and Brahmaputra rivers. Whilst the national meteorological services provided conventional deterministic forecasts a couple of days ahead, the local officials wanted forecasts with more lead time, to allow meaningful anticipatory actions to be taken ahead of possible flooding events. They needed reliable predictions a week or more ahead. With such forecasts, families could store several days' worth of food and safe drinking water. Cattle and poultry, crop seed and other belongings could be secured in safer higher locations. Most important, plans could be made for complete evacuations. Of particular concern were those who lived on river islands called "chars". It was difficult to make such plans when you had only a day or two of warning.

Webster showed it was possible to make skilful probabilistic flood forecasts for the Brahmaputra and Ganges as much as two weeks ahead, by coupling the ensemble predictions of rainfall from the European Centre for Medium-Range Weather Forecasts to hydrological models of these rivers. Flooding would occur when enough rainfall fell in the catchment basins of these rivers. In this

way, the combined weather/hydrological ensemble system would predict the probability that one or both of these rivers would flood.

However, could probabilistic predictions of flooding be used by the local communities? Would they be able to make sense of probabilities? It wasn't obvious. After all, when I was developing these ensemble prediction systems, I had been told repeatedly by weather forecasters that the British public would never understand the concept of probability (despite the fact that many of them bet on horses, where the "odds of winning" are well understood).

Answering these questions would need a field trial. With support from the US Agency for International Development (USAID)⁴, Webster began a pilot project to provide probabilistic forecasts to local villages in some of the flood-prone districts in Bangladesh.

The project was an enormous success. It ran for two years—2007 through 2008. There were two prolonged floods in 2007 and a third in 2008. For each of these there was a strong signal in the ensemble forecasts, allowing anticipatory action to be taken. The Asian Disaster Preparedness Center⁵ assessed the value of these probabilistic forecasts. Farmers with fish farms or a focus on fishing saved about \$130 per household (e.g., from fish farm protection). Mainly agricultural households saved about \$190 from early cropping. Households with significant livestock benefitted the most, with an average savings of around \$500 from advanced warnings to move livestock to higher ground. More general savings of \$270 per household arose from the protection of household assets. At the time, the average income in Bangladesh was \$470 per year and half the population existed on less than \$1.25 per day. As such, the savings were substantial indeed.

Webster needn't have worried that the farmers wouldn't understand probabilities. When he asked one farmer whether he could cope with probabilities, the farmer replied:

"That is fine! Only God knows 100 percent what will happen, and he is not telling and you are not God!"

With probabilities, the farmer realised, he had something more valuable than simple guesswork. Peter Webster's groundbreaking study showed that societies faced with environmental catastrophe are indeed ready, willing and able to accept and act on probabilistic forecast information.

The legacy of Webster's work is the approach referred to earlier as anticipatory action. This approach is transforming the way in which disaster preparedness agencies operate. In collaboration with the International Federation of Red Cross and Red Crescent Societies, it is being geared up for application around the world. Associated with such anticipatory action is a scheme called "forecast-based finance" grounded in three essential elements: disaster relief emergency funds, a forecast-based probabilistic trigger to release such funds and a pre-agreed action plan when the trigger conditions are met.

As an early example of this system in action, on July 4, 2020, fed by ECMWF ensemble forecasts, the European Commission's Global Flood Awareness System (GloFAS) predicted a high probability of severe flooding in Bangladesh (which occurred). The probability, combined with an independent assessment by the Government of Bangladesh's Flood Forecasting and Warning Centre, was high enough to trigger the United Nations Central Emergency Response Fund (CERF) to release \$5.2 million of funds to a number of local organisations, who then prepared to distribute assistance including cash, livestock feed, storage drums and hygiene kits. This was the fastest allocation of funds since CERF was established in 2005 and the first time it had been done before the peak floods

occurred. Ultimately 200,000 people were able to benefit from this anticipatory action.

In September 2021, opening a major conference on anticipatory action, UN secretary-general António Guterres said that CERF had invested \$140 million to scale up anticipatory action in twelve countries, noting that “anticipatory action protects lives”, and concluded that anticipatory action will become central to the UN’s agenda in the humanitarian sector going forward. This is a great legacy of Webster’s pioneering field trial and one that would have been impossible without reliable ensemble predictions.

Of course, communication issues remain critical. Most of us experience severe weather in one form or other from time to time, and we need to know when to take notice: the threat to life notwithstanding, such weather can lead to property damage, to travel delays and cancellations or to loss of electricity or water supplies. As a way to communicate the risk of a forecast event, the UK Met Office puts out “yellow”, “amber” and “red” warnings of severe weather based on what is called a “warning impact matrix” (see [Fig. 42](#)). The weather event E is characterised by its estimated impact and by its forecast likelihood. In the warning impact matrix, impact varies along the horizontal axis and likelihood varies along the vertical axis. In terms of the cost-loss model described earlier, we can imagine that moving along the horizontal axis, the loss, L , due to E gets bigger and bigger. Moving up the vertical axis, the probability, p of E gets bigger and bigger. Severe weather warnings are based on the product pL . Hence, for example, an amber warning could refer either to an event with medium impact but high probability or to an event with high impact but only medium probability. The probabilities that determine such warnings are taken from the ensemble forecast systems.

Of course, even when people are well informed, they do not always take action. It was reported that some people did not leave their shoreline homes ahead of the well-forecast, deadly tropical cyclone Idai that hit Mozambique in 2019, for fear their houses would be burgled. In such situations, anticipatory action could include the automatic provision of insurance coverage or of guards to protect houses.

Anticipatory action based on probabilistic thresholds isn't relevant only for flooding events or storms. There is a growing problem of developing and maintaining global food security. In over 80 percent of the world, farming is rain-fed, meaning that water for agriculture comes principally from rainfall. Many farmers around the world know the importance of planting and harvesting crops when there will be rainfall and when it is dry, respectively. Reliable probabilistic forecasts can help them decide on planting, cropping and harvest strategies in ways for which deterministic forecasts are useless.

Warning Impact Matrix

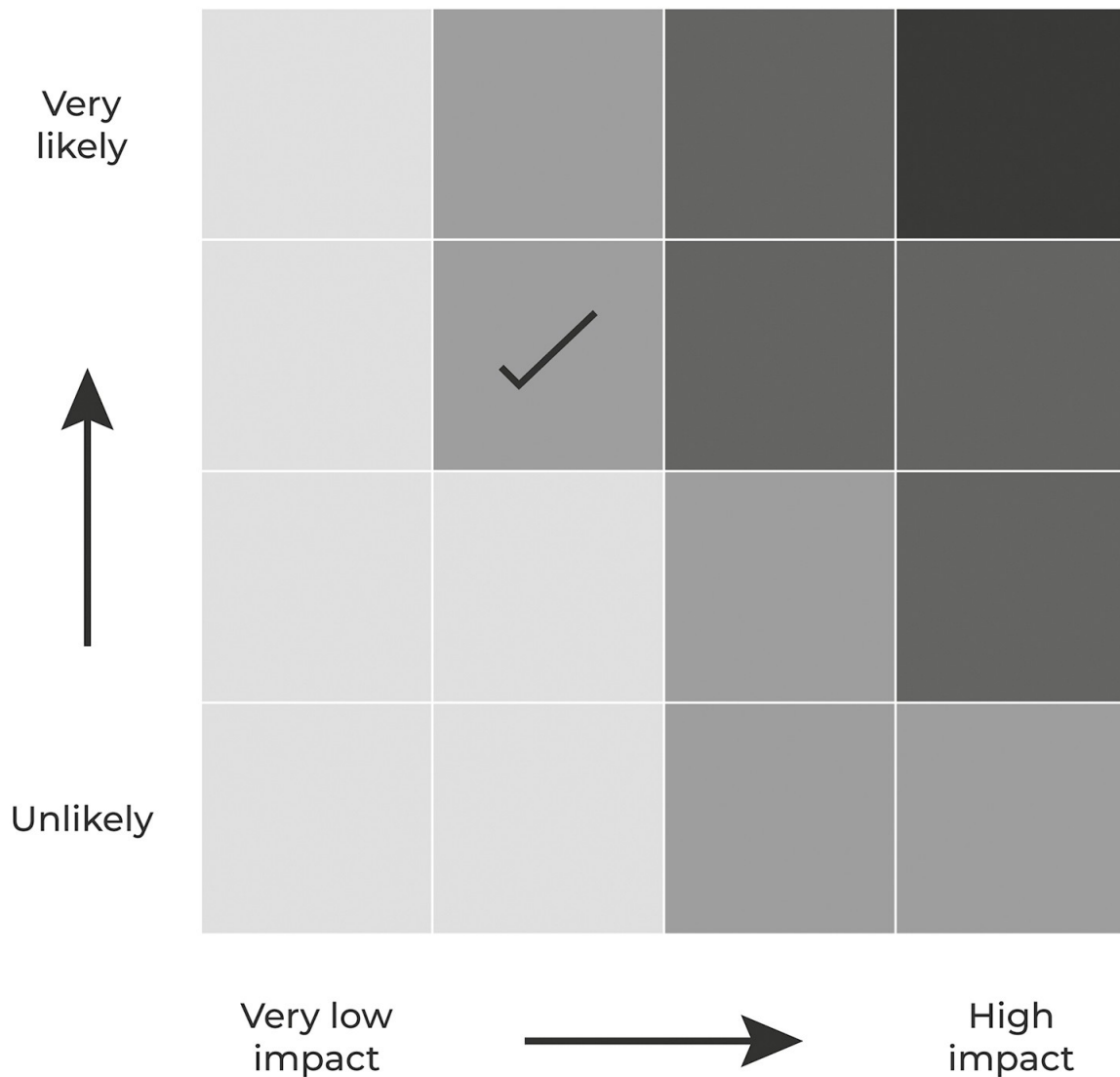


FIG. 42. Risk matrix used to determine severe weather warnings in the UK. Shown in the figure are boxes with four shades of grey, corresponding to “no warning”, “yellow warning”, “amber warning” and “red warning”—the latter being the most severe weather warning. The shading is determined by the product of impact and likelihood of occurrence. For example, an amber warning could mean either a high probability of a medium-impact event or a medium probability of a high-impact event. The probabilities are determined by ensemble forecast systems. The ticked box corresponds to a “yellow warning”.

CAN THE COST-LOSS MODEL help us to decide whether to take anticipatory action in the light of the threat of climate change?

Let's summarise two possible arguments for and against taking anticipatory action.

On the one hand:

The climate crisis is a manifestation of our overindulgent and profligate way of life. We sowed the wind; nature is now reaping the whirlwind. As written by James Lovelock in his *Revenge of Gaia*⁶,

"Like an old lady who has to share her house with a growing and destructive group of teenagers, Gaia grows angry, and if they do not mend their ways, she will evict them."

We have to rid ourselves of our dependence on fossil fuels as quickly as possible and live a simpler life more in harmony with nature.

On the other hand:

Decarbonising the economies of the world will stunt their economic growth, particularly in relatively poor parts of the developing world, preventing them from attaining the standards of living which the developed world achieved on the back of cheap carbon-intensive energy. And if we hinder the economic growth of these poorer countries, we will lock in their high human fertility rates for many years to come, exacerbating problems of growing population levels and further degrading the planet. All this economic pain is simply not worth it—the impact of unmitigated climate change on global gross domestic product will be minimal. And if climate change means we have to adapt to new norms of weather, so be it—we can afford it, it's not a big deal.

Perhaps the cost-loss model I've described could provide a framework for trying to address these issues objectively. Does the likelihood p of dangerous levels of climate change exceed the cost C of cutting greenhouse-gas emissions to zero divided by the damages L associated with climate change? If the costs were small enough or the losses large enough, it would make sense to take action even if the probabilities were small. But are they? If we could boil it down to a seemingly straightforward calculation like this, we'd have a clear-cut strategy on whether and how to deal with climate change.

This was a question addressed by probably the most authoritative study on the impact on the global economy of climate change: the Stern Review⁷, published in 2006, and authored by the distinguished economist Lord Nick Stern. Using estimated losses L and costs C based on gross domestic product (GDP), Stern's main conclusion was that the probability p of dangerous levels of warming is sufficiently great that it is certainly worth taking action now to cut our carbon emissions. But there were those who criticised the basic assumptions of the Stern Review. These critics argued that the science of climate change was not good enough to estimate p reliably, that L was overestimated (particularly when the future is discounted, as one has to do in any economic calculation about the future), and that C was underestimated.

Let's look at each of these issues in turn.

Whilst I fully agree that we could do a better job in estimating p —for example, by pooling human and computer resources to build much-higher-resolution global models (e.g., at a “CERN for climate change” as discussed in [Chapter 6](#))—there is no evidence that climate models have overestimated p . As discussed in [Chapter 6](#), climate projections from thirty or more years ago have done a good job in predicting observed rates of global warming.

Let's look at L —the avoidable damages caused by climate change. We need to be able to quantify L in some shape or form. If we can't put a figure in dollars or pounds on L , then we won't be able to determine whether the cost C of cutting emissions is worth it or not.

As we have seen, economists have tried to estimate the impact of L on global GDP. According to the Stern Review without action, climate change could reduce GDP by 5 percent or more per year. Others are not so pessimistic. The Nobel Prize-winning American economist William Nordhaus has estimated that 3°C (5.4°F) of warming would lead to only just over 2 percent reduction of GDP altogether. These are massive differences. Who is right?

One problem is that, as we discussed in [Chapter 8](#), economic models tend to be rather simple. The impact L in such models is typically determined by the global-mean temperature change predicted by the climate models. One could argue that they should take the regional changes of climate into account. However, as we have discussed, the rather coarse-resolution climate models do not simulate well the intensity of extreme weather events. On top of this, the economic-impact assessment models do not have the complexity of the agent-based models discussed in [Chapter 8](#), whose form allows them to be driven by regional changes in weather patterns. I hope this rather unsatisfactory state of affairs will change with the development of a CERN for climate change, discussed in [Chapter 6](#).

However, there is another issue. Let's think again about Ahmadul, our hypothetical representative farmer from Bangladesh. How do we quantify the impact of climate change in monetary terms on a representative of a community with a completely negligible impact on global GDP?

We are edging towards a topic which some people might feel is taboo. How much is a human life worth? This is

certainly a distasteful topic, but nevertheless it is one that has to be addressed if we are to make progress in this area. One could of course take the view that a human life is priceless. However, there were just short of 25,000 people killed on the UK roads in 2020. If we valued a human life as literally priceless, then it would make sense to impose a ten-mile-per-hour speed limit on all roads. However, most of us probably feel that the extra time it would take to travel is not justified, even if it does save lives. That is to say, we implicitly put a value, in terms of our time, on a human life.

Statisticians have known that it is important to come up with some value of a statistical life (VSL), as it is called, in order to argue either for or against regulatory legislation. In his book *Pricing Lives*⁸, W. Kip Viscusi, one of the pioneers in this field, discusses an issue which arose during the Reagan presidency in the 1980s: Should firms be required to label dangerous chemicals in the workplace? At the time, government agencies would estimate the value of life in terms of a person's lost earnings and associated costs if he or she died. On this basis, the US Office of Management and Budget argued that the cost of adding the labelling wasn't worth it, and therefore didn't recommend the legislation that required companies to provide labelling.

Viscusi argued that the value of life was not being calculated properly. Instead, he argued that VSL should be estimated from evidence of how much extra money individuals would require to accept an increased risk of death (or some life-changing disability which could make life a misery). Viscusi presented evidence from US-based statistical data at the time that workers were prepared to accept an annual wage premium of \$300 for a job where the extra annual worker fatality risk was 1/10,000. From this, we can use the cost-loss model described earlier to estimate a value of life. That is, if we write $C = pL$ where

now $C = \$300$ and $p = 1/10,000$, then we have $L = \$3$ million. That was in 1982. Adjusted for inflation we get a modern figure of VSL in the US as perhaps a little less than \$10 million.

However, we have to face the uncomfortable fact that VSL in the developing world is not as large as that in the US or other wealthy countries. Sadly, workers in developing countries will accept a smaller wage premium for this extra risk of death or disability. How can we adjust for this, so that we can get a figure that applies to both Ahmadul in Bangladesh and Jim in Seattle? Very roughly, a formula that is widely used defines VSL as 100 times the per capita GDP in the country of interest⁹. On this basis, we can value Ahmadul's life at 100 times the per capita GDP of Bangladesh.

Remember VSL is based on the monetary premium not only of death but of some life-changing disability. Based on the discussion in [Chapter 6](#), I would say that living in a 4°C (7.2°F) or greater warmer world, a hell on earth (HoE) as far as I can judge, is not too far off the equivalent of a life-changing disability. Simply to be conservative, if VSL is 100 times the per capita GDP, let's estimate the value of avoiding HoE at 50 times the per capita GDP.

How likely is it that we will experience at least 4°C of warming if we do nothing to cut emissions? According to what I have discussed, it is, I would say, around 0.3. It does depend critically on these cloud feedbacks about which our knowledge is currently rather limited. However, if we multiply the probability, $p = 0.3$, by a loss, L , corresponding to 50 times per capita GDP, we end up with the "risk", pL , of HoE at 15 times per capita GDP.

What is the cost of avoiding HoE? For many years the cost of decarbonising a country's economy was assumed to be a few percent of the country's GDP. However, according to analysis performed by the UK's Committee on Climate

Change in its Sixth Carbon Budget Methodology report¹⁰, it seems that the phenomenal drop in the cost of renewable energy (wind and solar) means that the cost could be as low as 1 percent of GDP¹¹. But let's assume 2 percent of GDP in general. Hence, for each person, this is a cost, C , of $1/50$ of GDP, i.e., per capita.

On this basis, C is 750 times smaller than the risk, pL , of HoE. It seems overwhelmingly worth it to cut our carbon emissions.

However, this conclusion could be undermined by the issue of discount rates. If you had a choice of receiving \$100 today or \$100 in ten years' time, you would likely prefer to receive it today (inflationary devaluation notwithstanding). For example, you might invest your \$100 in some business that you hope will make you several hundred dollars in ten years' time. If one estimates discount rates from the financial markets (e.g., based on financial assets such as shares), then one might discount future losses, L , by as much as 6 percent per annum.

Hence, some economists argue that we should compare C not with pL , but with pL discounted by this rate of around 6 percent per annum. On this basis, if L occurs sufficiently far in the future, pL will effectively be discounted to zero. Then no cost paid today will be worth it.

Does the application of a financial discount rate make sense? As far as I can see, the concept of human suffering makes a mockery of the notion of economic discounting based on financial assets like shares. Human suffering fifty years from now seems no less troubling to me than if it occurs five years from now, or indeed tomorrow.

Some people argue that if we are wealthier, as we may be in the future, then we will be able to cope better with the hell on earth of a 4°C warmer world. I don't buy this. One should look at pictures of relatively wealthy German couples crying on each other's shoulders after their houses

were swept away in the Rhine Valley floods of 2021. It makes you realise that even if in the coming hundred years we brought the standards of living of most of the developing world to those in Germany today, a 4°C warmer world would still be a hell on earth. Raising the per capita income in the developing world by a factor of ten, say, is not going to help much in the face of some devastating storm. And in any case, if the world is heading for a hell on earth, perhaps we will not be wealthier in the future. If that's the case, then perhaps the discount rate should be negative.

Since we have left the strict domain of science, we need to ask one further question: Who should be paying to mitigate the effects of climate change? Should Ahmadul be paying? Well, he played no role whatsoever in causing the problem in the first place. We in the rich, developed world have benefitted from cheap, carbon-rich energy. Perhaps it is time for us to pay some of that back.

Indeed, ethical and altruistic issues aside, it is in the developed world's interest to ensure that Ahmadul's life is not completely unbearable. This is because he and potentially billions like him have another option: rather than suffer and potentially perish from existential heatwaves, storm surges and extended drought, he and his family can try to migrate polewards, where the weather is likely to be more bearable. Historically, migration is how civilisations coped with local changes in climate. Nowadays, such mass migrations will be the source of major conflict, as we have discussed. If the developed world does not like this option, then it can try to ensure Ahmadul and his children can continue to live fulfilled lives where they are currently living. One can think back to the Marshall Plan, when the US provided financial help to war-torn Europe, not necessarily for altruistic reasons, but to

prevent Europe turning to communism. We need a climate Marshall Plan for the future¹².

Let's summarise.

If the local weather forecast predicts an intense storm with some significant probability, it's your decision whether to act on it or not. The underlying science of meteorology does not tell you that you must act if hurricane-force winds are predicted for tomorrow. It's not for the weather forecaster to tell you what to do. However, coupling the meteorology to a cost-loss model can help you make a decision, for example, about whether to avoid travelling on a day when hurricane-force winds are predicted with some significant probability.

We can look at climate change in a similar way. Does burning of fossil fuel lead to an increase in carbon dioxide concentration in the atmosphere? According to science, certainly yes. Is carbon dioxide a greenhouse gas? According to science, certainly yes. Will emissions of greenhouse gases increase the risk of dangerous levels of climate change? According to science, certainly yes. Should we therefore cut our emissions of greenhouse gases as quickly as possible? Science itself is agnostic on this last point. The activists who simply say, "Listen to the science", seem to have missed this point. The physicist Sabine Hossenfelder put it in a rather graphic way (apparently referring to something that drunken Germans do from railway bridges): science does not tell you not to pee onto high-voltage electricity lines; it tells you that urine is a good conductor of electricity.

Just as with weather prediction, a cost-loss analysis can help you make a decision about whether to take anticipatory action regarding climate change. However, this requires putting a value on things which do not have an unambiguous value, like living in a future hell on earth. Based on the way we value our own existence in other

PART III

Understanding the Chaotic Universe and Our Place in It

My own view is that to understand quantum nonlocality we shall require a radical new theory. This new theory will not just be a slight modification of quantum mechanics but something as different from standard quantum mechanics as general relativity is from Newtonian gravity. It would have to be something which has a completely different conceptual framework.

ROGER PENROSE¹

If a machine is expected to be infallible, it cannot also be intelligent.

ALAN TURING²

Two key ideas from Part I are applied to try to understand two of the most puzzling aspects of the universe: the world of quantum physics and ourselves. We start by assuming that the geometry of chaos applies to the universe as a whole. This leads to the crucial idea that certain counterfactual worlds, where we might have done something but didn't, could actually be inconsistent with the laws of physics. This helps make comprehensible certain long-standing quantum mysteries. Following this I

discuss the idea that the human brain makes constructive use of noise to model the world around us, and that this has enabled us to become the creative species we are. I speculate that the geometry of chaos can help explain two of our most visceral but puzzling experiences: having free will and being conscious. I conclude with a novel suggestion about the nature of God. A caveat: the discussion in Part III is much more speculative than Parts I and II.



11

Quantum Uncertainty

Reality Regained?

WE LEFT OUR ANALYSIS OF QUANTUM UNCERTAINTY IN [CHAPTER 4](#) in a state of confusion. Einstein had proposed that the quantum wavefunction describes an ensemble of possible worlds. As an illustration, he imagined a photon moving towards a point on a phosphorescent hemisphere. According to Einstein, in each ensemble member, the photon takes a different path to the hemisphere. We don't know which of the ensemble of possible worlds is the real world until we make some macroscopic measurement, such as observing a flash from the phosphorescent material. From Einstein's perspective, quantum uncertainty is epistemic—it reflects our own uncertainty, not the photon's uncertainty. Bohr, on the other hand, believed that quantum uncertainty was inherent to a quantum system—it is ontological in nature. The experiment Bell proposed and that was later successfully performed using entangled photons casts severe doubt on Einstein's ensemble interpretation and in so doing supports Bohr's position. Seemingly even the particles don't know where they are going until they get there. Whatever that means.

What to do? For years, the answer was to shrug one's shoulders and say, "Get over it". Epistemic or ontological—frankly, who cares? Such issues were considered a matter for philosophers with their fancy words, but not for physicists with hard equations to solve. The mantra was to "Shut up and calculate". And in astrophysics, condensed matter physics, quantum computing, high-energy physics and other cutting-edge areas of research, quantum mechanics works well without worrying about these seemingly esoteric issues. In climate physics, for example, one has to think about how photons absorb and scatter in the atmosphere. But one never worries about the nature of uncertainty in these radiative processes. One just gets on and solves the Schrödinger equation, or an approximation for it, for the problem at hand.

Nowadays, physicists are finally starting to question the mantra. There are a number of reasons for this.

Firstly, string theory, our leading candidate for unifying quantum mechanics and gravity (and hence for providing a so-called theory of everything) is beginning to lose its lustre. String theory utilises a concept called supersymmetry. If nature were supersymmetric, the difference between the elementary particles which carry forces, such as the photon, and the elementary particles which feel these forces, such as the electron, would not be fundamental. Supersymmetric theories predict new types of elementary particles (e.g., so-called photinos and selectrons). Although string theory does not predict the masses of these supersymmetric particles, many physicists hoped they would have been found by the Large Hadron Collider at CERN. So far, no such particles have been observed.

On top of this, it has become clear in recent years that over 80 percent of the matter in the universe is in the form of what is called dark matter: matter of which we observe only its gravitational pull, but about which we know

nothing else. Dozens of experiments have searched for different types of hypothetical particles that might make up dark matter, but none has found even a trace. Currently, no one knows what dark matter is made of, if indeed it is made of anything¹.

Dark energy is another mystery. It causes the universe's expansion to accelerate, seemingly consistent with observations. Quantum field theory—the extension of quantum mechanics needed to describe the elementary particles—actually predicts a possible value for this acceleration. However, the predicted value² is too large by a factor of 10^{120} . If the universe accelerated according to the value predicted by quantum field theory, then stars and galaxies would not have had enough time to form before the expansion of the universe ripped them apart. Hence, we would not exist. Quantum field theory's prediction of dark energy is sometimes called the worst prediction in the history of science³. It makes Michael Fish's weather forecast virtually perfect by comparison!

As a result of all this, some physicists are beginning to ask a question that few previously had dared ask: Is there something wrong with quantum mechanics itself? Perhaps we will not properly synthesise gravity with the other fundamental forces of nature until we have understood what is wrong. One world-renowned physicist who, over the years, has dared raise this possibility is Nobel laureate Roger Penrose—see the quote at the beginning of this part of the book.

Emboldened, let's return to the Bell experiment. It's worth recapping by showing [Fig. 26](#) again (see [Fig. 43](#)) and replicating the particles' so-called look-up table in [Table 2](#) (see [Table 3](#)).

[Fig. 43](#) shows the Bell experiment, in which Alice and Bob measure the spin of pairs of entangled particles. [Table 3](#) can be considered part of a hypothetical look-up table for

Alice's particles. The existence of such a look-up table is implied by a standard deterministic hidden-variable model of quantum physics of the type Einstein would have favoured. The rows of the table correspond to three possible orientations of the SG devices. Each column of the table represents the outcomes of spin measurements on, say, one of Alice's particles. In principle the number of rows and columns in the look-up table may be much larger than this. However, three rows suffice for our purposes.

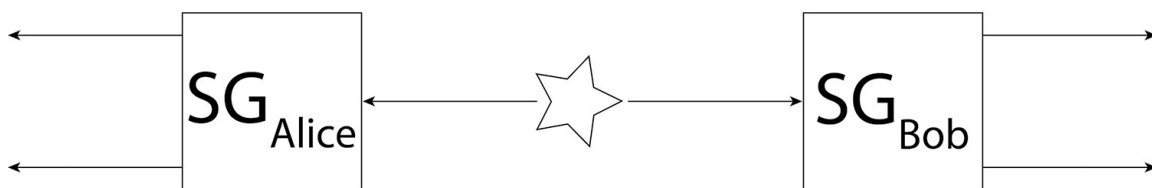


FIG. 43. The famous Bell experiment, using pairs of entangled particles moving in opposite directions. In this experiment, Alice's and Bob's SG devices can be oriented arbitrarily. However, if they are oriented in the same direction and Alice measures her particle to be spin up, Bob will measure his particle to be spin down.

1	+	-	-	+	+	+	-	+	+	+	+	+
2	-	+	-	+	-	-	-	+	-	-	-	-
3	-	+	-	-	+	-	+	-	+	+	+	+

TABLE 3. A hypothetical look-up table for the spin of 12 of Alice’s particles in a Bell experiment, for 3 possible choices of orientation of Alice’s SG device. The statistics of these +s and –s necessarily satisfy a mathematical inequality known as Bell’s inequality, described in the text. The look-up table for Bob’s 12 particles is exactly the same, except each + is replaced with a – and vice versa. Experimentally, Bell’s inequality is violated for certain choices of measurement orientation. This casts doubt on the existence of such look-up tables, and hence casts doubt on hidden-variable theories of quantum physics.

Recall the quantities A , B and C calculated from [Table 3](#), described in [Chapter 4](#). A is the number of columns where there is a + in the first row and a + in the second row. B is the number of columns where there is a – in the second row and a + in the third row. C is the number of columns where there is a + in the first row and a + in the third row. As mentioned (and proved in endnote 11 of [Chapter 4](#)), it is simple to show mathematically that $A + B$ will always be bigger than or equal to C , no matter how we fill in the entries to this table with +s and –s. This result is known as Bell’s inequality.

Let’s look again at how the experiment is performed to test whether this equality is satisfied in nature. In practice the experiments that are performed test a slightly more complicated version of Bell’s inequality. However, what I am about to draw attention to here applies equally to this more complex version.

For each particle pair in [Fig. 43](#), Alice and Bob have to decide, independently, whether to measure the spin of their particle in the “1”, “2” or “3” directions. We don’t need to worry what exactly these directions are, except that they are directions in which quantum mechanics predicts that Bell’s inequality is violated.

From the sub-ensemble of measurements where Alice chose 1 and Bob chose 2, we calculate the quantity A. To do this calculation, we must invoke the fact that the total spin of the two entangled particles is 0, which means that if Bob chose 2 and got +, then for this particle pair Alice would have got – if she had measured the spin of her particle in the 2 direction. Similarly, we can calculate B from the measurements where Alice chose 2 and Bob 3. Finally, we can calculate C from measurements where Alice chose 1 and Bob 3. I have shown some possible such results in [Table 4](#).

1	+	-	-	+					+	+	+	+
2	-	+	-	+	-	-	-	+				
3					+	-	+	-	+	+	+	+

TABLE 4. A partial look-up table for Alice based on (possible) results from a Bell experiment. For example, the first column is constructed from a pair of measurements where Alice measured + in the 1 direction and Bob + in the 2 direction (which means Alice would have measured – if she had chosen the 2 direction). Similarly, the twelfth column is constructed from measurements where Alice measured + in the 1 direction and Bob – in the 3 direction (which means Alice would have measured + if she had chosen the 3 direction).

The first point to notice in [Table 4](#) is that there are blanks. There are no columns which contain three entries,

as there are in [Table 3](#). It is easy to violate Bell's inequality from incomplete look-up tables. For example, from [Table 4](#), $A = 0$, $B = 0$, $C = 4$, which clearly violates Bell's inequality since here $A + B$ is less than C .

This demonstrates that estimates of $A + B$ and C based on experimental data which generate partial look-up tables, like [Table 4](#), don't necessarily invalidate hidden-variable theories. To be able to use the results of experiments to rule out hidden-variable theories, we need an additional assumption about the nature of the three experimental sub-ensembles of particle pairs from which A , B and C are estimated experimentally. This assumption is that the three sub-ensembles of particles are statistically equivalent to one another. If they have this property, then we can assume that the experimental estimates of A , B and C are the same as if we obtained A , B and C from a full look-up table, providing we have a big enough statistical sample (i.e., enough columns in the look-up table).

This assumption seems quite reasonable at first sight. Suppose we were performing a blind trial of a drug (e.g., a possible COVID-19 vaccine) for possible use in society. In the trial, one group of volunteers gets the drug and another a placebo. If we have devised the test properly, it shouldn't matter which group gets the drug and which group gets the placebo. For the drug trial to have validity, the two groups of volunteers should be statistically equivalent in terms of their ages, genders, ethnic origin, eye colour and other relevant characteristics.

When it comes to particles, however, it's not clear what we mean by saying that the sub-ensembles are statistically equivalent—the particles don't have genders, ages or eye colours to compare and contrast. In fact, the only knowable thing that unifies the members of one sub-ensemble over another is the type of spin measurement performed on the particles in that sub-ensemble (e.g., that Alice's particles were all measured in the 1 direction).

What is the key reason why physicists typically assume that the sub-ensembles are statistically equivalent? It is because of the conclusion that although Alice actually measured a particular particle in the 1 direction, she might just as well have measured it in the 2 or the 3 direction. For example, suppose Alice decided to measure a particle in the 1 direction on the basis that her grandmother was born in the first third of a month. Since the grandmother might just as well have been born in the last third of the month, so the argument goes, Alice might just as well have measured that same particle in the 3 direction.

In the quantum foundations community, asserting some physical link between the intrinsic properties of particles and whimsical determinants of measurement settings is viewed as implausibly conspiratorial. For this reason, it is typically assumed that the different sub-ensembles are indeed statistically equivalent. If we make this assumption, then we are faced with the uncomfortable fact that the violation of Bell's inequality can be explained only by giving up the notion of reality as something definite or by the equally ghastly notion of quantum action-at-a-distance.

Despite this entirely plausible argument in favour of statistical equivalence, I want to argue in this chapter that, based on the geometry of chaos, there may be an implicit physical linkage between a particle's hidden variables and the measurement settings, and this linkage is not in the least ridiculous or conspiratorial.

To understand this, let's rewind the clock to the previous millennium and imagine a counterfactual world where Alice's great-grandmother's pregnancy was for forty-two weeks instead of forty weeks. In this counterfactual world, Alice's grandmother was born in the last third of the month, and as a result Alice's particle was measured in the 3 direction, rather than the 1 direction.

A key question I want to ask is this: Is the world where the great-grandmother had a forty-two-week pregnancy

necessarily consistent with the laws of physics? We can put this a little more explicitly. Given the actual state of the universe, consider a mathematical state seemingly close to the actual state where Alice's great-grandmother's pregnancy was forty-two weeks long. Is this mathematical state of the universe necessarily consistent with the mathematical laws of physics?

Suppose for some reason—which we will come to—it isn't consistent. Then the counterfactual world where Alice actually measured in the 3 direction would also be inconsistent with the laws of physics. If this were so, then instead of treating as conspiratorially implausible a physical link between the intrinsic properties of Alice's particle with the actual measurement setting 1, the link would instead be wholly inevitable⁴!

Indeed, the only reason quantum foundations experts believe that the three sub-ensembles of particles are statistically equivalent is because of the implicit assumption that Alice and Bob might just as well have measured their particles in a different way from how they actually measured them. Take this away and the assumption of the statistical equivalence of the sub-ensembles disintegrates. And without that assumption, we do not have to assume spooky action-at-a-distance or indefinite reality to explain the experimental violation of Bell's inequality!

We tend to think about counterfactual worlds all the time, but, I suspect, we rarely ask the question of whether the counterfactual worlds we envisage might actually be consistent with the laws of physics. In [Chapter 9](#) we discussed how Archduke Franz Ferdinand's driver mistakenly took the wrong route. If he had taken the correct route, the archduke would not have been assassinated by Princip and the whole history of the twentieth century could have worked out differently. A

world where the archduke was not assassinated is a counterfactual world. Is such a world necessarily consistent with the laws of physics?

According to the laws of classical physics, it is. For example, in Lorenz's equations of chaos or the Navier-Stokes equation for a fluid, there is nothing preventing you making an arbitrarily small change to the starting conditions for these equations. Similarly, there is nothing in the classical laws of physics preventing you hypothesising a tiny change to the world—the flap of a butterfly's wings—that would have made the archduke's driver take a different route. As such, we can describe the laws of classical physics as “counterfactually definite”. Because of this, we frequently use the notion of counterfactuality to infer causality⁵. If I had not thrown the stone, the window would not have broken. Hence, because in reality I did throw the stone, I caused the window to break.

But this does not imply that some counterfactual worlds might be inconsistent with the laws of physics in an as-yet-to-be-discovered theory of quantum physics. Such a theory would be counterfactually indefinite and an example of what is called a superdeterministic⁶ theory. A good way to think about the difference between a deterministic model and a superdeterminist model—of the type I am proposing at least—is like this. In a deterministic model, the future is predetermined by the laws of physics from a given initial condition. The Navier-Stokes equation is an example of a deterministic model. Despite such determinism, we can envisage slightly altered initial conditions (e.g., a small change to a small eddy in the turbulent fluid). The deterministic evolution from the altered initial conditions, although different from what actually happened in the real world, is mathematically allowed in this deterministic model. By contrast, in a superdeterministic theory, one cannot assume that such altered initial conditions are

consistent with the theory's equations. And therefore one cannot assume that evolution from this altered initial condition is consistent with the theory's equations.

The assumption of counterfactual definiteness is not given the prominence it deserves in quantum foundations discussions because it's not immediately obvious how to formulate a model of quantum physics which has the property of counterfactual indefiniteness. One leading quantum expert who did understand the critical importance of counterfactual definiteness was the Russian Israeli mathematician Boris Tsirelson, who died in 2020. Tsirelson made a very important contribution to our understanding of Bell's inequality. He quantified the extent to which quantum mechanics violates the Bell inequality through the eponymous Tsirelson bound. It is important that any putative theory of quantum physics does not violate Bell's inequality beyond the Tsirelson bound. In his pedagogical description of Bell's inequality, Tsirelson states clearly why counterfactual definiteness is a vital assumption if a hidden-variable theory is to satisfy Bell's inequality⁷.

It turns out that counterfactual indefiniteness can also explain the puzzle of the sequential SG experiment of [Chapter 4](#). The key issue is whether a counterfactual experiment where we reverse the last two SG devices in [Fig. 25b](#)—keeping the particle and its hidden variables fixed—is consistent with our theory of physics. A theory where the counterfactual does not exist provides an explanation of what in quantum mechanics is called the “non-commutativity” of spin. In ordinary language, non-commutativity implies that we can't treat the sequential SG experiment as made of completely independent SG subcomponents—we can't simply reverse the order of the SG components with the same fixed particle; the whole is not simply the sum of its parts. An ensemble-based hidden-variable theory which has this type of counterfactual

indefiniteness can explain the sequential SG experiment. I describe one ahead.

It seems quite satisfactory that there may be a single common explanation to both the Bell experiment and the sequential SG experiment. Both illustrate an important property of the quantum world: its inherent indivisibility. This notion was emphasised in the title of David Bohm and Basil Hiley's influential book discussing hidden-variable theories of quantum physics ⁸: *The Undivided Universe*. This title, and the message it conveys—that the universe as a whole is much more than the sum of its parts—provides an important guide for the next part of the argument.

« »

WHAT FORM COULD THE laws of physics take which would render inconsistent with the laws certain key counterfactual quantum worlds? Here I claim that laws based on the geometry of chaos, such as we discussed in Part I, do this.

If we are to respect the inherent indivisibility of the quantum world as we've discussed, we need to think big, very big. We'll take Lorenz's insight and apply it, not just to a fluid-dynamical system on planet Earth but to the whole universe, with all of its stars, galaxies and clusters of galaxies. That is to say:

Conjecture A: The whole universe is a nonlinear dynamical system evolving on some fractal attractor ⁹ in cosmological state space.

Cosmological state space is just like trouser state space, or the Lorenz model's state space, but on steroids. Cosmological state space describes all the degrees of freedom of all the objects in the universe. It has a huge

number of dimensions. And like the Lorenz attractor, our cosmological attractor will have fractal gaps on all scales. The quantum counterfactual worlds that we have been discussing lie in these fractal gaps.

I call conjecture A the “cosmological invariant set postulate”. The phrase “invariant set” is just another mathematical way of saying “attractor”^{[10](#)}. The conjecture can be reframed in a different way:

Conjecture B: The laws of physics at their deepest describe the geometry of a fractal invariant set in cosmological state space.

The hypothetical counterfactual quantum worlds, if they do not lie on the invariant set, are inconsistent with conjectures A and B. This finally provides a possible mathematical way to understand the violation of Bell’s inequality without giving up on definite reality and without having to accept spooky action-at-a-distance.

How would we formulate this postulate mathematically? We can’t use ordinary “real” numbers to do this. As mentioned in [Chapter 2](#), real numbers are linked to Euclidean geometry, and the geometry of fractals is fundamentally different to Euclidean geometry. Instead, we need the mathematics of p -adic numbers—the numbers that pure mathematicians use to understand the “quantum” nature of mathematics—i.e., the whole numbers: 1, 2, 3... In the theory of p -adic numbers, a point which does not lie on its corresponding fractal is necessarily distant from a point which does, even though in terms of the familiar Euclidean distance, such points might look close together. That is to say, from the p -adic perspective, a change which moves a point off the cosmological invariant set is a large change, even though it may seem small from the familiar Euclidean perspective^{[11](#)}.

In a series of papers^{[12](#)}, I constructed a mathematical model for this cosmological invariant set described in these

conjectures. I call the general framework which underpins this model “invariant set theory”. What does the cosmological invariant set look like? Is it like the Lorenz or Rössler attractors of [Chapter 2](#)?

At present, the mathematics of invariant set theory tells us only about the fractal structure of the trajectories on the cosmological invariant set, not its global structure (see the very last sentences in this book). The fractal structure of trajectories is illustrated in [Fig. 44](#). Here, instead of a state-space trajectory being described by a single one-dimensional curve as in [Fig. 5](#), it is instead described by something rather akin to a piece of rope ([Fig. 44a](#)). That is to say, what looks from a distance like a single curve is actually a helix of trajectories twisted around themselves (cosmic DNA perhaps¹³). In turn, each of these smaller trajectories is itself a helix of trajectories twisted around themselves. If we take a cross-section through this rope-like trajectory segment, it has the fractal structure shown in [Fig. 12](#) for p -adic numbers, also shown in [Fig. 44b](#). When a quantum system interacts with the environment, in a process called quantum decoherence, the rope starts to unravel ([Fig. 44c](#)). Finally, the decohered trajectories cluster together ([Fig. 44d](#)). The clusters are associated with the discrete measurement outcomes (e.g., spin up/spin down) in standard quantum mechanics.

A key mathematical result is that the quantum counterfactual worlds described earlier, for either the sequential SG experiment of [Fig. 25](#) or the Bell experiment of [Fig. 43](#), necessarily lie in the gaps in between the strands of rope, and therefore cannot be characterised by p -adic numbers. These worlds do not lie on the invariant set. The key geometric ideas that lead to this result are described in the endnotes¹⁴. They make use of mathematical properties of spherical triangles and the distinction between rational and irrational numbers ¹⁵. The

key result means, by conjecture A, that such counterfactual experiments are inconsistent with the laws of physics. But this means that antecedent worlds in which the causes of these counterfactual experiments occur similarly cannot lie on the invariant set. This is why we can be sure that the counterfactual world where Alice's great-grandmother's pregnancy lasts two weeks longer is inconsistent with the laws of physics—not directly because of processes happening in the last millennium, but indirectly because these processes were the cause of a particular quantum measurement setting in this millennium. There is no way that Alice's great-grandmother could know this, not least because the invariant set is, by the discussion in Part I of this book, non-computable—there is no algorithm for determining whether a point in state space lies on a fractal set or not.

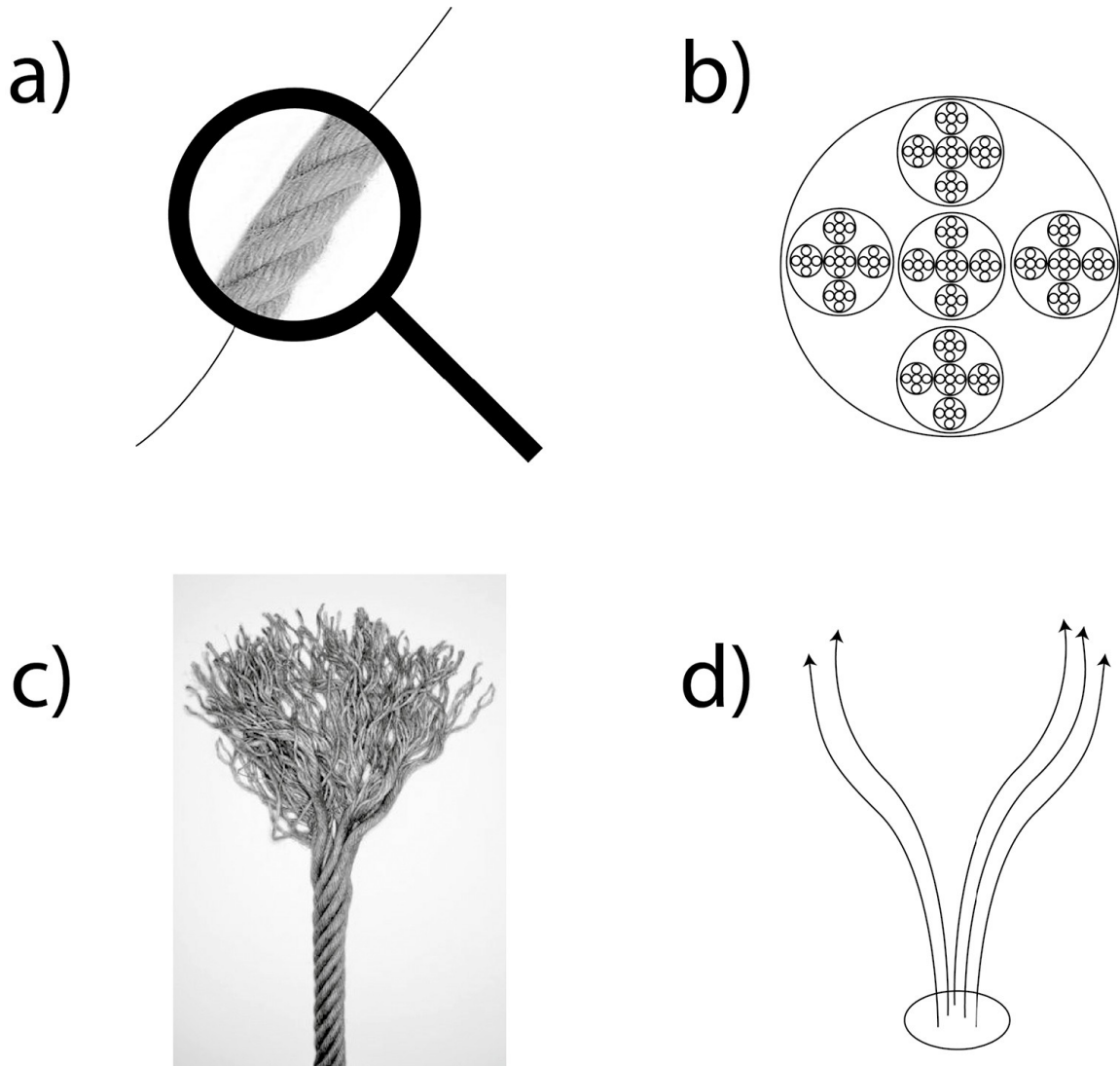


FIG. 44. a) Instead of a trajectory describing a simple curve in state space (c.f., [Fig. 5](#)), in invariant set theory, a trajectory, on magnification, resembles a piece of rope, comprising a helix of smaller-scale state-space trajectories, each trajectory of this helix itself comprising a further helix and so on. b) A cross-section through the helix is a geometric representation of the p -adic numbers, discussed in [Chapter 2](#). c) When a system interacts with the environment, the helix “decoheres” into its smaller strands. d) Finally, the decohered strands evolve into discrete clusters, a nonlinear process which we associate with the phenomenon of gravity. The clusters correspond to measurement outcomes in quantum mechanics.

In a nutshell, failure of counterfactual definiteness is exactly what is necessary to reconcile the violation of Bell’s inequality with Einstein’s ensemble interpretation, but

without giving up on the notion of a definite reality, and without invoking the spooky action-at-a-distance that he hated so much.

The same reasons (the ideas in note 14) explain why when we try to determine which path a photon took through the two slits in [Fig. 22](#), the interference pattern disappears. Here, a counterfactual world where we measure which slit the particle took, keeping the interference pattern fixed, lies off the invariant set.

There is another way to understand invariant set theory. Quantum mechanics arose from a bold speculation by the German physicist Max Planck: that light energy does not vary continuously, but in discrete packets which he called “quanta”. In order to accommodate this idea, the state space of quantum mechanics (quantum trouser space, if you like) has to be of a particular type. We call it Hilbert space, named after the mathematician David Hilbert, whose “halting problem” was discussed in [Chapter 2](#). Quantum trouser space is itself continuous—you can move from one point in Hilbert space to another through a series of infinitesimal steps. In invariant set theory, we take Planck’s idea one step further—we discretise Hilbert space. For this to work, the discretisation has to be done in a very specific way¹⁶. The gaps in the discretised Hilbert space correspond to counterfactual measurements which might have been performed on a quantum system but weren’t.

Not all counterfactual worlds are inconsistent in invariant set theory—there are plenty of neighbouring counterfactual trajectories which also lie on the invariant set. According to conjecture B, the laws of physics describe what happens along the real trajectory using laws built on the collective geometric properties of both real and counterfactual trajectories. From this point of view, we can interpret the two-slit experiment of [Fig. 22](#) as follows: even though in reality the photon travelled through only one slit

in [Fig. 22](#), there exist counterfactual trajectories on the invariant set where the photon travelled through the other slit. If the real trajectory is the yin¹⁷, then the counterfactual trajectories are the yang. Like the counterweight on the crankshaft in a piston engine, the counterfactual trajectories ensure the smooth running of the real quantum world. The ensemble of counterfactual and real trajectories—collectively describing the geometry of the invariant set—gives the appearance of an interfering quantum wave in space-time.

Quantum mechanics treats the ontology of the real and counterfactual trajectories as the same. That's why the notion of reality (a particle in two places at once) is so incomprehensible in quantum mechanics. In invariant set theory, by contrast, real and counterfactual trajectories have a distinct but interdependent ontology: they are not the same, but equally one cannot ignore the counterfactual trajectories if one wants to understand evolution on the real trajectory¹⁸. In the next part of this chapter, we will conclude that the counterfactual trajectories are in fact real trajectories, but for earlier or later epochs of the universe. Hence, according to invariant set theory, quantum physics ties together real-world evolutions of multiple different epochs of the universe. Not only is the spatial universe undivided, so too are the epochs of the universe.

The yin and yang of invariant set geometry can also help explain the power of quantum computers over traditional classical computers. The latter process information on only our own single classical trajectory, whilst the former additionally process information on the many counterfactual trajectories in the neighbourhood of our trajectory on the invariant set. Later on I will explore the possibility that this quantum yin and yang can also explain our visceral feeling of free will and consciousness.

Is there an experimental test for the cosmological invariant set postulate? I am working on this with physicists at the University of Bristol¹⁹. The key idea is that invariant set theory is based on finite mathematics (the “ p ” of “ p -adic” is a finite whole number). This has a number of consequences, one of which is that the number of ways you can vary a quantum system comprising n elemental systems (called qubits) will not be as large as predicted by quantum mechanics as n approaches the logarithm of p ²⁰. The quantum computing industry has indeed found it difficult to entangle a large number of qubits arbitrarily because qubits can easily interact with the noisy environment. Here I am suggesting that this obstruction may be inevitable once a large enough number of qubits are mutually entangled. It may be that this idea is linked to the notion of gravitational decoherence, discussed further on.

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DOES WHAT WE KNOW about the large-scale structure of the universe support the cosmological invariant set postulate (conjectures A and B)?

The universe evolved from a state of extremely high density and temperature known as the Big Bang. We know this partly for theoretical reasons. Not least, Roger Penrose won the Nobel Prize for work (with Stephen Hawking) showing that an initial cosmological space-time singularity, a point in space-time where the laws of general relativity actually break down, is a robust prediction of general relativity theory. We also know from the observational work of American astronomer Edwin Hubble in the 1930s that the universe is expanding, and therefore began from a very

condensed state. The cosmological microwave background, discovered in 1965 by American radio astronomers Arno Penzias and Robert Wilson, is a relic of the condensed Big Bang.

What is the fate of the universe? Will it finally collapse into a “Big Crunch”? If it does, can we then envisage that the current period of evolution of the universe from the Big Bang is just one epoch of a multi-epoch universe, where time has no beginning and no end, as in [Fig. 45](#)? The cosmological invariant set postulate requires the universe to evolve through these multiple epochs. Neighbouring trajectories on the invariant set correspond to either future or past epochs of the universe, evolving similarly but not identically to ours.

But there is a problem with such a scenario. It is conventionally believed from observations of distant supernovae that the universe is not only expanding but accelerating. Such acceleration is linked to the notion of dark energy. If the universe continues to accelerate, then it will never recollapse to a Big Crunch and hence to a multi-epoch universe. In such a situation we are not on a fractal invariant set but on some transient adjustment to a profoundly uninteresting invariant set devoid of stars and life.

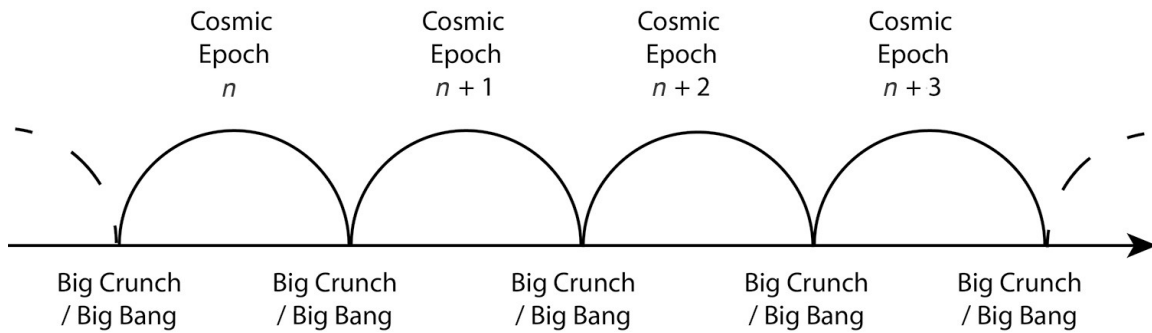


FIG. 45. A multi-epoch universe evolving from repeated Big Bangs and Big Crunches. A key question one could ask is this: What sort of geometry do these repeated epochs trace out in cosmological state space? I argue that such a universe could evolve on a fractal invariant set in state space, perhaps like the Lorenz attractor but in a much-higher-dimensional state space. If this is correct, it may explain some of the deepest mysteries of quantum mechanics, such as quantum nonlocality. This would contradict the notion of methodological reductionism, that to understand the mysteries of the universe, we necessarily need to probe smaller and smaller scales. Here, quantum nonlocality is explained by the structure of the universe on the largest conceivable scales.

Fortunately, there are three scenarios where this possibility does not happen. The first is that this dark energy will eventually reverse, aiding the ultimate collapse of this epoch of the universe²¹. The second is linked to a proposal by Roger Penrose that all the mass in the universe will end up in the form of massless particles like photons. Such massless particles have no concept of spatial or temporal scale. As such, the asymptotic end state of the universe can be seamlessly patched onto a condensed Big-Bang state of a new epoch of the universe, through a mathematical procedure known as conformal scaling²².

In view of the discussions in this book, I find the third possibility the most interesting. The notion that the universe is accelerating depends on an assumption—the cosmological principle—that the universe is homogeneous on the very largest scales. In this book we have discussed nonlinear systems (from turbulence, to networks of individuals, to economic systems, to fractals) which exhibit

power-law behaviour. Such power-law behaviour is indicative of structures that look the same on different scales. The ubiquity of such power-law behaviour might indeed extend to the universe on the largest scales. Recent discoveries of unexpectedly large structures in the universe suggest that this could be the case²³. If this is so, then perhaps the expansion of the universe is not accelerating—it's just an illusion²⁴. Then it certainly becomes possible that the universe will ultimately collapse to a “Big Crunch” singularity.

I recall as a child watching British cosmologist Fred Hoyle on TV arguing for his steady state theory of cosmology. In this theory, there is no Big Bang. Hoyle argued that the notion of a beginning of time was a rather inelegant concept. It somehow suggested the need for a god to provide just the “right” initial conditions from which the universe could start. Indeed, Hoyle coined the phrase “Big Bang” as a pejorative way to describe what he saw as an artificial beginning of the universe. However, Penzias and Wilson's observations rather killed Hoyle's theory. On the other hand, if the Big Bang merely represents the beginning of our current epoch, then Hoyle's idea that there is no beginning to time might be right after all.

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LET'S DISCUSS SOME CONSEQUENCES of the cosmological invariant set postulate.

For example, it can shed light on one deep problem in contemporary physics: the origin of the second law of thermodynamics. When we drop a glass onto the floor, it smashes into small pieces. We never see the time reverse of this. This is attributed to the second law of

thermodynamics: that entropy, a measure of disorder in the universe, increases with time. But for entropy to increase with time, it must have been small at the time of the Big Bang. Why should this be?

We can see the phenomenon of increasing entropy in the dynamics of simple chaotic systems. [Fig. 10](#) showed how a simple ring of points in state space evolves into a more complex irregular banana/boomerang shape. If we continued the simulation, we would see the banana shape deform further and become less and less like the original small ring. Eventually the distorted ring would spread across the whole fractal attractor. We can define a quantity which measures the distortion of the ring and, like the entropy of a thermodynamic system, it would typically increase with time, reaching a maximum when the ring of points has spread right across the entire attractor.

This characterises well the increase in entropy when we drop a glass and it smashes into a hundred small pieces. Think of the initial ring in [Fig. 10](#) as describing an ensemble of initial conditions in each of which we have a glass. For all practical purposes, the glass looks the same in all initial members of the ensemble, though the detailed arrangement of atoms in each of the ensemble of glasses is different. Because of these different arrangements of atoms, each glass will shatter in completely different ways and the shards of glass will spread out to form completely different patterns. The ensemble will have spread out in a chaotic fashion. Chaotic and thermodynamic systems have much in common.

However, and this is the key point, the top left of [Fig. 10](#) shows that there are regions of state space where there is very little change in the shape of the initial ring. Indeed, in these regions, the initial ring can actually get smaller. These regions denote stable parts of the attractor—where, in this analogy, dropping a glass wouldn't lead to noticeably

different configurations of smashed glass because the glass wouldn't shatter.

Now, if the universe is evolving on a cosmological attractor, then we would similarly imagine that there would be specific regions of state space where an initial ring (of universes) would get closer to each other as time progressed. In such regions, the entropy of the universe would be decreasing. Put another way, the heterogeneous geometry of the invariant set leads naturally to a picture where in most regions of state space entropy is increasing, but in some regions, entropy is actually decreasing.

For this to be relevant to understanding the low entropy of the universe at the time of the Big Bang, then we would expect regions of state space where entropy is decreasing to correspond to periods when the universe is gravitationally collapsing towards a Big Crunch. If this is so, then the entropy at the interface between one Big Crunch and the next Big Bang would be at a minimum²⁵.

What are the implications of this? Could it be that what we call "gravity" is actually a phenomenon associated with the heterogeneous geometry of the cosmological invariant set? In particular, is "gravity" a manifestation of the tendency of trajectories to cluster together on the invariant set, as shown in [Fig. 44d](#)? If this is so, then gravity will be associated with the quantum measurement process. This has already been suggested independently by Roger Penrose and Lajos Diósi²⁶. Trying to establish whether gravity really does play a role in quantum measurement is in my view one of the most important unresolved experiments in contemporary physics. Because such clustering is preceded by a "decoherent" unravelling of the trajectories on the invariant set ([Fig. 44c](#)), invariant set theory supports the idea that gravity and decoherence are inherently linked.

Whether thinking about quantum decoherence or the entropy of the universe, we are being led to the idea that gravity is of a completely different character to the other forces of nature. In some sense, we know this already—in general relativity theory, gravity is not a force at all, but a manifestation of the curvature of space-time. If this is correct, it suggests it is misguided to try to unify gravity with the other forces of nature using the principles of quantum field theory. If the cosmological invariant set postulate is correct, quantum field theory is not the path to unification. Indeed, going further, I would argue that trying to “quantise gravity”, which is the way most physicists think about the problem of synthesising our theories of quantum and gravitational physics, is rather putting the cart before the horse. Instead of a quantum theory of gravity, we might make more progress trying instead to develop a gravitational theory of the quantum²⁷.

If such an idea is correct, then there is no such particle as a graviton—the particle associated with quantum field theories of gravity, like string theory. That’s a prediction, albeit not one that will be directly testable for some time yet.

We noted before that quantum field theory implies, when coupled to Einstein’s theory of general relativity, an acceleration of the universe that is 120 orders of magnitude bigger than what is observed. In invariant set theory, there is no reason to suppose the effects of quantum physics would have a direct impact on the large-scale acceleration of the universe. In this sense, invariant set theory appears much closer to what is observed than is quantum field theory.

And what of dark matter? In Einstein’s theory of general relativity, the curvature of space-time is determined by the distribution of matter in space-time. It is possible to modify the field equations of general relativity so that the

curvature of our space-time is determined not only by the distribution of matter in our space-time but also by the distribution of matter in space-times associated with the neighbouring trajectories on the cosmological invariant set. If this idea is correct, then there may be no particle of dark matter; instead, dark matter would simply be the effect on space-time curvature from matter in neighbouring space-times on the invariant set.

And it is plausibly the case that so-called space-time singularities which form in black holes and at Big Bangs in standard classical general relativity would not occur if the curvature of space-time is influenced by matter in neighbouring trajectories on the invariant set. Solving the singularity problem has long been seen as a sine qua non for any putative unified theory of quantum and gravitational physics.

One of the key philosophies that has underpinned the development of theoretical physics over the twentieth century and into the twenty-first is that “smaller is more fundamental”—sometimes called “methodological reductionism”. Particle physicists want to build a bigger particle collider under Lake Geneva so that particles can be accelerated to higher speeds, and therefore to greater energies, to allow the physics of smaller scales to be probed. It’s assumed that we necessarily need to probe these smaller scales to improve our understanding of nature. My own belief is that methodological reductionism is a flawed philosophy, and that fundamental physics is going 180° in the wrong direction because of that. See the caption for [Fig. 45](#).

In conclusion, we do not necessarily have to accept either indeterminism or action-at-a-distance in describing the world of quantum physics. Using the geometry of chaos, Einstein’s picture of an ensemble of deterministic worlds may be right after all. If this is so, we conclude that we do live in a world in which elementary particles, and

indeed the notion of reality, are certain and definite. Perhaps this picture of the world will provide a better basis for synthesising quantum and gravitational physics. On the other hand, before these ideas will be accepted, we need clear-cut experiments in favour of invariant set theory over quantum mechanics. Until then, one should of course be quite doubtful about the ideas I have discussed in this chapter.

12

Our Noisy Brains

IN [CHAPTER 11](#), I ARGUED THAT ULTIMATELY THE LAWS OF PHYSICS may well be deterministic, and that these laws describe the geometry of the attractor/invariant set of the highest-order system imaginable—the universe as a whole. In [Chapter 3](#), we noted that if we want to model a high-order deterministic system with a low-order deterministic system, we should truncate the high-order system and add noise to represent the parts of the system we cannot model explicitly. We also observed that in many nonlinear situations, noise is a useful resource and not the nuisance we often imagine it to be; noise can amplify weak signals, for example. In this chapter I propose that, consistent with this, the human brain makes constructive use of noise in trying to create a low-order model of the high-order world around us. I will argue that such noise is a key element in making us the creative, innovative beings we are. However, unlike in the earlier examples I discussed, noise is not made by mathematical pseudorandom-number generators; it is made in hardware. The presence of such noise appears to be a consequence of the brain's extraordinary energy efficiency—that it can run on just 20W of power.

THE GREEK PHILOSOPHER EUCLID is commonly attributed with giving the first proof that there are infinitely many prime numbers. A prime number is a whole number greater than 1 that is divisible only by itself and 1; numbers such as 2, 3, 5, 7, 11 and 13 are prime. Euclid's proof goes as follows:

1. Suppose there are only finitely many prime numbers.
2. Hence there is a largest prime number. Call it P .
3. Form the product $2 \times 3 \times 5 \times 7 \times 11 \times 13 \times \dots \times P$ of all the prime numbers, and add 1 to the result. Call this number Q .
4. By construction, Q is not divisible by any of the prime numbers up to P , since such division leaves a remainder of 1.
5. But all numbers, including Q must be divisible by some prime number (possibly Q itself).
6. Hence there must exist a prime greater than P .
7. Hence the supposition in step 1 must be false.
8. Hence there are infinitely many prime numbers.

For example, suppose we thought that 5 is the largest prime number. Then Euclid says—think about $Q = 2 \times 3 \times 5 + 1 = 31$. Now, 31 is not divisible by 2 or 3 or 5—there is always a remainder of 1. Hence there must be a larger prime number than 5. In this case, 31 is itself a prime number.

What is the creative step in this proof? Well, step 1 is based on a commonplace technique in mathematics called *reductio ad absurdum* assume the opposite to what you want to prove and show that this leads to a contradiction.

Reductio ad absurdum was, at some stage in the history of mathematics, a creative step. It was known to be one of Euclid's favourite tricks. Hence, we can imagine that he simply pulled this out of his bag of useful tricks when he started to construct his proof. As such, it is not the key creative step for this particular proof. Step 2 is manifestly not a creative step—it follows immediately from step 1. Rather, the creative step is step 3. Once Euclid had the idea of forming the number Q steps 4–8 followed in a fairly deductive, mechanical way.

Let's try to create a computer that spontaneously creates mathematical theorems. We'll load up two "tricks" taken from Euclid's proof. The first is reductio ad absurdum assume the opposite of what you want to prove and show that this leads to a contradiction. The second trick will be step 3 of Euclid's proof: if you have some numbers, multiply them together and add 1.

So, have we made our computer creative? Well, let's see if it can come up with another classical mathematical theorem from ancient Greece: Pythagoras's proof that the square root of 2 is an irrational number, i.e., cannot be written as a fraction a/b (such as $983/765$) where a and b are whole numbers.

The computer can reasonably start by first using the reductio ad absurdum trick: assume the square root can be written as a fraction a/b . Perhaps the computer knows the rules of arithmetic and assumes that a and b have no common factors.

It now needs to find a contradiction. It has one more trick up its sleeve—it's now time to apply it: multiply a and b together and add 1. But that's a complete dead end. In fact, to find Pythagoras's proof, the computer needs a new trick (which is to square the numbers a and b)¹. But that trick is not in its library of tricks.

The computer hasn't exactly got nowhere—using the *reductio ad absurdum* trick is a good first start. But it's as if it has climbed a minor peak on the route to the main goal, and then has simply got stuck not knowing where to go from there.

We could add the new trick into the computer's memory, so that it now has three tricks to play with. With a little bit of extra help, that might enable the computer to prove that the square root of 2 is irrational. But are these tricks going to be sufficient to prove some other theorem, like the Navier-Stokes Millennium Prize problem mentioned in [Chapter 3](#)?

Of course, lots more tricks will be needed up one's mathematical sleeves to have any chance of cracking one of these big problems. However, the very fact that these are unsolved problems raises a rather interesting thought. Perhaps we will always need new tricks to prove important new theorems: having some fixed memory bank of tricks is never going to be good enough. Perhaps there is simply no end to the tricks needed to prove all mathematical theorems. This is (effectively) what Gödel's incompleteness theorem, which we met in [Chapter 2](#), suggests. If this is so, computers may have a hard job coming up with really interesting new mathematical theorems.

How did Euclid get this idea of forming the number ϕ ? He didn't tell us. However, other mathematicians have told us how their eureka moments have arisen, and something very interesting emerges from them.

Here is what Henri Poincaré (whose work was discussed in [Chapter 1](#)) said about the moment when he had a creative flash of insight²:

Just at this time, I left Caen, where I was living, to go on a geologic excursion under the auspices of the School of Mines. The incidents of the travel made me

forget my mathematical work. Having reached Coutances, we entered an omnibus to go some place or other. At the moment when I put my foot on the step, the idea came to me, without anything in my former thoughts seeming to have paved the way for it, that the transformations I had used to define the Fuchsian functions were identical with those of non-Euclidian geometry. I did not verify the idea; I should not have had time, as, upon taking my seat in the omnibus, I went on with a conversation already commenced, but I felt a perfect certainty. On my return to Caen, for convenience sake, I verified the result at my leisure.

Here is how the critical idea which led to Roger Penrose's Nobel Prize-winning work on space-time singularities came to him³. Penrose was walking to work with a colleague and they were deep in conversation. They crossed the road and stopped talking to watch out for traffic. When they got to the other side, they started talking again. When the colleague had left, Penrose had a strange feeling of elation, but couldn't work out why. He went through all the things he had done that day, including what he had for breakfast. Finally, he thought about the moment when he was crossing the road and realised that in that moment he had had the crucial idea about how to characterise extreme gravitational collapse in a generic way. This idea led to the 1965 paper that was the basis of his Nobel Prize.

Let me quote Andrew Wiles, perhaps the most famous mathematician alive today, whose work was mentioned in [Chapter 2](#). Here is what Wiles says⁴ about creative moments:

In particular, when you reach a real impasse, when there's a real problem that you want to overcome, then the routine mathematical thinking is of no use to you. Leading up to that kind of new idea there has to be a long period of tremendous focus on the problem without any distraction. You have to really think about nothing but that problem—just concentrate on it. Then you stop. Afterwards there seems to be a kind of period of relaxation during which the subconscious appears to take over and it's during that time that some new insight comes.

And it's not just creativity in mathematics and physics research. Comedian John Cleese, part of the famous Monty Python team, makes much the same point about artistic creativity—it occurs when you relax and let your unconscious mind wander⁵.

In all these cases, the key creative insight comes during a moment of relaxation. Now, few of us have been fortunate enough to have such moments of creativity as those mentioned earlier. However, perhaps an example we can all relate to (particularly those of us of advancing years) is when we have completely forgotten some person's name. We can see their face in our heads, or perhaps even in reality. Perhaps we know them extremely well. Simply trying harder and harder to remember the name doesn't work. Often the best thing to do is stop trying and just let your mind relax. Then, out of nowhere, the name will suddenly and mysteriously pop out of the ether.

But what's going on here from a physical perspective? If we want to create genuinely creative AI systems, should we tell the AI program to stop working flat out on some problem and just idle for a few minutes, hoping for inspiration?

That seems a silly idea, for the simple reason that digital computers are ultimately just on-off switches. It's possible to express arithmetic operations such as $1 + 1$ or 24×37 in terms of switches; this is called Boolean logic, named after Victorian mathematician George Boole. And modern computers perform these switching processes amazingly quickly. However, "relaxing" and doing the switching more slowly, for example, is clearly not itself going to make the computer creative.

How, by contrast, do our brains add $1 + 1$ and make 2? Perhaps the answer is so ingrained that we just know that $1 + 1 = 2$. So, let's try something harder: What's 24×37 ? How would I go about answering that?

At a very young, preschool age, I would go for walks in our local woods with my father. We would take a path up to a high point and admire the vista across the neighbouring countryside. We would then walk down via a different route, one which was a little steeper and had steps cut into the path. There were about twenty steps in all. As we went down the steps, we would both count them out loud: one, two, three, four... I learned to count by repeating the numbers as we walked down the steps week after week until they were embedded in my head.

And then, at my elementary school, the "times tables" were drummed into my head. After a while I could effortlessly recall that "four sevens are twenty-eight". So, if I was later asked by my teacher what 4×7 is (on pain of having to write out the 7 times table 100 times), I wouldn't set up a biological circuit of Boolean switches in my brain; I would simply access my memory and quickly reply "twenty-eight". The external sound "four times seven?" automatically triggered the rather mindless response "twenty-eight!" in my head.

Later in school I learned the mechanics of long multiplication. As an adult, it's something I hardly ever use, and although the technique is in my memory somewhere,

it's not as readily accessible as it was when I was seven or eight years old.

And so, if I were forced to multiply 24 by 37, I wouldn't use Boolean logic at all. I would simply try as best I could to retrieve the relevant patterns from memory: the patterns that describe the algorithm for doing long multiplication, then the pattern associated with the "times tables". Finally, I would need some "scratchpad" memory (of which I definitely don't have enough these days) for storing the intermediate steps. All in all, it's a totally different process to that of a digital computer.

The effort of retrieving data from memory takes energy. If a friend insisted I do this multiplication calculation when walking along the road with him (don't ask me why), then I would have to stop talking, stop walking, close my eyes, possibly put my hands over my ears and do nothing else but focus on the problem at hand. Presumably, I do this in order to channel all the available energy into performing accurately the required data transfers from memory. To do this, I shut down all other tasks that may be consuming energy and focus all available energy to the task in hand.

Of course, this is an unusual state to be in. Before my friend asked me to multiply these numbers together, I was happily multitasking: walking down the road (unthinkingly putting one leg in front of the other), chatting about the latest news, looking up and admiring the clouds in the sky and the leaves on the trees, listening to the birds chirping and wondering what I might have for lunch. In this state, the available energy being fed to my brain was being used for a multitude of tasks.

In his renowned book *Thinking, Fast and Slow*, the psychologist Daniel Kahneman has famously described the brain in a binary way rather similar to this. Most of the time (when walking, chatting, looking around) it operates in a mode he calls "System 1". System 1 denotes a rather

fast, automatic, effortless mode of operation. By contrast, with eyes closed, focussing on one specific task, the brain is in what Kahneman calls System 2, a much slower, more deliberate mode of operation.

This notion of a binary mode of operation had in fact been described some years earlier by cognitive scientist Guy Claxton in his book *Hare Brain, Tortoise Mind*⁷. Here “tortoise mind” describes a more playful, leisureful and dreamy state, somewhat akin to Kahneman’s System 1. By contrast, “hare brain” describes the type of thinking that requires reason, logic and deliberate thinking, as in Kahneman’s System 2. The key point of Claxton’s book, one that seems completely consistent with the experiences described earlier, is that the leisurely tortoise mind, for all its apparent aimlessness, is just as intelligent as the more logical hare brain.

Here I want to think about energy as a way of describing this bimodal operation of the brain.

Every second, the brain consumes about 20 joules of energy. That is, it uses the same amount of power, 20W, as an electric light bulb. This is tiny compared with the power used by a supercomputer that runs a weather forecast model—about 20MW, six orders of magnitude more. It seems from studies of blood flow into the brain that there isn’t much variation in the total amount of power consumed whether we are in System 1 or System 2. However, the fact that we can multitask when in System 1 suggests that, in this mode, power is spread between many active tasks, so that power per active task is especially low. By contrast, when operating in System 2 mode, all the available power is perhaps focussed on a single task, making the power per active task that much higher. For this reason, I will refer to the two modes as the low-power mode and the power-intensive mode.

Now, there are some 80 billion neurons in the brain. Information propagates along the slender axons of a neuron (see [Fig. 46](#)) by means of spiky electric signals. However, the axons are so long that these electric signals would have completely dissipated long before they got to the end of an axon, were it not for “protein transistors” which boost the spikes at regular intervals along the axon.

These protein transistors comprise small channels in the membrane, which otherwise act as an insulating sheath around the axon. Electrically charged ions—essentially atoms such as sodium or potassium with the electron in their outer shell stripped off—flow through these channels in response to the voltage fluctuations from the signal propagating along the axon. By being correlated with the signal propagating along the axon, the flow of ions can amplify the signal.

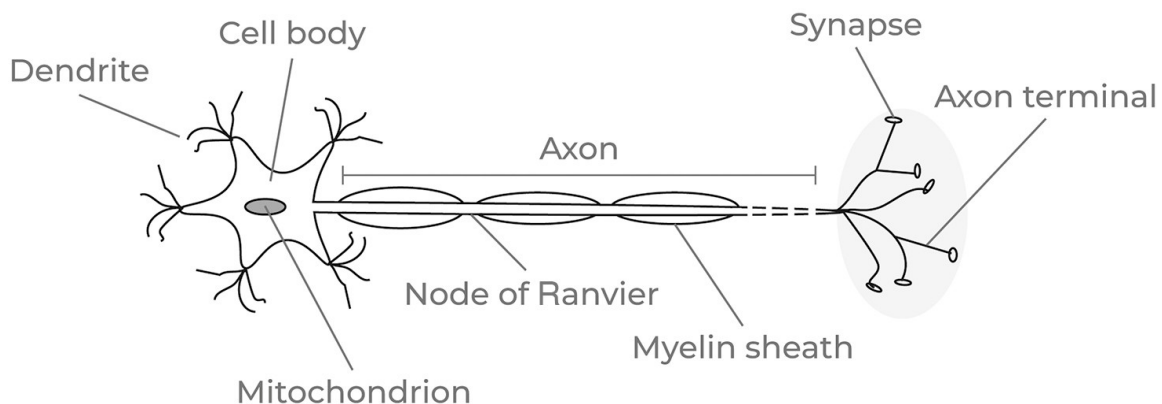


FIG. 46. The different parts of a neuron. Electric “spike” signals are carried along the axon. The signals are amplified by “protein transistors” associated with flows of ions across voltage-gated channels known as “nodes of Ranvier”.

As I mentioned in [Chapter 3](#), if we start to reduce the voltage across regular silicon transistors, they start acting unreliably and the switching process starts to become susceptible to thermal noise. Could it be that in the low-power mode, there is insufficient energy to ensure that the protein transistors operate completely deterministically?

The answer seems to be yes. There is evidence that slender axons with a diameter less than about 1 micron (10^{-3}mm) are indeed susceptible to noise⁸. Now, the median size of axons in the human brain is about 0.5 microns⁹. Hence, particularly in situations when the available 20W of power is being spread around the 80 billion neurons in the brain—i.e., in the low-power mode when the brain is multitasking—the operation of any one of the active neurons will be susceptible to noise.

There are advantages and disadvantages to such slender neurons. A disadvantage is that an electrical signal travels more slowly along an axon as the axon diameter decreases. If it is vital to have a super-fast reaction time to evade predators, then a decrease in axon diameter may be an evolutionary dead end. On the other hand, if one can take advantage of social groupings to fend off predators, then the disadvantage of slower reaction times may be more than compensated by the extra processing power arising from being able to cram more neurons into a given space.

If fast reaction time is not vital, could a susceptibility to noise arising from neuron miniaturisation have an evolutionary advantage? Let's look at some of the processes discussed in [Chapter 3](#), in which noise was found to be beneficial, and see if these might apply to the way the brain works.

For example, consider the process of stochastic rounding—using noise to truncate a signal comprising many bits. With reference to [Fig. 17](#), if stochastic rounding was being performed on the raw visual data that was being received by our eyes, and the truncated data was sent along our neurons to the part of the brain responsible for our cognitive perceptions, then perhaps we would be able to perceive the full grey shading, even though the brain was receiving only the 1-bit black-and-white bits.

The simulated annealing algorithm appears to describe well how we humans go about finding a solution to a problem. We start by applying some well-known tricks (like *reductio ad absurdum*) that have helped us in the past. This gets us some way to the solution, but then we become completely stuck. We rack our brains for new ideas that will help us out of the rut. If the ideas take us forward, we accept them. However, in the initial phase we are quite prepared to go back to step 0 if it will help us find a better route to the final solution. In these early stages, we are potentially prepared to accept any old idea, however crazy it might seem. But when we have made a lot of progress and feel that we are getting close to the final solution, we'll be much more discriminating with the ideas that come to us, accepting only the really promising ones for further analysis. Remarkably, the simulated annealing algorithm also describes quite well the process of evolution by natural selection.

However, in order to generate random new ideas, the brain must be susceptible to noise, and this appears possible only when the brain is in its low-power mode. This could explain why eureka moments occur when we are relaxing and not concentrating hard on a problem. It is in such a relaxation mode that the presence of noise can help us jump out of a cognitive roadblock and in this way advance our understanding.

By contrast, when the brain operates in the power-intensive mode, i.e., when other units are disabled to allow as much of the 20W of power to be focussed, then it appears there is enough energy available to the active neurons that they operate deterministically. This is when we can check whether the crazy ideas the low-power mode provides us are actually any good.

The theoretical physicist Michael Berry has dubbed creative ideas which seemingly arise out of nowhere "claritons"—mysterious particle-like units of creativity. It

would seem that the clariton is a product of the low-power mode. However, Berry notes that more often than not, the appearance of a clariton is, in the cold light of day, annihilated by an “anti-clariton”. It would appear that the anti-clariton is the product of the power-intensive mode of thinking, triggered when you apply some logical thinking to the idea that has popped out of the blue. It’s just as well we have the capability to check all the crazy ideas that come out of our head.

If these ideas are right, then the process we call creativity—what otherwise might be called inductive reasoning¹⁰—involves a synergy between the low-power and the power-intensive modes: in other words, a close interplay between stochasticity and determinism. Being susceptible to noise, the low-power mode is producing a stream of sometimes bizarre, sometimes useful ideas. The more deterministic power-intensive mode then filters these, rejecting the more bizarre ones and retaining the ones that stand up to scrutiny. Like the simulated annealing algorithm, the closer we come to the final solution, the more discriminating we are about the crazy ideas that we will pursue.

In *The Emperor’s New Mind*¹¹, Roger Penrose famously asked what it is that makes us capable of understanding Gödel’s theorem, the famous twentieth-century theorem described in [Chapter 2](#), that there are mathematically sound statements that assert their unprovability. He makes the point that if our brains worked by algorithm, we would not be able to understand the concept that there are mathematical truths that cannot be proved algorithmically. The fact that we do understand this could perhaps be explained if random noise, something not describable by algorithm, is a necessary element of cognition. Of course, you may complain that in [Chapter 11](#) I was arguing that even the most fundamental type of noise, quantum noise, is

not truly random, but is associated with some kind of chaotic determinism described by the cosmological invariant set postulate. However, as discussed in [Chapter 2](#), a fractal invariant set is itself algorithmically undecidable. Hence, if the brain is susceptible to this most fundamental type of noise, then it truly is not fully describable by algorithm.

I'm not the first to appeal to randomness in this way. In his influential paper "The Imitation Game"¹² Alan Turing himself commented, "It is wise to include a random element in a learning machine", and cited yet another simple example where noise can be used to find the solution to a problem efficiently.

In this context, it is interesting to note that if the miniaturisation of transistors continued, quantum tunnelling effects would make the transistors unreliable as switches. But if we take the view that noise is a necessary resource for creativity, then taking at least some of the transistors into this noisy quantum regime, and at the same time reducing the voltage across them, could be an important step in making a creative AI system capable, for example, of coming up with interesting new mathematical theorems.

Perhaps future AI devices will comprise two classes of chip: the conventional power-intensive chips, performing computations deterministically, where determinism is vital (you don't want a computer to send money to an incorrect bank account because it got the last digit of the account number wrong), and a new class of low-power noisy chips, performing computations stochastically. Data transfer between these chips and to memory would be performed at low numerical precision, using the stochastic rounding ideas we've discussed.

LET'S LOOK AT SOME possible practical consequences of regarding the brain as a noisy energy-efficient organ.

Every day we have to make decisions, some 35,000 of them according to some sources. Most are unimportant—whether to watch TV this evening or read a book—and barely require much effort. A few are important and require careful thought before we can decide what to do.

If we simply characterise decisions as important ones and unimportant ones, then probably most of us follow a simple strategy in making decisions: let the low-power mode make the many unimportant decisions in our life, and our analytic power-intensive mode make the fewer difficult decisions. Hence, we clearly don't weigh the pros and cons of putting the left foot forward, having just put the right foot forward, whilst walking down the street. By contrast, for the agonising decisions, we typically set out the pros and cons as carefully as we can. We think hard about each of these, and how much weight to give them. The power-intensive mode is hard at work for these difficult types of decisions.

This partition of decisions to the low-power and power-intensive modes makes good use of the available energy in the brain. It allows us to process those thousands of decisions we make each day rather efficiently. It means that the default mode the brain uses for making a decision is the low-power mode. We might call this strategy "lazy", but if we dealt with each decision using our power-intensive mode, we would hardly get to decision fifty before it was time to go to bed at the end of the day. Defaulting most decisions to the low-power mode is manifestly a rational strategy.

There are some consequences of this. One is that it makes us appear irrational, even though I believe we are not being irrational. Here is a now famous example of our seeming irrationality. A bat and ball cost \$1.10. The bat cost one dollar more than the ball. How much does the ball cost?

The obvious answer, the one that most people intuitively respond with, is 10 cents. However, this is not the correct answer. If the ball cost 10 cents, then the bat would cost \$1.10 and the total cost would be \$1.20. The correct answer is 5 cents.

According to Kahneman, in his book *Thinking, Fast and Slow*, more than 50 percent of students at Harvard, MIT and Princeton gave the incorrect answer. Kahneman found these results shocking—seemingly proving we are not the rational species we think we are. But this conclusion is wrong in my view. These are the top students in the US. They were selected by highly competitive exams and interviews. There must be something deeper going on.

Some commentators feel that these examples are too artificial for any useful conclusions to be drawn. However, I disagree. I believe the key to resolving this dilemma is to think about energy. The question of how much energy it is rational to assign to making a decision, or to answering a question, is surely going to be important when energy per second is limited to 20W.

Suppose I came up to you in the street and asked you the bat and ball question. How much energy would you be willing to expend answering it? If you were a student walking to the local examination hall to take some exam, you might well decide not to expend any energy at all. You don't want to exhaust your brain before you have even started the exam. In this circumstance the rational answer is no answer, and you politely respond that you are sorry but you don't have time to answer the question.

On the other hand, suppose it's your professor asking the question during one of your classes: then you might decide that you should expend a little energy answering it. However, if she has told you that the answers will be anonymised, and that there are no credits for answering correctly, then perhaps you might conclude that there is no good reason to expend a lot of energy answering the question, not least because you have other homework assignments to complete, for which you will get credits. Under these conditions, your answer to the bat and ball question may well be 10 cents. It's the rational answer, given the rational decision to expend only a small amount of energy answering the problem. That is to say, it is perfectly rational to give the wrong answer.

By contrast, if it mattered to you that the answer should be right—for instance, if it was a question on your exam—it would be rational to utilise all available energy to try to answer correctly. You analyse the question carefully in power-intensive mode. Now 5 cents is the rational answer.

The problem with a strategy based on maximising energy efficiency is that it can lead to a bias in the use of the low-power mode on decisions to problems that lie on a borderline between being important and unimportant. As a result, we end up bitterly regretting some decisions which we made too glibly, not giving them enough care and attention. I can think of examples in my own life.

So, what is the solution to this type of problem? I believe that it is to become aware that for many of us, our brain has a tendency to systematically downgrade the importance of decisions that lie near the interface between being important and unimportant by sending them to the low-power mode to sort out. If we are aware of this, then we can potentially become more sensitive to the small alarm bells that might ring, and not simply dismiss them without further thought. We can then start to engage the power-intensive mode a little more on the problem.

Essentially, it's about making a little more use of that all-important synergy between the low-power and the power-intensive modes—the synergy that makes us the intelligent, creative species we are.

On the other hand, there are some decisions that are obviously important, when we know we have to engage our power-intensive analytic mode. The problem here is a different one. It is that being overly analytic can paralyse us with indecision. Like Buridan's donkey, stuck midway between a bale of hay and a pail of water, we cannot figure out which way to turn. This was me when I was trying to decide whether to change fields at the end of my PhD studies.

Perhaps here we can learn from the eureka moments of those eminent scientists. The creative steps on the route to their breakthroughs came from a subtle interplay between the two modes. The power-intensive mode had taken the creative process only so far. The crucial eureka moment actually occurred in low-power mode, it seems, precisely because the brain was susceptible to noise.

Learning from the eureka moments suggests that we should certainly start by listing the pros and cons and thinking hard about them, maybe for several days, for these difficult and important decisions. Indeed, it makes sense to try to construct an ensemble of worlds where these different pros and cons are realised. One doesn't need to actually derive probabilities from such an ensemble. Merely constructing "storylines" portrayed in individual ensemble members is itself an important step. In fact, the idea of ensemble members as plausible "storylines" is becoming important in climate science, in situations where probabilities are difficult to compute¹³.

But then we should try to get on with the rest of our lives, as far as this is possible. Like with the eureka moments, the hope would be that in moments of relaxation,

our instinctive side will tell us what is the right decision. Again, the idea is to make constructive use of noise in low-power mode to provide new perspectives which our more analytic, deterministic power-intensive mode is failing to give us.

The order in which we engage the two cognitive systems is crucially important here. If we engage the low-power mode first and come to an instinctive preliminary conclusion, this could colour and distort the way in which we do the analytic thinking when engaging the power-intensive mode. This is where we are susceptible to “confirmation bias”: we simply find evidence that supports our gut instinct and disregard the rest. The power-intensive mode is no longer the dispassionate, disinterested analyser of information. Rather, we should engage the low-power mode after we have got as far as we can with the power-intensive mode. That’s the way the great eureka moments were made.

For example, if you start with a gut feeling that climate change is for woke tree huggers and that its importance as a problem facing humanity is overblown, then you will simply seek out naysayer websites and other sources of information that support this instinctive view. Conversely, if you come to the problem from the intuitive perspective that we have polluted the planet with our carbon emissions and are reaping the inevitable catastrophic consequences, then similarly, you are likely to read only alarmist websites about climate change. This is why I think that the IPCC assessment reports are probably the best disinterested source of information on climate science—though, unfortunately, they are not the easiest to read. The scientists I have worked with on IPCC are typically not politically motivated—they merely want to present their science as accurately and as accessibly as possible.

If there is a bottom line here, it is that in decision-making, whether with relatively unimportant decisions or

important decisions, we should be aware of the extraordinary capabilities we have when we make use of that interplay between the low-power stochastic and power-intensive deterministic modes of our brain—the modes that appear to be central in making us the creative, imaginative species that we are.



13

Free Will, Consciousness and God

IT SEEMS THE BRAIN IS AN ENERGY-EFFICIENT ORGAN. LET'S TAKE this argument a bit further. Apart from noise, what other physics might arise in the brain as a consequence of its energy efficiency?

Let's return to the issue of signal propagation along the long, slender axons of neurons. As mentioned, these signals have to be boosted by protein transistors as they propagate from one end of the axon to the other. These transistors are powered by the flow of ions in the neuronal membrane walls of the axons.

Scientists from the University of Salzburg¹ conclude that there isn't enough available energy to explain these ion flow rates if the physics of ion transfer is entirely classical. Instead, they have proposed a quantum process that will do the job using much smaller amounts of energy. The idea is to think of an ion not as a little particle but instead as an extended quantum wavefunction. The front of the wavefunction can then manipulate the electrical potential in the ion channel in such a way as to allow the rest of the wavefunction to propagate through efficiently and without hindrance. The timescale for this process to work has to be very small indeed and is entirely consistent with the very

fast quantum decoherence timescales that operate in the brain.

If this is correct, then we can't think of the brain as if it were a conventional classical computer. It must be some kind of noisy, classical/quantum-hybrid computer. What does this mean in practice? The hallmark of a quantum computer is that it can perform certain types of calculations much faster than a conventional classical computer. This is called quantum supremacy and has already been established for somewhat arcane and useless types of computation². In [Chapter 11](#), I described briefly how invariant set theory could explain this speed-up: if the laws of quantum physics describe the geometry of the invariant set, then a quantum computer, driven by these geometric laws, can access information in the neighbouring trajectories on the invariant set. By contrast, a classical computer can access information only on a single state-space trajectory, the one that corresponds to the world of classical physics.

As we humans don't have the prodigious computing capability of quantum computers, we are manifestly not pure quantum computers. However, perhaps we nevertheless have some residual capabilities associated with a weak cognitive perception of these neighbouring trajectories on the invariant set—what could be called a “sixth” sense. This might help explain some of the differences between us and conventional computers. This idea is speculative. But let's run with it for a while and see where it gets us. How might such a sixth sense manifest itself?

I am not the first to propose a role for quantum physics in the brain. Roger Penrose and Stuart Hameroff³ have proposed microtubules inside neurons as a place where quantum cognition takes place. Perhaps that's right, though I'm not enough of an expert to have a firm view on

this. In any case, I don't think that anything I discuss ahead is inconsistent with Penrose and Hameroff's basic proposal. Rather, I believe what I am proposing builds on these ideas.

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THE TOPIC OF FREE will is one that has vexed philosophers for millennia. The twelfth-century Persian poet Jalāl al-Dīn Rūmī wrote⁴, “There is a disputation that will continue until mankind is raised from the dead, between the necessitarians and the partisans of free will.” Here a necessitarian is a determinist, one, like me, who believes that the laws of physics are ultimately deterministic in form.

Over the centuries, debate on free will has focussed on whether determinism is compatible with free will. Compatibilists believe that if free will is defined in terms of an absence of constraints preventing you from doing as you want to do, then a belief in free will is indeed compatible with a belief in determinism. From this perspective, free will is not a contentious topic. For some decisions—what holiday to take, what car to drive, what house to live in—I am not a free agent; financial constraints limit my choice. And of course, I am not free to break the laws of physics. I am not free to flap my arms up and down and fly away from an unpleasant situation. On the other hand, I play golf on Sunday mornings of my own free will—there are no constraints that I am aware of that make me so do.

However, incompatibilists believe that free will cannot be explained in this way. For them, free will somehow denotes an ability “to have done otherwise”. Such an ability would appear to be incompatible with determinism since it seems to imply that the world could evolve in multiple ways

from a given state—one way in which I did what I did, and one or more other ways in which I did otherwise.

I don't like the idea of indeterminacy at a fundamental level, even though, as mentioned, noise is necessary for modelling complex systems at a practical level. But that doesn't make indeterminism wrong. In fact, one of the key arguments that incompatibilists bring to the table has, in the past, made me question my belief in determinism. The argument is that determinism undermines any claim that we are moral beings. The philosopher Peter van Inwagen has summarised the argument like this⁵:

"If determinism is true, then our acts are the consequences of the laws of nature and events in the remote past. But it is not up to us what went on before we were born; and neither is it up to us what the laws of nature are. Therefore, the consequences of these things (including our own acts) are not up to us."

Imagine a man, convicted of murder, about to be sentenced. When the judge asks the man if he has any final comments before he passes sentence, the convict says: "But I had no choice in whether to commit the crime. It was already determined that I would commit the crime at the time of the Big Bang". Is this enough for the judge to set him free?

You see the problem. Do we excuse Hitler for his crimes against humanity because it was predetermined in the cosmic initial conditions that he would be a mass murderer? Compatibilist philosopher Daniel Dennett argues⁶ that we simply do not exempt someone from blame, or indeed praise, because we think he could not do otherwise. Dennett cites Martin Luther, who initiated the Protestant Reformation declaring, "Here I stand, I can do no other." According to Dennett, Luther was taking full responsibility for acting of his own free will, whatever free will means.

Despite being a compatibilist like Dennett, I don't think the moral argument against determinism can be simply swept away as Dennett states. This is an issue about which there is a vast literature to which I can't do justice here. Instead, I am simply going to try to add something to the discussion, based on ideas from the geometry of chaos.

Indeed, I want to focus on a question which I don't see addressed much in discussions of free will, but which I believe is important. Why is it that people feel so strongly, so viscerally, that they could have done otherwise? I feel this every Sunday morning when I play golf. I drive home almost always disappointed with my efforts. I chastise myself with thoughts like "If only I hadn't rushed my swing on the sixth, I wouldn't have hit the ball out of bounds on the left." And yet I am an unashamed determinist, for whom the notion that I could have done otherwise—that I could have not rushed my swing on the sixth—makes no sense. It's as if my power-intensive mode, my careful analytic mode of thinking, says, "The idea that I could have done otherwise makes no sense." However, my low-power intuitive mode of thinking says, "Oh yes it does!" So, what is going on here? Computers, I suppose, do not have these feelings. Why do we?

This problem is not solved by indeterminism. If it were randomness in the laws of physics that made me do what I did, then that wouldn't explain why I feel so strongly that I could have done otherwise. It seems the problem rather transcends the issue of determinism vs. indeterminism.

Perhaps there is a simple explanation. We have memories of past occasions when, in similar circumstances, we did otherwise—when we didn't rush our golf swings, for example. Does the feeling that we could have done otherwise merely reflect such memories of previous, similar occasions? Whilst this is a possible explanation, I am somehow not convinced. More importantly, I don't think it would convince those who have a strong belief that they

could have done otherwise. As such, I don't think this explains the very visceral nature of a belief in free will.

Here is another possible way of looking at the issue. Suppose our cognition is influenced by the yin and yang of quantum physics, as described in [Chapter 11](#). In particular, suppose we have some cognition of neighbouring⁷ counterfactual worlds on the invariant set (the strands of rope in [Fig. 44a](#) that are close to the strand of rope corresponding to our current reality). These neighbouring trajectories correspond to worlds in which we did otherwise than what we actually did. With a weak perception of these neighbouring worlds, the brain would interpret them as implying an ability to have done otherwise. My suggestion is that it is our weak perception of these neighbouring worlds—related to the fact that energy efficiency demands that quantum physics does operate in the brain—gives us this extraordinarily visceral feeling that we could have done otherwise, even though we couldn't have actually done otherwise. A purely classical computer could never experience this, of course. A quantum computer experiences it much more strongly, I would contend, every time it performs a nonclassical computation.

There is a corollary to all this. It could be that our perception of these neighbouring worlds is sufficiently weak that we are unable to distinguish between neighbouring trajectories that lie on the invariant set and hypothetical trajectories that lie in the fractal gaps of the invariant set. If this is right, our brain extends the feeling “I could have done otherwise” to all counterfactual worlds, irrespective of whether they do or do not satisfy the invariant set postulate. As discussed in [Chapter 11](#), in quantum contexts, counterfactual reasoning can sometimes be flawed reasoning. However, perhaps our sixth sense isn't sufficiently well honed to distinguish neighbouring

trajectories on and off the invariant set. This is why, in my view, quantum physics can sometimes seem intuitively weird. Theories of counterfactual causation⁸—models of the world which permit more or less unlimited use of counterfactual reasoning—should therefore be treated with extreme caution. Even though they seem reasonable, they may not be, especially when being used to describe quantum physics.

Let's return to the more important conundrum, the difficulty of explaining how, under determinism, we can have a sense of moral responsibility. Recall the man convicted of murder, about to be sentenced. He has just said to the judge that he had no choice in whether or not to commit the crime, simply because it was already determined what he would do at the time of the Big Bang.

However, the judge (let's say) has read this book and replies to the accused:

"I'm afraid that what happened at the time of the Big Bang isn't relevant. Your actions are determined by the geometry of the cosmological invariant set."

The convict immediately responds, as the judge suspected he would:

"All right then. If the cosmological initial conditions didn't make me do it, then the invariant set made me do it. I don't care, the argument is still the same."

But now the judge can deliver the knock-out blow:

"But it's not the same! You are part of the invariant set. Your behaviour determines the geometry of the invariant set as much as the geometry of the invariant set determines your behaviour. That is to say, you do have some moral responsibility for what you did, determinism notwithstanding!"

The convict is dumbfounded. What on earth does the judge mean?

There is something very important here that needs unpacking. In classical physics, e.g., as discussed in [Chapters 1](#) through 3, we treat the initial conditions as completely separate from the dynamical laws of physics that evolve systems in time. Compared with other states of the universe, the Big Bang initial conditions are considered fundamental in determining, alongside the dynamical laws, how the universe will evolve. This is why Inwagen refers to conditions in the remote past.

However, in invariant set theory, the initial conditions and the dynamical laws are not independent of each other: they are both subservient to the geometry of the fractal invariant set. Because of this, the Big Bang initial conditions are no more fundamental than any other state on the invariant set. As such, the invariant set is timeless: no point on the invariant set, corresponding to a state of the universe at a particular time, is more fundamental than any other point.

An aphorism for Einstein's theory of general relativity was formulated by American physicist John Archibald Wheeler. He said of general relativity that "[the geometry of] space-time tells matter how to move, and matter tells space-time how to curve." I like to think of the invariant set postulate in a very similar way. That is, we can say that the geometry of the invariant set tells matter how to evolve in time, whilst matter tells the invariant set how to shape itself. But we are part of the matter that evolves in time on the invariant set, and how we evolve in time in part describes our behaviour. Hence, we can say in a self-referential way, the invariant set geometry determines how we behave, whilst our behaviour determines the geometry of the invariant set.

This is the duality to which the judge is referring. The argument "the invariant set made me do it," whilst true, is incomplete. The other part of the argument is that what we do determines the geometry of the invariant set. That may

sound like we have abandoned the notion of determinism. But in fact, all we are abandoning is the notion that our behaviour is that of a computational automaton. We can appeal to the discussion in [Chapter 2](#), that a fractal invariant set is algorithmically undecidable. As a result, any attempt to emulate our behaviour computationally must be inherently stochastic. In this sense we are not mindless automata—we do have moral responsibility. This is entirely consistent with Penrose’s assertion that conscious minds are not algorithmic.

This could be a plausible explanation of why determinism does not absolve us from moral responsibility.

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LIKE FREE WILL, THE meaning of consciousness continues to baffle us. And also like free will, the literature on this topic is vast and would take a separate book to review properly. In recent years, for example, a theory of consciousness based on what is called “integrated information theory” has developed⁹—the more so-called integrated information a system has, the more the system is said to be conscious. As in the case of free will, I do not seek to challenge these theories, but merely ask whether the geometry of chaos could add to them.

Here I want to focus on two aspects of this problem. The first is how the geometry of chaos might help define consciousness in an objective way. The second is to know how, even in principle, something like consciousness could emerge from a collection of otherwise inanimate elementary particles. For many people it seems implausible that the works of Mozart, Shakespeare and Einstein could emerge from such a collection of particles. Following ideas formulated by Plato and Descartes, maybe there really are

two different worlds: the physical world and the spiritual world. As a physicist, I would find it hard to accept such a notion since it would imply there is a core aspect of the world that cannot be studied by the scientific method. Of course, this might be the case, but I am not currently inclined to give up the idea that science can explain such matters. Nevertheless, I completely understand why the idea of emergence of conscious creative creatures from elementary particles seems completely implausible. As one scientific colleague put it, do the electrons and protons in the brain somehow know that Mozart was a superior composer to Mendelssohn (or the Beatles to the Rolling Stones, in my currency)?

However, perhaps this scepticism about the emergence of consciousness from elementary particles reflects not so much the domain of applicability of the scientific method as scepticism about methodological reductionism—that “smaller” is necessarily “more fundamental”, which is certainly part of the credo of contemporary science. In [Chapter 11](#), I argued that such methodological reductionism is leading us into problems with trying to understand the nature of elementary particles. To understand these particles, I argued, it is necessary to consider, as equally fundamental, the structure of the universe on the very largest scales of space and time. That is a key message of the invariant set postulate.

Could the same type of top-down argument also help explain the mystery of consciousness? Well, let’s try to be a little more concrete. Rather than ask, “What is consciousness?,” let me instead ask, “What does it mean to be conscious of something specific?”

If I look up from my laptop screen, the first thing I notice is a cactus plant in a white pot which my son bought my wife and me as a small present to mark our wedding anniversary (how nice of him!). The plant pot sits on the windowsill just behind my desk. Next to it is a small clock

also sitting on the windowsill, a short distance from the cactus.

When I am focussing on my laptop screen, trying to think what to write next, I am not at all conscious of either the cactus or the clock, even though they are in my field of vision. That is to say, even though photons from the sun are bouncing off the cactus and entering my brain, I am not aware of the cactus. In terms of the model my brain has of the world, when I am focussing on the laptop screen, the world comprises simply the laptop screen and an undifferentiated set of other objects. As far as this model is concerned, neither the cactus nor the clock has an existence independent of each other, or of the rest of the world.

On the other hand, if I stop writing and look straight ahead, perhaps hoping for some inspiration, then, and only then, I may become conscious of the cactus. My model of the world now includes a cactus, and perhaps a clock, and the rest of the external world.

What is going on? It seems to me that it is only when I am conscious of the cactus and the clock that I sense they have an existence independent of each other, and of the other objects in the universe. What does this mean? Well, as with so much else in this book, let's think about the cactus and the clock as two objects in "cactus-clock" state space. If we are attributing an existence to the cactus that is independent of the clock, it means that it is possible, in principle at least, that the position of the cactus can be altered independently of the clock's in "cactus-clock" state space. Now, when I (very occasionally) clean the windowsill in my study, I will move the cactus relative to the clock. Perhaps my sense that the cactus has an independent existence from the clock comes from a memory of moving the cactus relative to the clock during such window-cleaning moments.

However, perhaps there is another explanation. Perhaps my sixth sense, the one that gave me such a visceral feeling of free will, could also explain this notion of the cactus as an object with an independent existence. I argued that our visceral feeling of free will arises from a perception of trajectories on the invariant set that are close to the trajectory on which we evolve. On these nearby trajectories, I did indeed “do otherwise”. In a similar way, on the nearby trajectories on the invariant set, the position of the cactus may indeed vary relative to the clock. Perhaps the difference in relative position on the neighbouring trajectories is tiny, but nevertheless is enough for me to sense that the cactus and the clock are independent objects, both relative to each other and to the external world more generally. It’s the quantum yin and yang of real and counterfactual trajectories on the invariant set once again.

I am reminded of a feature of smartphone cameras that lets the camera take a tiny piece of movie—just a fraction of a second—before fixing on the still photo shot. When you see the photo, you also see the short movie sequence preceding it. Even though the movie lasts for only a fraction of a second, it makes the objects in the static photo seem much more lifelike. It’s a kind of augmented reality that the photo alone does not have.

In short, I am suggesting that to be conscious of an object is to be aware that the object has an existence independent of the rest of the world. I am speculating that this awareness is itself a consequence of two claims: that for reasons of energy efficiency quantum physics does play a role in cognition, and that the laws of quantum physics at their most fundamental describe the geometry of the cosmological invariant set. This awareness gives us a visceral augmented sense of reality that mere memories do not have.

With this in mind, let us return to the question of what distinguishes us from AI systems. We concluded in the previous chapter that noise in the brain is a necessary part of the answer. But is it sufficient? Is consciousness a necessary part of understanding, as Penrose argues it must be? I don't know for sure, but if it is, then developing creative AI will require a close synergy between quantum and classical computing, in addition to the synergy discussed in the previous chapter between power-intensive and low-power classical computing.

If this is so, then we have a long way to go before we create truly intelligent machines. "The singularity", the day when machines completely take over, is likely not a few years away, as some commentators believe; it is many decades, potentially centuries away. We will need our creative brains for quite some time yet.

« »

IN THE LAST PART of this final chapter, I want to discuss something that would be taboo in a scientific paper: to relate the ideas I have been discussing on uncertainty and the geometry of chaos with the concept of spirituality and religion.

The idea that there is more to us humans than a collection of inanimate atoms is surely an understandable response to the notion that the laws of physics are uncaring mathematical equations and the universe vast and impersonal. Perhaps if one could show how a human being is an emergent property of the Schrödinger equation, then one would have to admit that spirituality is merely a prop for coping with the challenges of life, with nothing deeper underpinning the Schrödinger equation. But science cannot show such a thing. Not by a long way.

From spirituality comes the concept of a caring god, a supreme and yet personal creator. Many of those who believe in such a god do so because they seek a purpose to life. The idea that each of us is merely a random thermodynamic fluctuation with a mean lifetime of some three score years and ten can make our lives seem rather pointless. Indeed, according to contemporary cosmological models with fixed dark energy, humanity is restricted to an infinitesimally short blip near the beginning of the universe, after which nothing happens at all. It's as if all the wonders of nature, the works of Einstein, Shakespeare, the Beatles and Beethoven, literally count for nothing—they are an irrelevance for the utterly dilute state of the universe that will dominate almost all of its existence. As a dispassionate scientist, perhaps I should simply shrug my shoulders and say that if that's the way it is, then that's the way it is. But it's such a depressing scenario. If we were truly certain that this is the state to which the universe will evolve, then why bother doing anything? Why bother having children—it's only prolonging the agony. Why bother to write this book, other than to pass the time of day? It's no wonder that many people seek a more meaningful reason for living.

The concept of a god, with whom we will spend eternity after we die, is somewhat comforting, though the idea of spending an infinite amount of time in perfect happiness does sound like it might get slightly boring after a while (there are only so many rounds of golf you can play in which you and your playing partners score twenty under par before the game gets tedious). But leaving these details aside, the concept of an afterlife is especially appealing as it provides an opportunity to meet loved ones who departed life before you do.

As scientists we can mock such views for their naivety or for their quaint anthropomorphic character (God as a wise old man with a white beard, for example). However, the

fact of the matter is that there are many profound uncertainties about the nature of physical reality. We have discussed some of them here. How can scientists dismiss views about religion and spirituality when their own theories are riddled with such deep uncertainties?

In science we insist on evidence and revise beliefs when evidence shifts. In religion one is asked to have faith without compelling evidence. From this point of view, science and religion seem incompatible. However, it's often not as straightforward as this. Scientists and nonscientists alike ponder the meaning of their lives. Are we just irrelevant random fluctuations in a universe destined for some bland heat death, or do our lives have some deeper purpose? Indeed, could we collectively play an important role in the evolution of the universe? Science cannot answer these questions and so people look elsewhere for answers. With this in mind, perhaps the geometry of chaos might also provide new insights into these questions and in so doing might help bridge the rift that exists between science, spirituality and religion.

For example, in the cosmological invariant set model, there is an afterlife. We and our loved ones will be coming back in future epochs of the universe on trajectories in state space close to the present one, but none precisely the same as the present one. We will have an opportunity not to make the same mistakes again. Equally, we will likely mess up some of the successes we have made of our lives this time round. Perhaps it will be better next time round, perhaps it will be worse. Who knows? That makes it all quite exciting and something to look forward to (arguably much more exciting than a future of perfect bliss and happiness). For each epoch there will be such an afterlife: there is no ending to the number of trajectories on the cosmological invariant set.

Indeed, as someone who rejected the Catholic religion of his childhood¹⁰, I tend in my more metaphysical moments to think of the cosmological invariant set as a logical alternative to the god of my early years. Each point in the set is a synthesis of everything that is, has been and will be, indeed of everything that might be, might have been and could be in the future. It is an omniscient structure: it knows which states of the universe are physically real and which are not, even though we, constrained by the notion of algorithmic decidability, cannot know this. In addition, the invariant set lies completely outside the constraints of time: as an entity it is literally timeless¹¹. Does the invariant set listen if we pray to it? Well, our hopes, aspirations and indeed prayers are part of the self-referential process discussed earlier in this chapter, the process that gave us moral responsibility, so in some sense the answer is yes.

As illustrated in [Fig. 6](#), the Lorenz attractor looks like a butterfly. This is coincidentally apposite, as by studying this attractor, Lorenz discovered the butterfly effect. If we could finally construct a picture of the global fractal geometry of the cosmological invariant set, something I have not been able to do, and indeed something that will not be possible to do precisely because such a set is algorithmically undecidable, what would it look like? Well, as a multidimensional object it would have many different guises depending on which dimensions in state space we decide to probe. To what would these different guises correspond?

Perhaps the multifaceted face of God?

It feels like an intriguing idea. But of course I have my doubts.

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BASIC
BOOKS

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TIM PALMER is a Royal Society research professor in the Department of Physics at the University of Oxford. Following a PhD in general relativity theory, Palmer spent much of his career working on the predictability and dynamics of weather and climate, developing probabilistic ensemble prediction systems across a range of weather and climate timescales. For this work he was elected a fellow of the Royal Society and an international member of the United States National Academy of Sciences. He was also a lead author for the IPCC climate assessment reports. In addition, Palmer researches the foundations of quantum physics and the role of noise and quantum processes in the brain. For this he was elected a member of the Foundational Questions Institute and the Association for Mathematical Consciousness Science.

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Notes

Epigraphs

- [1.](#) Feynman (1999).
- [2.](#) Gleick (1992).

Author's Note

[1.](#) In my first year of graduate research, I was supervised by one of Sciama's earlier students, John Miller, who tried to get me interested in solving the equations of general relativity by computer. At the time, ironic in the light of my later research in weather and climate, this didn't interest me at all—I wanted to do something more theoretical. For the remaining years of my graduate research, I was supervised directly by Sciama.

[2.](#) My main contribution to general relativity theory was the formulation of some of the first quasi-local descriptions of gravitational energy momentum in general space-times. See Palmer (1978a, 1978b).

- [3.](#) Isham et al. (1975).
- [4.](#) Paltridge (1978).
- [5.](#) Gleick (1987).
- [6.](#) Palmer (1995).
- [7.](#) Palem (2014).
- [8.](#) Palmer (2015).

Introduction

1. Who was this woman who called the BBC Weather Centre? It seems she had a sister living in France. The sister had been listening to the weather forecasts from the French Weather Service, Météo-France. A few days before the hurricane hit, Météo-France was warning of an exceptional gale in the English Channel region. Then, mysteriously, as the time got closer to the expected arrival of these gales, the warnings from Météo-France stopped. What was going on? It was the butterfly effect at work. For all its sophisticated use of supercomputers and satellite technology, weather forecasting in the 1980s had not embraced the concept of uncertainty. Like with the astrologers, there were no attempts to estimate error bars for the forecasts.

2. BBC News, January 6, 2017: Crash was economists' "Michael Fish" moment, says Andy Haldane. www.bbc.co.uk/news/uk-politics-38525924.

3. Kay and King (2020).

4. Oxford Learner's Dictionary, n.d.: doubt. www.oxfordlearnersdictionaries.com/definition/english/doubt_2.

Part I: The Science of Uncertainty

1. Stewart (1997).

Chapter 1: Chaos, Chaos Everywhere

1. To construct an ellipse, first choose two fixed points in the plane. These are called the foci. Then the ellipse is defined as the set of points where the sum of the distances between each point and the two foci is a fixed constant.

2. Wikipedia, accessed February 8, 2022: Laplace's demon. https://en.wikipedia.org/wiki/Laplace%27s_demon.

3. For example, an ellipse can be defined as the locus of all points which satisfy the formulas $x(t) = a \cos t$ and $y(t) = b \sin t$, where $\cos$ and $\sin$ are standard trigonometric functions.

4. Tardivel, S. (@simon_tardivel), January 10, 2020, 3:14 p.m.: Par exemple.... Twitter. https://twitter.com/simon_tardivel/status/1215728659010670594.

5. Tremaine, S., 2011: Is the solar system stable? Institute for Advanced Study. www.ias.edu/ideas/2011/tremaine-solar-system.

6. Indeed, in November 2021, NASA launched a mission to deliberately strike an asteroid with a spacecraft to test whether we can divert a small asteroid which might otherwise hit Earth.

7. A fluid can be a liquid or a gas.

8. In fact, a model which is a finite truncation of the Navier-Stokes equation will eventually repeat itself, unless it is perturbed with random noise.

9. Lorenz (1993).

10. “Angela”, September 2020: Chaotic waterwheel by Harvard Natural Sciences Demonstration. Go Science Girls. <https://gosciencegirls.com/chaotic-waterwheel-harvard-natural-sciences-lecture>.

11. Lorenz (1963).

12. May (1976).

13. $x_{n+1} = ax_n(1 - x_n)$. Here x_n denotes the population of the n th generation, “normalised” so that $x_n = 1$ denotes a situation where the population is as large as it can possibly be given the finite environmental resources, and $x_n = 0$ denotes a situation where the population has crashed to zero. The number a is a constant. When $a > 3.57$, the population varies chaotically and is impossible to predict for a long time ahead.

[14](#). There is a field called “quantum chaos” which tries to understand how classical chaos can emerge from quantum mechanics.

Chapter 2: The Geometry of Chaos

[1](#). However, if we measure positions and velocities relative to one of the bodies (e.g., the sun), that reduces the dimension of state space to 12. And it turns out that there are various additional constraints implied by Newton’s laws of motion which reduce the dimension of state space to just 7.

[2](#). Technically the state space of the Navier-Stokes equation is infinitely dimensional. However, this assumes that scales can become infinitesimally small. In practice the Navier-Stokes equation would break down due to quantum effects before this limit. However, even if we assume the equation applies to only macroscopic non-quantum scales, the dimension of the state space of weather could be as large as 10^{50} , certainly much, much larger than could ever be represented explicitly in a weather model.

[3](#). Actually, there is a subtlety here. If you use a computer to solve Lorenz’s equations, then in fact the trajectories will be closed 1-dimensional curves. This is because on a computer, the states X , Y and Z are represented by a finite number of binary digits, or bits, as they are known in computer science jargon. In scientific computation one typically represents variables like X , Y and Z with 64 bits. With this finite representation, the Lorenz attractor will be a rather complicated closed 1-dimensional curve. If instead you allow X , Y and Z to be represented with 128 bits, the 1-dimensional curve will be longer before it again joins up to be a closed 1-dimensional curve. Go to 256 bits and the closed curve will be longer still. What happens in the limit when we imagine a hypothetical

computer where X , Y and Z can be represented by an unlimited number of bits? Being based on calculus, the mathematical forms of Lorenz's equations do not themselves put any restrictions on the number of bits used to represent X , Y and Z .

4. Vallin (2013).

5. The so-called Hausdorff dimension is equal to $\ln(2)/\ln(3)$ where $\ln$ denotes the logarithm to the natural base of e . This is an irrational number.

6. Who got his inspiration from the work of L. F. Richardson, who showed that the coastline of Britain approximated well to a fractal—see Chapter 9.

7. Rössler (1976).

8. During World War II, Cambridge mathematicians Mary Cartwright and J. L. Littlewood (the distinguished number theorist) had found hints of chaos in what are called the van der Pol oscillator equations. In 1960, MIT mathematician Norman Levinson had contacted celebrated topologist Steve Smale about this, since it appeared to contradict a conjecture of Smale's about the behaviour of dynamical systems. By way of clarification, Smale realised that there may be dynamical systems exhibiting the type of folding geometry described in Fig. 9. Smale's fractal construction is called Smale's horseshoe. However, until Lorenz, Smale did not know of specific equations which, based on Newton's calculus, would generate this type of geometric structure.

9. Stewart (1997).

10. See Stewart (2000).

11. If $dX/dt = F(X)$ is a nonlinear dynamical system, so that F is at least quadratic in X , then the so-called Jacobian operator (dF/dX) in the linearised form of the dynamical system $d\delta X/dt = (dF/dX)\delta X$ depends at least linearly on X . In this way, the growth of small perturbations depends on the initial conditions.

12. The Liouville equation for a time-reversible Hamiltonian system can be written in the form $\partial \rho / \partial t = -[\rho, H]$ where $[\dots]$ is the so-called Poisson bracket and H the Hamiltonian. In this form the Liouville equation is formally very similar to the Schrödinger equation of quantum theory.

13. Epstein (1969).

14. Here a constant “forcing term”, C , has been added to the original Lorenz equations, giving

$$\begin{aligned} dX/dt &= \sigma(Y - X) + C, & dY/dt &= X\rho - Z - Y + C, & dZ/dt \\ &= XY - \beta Z \end{aligned}$$

15. In pure mathematics, it is usually a prime number—a whole number greater than 1 only divisible by itself and 1.

16. Technically, a 2-adic “integer”.

17. Technically, the 5-adic “integers”.

18. There are no integers X , Y and Z greater than 2, with $X^n + Y^n = Z^n$.

19. Quarmyne, Nyani, June 28, 2016: The oracle of arithmetic. [Quanta Magazine. www.quantamagazine.org/peter-scholze-and-the-future-of-arithmetic-geometry-20160628](http://www.quantamagazine.org/peter-scholze-and-the-future-of-arithmetic-geometry-20160628).

20. E.g., Penrose (1989).

21. Penrose (1989). Some have argued that since non-computability concerns properties of infinite sets, it should play no role in physics. I disagree with this. Non-computability can be thought of as a limiting case of finite systems that are called computationally irreducible. Computational irreducibility is a concept proposed by Wolfram (2002). The idea is that you can’t fully describe the behaviour of a computationally irreducible system using a computationally simpler model of that system. In such a situation, the computationally simpler system will always be an incomplete representation of the full system. Non-computability can be thought of as describing the

incompleteness of a system whose simpler representation is algorithmic but otherwise as complex as we like.

[22](#). Penrose (1989).

[23](#). Dube (1993).

[24](#). Barnsley (2006).

[25](#). However, Blum et al. (1998) were able to show directly that the problem of finding an algorithm which will halt if a chosen point in state space lies on a fractal attractor, like the Lorenz attractor, is algorithmically impossible.

Chapter 3: Noisy, Million-Dollar Butterflies

[1](#). Because of the rotation of the Earth, Kolmogorov's theory has to be modified somewhat to describe large-scale turbulence in the atmosphere.

[2](#). Inspired by a satirical poem by Jonathan Swift about the politics of his day.

[3](#). Woollings (2019).

[4](#). The Navier-Stokes equation is often referred to in the plural, i.e., as "equations". However, it's essentially the same equation for each of three different directions in physical space. For conceptual simplicity I will describe it in the singular.

[5](#). Lorenz (1969).

[6](#). Palmer et al. (2014).

[7](#). This means that the two horizontal sides have a size of 100 km. The vertical side of the volume would be much smaller than this, say 1 km.

[8](#). The Navier-Stokes equation is not the only equation you need to forecast the weather, but it is certainly central.

[9](#). The process of using observations to construct the initial state for a weather model is called "data assimilation". It is in fact a mathematically complex procedure.

[10.](#) In fact, a doubling of resolution in this way will typically take up to sixteen times more computing resources: eight times because a volume is three dimensional, and on top of this the finite time steps over which the solution is calculated have to be reduced to prevent the model from becoming computationally unstable.

[11.](#) Clay Mathematics Institute, n.d.: Millennium problems. www.claymath.org/millennium-problems.

[12.](#) One problem, the famous Riemann hypothesis, is common to both lists.

[13.](#) Palmer (2019).

[14.](#) Palmer (2015).

[15.](#) Paxton et al. (2022).

[16.](#) Since writing this, I have discovered that a company called Graphcore is producing a computer processor with just such a capability.

[17.](#) This effect was first described in Kwasniok (2014).

Chapter 4: Quantum Uncertainty: Reality Lost?

[1.](#) Indeed Richard Feynman said it was the only quantum mystery, though not all quantum foundations researchers agree with him on this (Catani et al., 2021).

[2.](#) Their square roots to be more precise.

[3.](#) Now discovered by the Laser Interferometer Gravitational-Wave Observatory (LIGO).

[4.](#) That is to say, many researchers think Einstein was referring to entanglement when complaining about “spooky” quantum action-at-a-distance. This example shows that this is not so.

[5.](#) Becker (2018).

[6.](#) Originally the experiment was done using silver atoms with a single electron in their outer shell. It is much more difficult to do the experiment with free electrons. However,

for conceptual purposes, let's imagine the particles are free electrons.

7. We could hypothesise, for example, that the intermediate SGx device somehow randomises the hidden variables of the particles.

8. Bell (1993).

9. The European Organization for Nuclear Research, home of the Large Hadron Collider.

10. Bell (1964).

11. Here is a proof of Bell's inequality, taken from Rae (1986).

Let $n(+, -, .)$ denote the number of times in a table, like Table 2, that there is a + symbol in the first row, a - symbol in the second row, and either symbol in the third row. Similarly, for example, let $n(-, ., +)$ denote the number of times there is a - symbol in the first row, either symbol in the second row, and a + symbol in the third row.

Generalising in an obvious way we have

$$A = n(+, +, .) = n(+, +, +) + n(+, +, -)$$

$$B = n(., -, +) = n(+, -, +) + n(-, -, +)$$

$$C = n(+, ., +) = n(+, +, +) + n(+, -, +)$$

Adding the first two equations together we get

$$A + B = n(+, +, +) + n(+, +, -) + n(+, -, +) + n(-, -, +)$$

Now the first and third terms of this equation, added together, give C. Hence,

$$A + B = C + n(+, +, -) + n(-, -, +)$$

Since the last two terms are never negative

$$A + B \geq C$$

QED.

12. In practice the experiment is performed with photons. The SG devices correspond to photon polarisers.

13. For example, if a source sends out a red ball in one direction and a blue ball in an opposite direction, and I see the red ball coming towards me, then I will know for sure

that the blue ball is going away from me, even if I don't see it.

[14](#). Clauser et al. (1969).

[15](#). There is now an overwhelming body of experimental results on the violation of Bell's inequality, starting with Clauser and Horne (1974) and Aspect et al. (1981).

Part II: Predicting Our Chaotic World

[1](#). Pyarelal, 1958: Mahatma Gandhi: The Last Phase Navajivan Publishing House.

[2](#). Krebs et al. (2021).

Chapter 5: The Two Roads to Monte Carlo

[1](#). Oceans also typically warm less at the surface than land because ocean eddies can mix heat from the sun down to the subsurface much more effectively than is possible on land.

[2](#). Credit also has to be given to Austrian meteorologist Felix Exner, who also developed prediction schemes using physically based models around this time.

[3](#). Peter Lynch once sent me an email in which he had set the history of meteorology to verse. He asked for further contributions. I sent him this about Lewis Fry Richardson:

Although LFR was quite bright
Some said that his forecasts were sh*te.
Till Lynch went and built a
Smart digital filter
Which made the predictions just right!

in the hope it might be included in his anthology. Alas, I don't think it made it because of an injudicious choice of word in line two. I include it here as a way of

acknowledging Peter's work in this area (Lynch and Huang [1992]).

4. It is a sad fact that the main use of the most powerful supercomputers today is nuclear weapons testing. I argue later that these top-of-the-range supercomputers should be dedicated to weather and climate prediction.

5. I can recall being at a scientific meeting where we were discussing a recently discovered phenomenon: when in some years the upwelling got stronger than normal rather than weaker. We named this phenomenon the Anti-El Niño event, until a colleague pointed out that if El Niño stood for “the Christ Child”, then Anti-El Niño would presumably stand for “the Antichrist Child”! After that, we switched to the term “La Niña”, and the name has stuck.

6. Horel and Wallace (1981).

7. On the other hand, with Europe so far from the tropical Pacific, these teleconnections can be quite noisy, akin to trying to hear someone speaking at the opposite end of a room at a party full of noisy people.

8. One of the achievements I am most proud of during this period was working with fluid dynamicist Michael McIntyre, from Cambridge, and discovering the largest breaking waves in the atmosphere. These occur in the northern hemisphere stratosphere and much less in the southern hemisphere stratosphere—due primarily to the different distribution of mountains in the two hemispheres—and this provides an explanation of why the ozone hole was discovered in the southern hemisphere and not the northern hemisphere.

9. Shukla (1981).

10. Murphy and Palmer (1986).

11. From *My Fair Lady*, a 1964 movie based on a 1956 stage musical, itself based on George Bernard Shaw's stage play *Pygmalion*

12. The development of this system was the work of many colleagues: see this book's acknowledgements.

[13.](#) There is always a chance of tossing 10 heads in a row with a coin whose heads and tails are equally likely. However, the more often this happens, the more likely it is that the coin is biased.

[14.](#) Cane et al. (1986).

[15.](#) Palmer et al. (2004).

Chapter 6: Climate Change: Catastrophe or Just Lukewarm?

[1.](#) My thanks to Peter Webster for this suggestion.

[2.](#) Oreskes and Conway (2011).

[3.](#) This is not actually the mechanism that makes air inside a greenhouse warm. There the sun shines through the glass and warms the ground, which heats the air. The glass is a physical barrier preventing the currents of warm air from escaping from the greenhouse. But the term has stuck, so we have to live with it.

[4.](#) Detecting photons in what would be called the broadband infrared part of the electromagnetic spectrum.

[5.](#) Atmospheric circulations, e.g., in the form of thunderstorm clouds, help adjust the surface temperature to the greenhouse gas warming aloft.

[6.](#) A typical column of air might contain about 2.5 mm (0.10 inches) of water vapour, but only 0.1 mm (0.004 inches) of cloud liquid or ice water.

[7.](#) Such a negative cloud feedback effect, because it cools the surface but warms the mid-atmosphere, could have an undesirable effect on the hydrological cycle, i.e., on rainfall around the world.

[8.](#) Though the relationship between the degree of subsidence and depth and extent of cloud amount is complex. See van der Dussen, J. J., S. R. de Roode and A. P. Siebesma, 2016: How large-scale subsidence affects stratocumulus transitions. *Atmospheric Chemistry and Physics* 16, 691–701. doi: 10.5194/acp-16-691-2016.

9. Since the strength of the Walker circulation also depends on sea surface temperature in the tropical Pacific, knowing how marine stratocumulus clouds react to increasing greenhouse gases also depends on how the tropical Pacific Ocean reacts to increasing greenhouse gases. Climate is a very interactive system.

10. Suki Manabe won the Nobel Physics prize, along with fellow climatologist Klaus Hasselmann, in 2021. It was a great moment for climate scientists like me to see our field recognised as part of mainstream physics.

11. Manabe and Wetherald (1975).

12. The warming values illustrated are those that arise in the models when the climate system has come into balance with the extra amount of carbon dioxide in the atmosphere.

13. See Zelinka, M., April 2, 2021: Effective climate sensitivity. Github.
https://github.com/mzelinka/cmip56_forcing_feedback_ecs/blob/master/ECS6_histogram_gavin1.pdf.

14. Roe and Baker (2007).

15. It is “fat” compared with a Gaussian or Bell-shaped probability distribution.

16. Although my curve suggests substantial non-zero probabilistic values greater than 6°C (10.8°F), such values are not supported by paleoclimatic data from the distant past and should likely be discounted.

17. Mandelbrot (2008).

18. See Roe and Baker (2007). There are other feedback effects which I have not discussed here. However, the most important is the cloud feedback.

19. And the related effect of aerosols.

20. These scenarios take other factors into account, like changes in land use and in aerosol emissions.

21. Hausfather et al. (2020).

22. Dockrill, P., November 5, 2015: Middle East may be uninhabitable this century due to deadly heat, study finds.

Science Alert. www.sciencealert.com/middle-east-may-be-uninhabitable-this-century-due-to-deadly-heat-study-finds.

23. The polewards shift of the jet streams is predicted by many contemporary climate models, but is one of the regional predictions about which we are not completely confident. I discuss the need for developing much- higher-resolution models later in the chapter.

24. It's not a priori obvious that increased carbon dioxide concentrations will make such heatwaves more or less likely. Although increased carbon dioxide concentrations will warm the air, they could conceivably make the circulation patterns that lead to heatwaves less likely. This is one reason why it is vital to develop reliable physics-based models to study such questions.

25. Allen, M., 2003: Liability for climate change. *Nature* 421, 891–892. www.nature.com/articles/421891a.

26. Currently we put more computing resources into simulating nuclear weapons than we do into simulating climate change.

27. A phenomenon called Rayleigh-Bénard convection.

28. Palmer and Stevens (2019).

29. Turing (1950).

30. Of course, we should coarse-grain the real-world data onto gridboxes the same size as the model before doing the comparison.

31. National Academies of Sciences, Engineering, and Medicine, 2021: Reflecting sunlight: Recommendations for solar geoengineering research and research governance. The National Academies Press. www.nap.edu/catalog/25762/reflecting-sunlight-recommendations-for-solar-geoengineering-research-and-research-governance.

32. Hossenfelder, S., and T. Palmer, December 9, 2021: An international institute will help us manage climate change. *Scientific American*

www.scientificamerican.com/article/an-international-institute-will-help-us-manage-climate-change.

[33](#). Deser, C., 2020: Certain uncertainty: The role of internal climate variability in projections of regional climate change and risk management. *Earth's Future* 8. doi: 10.1029/2020EF001854.

Chapter 7: Pandemics

[1](#). Barabási (2016).

[2](#). Mandelbrot (2008).

[3](#). Herrmann and Schwartz (2020).

[4](#). MRC Centre for Global Infectious Disease Analysis, July 19, 2021: Effective climate sensitivity. Github. <https://github.com/mrc-ide/covid-sim>.

[5](#). Edeling et al. (2021).

[6](#). In writing this book, I was told that in some circles, the words “doubt” and “ignorance” tend to go hand in hand. I tend to think of ignorance as a blanket term for a complete lack of knowledge. However, that is not the way it is used in epistemological philosophy, and that the Rumsfeld quote is a way to distinguish “recognised ignorance” (known unknowns) from “total ignorance” (unknown unknowns).

[7](#). Edeling et al. (2021).

[8](#). Based on models from Imperial College, the London School of Hygiene and Tropical Medicine, and Warwick University.

[9](#). Cramer, E. Y., E. L. Ray, V. K. Lopez, J. Bracher, A. Brennen et al., 2021: Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the US. medRxiv.

www.medrxiv.org/content/10.1101/2021.02.03.21250974v1.

[10](#). Vallance, P., December 23, 2021: It's not true Covid modellers look only at worst outcomes. *The Times*

www.thetimes.co.uk/article/its-not-true-covid-modellers-look-only-at-worst-outcomes-5c9pcpdwr.

11. Maher, M. C., et al., January 11, 2022: Predicting the mutational drivers of future SARS-CoV-2 variants of concern. Science Translational Medicine doi: 10.1126/scitranslmed.abk3445.

Chapter 8: Financial Crashes

1. BBC News, January 6, 2017: Crash was economists' "Michael Fish" moment, says Andy Haldane. www.bbc.co.uk/news/uk-politics-38525924.

2. Kay and King (2020).

3. Bank of England, November 2021: Bank of England monetary policy report. www.bankofengland.co.uk/-/media/boe/files/monetary-policy-report/2021/november/monetary-policy-report-november-2021.pdf.

4. Gabaix et al. (2008).

5. Gleick (1987).

6. Hossenfelder (2018).

7. Bookstaber (2017).

8. Thurner et al. (2012).

9. Emergent properties from Sabuco's agent-based model resemble those of the so-called Goodwin class struggle model of endogenous economic fluctuations.

10. Poledna et al. (2020).

11. Gill et al. (2021).

12. Indeed, perhaps things are changing. In 2018 Andy Haldane published a paper (Haldane and Turrell [2018]) assessing the value of agent-based models. He and his coauthor concluded that agent-based models are complementary to existing approaches and are suited to answering macroeconomic questions where complexity,

heterogeneity, networks and heuristics play an important role.

Chapter 9: Deadly Conflict and the Digital Ensemble of Spaceship Earth

[1.](#) Department of Peace and Conflict Research, n.d.: ViEWS: A political violence early-warning system. Uppsala University. www.pcr.uu.se/research/views.

[2.](#) Ferguson, S., April 14, 2020: Apollo 13: The first digital twin. Siemens Blog <https://blogs.sw.siemens.com/simcenter/apollo-13-the-first-digital-twin>.

[3.](#) Dasgupta, S. P., 2021: Final report—The economics of biodiversity: The Dasgupta review. HM Treasury. www.gov.uk/government/publications/final-report-the-economics-of-biodiversity-the-dasgupta-review.

[4.](#) European Commission, last updated February 1, 2022: Destination earth. <https://digital-strategy.ec.europa.eu/en/policies/destination-earth>.

[5.](#) The UK decided not to contribute to Digital Europe, the European Commission's programme which funds DestinE, after leaving the EU.

Chapter 10: Decisions! Decisions!

[1.](#) The value averaged over all cost-loss ratios is equal to a common skill score used to evaluate probability forecasts: the so-called Brier skill score.

[2.](#) Cyclone Bhola in 1970.

[3.](#) His book (Webster [2020]) is a masterpiece.

[4.](#) President John. F. Kennedy created the United States Agency for International Development to lead the US government's international development and humanitarian efforts.

[5](#). Webster et al. (2010).

[6](#). Lovelock (2006).

[7](#). See Stern (2015).

[8](#). Viscusi (2018).

[9](#). This would actually translate to a VSL less than \$10 million in the US.

[10](#). Committee on Climate Change 6th Budget Report (2021).

[11](#). Although the price of solar and wind power has dropped and will likely continue to drop, these sources of energy are subject to the vagaries of weather. Of particular relevance in the UK are the persistent high-pressure systems which can form over northern Europe and last for weeks, or even months. At the time of writing, the wind turbines in the North Sea have, for many months, been producing significantly less electricity than expected due to a “wind drought” caused by such persistent atmospheric high pressure. This has in part contributed to a big increase in the price of gas. Although the current spell of weak winds is surely temporary, we cannot be certain that these wind droughts won’t get more frequent in the future due to climate change (as part of a phenomenon known as “global stilling”)—such uncertainty is due to the inadequacy of current-generation climate models on the regional scale, as discussed in Chapter 6. For this reason, an over-reliance on wind energy is fraught with risk. Such risk can be mitigated by developing plentiful energy storage capability, and one idea being considered is storing hydrogen in underground caverns. However, we will need plentiful hydrogen to decarbonise transport, home heating and industry (e.g., cement manufacture), and diverting hydrogen to storage may end up meaning we do not have enough capacity in total. For this reason, I am personally in favour of a much stronger role for modern nuclear energy in the renewable mix, even though at present nuclear power is more expensive than wind and solar. In fact, the

high temperatures produced by nuclear reactors can be used to coproduce hydrogen along with electricity. At the time of writing, China is about to switch on a new type of nuclear reactor based on the thorium cycle. Thorium-based reactors have few of the disadvantages of the traditional uranium (U_{235}) reactors. Putting all this together, the estimate that decarbonisation will cost only 1 percent of GDP may turn out to be an underestimate. However, this does not undermine the conclusion that, based on a cost-loss analysis based on VSL, decarbonisation is worth it.

[12](#). Palmer (2020a).

Part III: Understanding the Chaotic Universe and Our Place in It

[1](#). Penrose (1997).

[2](#). Turing (1947).

Chapter 11: Quantum Uncertainty: Reality Regained?

[1](#). Another possibility, discussed further ahead, is that the theory of general relativity needs a little modification.

[2](#). Wikipedia, accessed February 20, 2022: Cosmological constant.

https://en.wikipedia.org/wiki/Cosmological_constant.

[3](#). In practice it can be assumed that there is another term in Einstein's field equations for general relativity, known as the cosmological constant, that almost but not quite cancels this quantum acceleration. Although adding the cosmological constant might technically solve this seeming incompatibility of dark energy and quantum mechanics, it leaves a feeling that something isn't right.

[4](#). At the very end of his paper "Free Variables and Local Causality" (collected in Bell [1993]), John Bell writes, "Of

course it might be that these reasonable ideas about physical randomisers are just wrong—for the purpose at hand. A theory may appear in which such conspiracies inevitably occur, and these conspiracies may then seem more digestible than the nonlocalities of other theories. When that theory is announced, I will not refuse to listen, either on methodological or other grounds.” Sadly, Bell was not able to listen when my paper (2020b) was published; he died in 1990. How I would have liked to discuss this material with him.

5. Pearl (2009).

6. Hossenfelder and Palmer (2020).

7. Tsirelson wrote about Bell’s inequality in the “Counterfactual definiteness” section of a Knowino article titled “Entanglement (Physics)”: [www.tau.ac.il/~tsirel/dump/Static/knowino.org/wiki/Entanglement_\(physics\).html#Counterfactual_definiteness](http://www.tau.ac.il/~tsirel/dump/Static/knowino.org/wiki/Entanglement_(physics).html#Counterfactual_definiteness).

8. Bohm and Hiley (1993).

9. Like the Lorenz attractor, I assume, the cosmological attractor lies in a bounded region of cosmological state space.

10. This is a mathematical way of saying that if a state lies on the attractor now, then the evolved state will always lie on the attractor in the future, and has always lain on the attractor in the past. Conversely, if a putative state does not lie on the attractor now, then it will never lie on it in the future, and has never lain on it in the past. In this sense the points of the attractor are invariant under the dynamical laws which evolve states from one time to another.

11. It is for this reason my model is not “fine-tuned”, an objection sometimes raised about this model.

12. Palmer (1995, 2009, 2020b, 2021, 2022).

13. Curiously, it becomes a double helix if one includes antimatter effects.

14. I will describe some of the key ideas in this footnote. Whether we are discussing Bell’s experiment or a

sequential SG experiment with three SG devices, we are dealing with look-up tables with three directions which I will write as “1”, “2” and “3”. These three directions can be represented by three points on the celestial sphere (the sphere on which the ancients used to think the stars all lay). Consider the spherical triangle whose vertices are these three points. In invariant set theory, the angle between two arcs of such a triangle must be a rational angle of the form $(n/p) \times 360^\circ$. In addition, the cosine of the angular arc length of any of the sides of this triangle must be a rational number of the form m/p . Here, m , n and p are all whole numbers, and cosine is the elementary trigonometric function taught in high school. It turns out that, using the so-called cosine rule for spherical triangles, together with Niven’s theorem from number theory, it is impossible to satisfy these conditions for such triangles if p is a prime number greater than 11. The situation is similar to Penrose’s “impossible” triangle—the tribar. That is to say, it is possible to implement the constraints of invariant set theory for any two sides of the spherical triangle, but not the full three.

In the case of Bell’s experiment, this has the following implications. Suppose Alice measures her particle’s spin in the 1 direction and Bob measures his particle’s spin in the 2 direction. By requiring that the cosine of the angular distance between the 1 vertex and the 2 vertex of our spherical triangle be rational—no problem with that—Alice can deduce that had she measured her particle in Bob’s 2 direction, the spin would have been the opposite to what Bob observed. So far so good. In addition, Bob can assert counterfactually that although he actually chose the 2 direction, he might have chosen the 3 direction. This assertion requires the cosine of the angular distance between the 2 and 3 vertices to be rational. Again, no problem with that. But now Niven’s theorem kicks in to say that the cosine of the angular distance between the 1 and 3

vertices—of the third side of the triangle—cannot be rational. This means that Alice cannot deduce that had she measured her particle in Bob's counterfactual 3 direction, she would have got the opposite result to Bob's. Following the logic of this is a bit of a mental stretch, but it implies that it is impossible for Alice and Bob's particles to have complete look-up tables for all three measurement directions. And without complete look-up tables, it is impossible to assert that invariant set theory must satisfy Bell's inequality, even though invariant set theory is deterministic and locally causal.

In the case of the sequential SG experiment, the "impossible triangle" rule means that a counterfactual experiment in which we reverse the order of the last two SG devices (keeping the particle fixed) similarly cannot lie on the invariant set. The latter situation illustrates what in quantum mechanics is called the non-commutativity of spin. It means that we cannot treat the individual SG devices as single independent apparatuses.

See Palmer (2022) for more details.

15. And so, as Galileo said, the language of nature is written in the geometry of triangles.

16. Making use of number-theoretic properties of trigonometric functions, c.f., Niven's theorem.

17. In ancient Chinese philosophy, yin and yang is a concept that describes how obviously opposite or contrary forces may actually be complementary, interconnected and interdependent in the natural world, and how they may give rise to each other as they interrelate to one another (from Wikipedia, accessed April 3, 2022: Yin and yang. https://en.wikipedia.org/wiki/Yin_and_yang).

18. At least until the trajectories decohere by the process illustrated in Fig. 44c.

19. Hance et al. (2021).

20. A crucial question is, What actually is the numerical value of p ? There are good reasons to suppose p will

depend on the energy, E , of the quantum bits (qubits) being described. For example, if each qubit is associated with some energy, E , then, using the Planck-Einstein relationship $E = \hbar \omega$, where $\hbar$ is Planck's constant, we can naturally associate a timescale $\hbar/E$ with that qubit. My proposal is that the dimensionless number p (for that qubit, and hence for the helix in that subset of state space) is the ratio $\hbar/E$ to the Planck time $\sqrt{\hbar G/c^5}$ where G is the gravitational constant and c is the speed of light. This has two consequences: 1) The larger is E the smaller is p . That is to say, the more energetic they are, the fewer the qubits can be entangled together before they become inherently decoherent. 2) The larger is G the smaller is p i.e., the stronger is gravity, the fewer the qubits that can be entangled together before they become inherently decoherent. This means that my proposal for decoherence (when the number of entangled qubits exceeds the logarithm to base 2 of p) will broadly coincide with the Penrose/Diósi criterion for gravitationally induced decoherence. This makes sense if gravity is a manifestation of the heterogeneity of the invariant set geometry. For example, if heterogeneous clustering is very weak, you would need a very large number of trajectory segments (large p) to diagnose the existence of discrete clusters. Conversely, if clustering is very strong, you would see it with relatively few trajectories (small p).

[21](#). Steinhardt and Turok (2007).

[22](#). Penrose (2010).

[23](#). Wood, C., December 13, 2021: Cosmologists parry attacks on the vaunted cosmological principle. Quanta Magazine www.quantamagazine.org/giant-arc-of-galaxies-puts-basic-cosmology-under-scrutiny-20211213.

[24](#). Colin et al. (2019).

[25](#). Corresponding to a “Janus point” in the language of Julian Barbour—see Barbour (2020).

- [26.](#) See, e.g., Penrose (2004).
[27.](#) Palmer (2017).

Chapter 12: Our Noisy Brains

[1.](#) What you do is write $2b^2 = a^2$ and deduce from this that a^2 is an even number and hence a is an even number—if a was odd, then its square would be odd. This means we can write $a = 2c$ and substituting into $2b^2 = a^2$ one can deduce that b must be an even number. But this contradicts the assumption that a and b have no common factors, and hence the original assumption is false.

[2.](#) Poincaré (1904).

[3.](#) Nobel Prize Outreach, accessed February 19, 2022: Roger Penrose—Interview. The Nobel Prize. www.nobelprize.org/prizes/physics/2020/penrose/interview.

[4.](#) Singh (1997).

[5.](#) Cleese (2021).

[6.](#) Kahneman (2011).

[7.](#) Claxton (1998).

[8.](#) Faisal et al. (2008); Rolls and Deco (2010); Palmer and O'Shea (2015).

[9.](#) Liewald et al. (2014).

[10.](#) And perhaps abductive reasoning too: Wikipedia, accessed February 20, 2022: Abductive reasoning. https://en.wikipedia.org/wiki/Abductive_reasoning.

[11.](#) Penrose (1989).

[12.](#) Turing (1950).

[13.](#) Shepherd et al. (2018).

Chapter 13: Free Will, Consciousness and God

[1.](#) Summhammer et al. (2018).

2. Wikipedia, accessed February 19, 2022: Quantum supremacy.

https://en.wikipedia.org/wiki/Quantum_supremacy.

3. Wikipedia, accessed February 19, 2022: Orchestrated objective reduction.

https://en.wikipedia.org/wiki/Orchestrated_objective_reduction.

4. See Kane (2002).

5. van Inwagen (2002).

6. Dennett (2002).

7. These would be p-adic neighbourhoods as shown in Fig. 44.

8. Pearl (2009).

9. Wikipedia, accessed February 19, 2022: Integrated information theory.

https://en.wikipedia.org/wiki/Integrated_information_theory

10. I was taught by Jesuits. I still recall being taught one of their proofs of God. “Hands up, those who have been to Australia,” said the priest. No one put their hand up. “Hands up, those who doubt the existence of Australia.” No one put their hand up. “There you are,” concluded the priest. “You do not physically need to experience something to believe in it.” If any Jesuits read my book, even though they won’t agree with my comments about God, they will probably recognise the type of logic I have employed in the book. In this respect, their teaching style has rather profoundly rubbed off on me.

11. As is the so-called Wheeler-DeWitt equation for quantum cosmology.